

Characterization of High Power Devices

**A Short Course Covering
Component-Level Modeling and
Measurement, Circuit
Design and Analysis and System Modeling**



Characterization of High Power Devices

Abstract - Device characterization is an essential process in many aspects of research, development and testing of RF and microwave devices. In this course, we will explore various interconnected topics of device characterization that form the amplifier design flow. Topics include pulsed IV and S-parameters for compact model extraction, load pull for model validation and measurement, amplifier design and IC stability analysis, X-parameter modeling and system-level simulations. Instructors from Maury Microwave, Agilent Technologies and AMCAD Engineering will provide instruction and demonstrations.



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Maury Microwave



AMCAD Engineering
Advanced Modeling for Computer-Aided Design

Characterization of High Power Devices

Load Pull

S-Parameters

IV Curves

Compact Models

Amplifier Design

Circuit Simulation

Harmonic Balance

Amplifier Stability

X-Parameters

What do they mean? Are they somehow related?



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AMCAD Engineering
Advanced Modeling for Computer-Aided Design

System Design from Compact Models

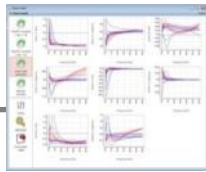
Component level

Circuit level

System level



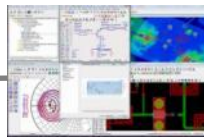
Pulsed IV and RF
measurements



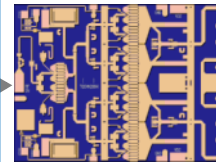
Compact FET
model extraction



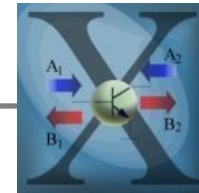
VNA based
load pull



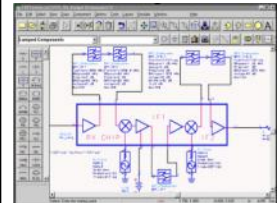
FET Compact
model Validation
& Refinement



IC Design &
Stability analysis



IC X-Parameter
model



Simulation at
system level

TOOLS

PIV
PNA-X
IVCAD

IVCAD
ICCAP

TUNERS
PNA-X
IVCAD

ADS

ADS
IVCAD
STAN

ADS
(PNA-X)

ADS

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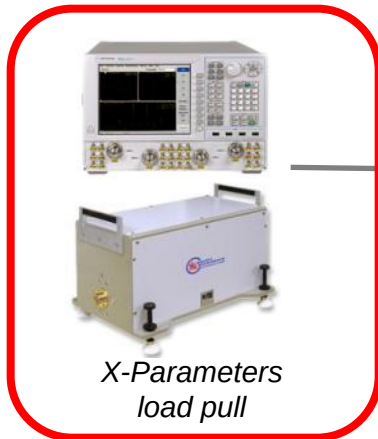
Maury Microwave



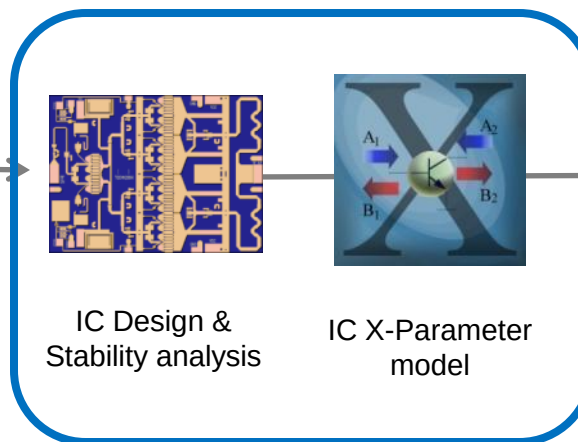
AMCAD Engineering
Advanced Modeling for Computer-Aided Design

System Design from X-Parameters

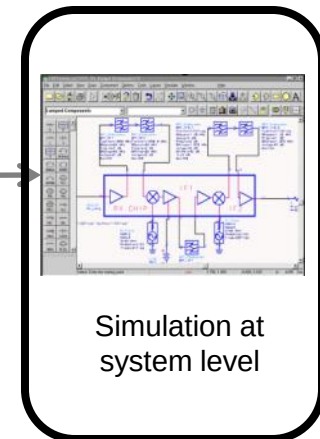
Component level



Circuit level



System level



TOOLS

TUNERS
PNA-X NVNA
IVCAD

ADS
IVCAD
STAN

ADS
(PNA-X)

ADS



Agilent Technologies



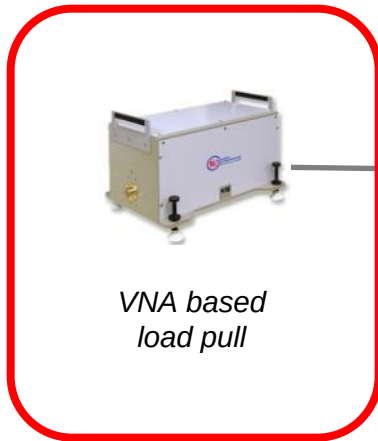
Maury Microwave



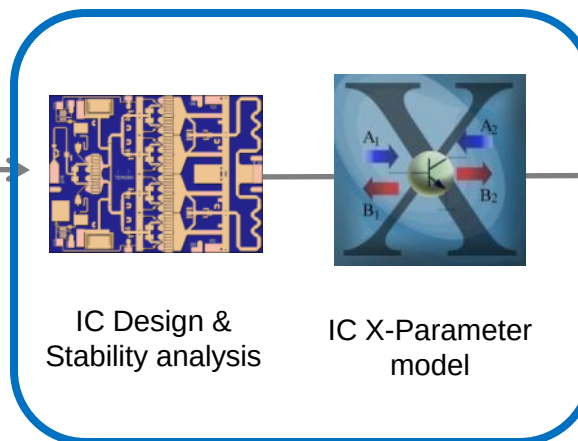
AMCAD Engineering
Advanced Modeling for Computer-Aided Design

System Design from Measurements

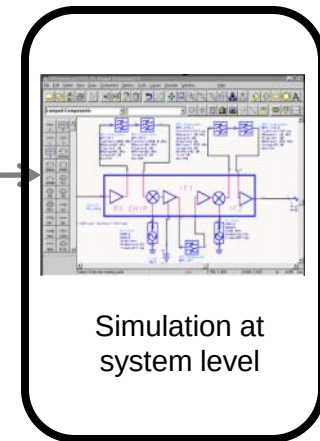
Component level



Circuit level



System level



TOOLS

TUNERS
PNA-X (or PSG/PSA)
IVCAD

ADS
IVCAD
STAN

ADS
(PNA-X)

ADS

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Maury Microwave



AMCAD Engineering
Advanced Modeling for Computer-Aided Design

Design flow entry points

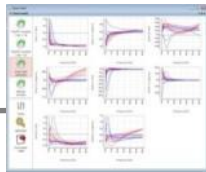
Component level

Circuit level

System level



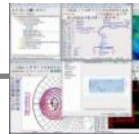
Pulsed IV and RF measurements



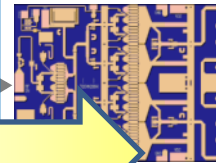
Compact FET model extraction



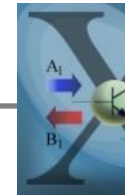
VNA based load pull



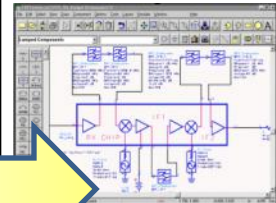
FET Compact model Validation & Refinement



IC Design & Stability analysis



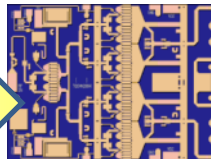
IC X-Parameter model



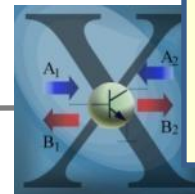
Simulation at system level



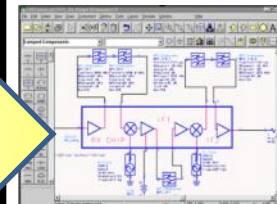
VNA based load pull



IC Design & Stability analysis



IC X-Parameter model



Simulation at system level

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- Instructor Introduction
- Pulsed IV/RF and Compact Modeling
- Load Pull
- Amplifier Design in ADS
- Stability Analysis
- X-Parameters
- System-level Simulations

Instructor – Gary Simpson



Gary Simpson received his Bachelor degree from DeVry Institute of Technology in 1972, and his Masters degree from Arizona State University in 1978. He has been involved with microwave measurements since 1973, starting with device characterization through manual load pull on microwave power transistors at his first job at Motorola. He has been with Maury Microwave since 1982, where he began developing components and fixtures for microwave measurements, including network analyzer calibration standards and techniques. Gary is a pioneer in device characterization systems; in 1987 he developed one of the earliest automated slide-screw tuners for advanced load pull measurements. Since then, he has been responsible for much of the on-going development of device characterization techniques, methodologies and systems. He is currently Chief Technical Officer at Maury Microwave Corp.

Instructor – Tony Gasseling



Tony Gasseling received his PhD from University of Limoges in 2003. The topic of his PhD was “A new characterization technique of "Four hot S parameters" for the study of nonlinear parametric behaviors of microwave components”. Throughout his education, Tony focused on advanced device characterization techniques with emphasis on transistor modeling. In 2004, with the support of the European Social Fund, he launched AMCAD Engineering, a spin-off of the XLIM Laboratory (Limoges-FRANCE). Today, AMCAD Engineering employs 10 PhDs to support a strong innovation in the field of testing solutions for RF and microwave circuits.

Instructor – Stéphane Dellier



Stéphane Dellier received his MSc degree and PhD degree in electrical engineering from XLIM laboratory, University of Limoges, France, respectively in 2001 and 2005. His PhD research is focused on microwave circuits design.

In 2004 he co-founded AMCAD Engineering, company providing new RF and microwave solutions to semiconductor professionals. He is currently project leader at AMCAD Engineering focused on the development of IVCAD software platform for characterization and modeling of RF devices.

Instructor – David Ballo



David Ballo is a Senior Application Engineer with 33 years of experience at Agilent Technologies' Component Test Division in Santa Rosa, California. After graduating from the University of Washington in Seattle, he spent ten years in R&D designing analog and RF circuits for signal analyzers. Since then, he has worked on developing and presenting seminars and papers, and writing application notes and technical articles on a wide variety of network- and spectrum-analyzer measurement topics.

Instructor – Al Lorona

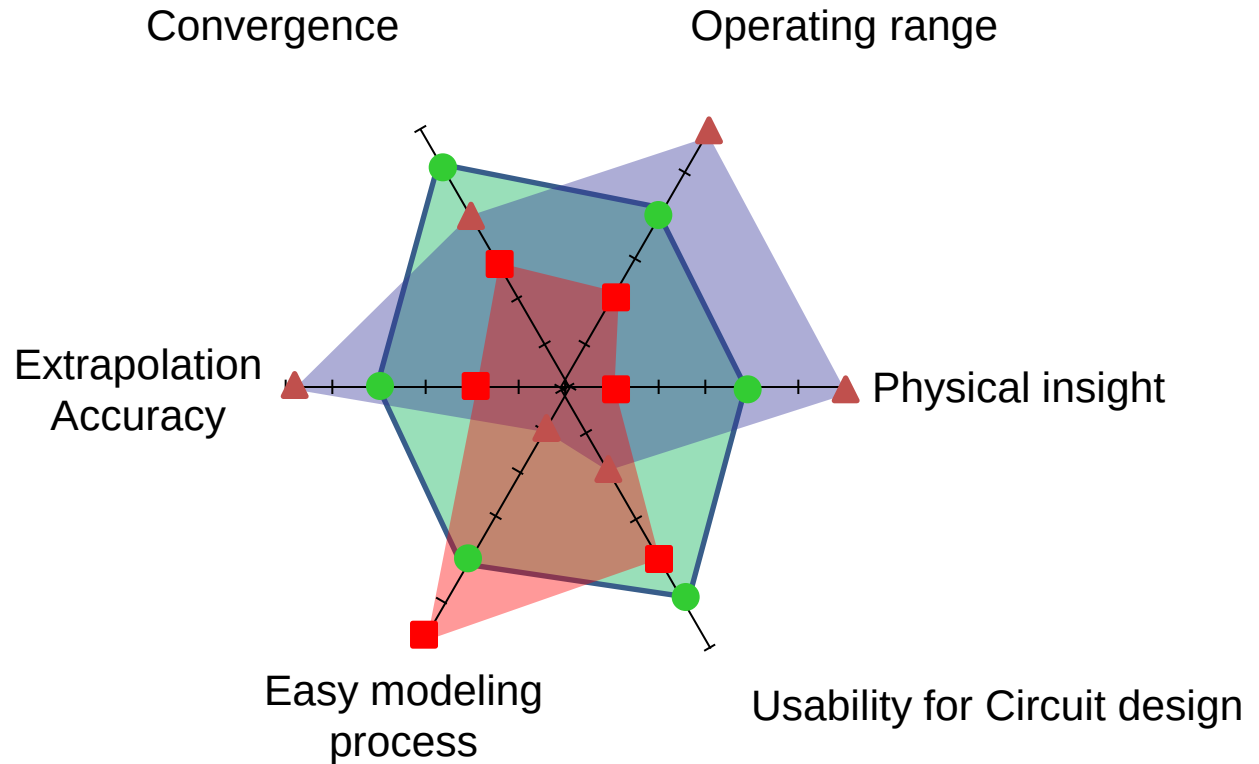
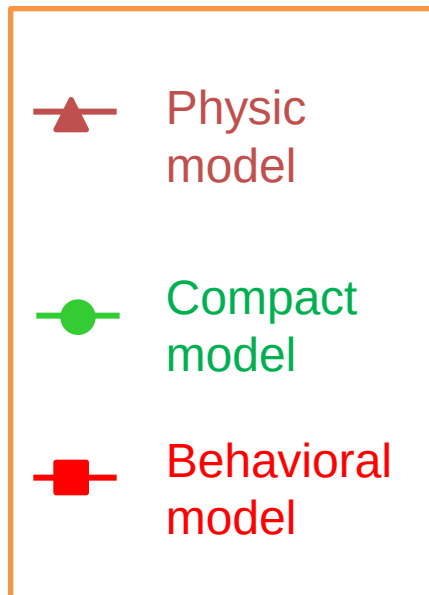


Al Lorona is an Application Engineer who helps customers use SystemVue and other Agilent Technologies EDA products more effectively and creatively. With 24 years of experience at Hewlett-Packard and Agilent he is a seasoned presenter, teacher and sales team member. Al is based in southern California.

- Instructor Introduction
- Pulsed IV/RF and Compact Modeling
- Load Pull
- Amplifier Design in ADS
- Stability Analysis
- X-Parameters
- System-level Simulations



Large-Signal Transistor Models



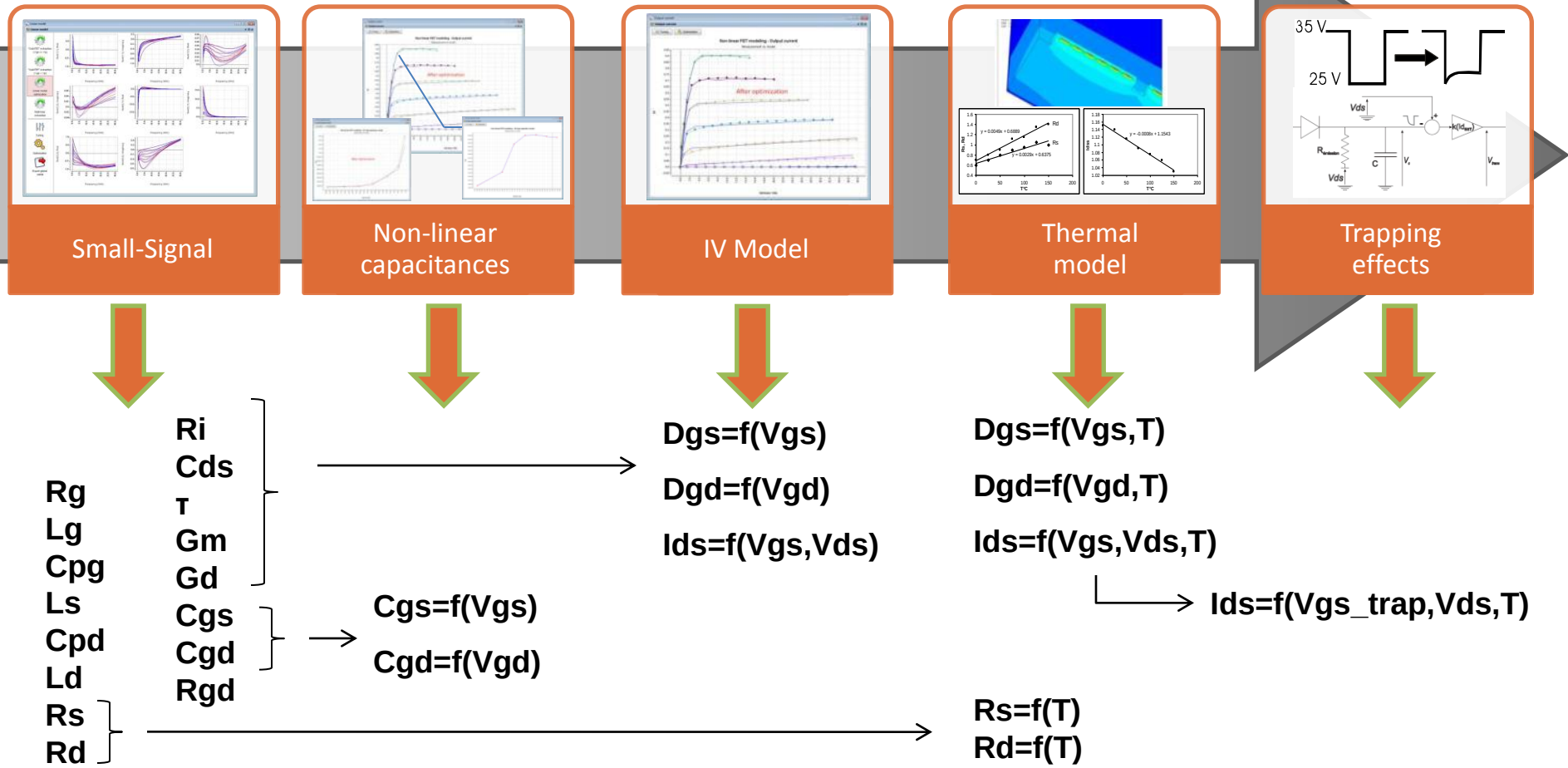
Commercial compact FET models

- Mostly used models for GaN HEMTs

FET models	Number of parameters	Electro-thermal effect	Trapping Effects	Original Device Context
Curtice3 [1]	59	No	No	GaAs FET
CFET [2]	53	Yes	No	HEMT
EEHEMT1 [3]	71	No	No	HEMT
Angelov [4]	80	Yes	No	HEMT/MESFET
AMCAD HEMT1 [5]	65	Yes	Yes	GaN HEMT

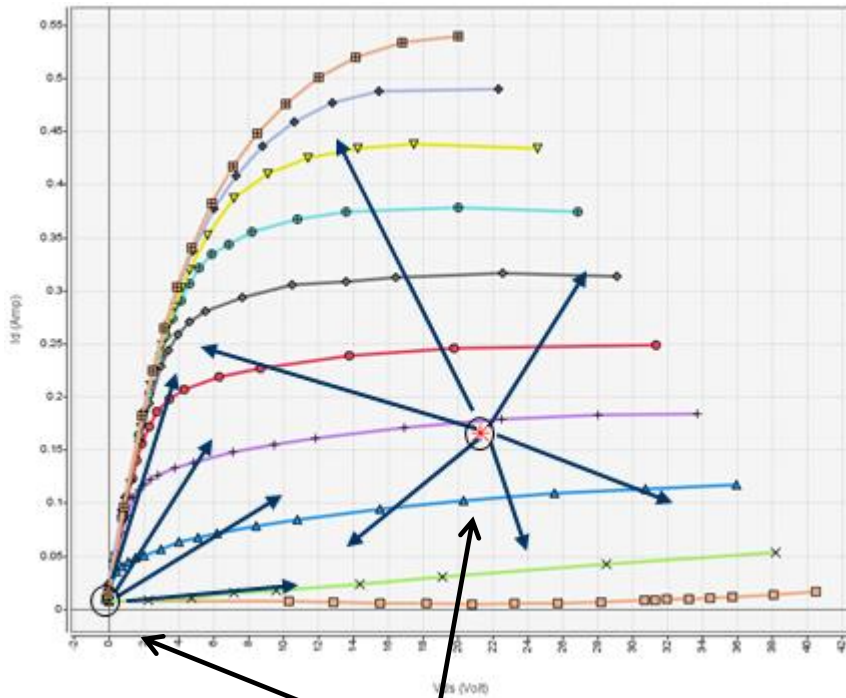


Compact FET model extraction flow



Various effects are successively added

Pulsed IV measurements

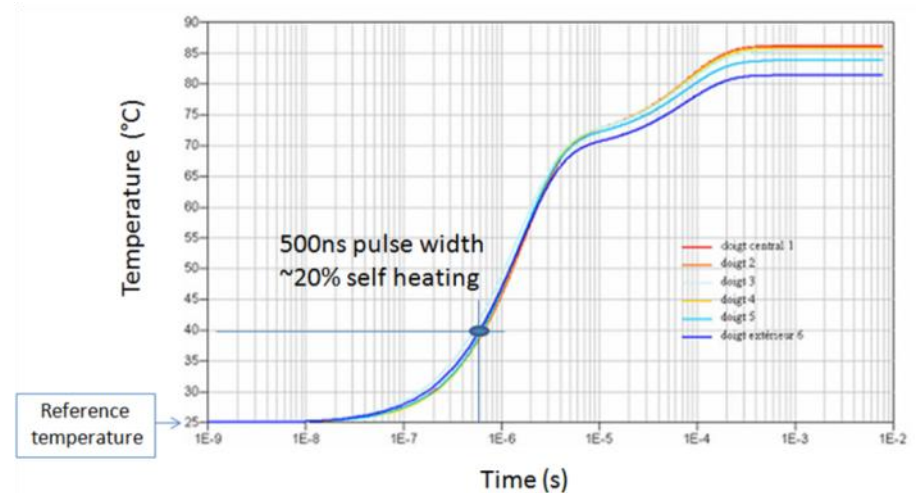


Several quiescent bias point

Short pulse : Quasi-isothermal conditions

Low duty cycle : Constant mean temperature

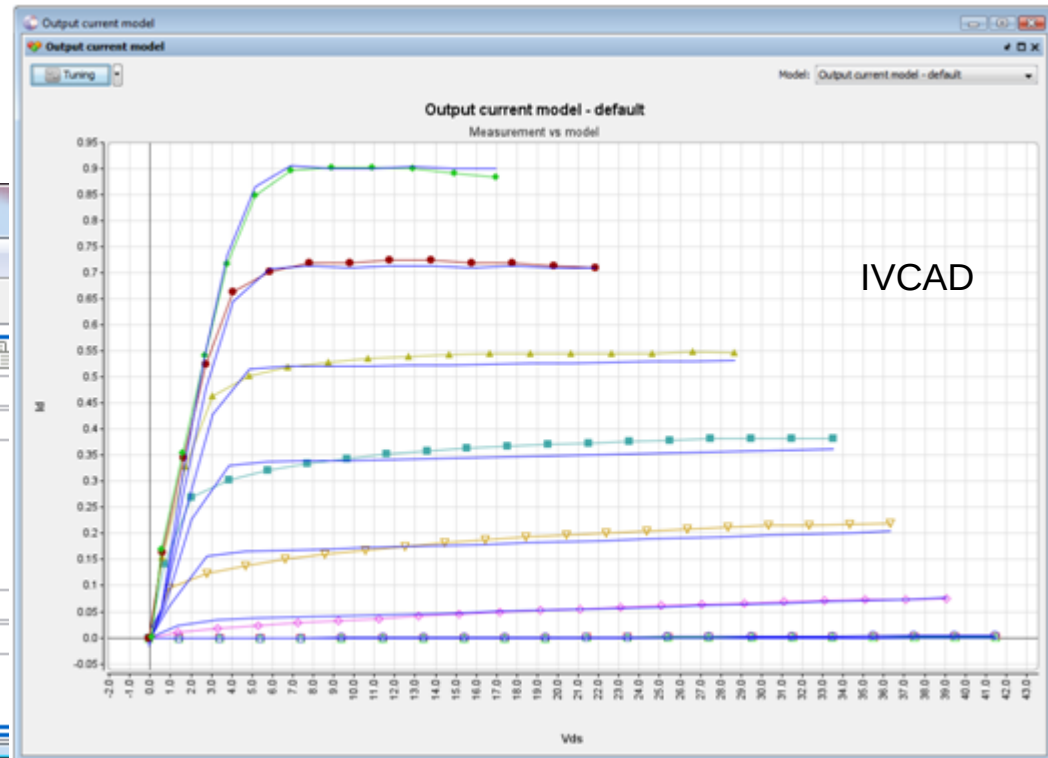
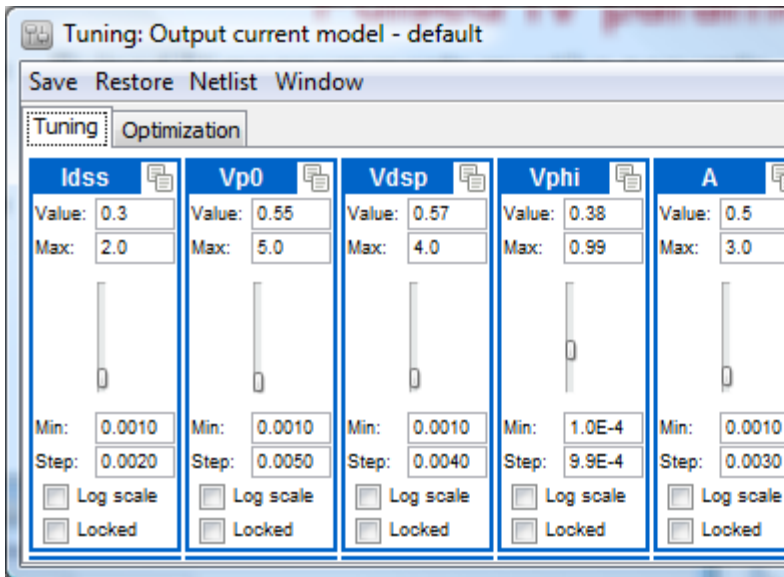
Quiescent bias point : Thermal conditions fixed



Pulsed Measurements = quasi isothermal operating conditions

Pulsed IV measurements

- Pulsed IV measurements must be accurate from low to high voltage/current values
- Accurate IV data =
 - Reliable current source
 - Transconductance
 - Leakage current
 - Ideality factor schottky diode

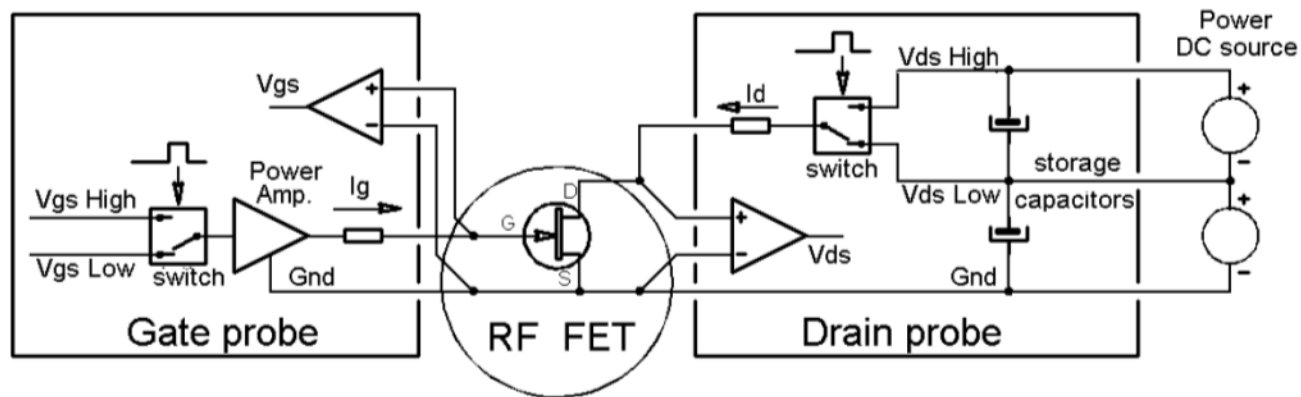
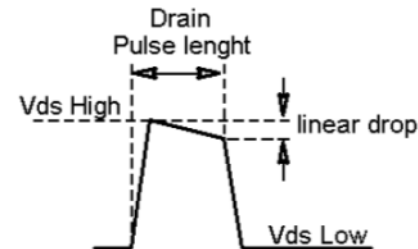
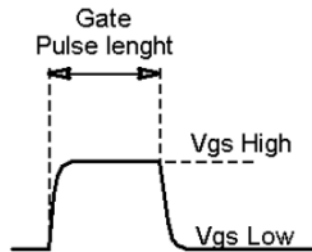




Pulsed IV measurements

How to get accurate pulsed IV measurements ?

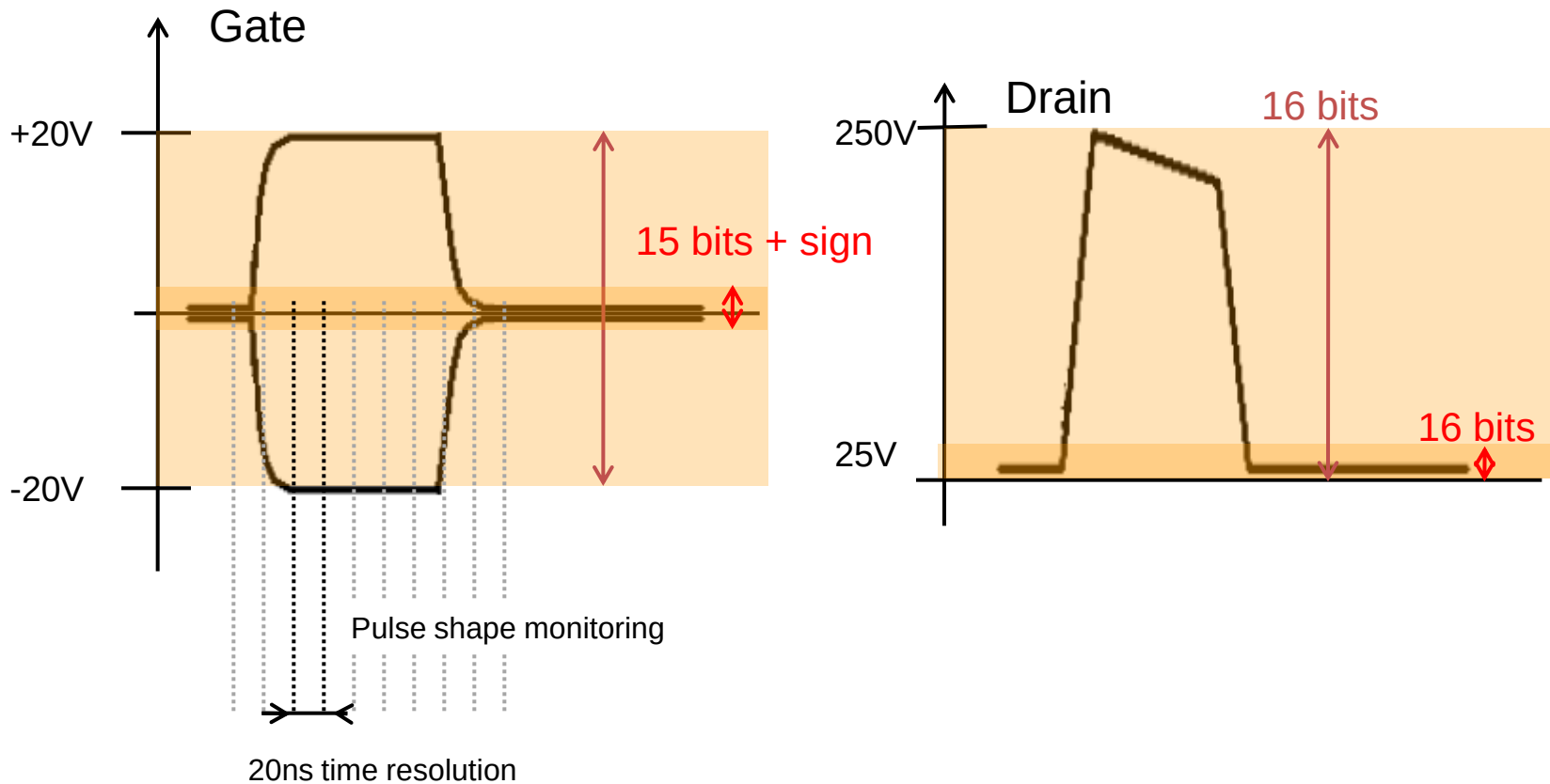
➤ PIV system





Pulsed IV measurements

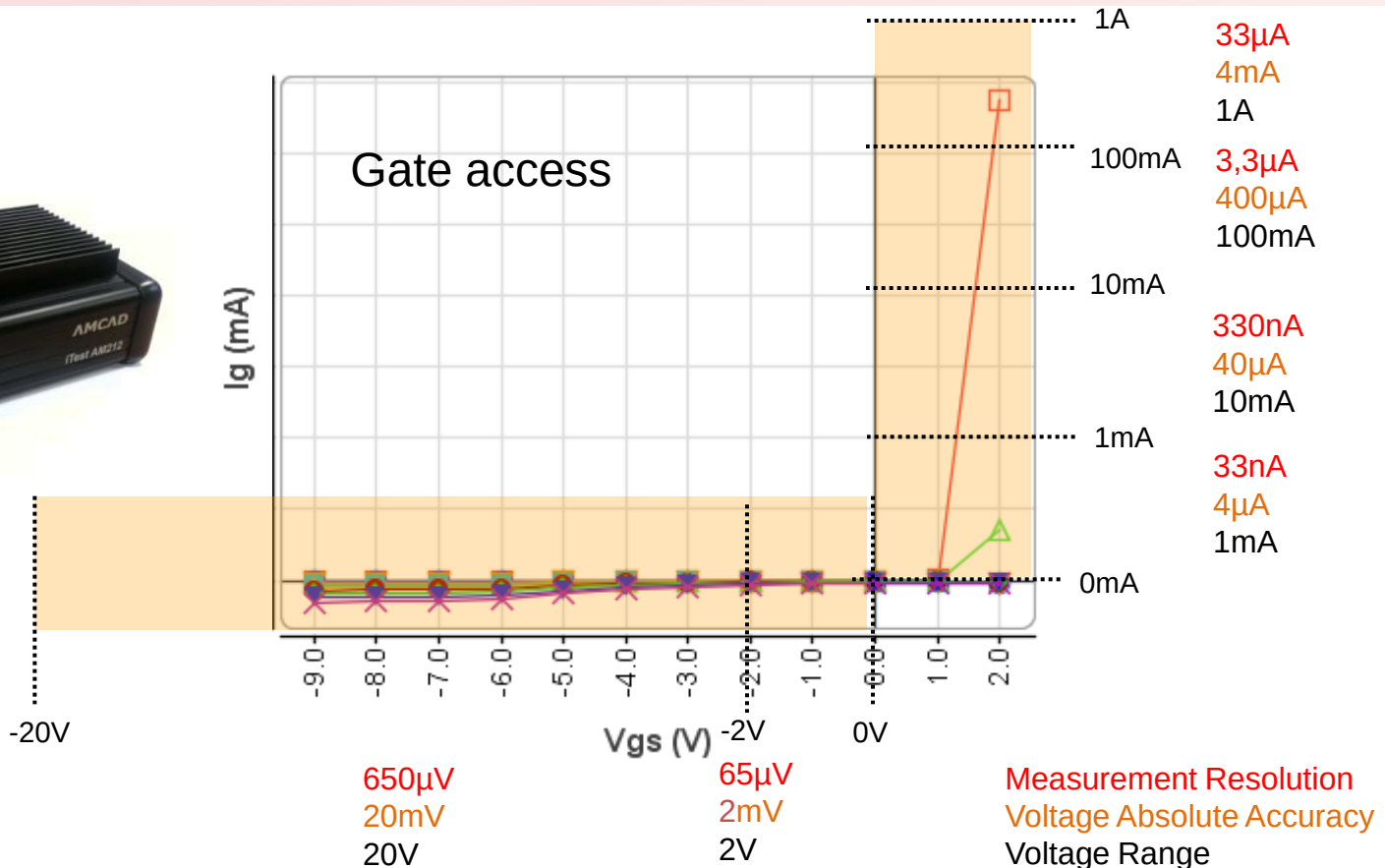
How to get accurate pulsed IV measurements ?



Pulsed IV measurements

How to get accurate pulsed IV measurements ?

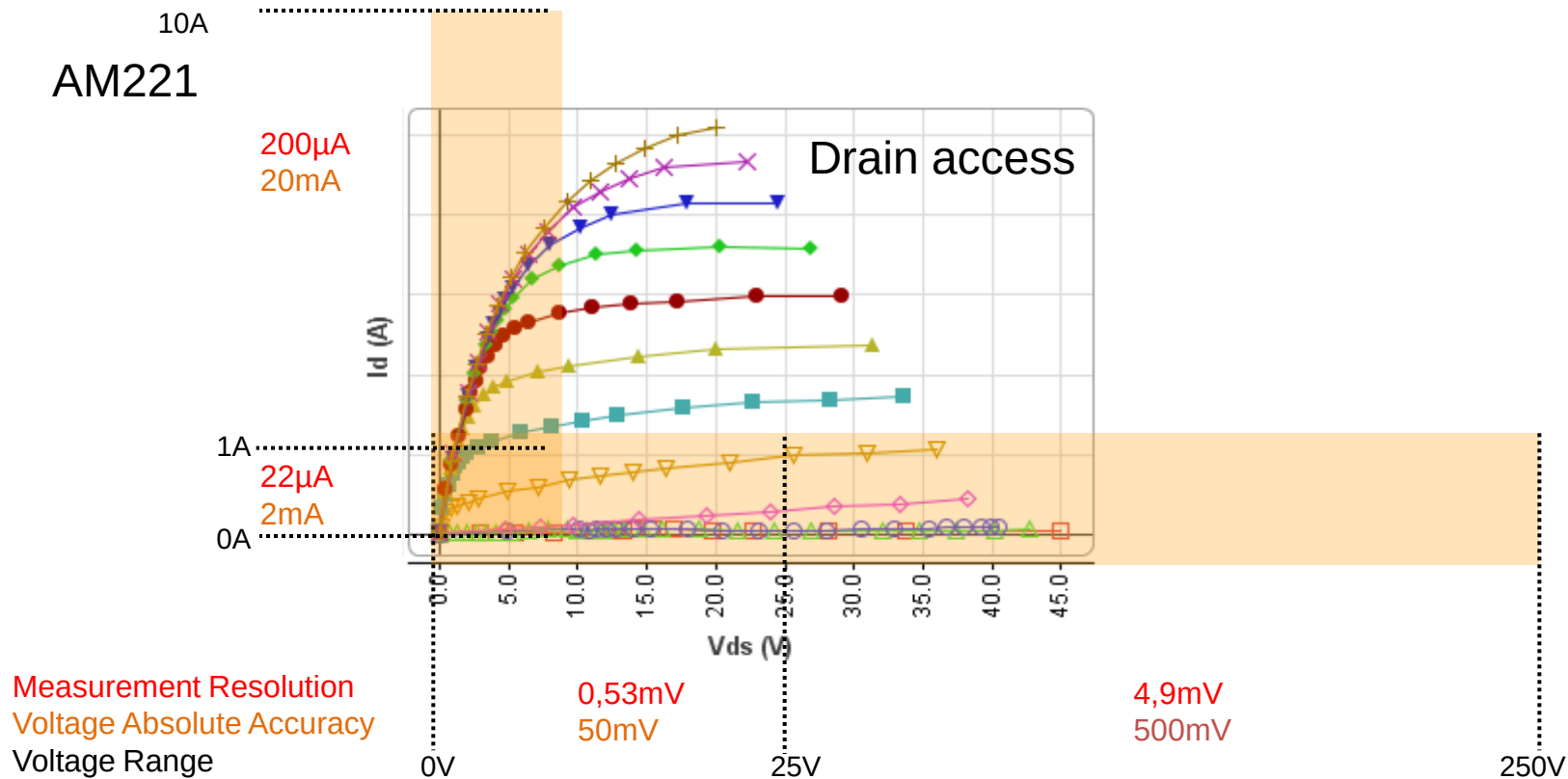
AM212





Pulsed IV measurements

How to get accurate pulsed IV measurements ?





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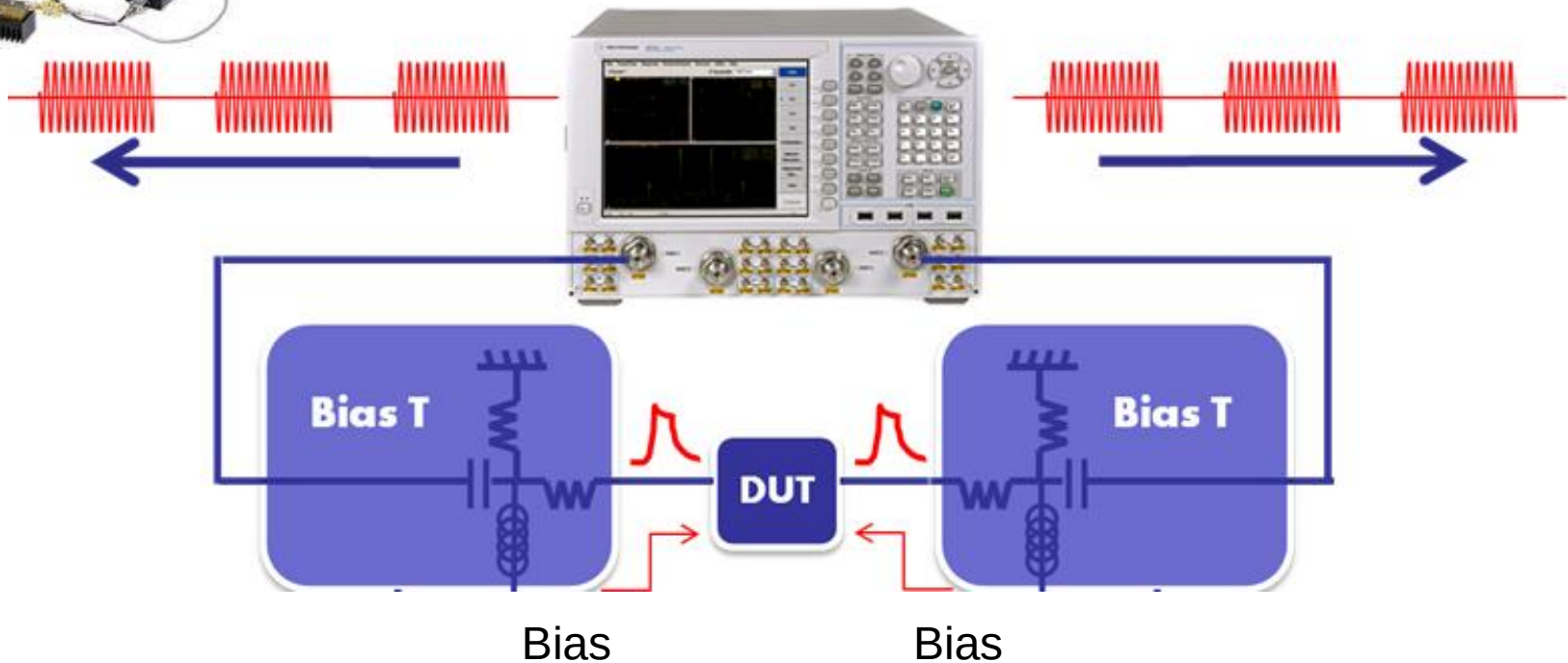


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Pulsed S-parameter measurements



Pulsed S parameter measurements



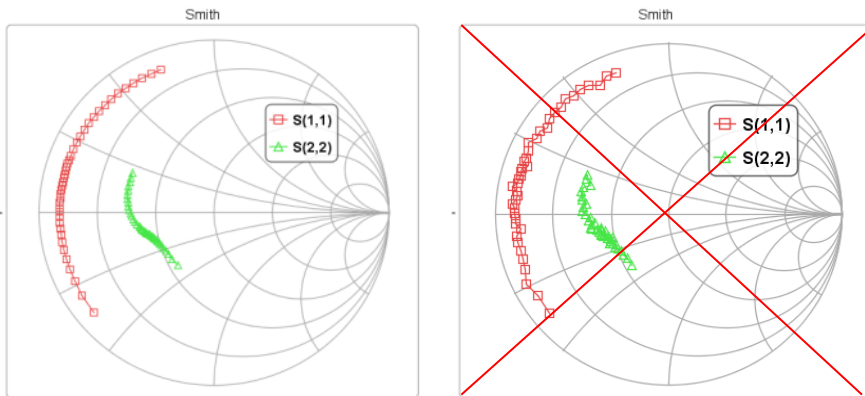
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Pulsed S-parameter measurements

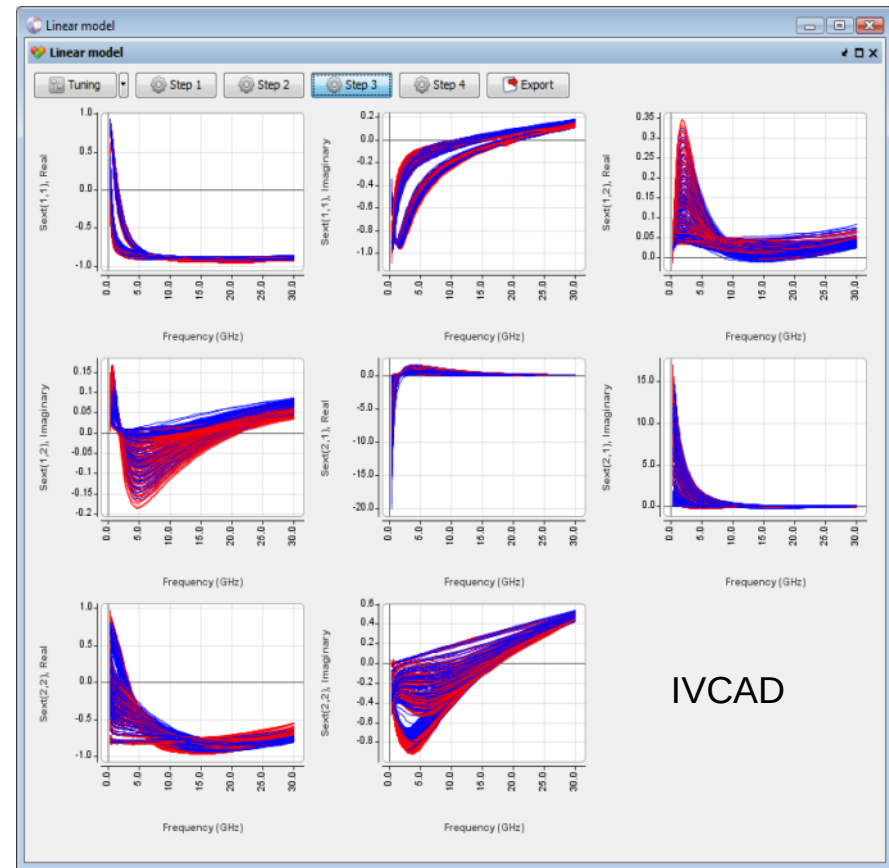
The first & most important point :

- Pulsed S parameter measurements must not be noisy
- Small S2P measurement variation = strong influence over the linear model extraction : optimization algorithm

Requirements :



Dynamic range in pulsed mode > 90dB for Duty Cycle ~ 5%



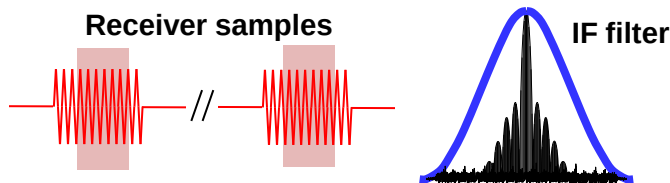


Pulsed S-parameter measurements

Pulsed S-parameter measurements must not be noisy at low duty cycle with narrow pulse width

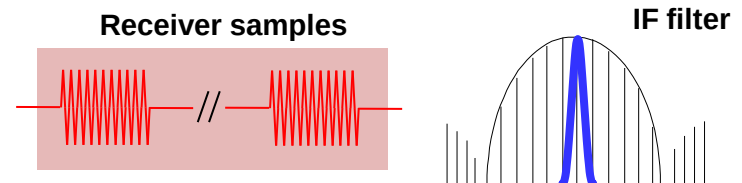
Pulse detection methods

Wideband detection



- No pulse desensitization
- Increased noise with narrow pulse width due to wider IF bandwidth
- Limited pulse width by maximum available IF bandwidth

Narrowband detection



- Narrower minimum pulse width than wideband pulse
- Reduced dynamic range with low duty cycle due to pulse desensitization by $20 \cdot \log(\text{duty cycle})$



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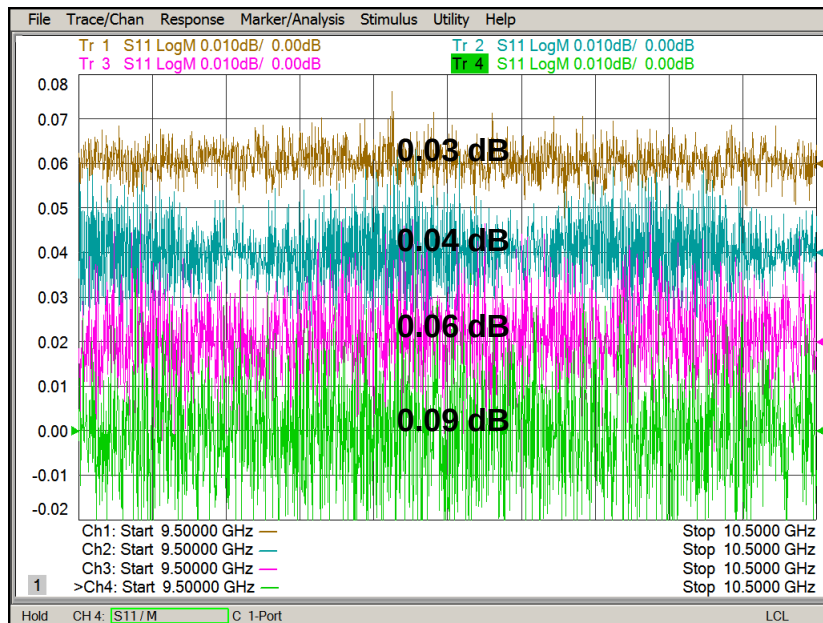


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PNA/PNA-X Noise reduction techniques and performances

Peak-to-peak noise with wideband detection at 10% duty cycle

No averaging in calibration and measurements



Averaging 20 times in calibration and measurements

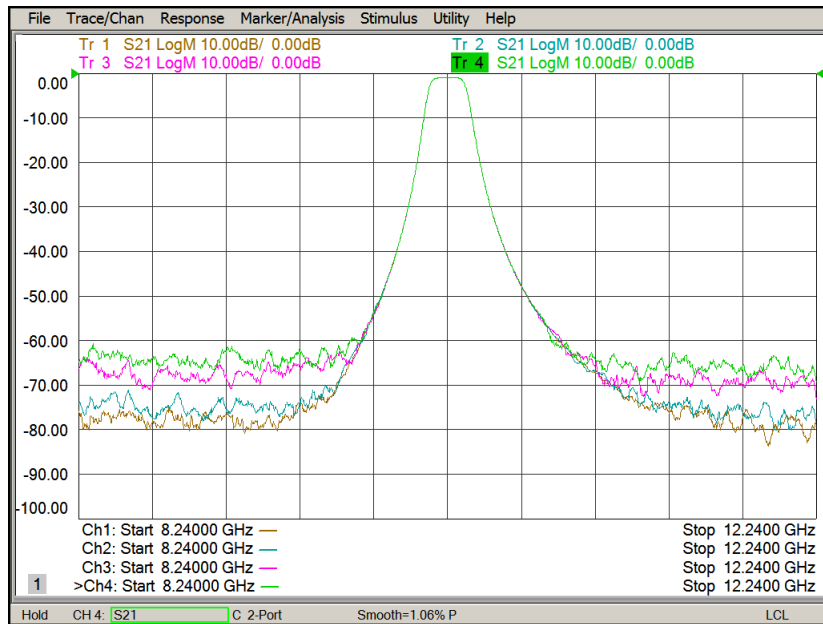


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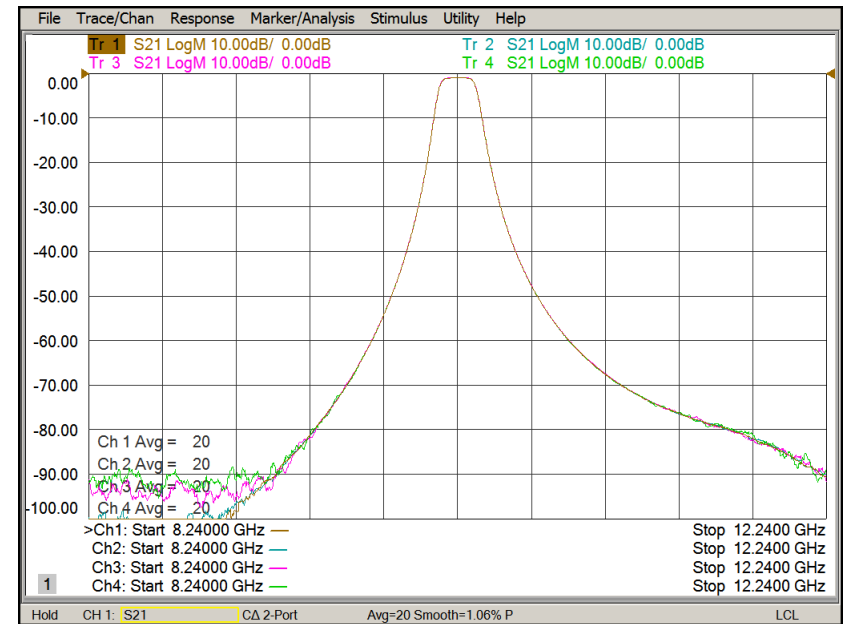
PNA/PNA-X Noise reduction techniques and performances

Dynamic range with wideband detection at 10% duty cycle with 10 us, 5 us, 1 us, 500 ns pulse width

No averaging in calibration and measurements, 1% smoothing on



Averaging 20 times in calibration and measurements, 1% smoothing on





PNA/PNA-X Noise reduction techniques and performances

Peak-to-peak noise at 10% duty cycle

Wideband detection with 20 times averaging in calibration and measurements



Pulse width

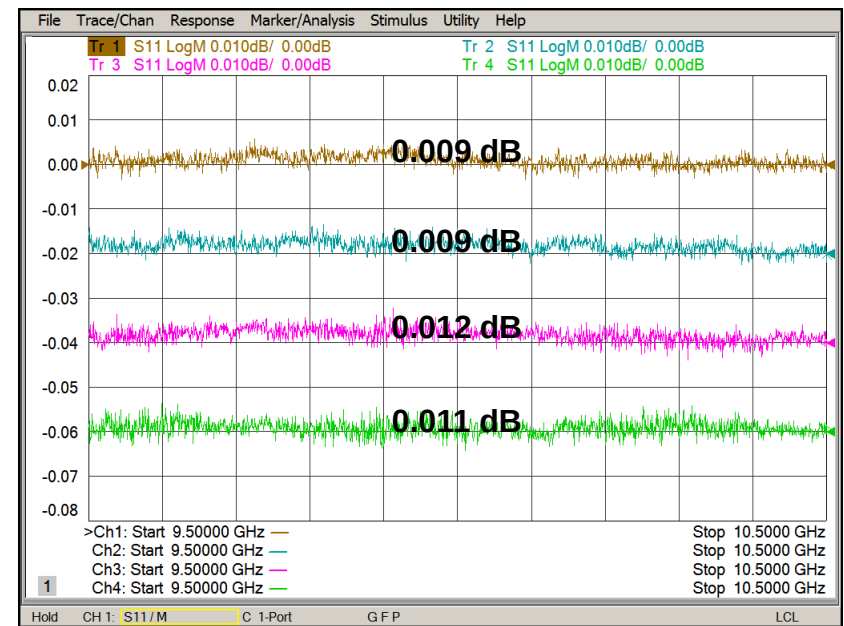
10 us

5 us

1 us

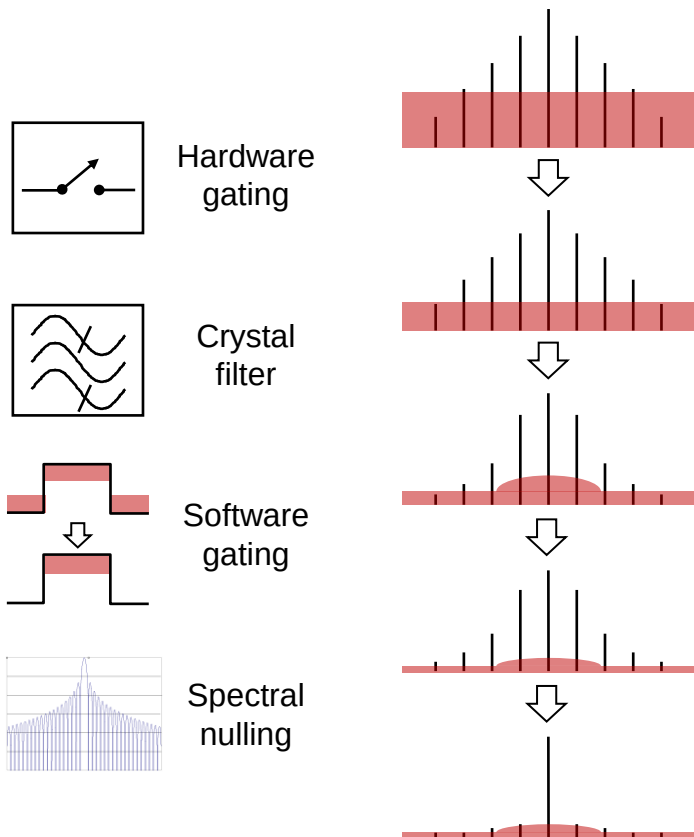
500 ns

Narrowband detection with no averaging in calibration and measurements

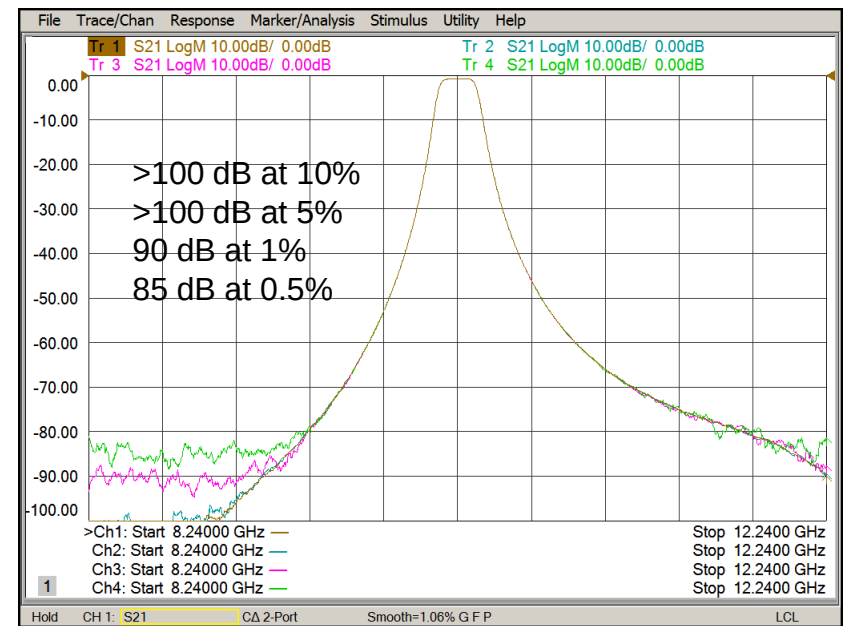


PNA/PNA-X Noise reduction techniques and performances

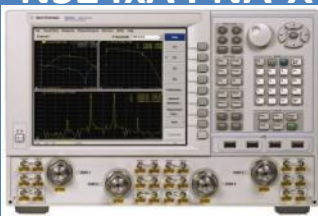
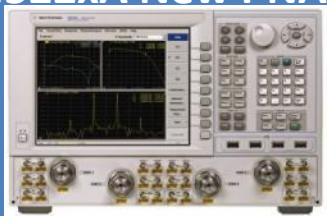
Dynamic range with narrowband detection at 500 ns pulse width



No Averaging, 1% smoothing on, 500 Hz IF bandwidth



VNA performance comparisons

	E836x Legacy PNA	N524xA PNA-X	N522xA New PNA
			
Pulse generator	External	Internal/External	Internal/External
Pulse modulator	External	Internal/External	Internal/External
Wideband detection			
Max BW/Min PW	35 kHz / 50 us	15 MHz / 100 ns	15 MHz / 100 ns
High level noise*	0.006 dBrms	0.002 dBrms	0.002 to 0.003 dBrms
Dynamic range**	114 to 123 dB	124 to 129 dB	127 dB
Narrowband detection			
Min IF gate width	20 ns	<20 ns	<20 ns
Dynamic range***	85 dB	<105 dB	<105 dB

* Specified as trace noise magnitude, at 20 GHz, at 1 kHz IF bandwidth

** Specified performance at 20 GHz, at 10 Hz IF bandwidth

*** Measured performance at 10 GHz at 10 Hz IF bandwidth, 1% duty cycle

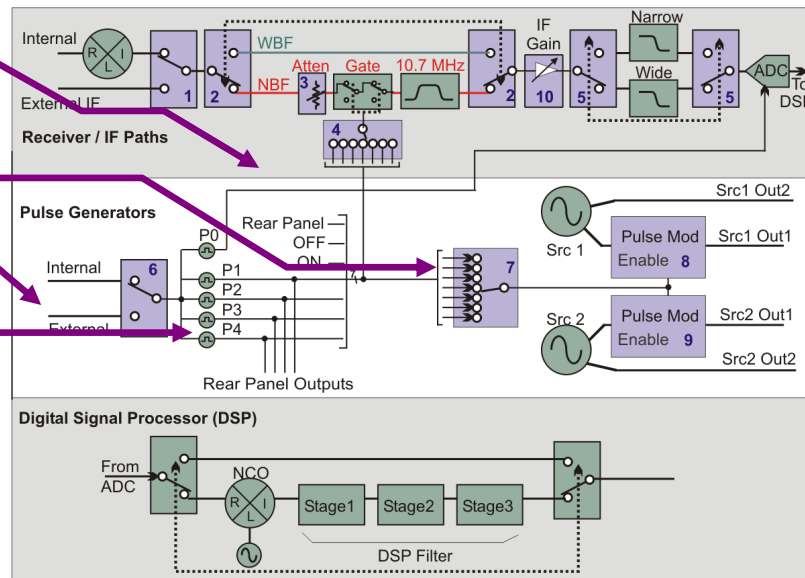
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PNA/PNA-X internal pulse access

- Internal or external master pulse (PULSE SYNC IN)
- Synchronized data acquisition (P0)
- Synchronized internal pulse generators (P1 – P4) with independent delay and width
- Internal or external drive for modulators or receiver gates

N1966A Pulse I/O adapter

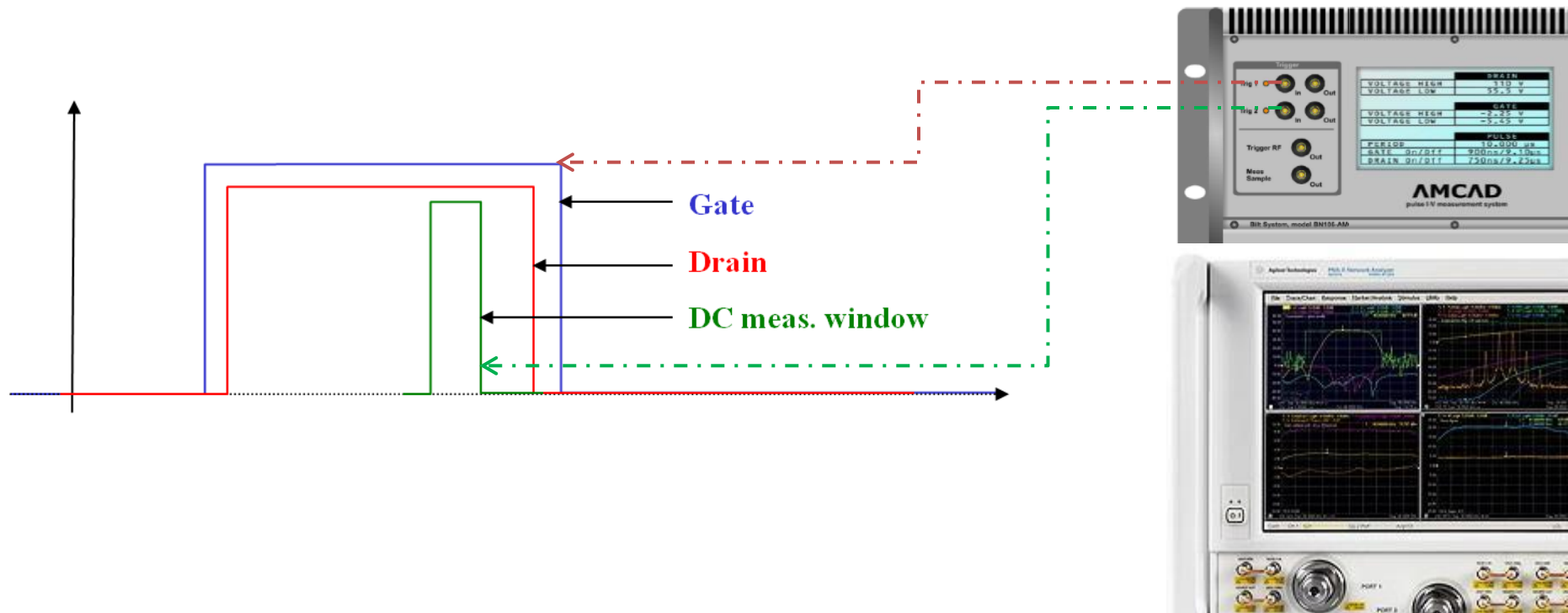


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Pulsed IV/RF parameter measurements

How to get accurate pulsed IV measurements ?

Synchronisation between Pulse IV and pulse S parameters

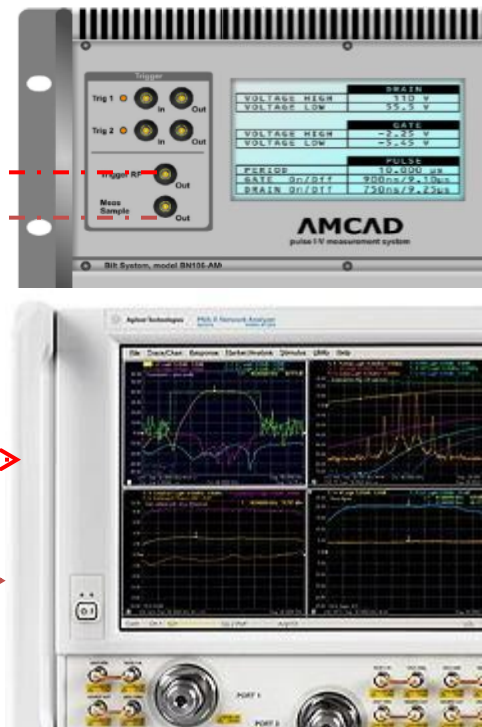
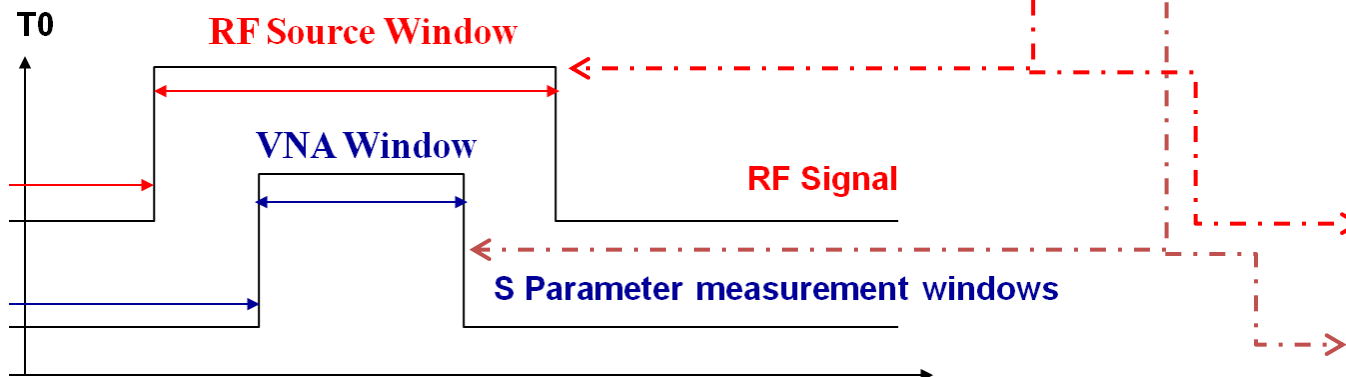




Pulsed S-parameter measurements

How to get accurate pulsed IV measurements ?

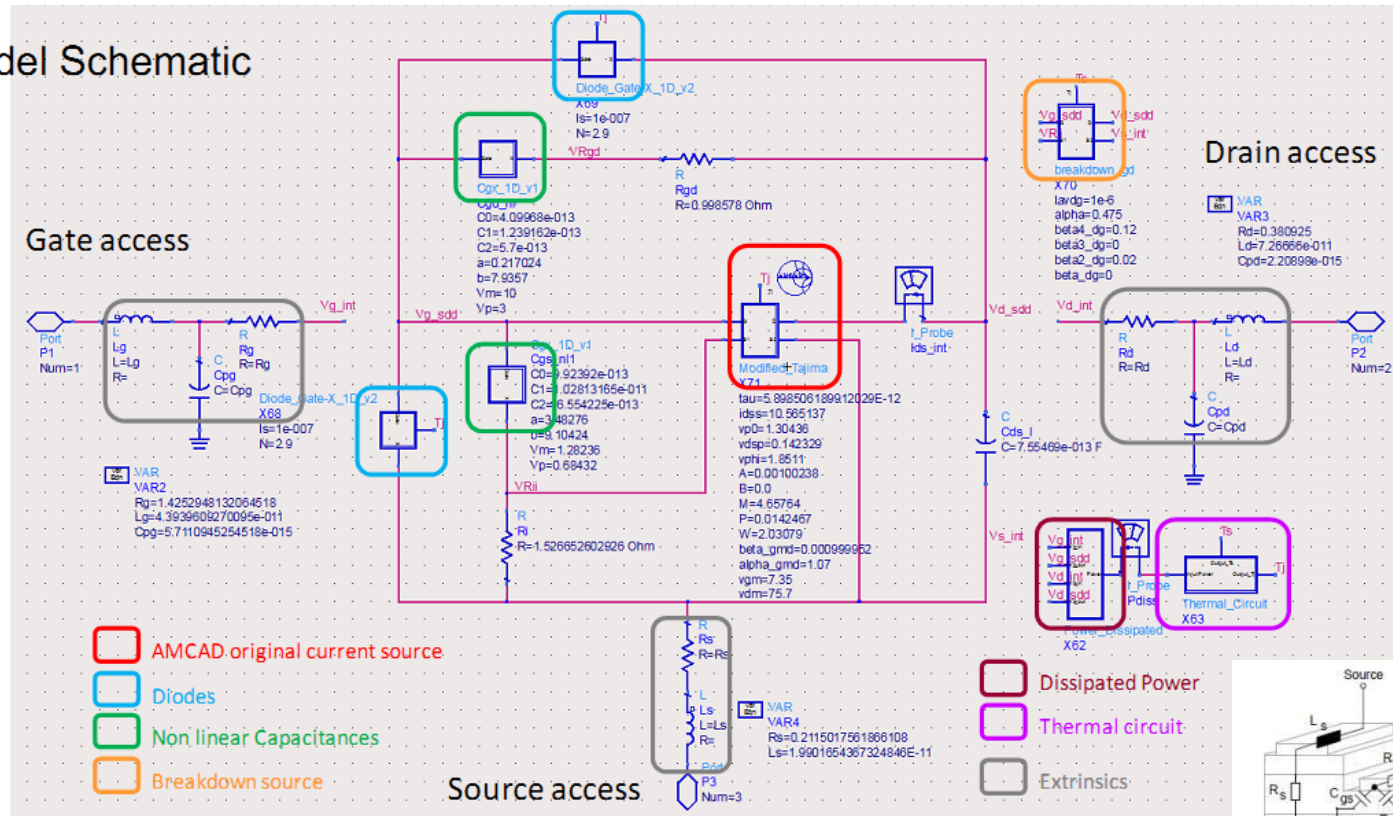
Synchronisation between Pulse IV and pulse S parameters



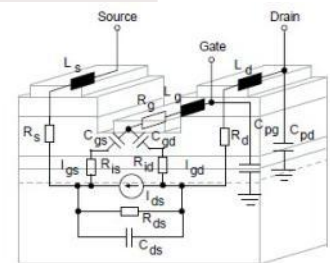


Parameter extraction methodology

Model Schematic

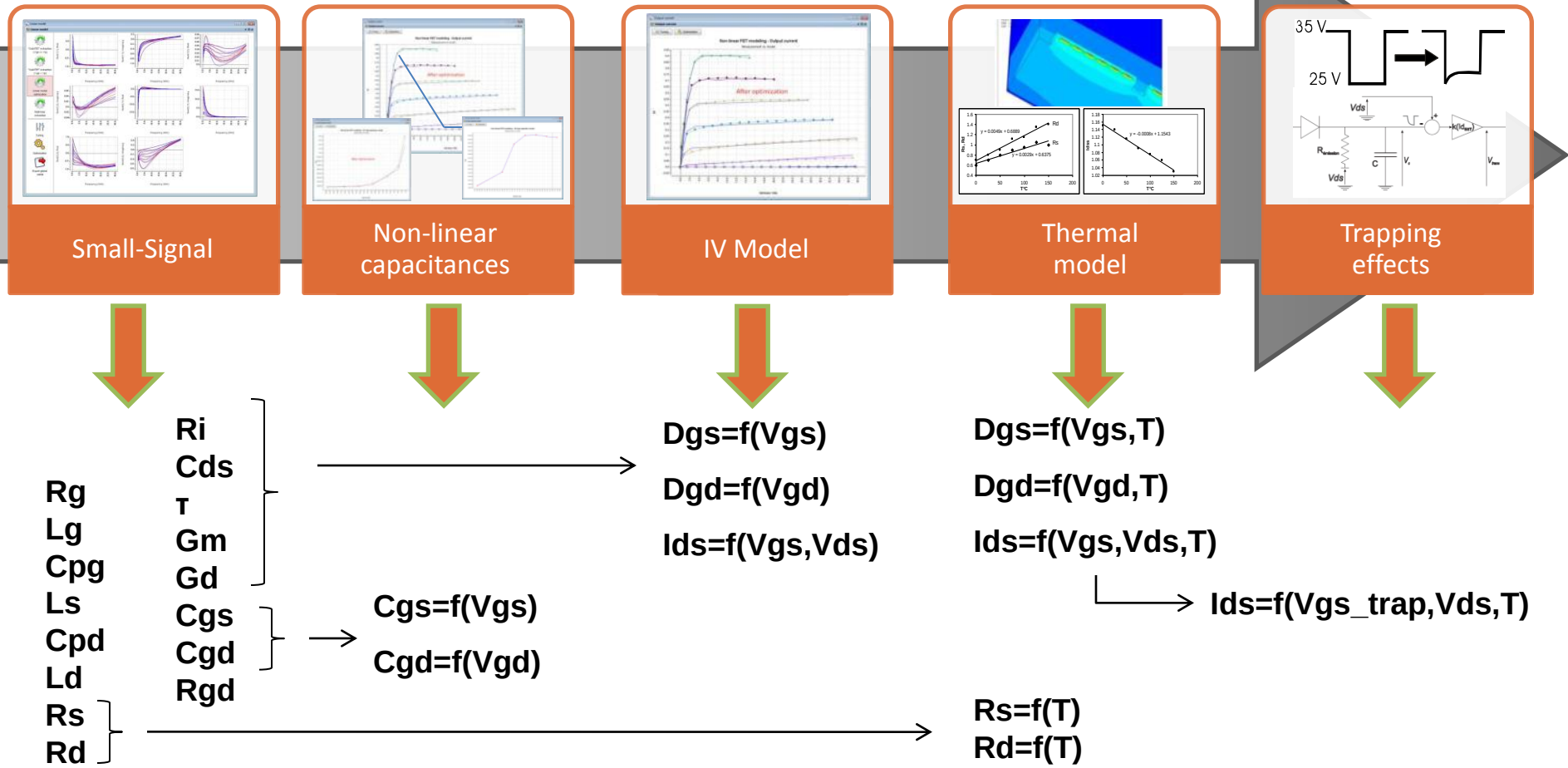


- Based on 18-element AMCAD original current source
- Compatible with Agilent ADS





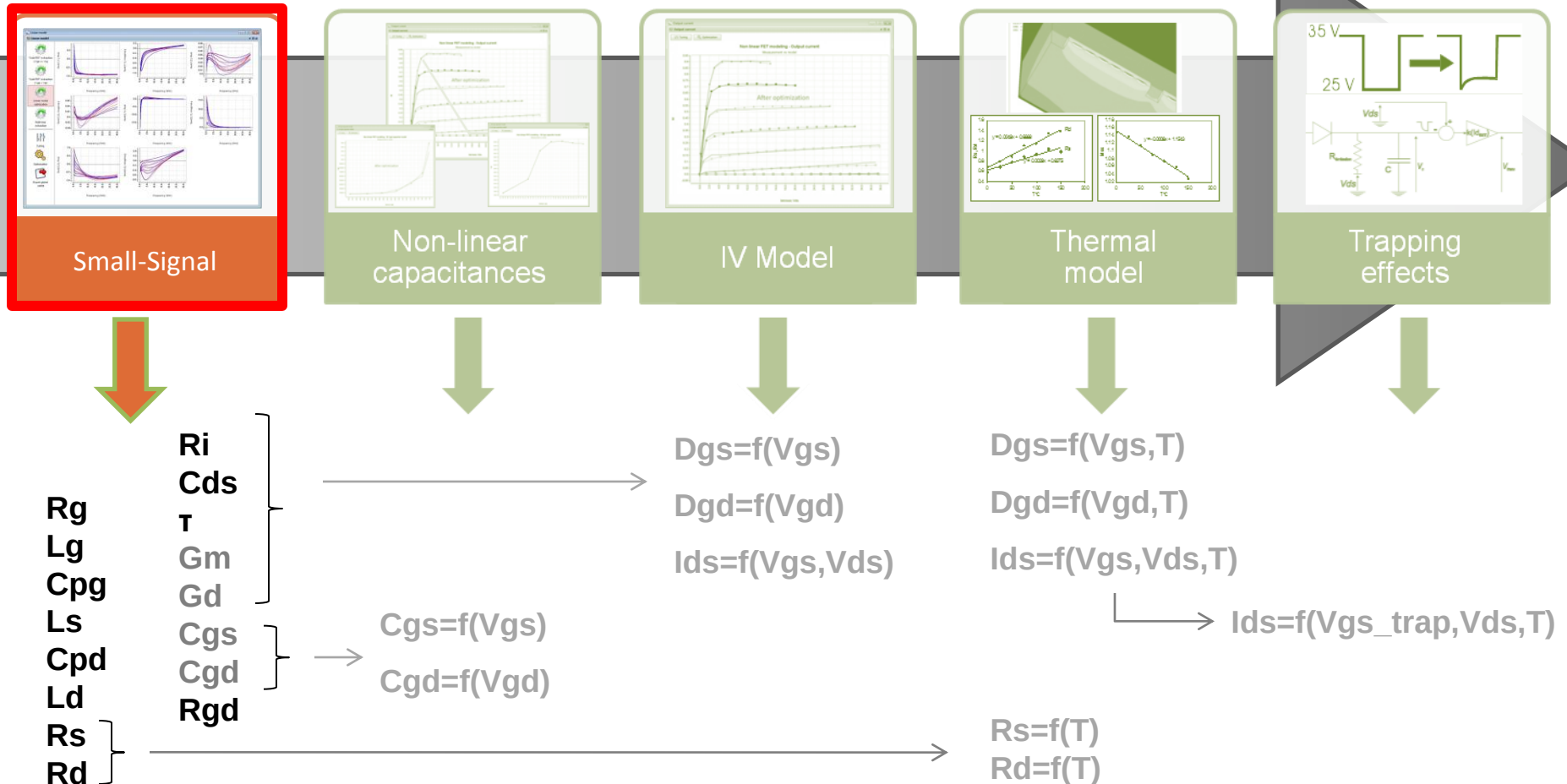
Compact FET model extraction flow



Various effects are successively added



Compact FET model extraction flow

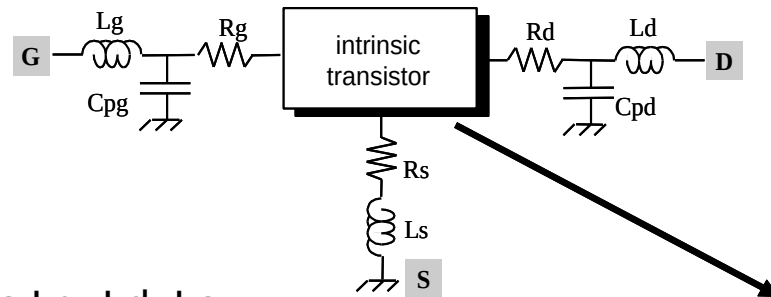


Various effects are successively added



Small signal FET modeling

- Extraction of extrinsic and intrinsic parameters:

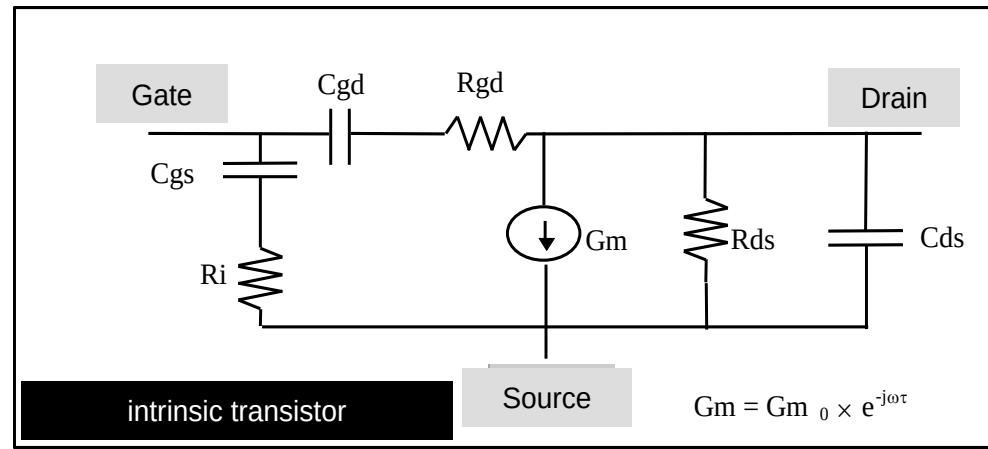


- Extrinsic parameters

- pad capacitances C_{pg} , C_{pd}
- port metallisation inductances L_g , L_d , L_s
- port ohmic resistances R_g , R_d , R_s

- Intrinsic parameters

- channel capacitances C_{gs} , C_{gd}
- voltage-controlled current source with transconductance g_m and transit time delay τ
- ohmic resistances R_i , R_{gd}
- output capacitance C_{ds} and resistance R_{ds}





Small signal FET modeling

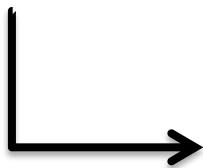
- Extraction of extrinsic and intrinsic parameters:
- Foundry parameters (configuration window)

R_{square} (Ω/sq) - square resistance

L_{DS} (m) - drain-source distance

Z_{total} (m) - total gate development

Foundry parameters (linear model)	
R_{square} :	450.0 Ω/sq
L_{DS} :	3.0E-6 m
Z_{total} :	0.0010 m



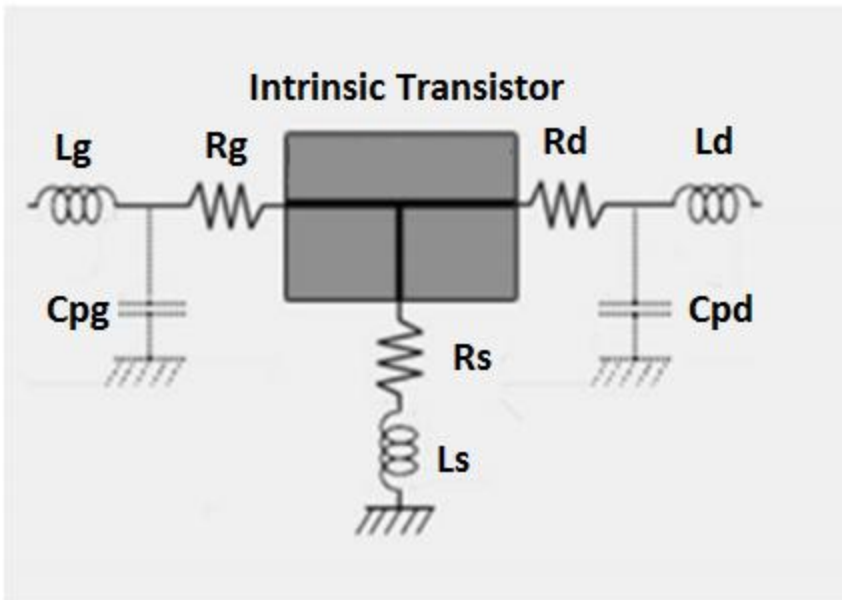
Calculus of Channel resistance

$$R_c = R_{\text{square}} * (L_{\text{DS}}/Z_{\text{total}})$$

Small signal FET modeling

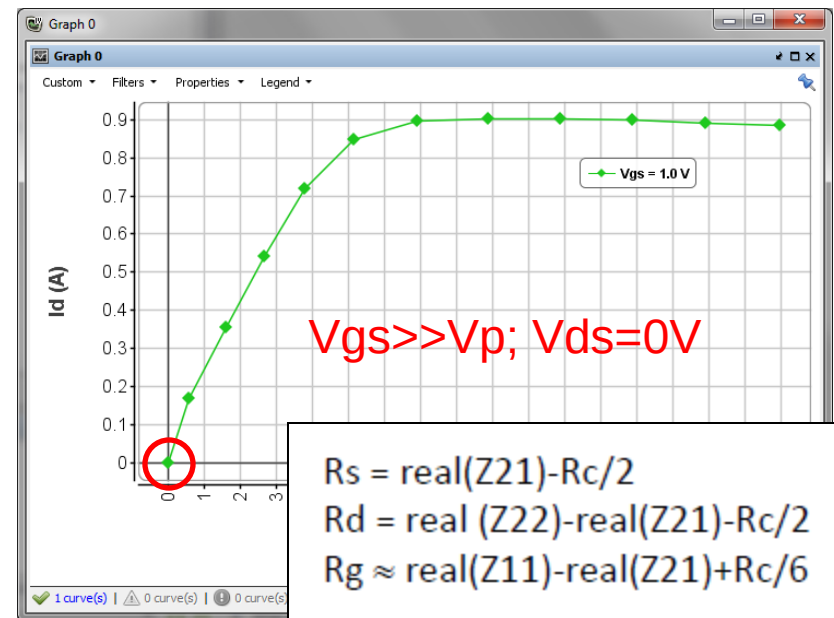
- Extraction of extrinsic and intrinsic parameters:

Step 1: Schottky junction conduction at $V_{ds}=0V$ (Optional)



Theory: Explicit computation of the extrinsic parasitic elements

R_g , R_d , R_s , L_g , L_d , L_s



$$R_s = \text{real}(Z_{21}) - R_c/2$$

$$R_d = \text{real}(Z_{22}) - \text{real}(Z_{21}) - R_c/2$$

$$R_g \approx \text{real}(Z_{11}) - \text{real}(Z_{21}) + R_c/6$$

$$L_s = \text{Im}(Z_{21})/W$$

$$L_d = (\text{Im}(Z_{22}) - \text{Im}(Z_{21}))/W$$

$$L_g = (\text{Im}(Z_{11}) - \text{Im}(Z_{21}))/W$$

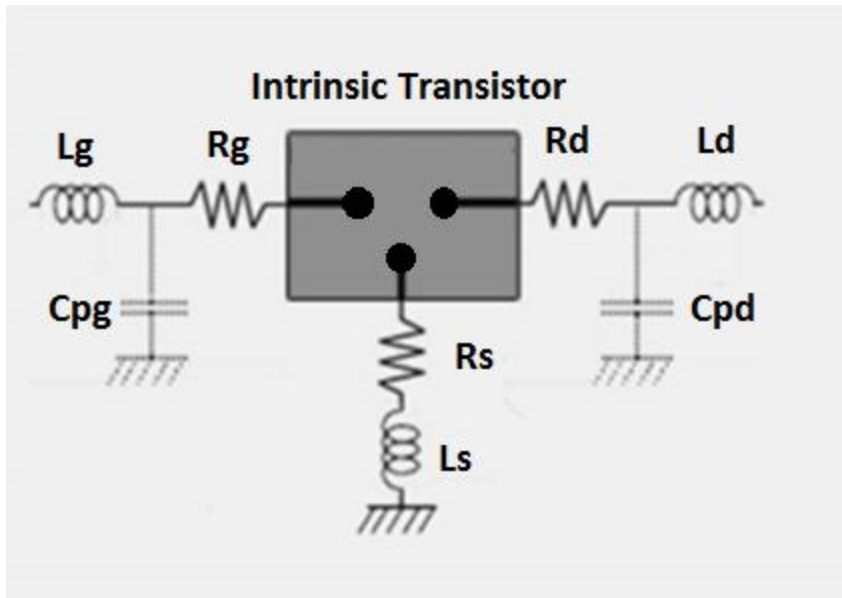


Small signal FET modeling

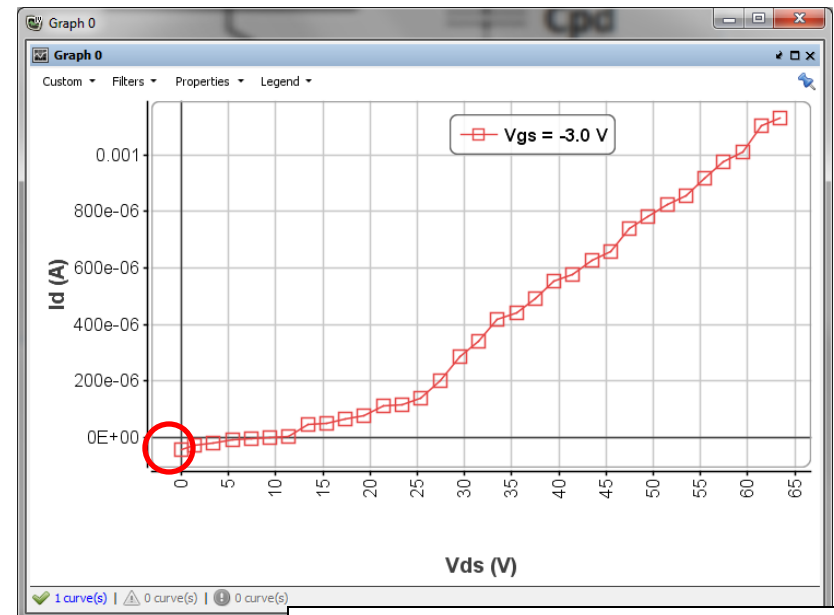
- Extraction of extrinsic and intrinsic parameters:

Step 2: Channel pinch-off conditions (Optional)

$V_{gs} \ll \text{Pinch-off}; V_{ds} = 0V$



Theory: Explicit computation of the extrinsic parasitic elements : **Cpd** and **Cpg**



$$C_{pd} = (\text{Im}(Y_{22}) + \text{Im}(Y_{21}))/W$$

$$C_{pg} = ((\text{Im}(Y_{11}) + 2 * \text{Im}(Y_{21}))/W$$



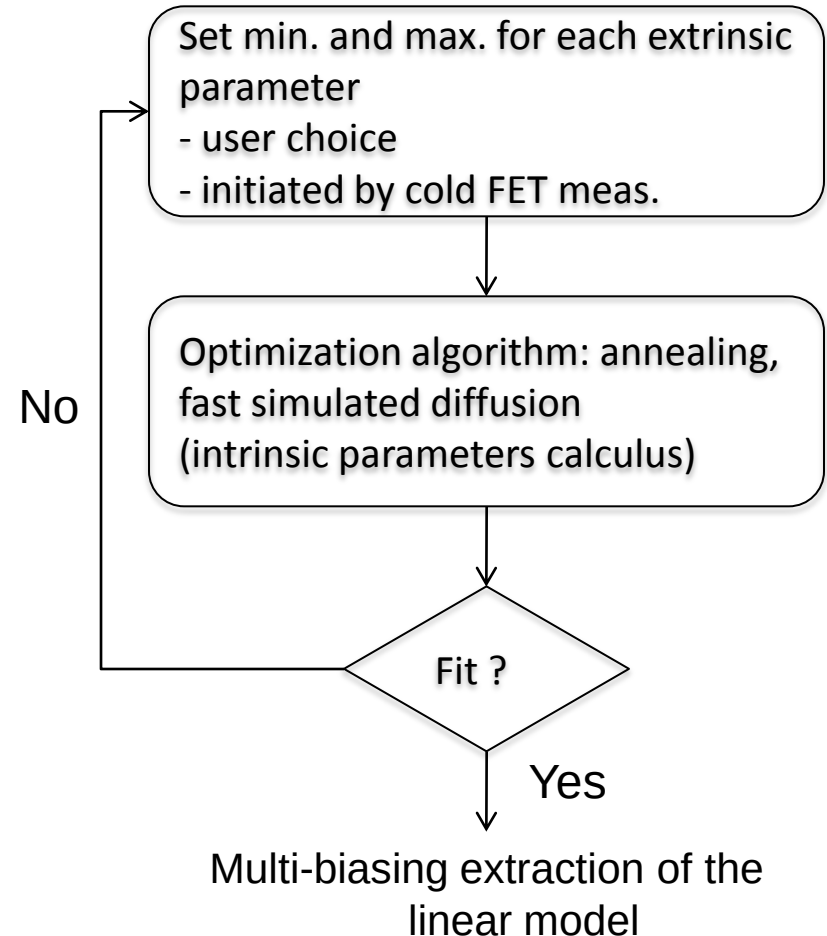
Small signal FET modeling

Step 3 – Linear model optimization

- Extraction of extrinsic and intrinsic parameters:

There is only one set of extrinsic parameters for which intrinsic parameters are independent from the frequency

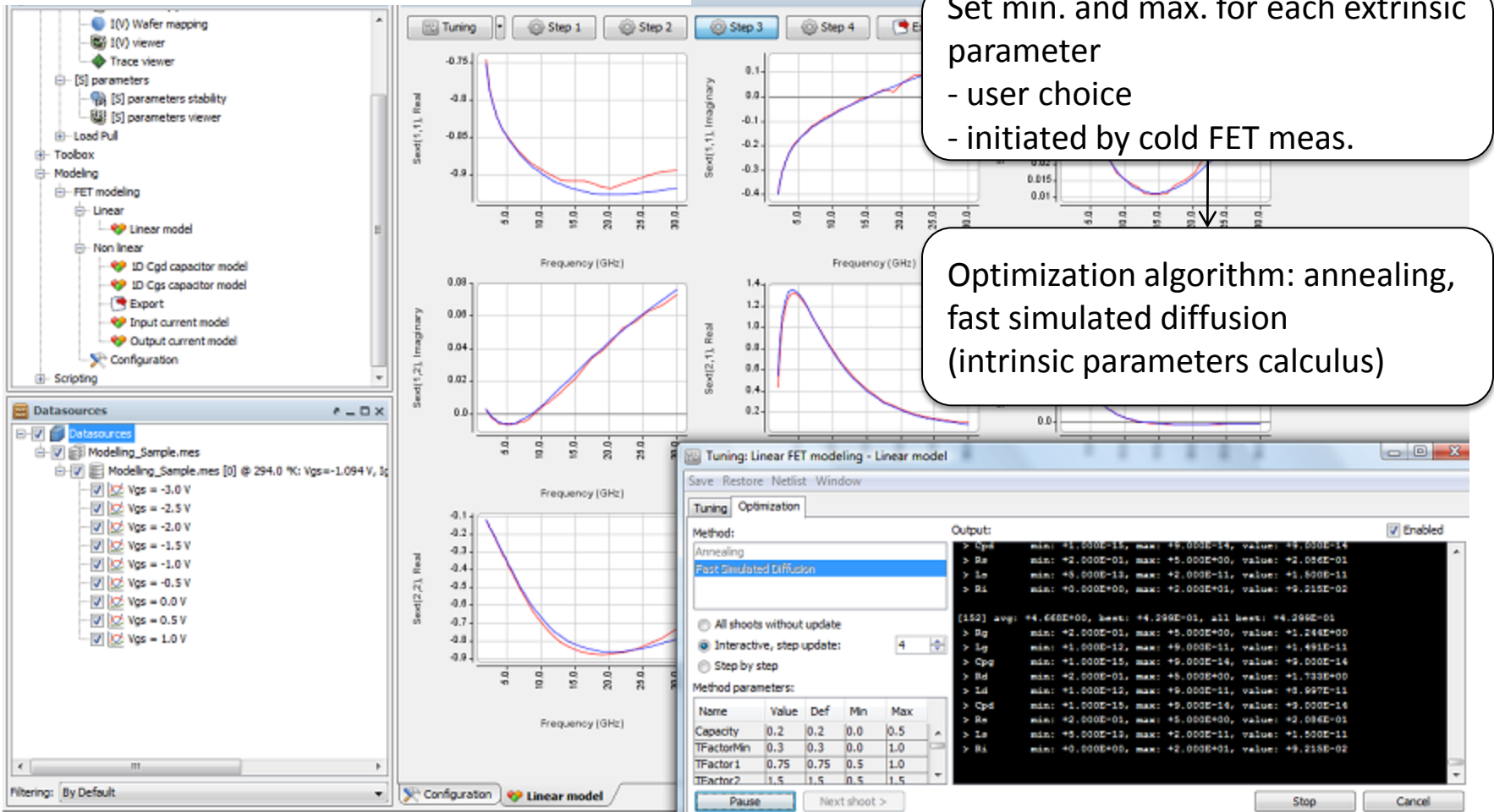
For a given set of extrinsic parameters, intrinsic admittance matrix of the device is extracted from measured [S] parameters





Small signal FET modeling

Step 3 – Linear model optimization



Set min. and max. for each extrinsic parameter

- user choice

- initiated by cold FET meas.

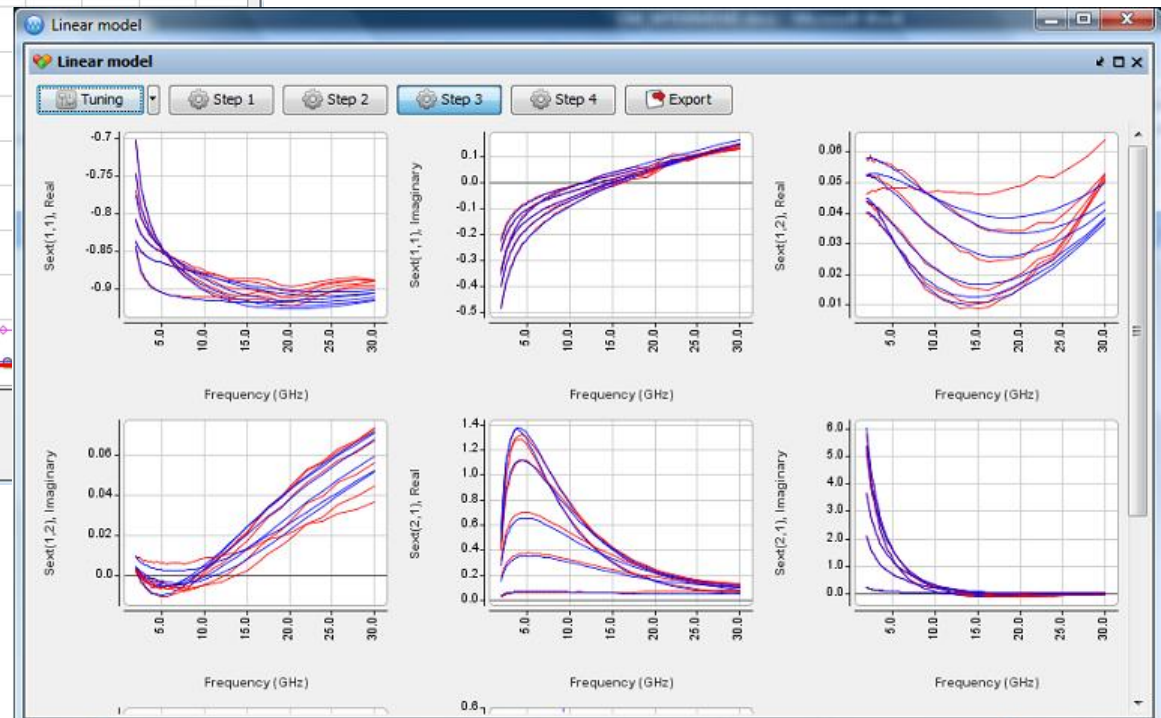
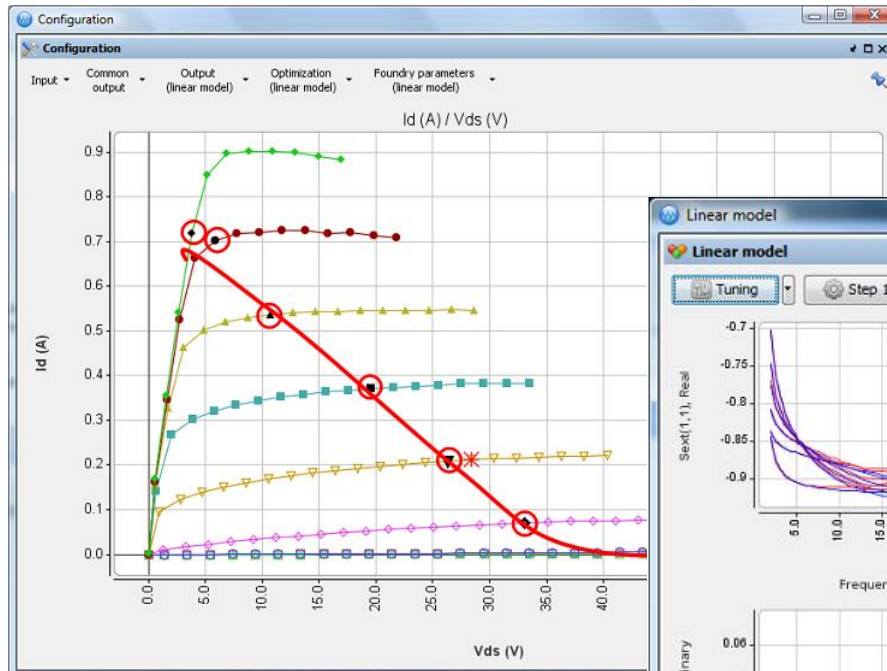
Optimization algorithm: annealing,
fast simulated diffusion
(intrinsic parameters calculus)



Small signal FET modeling

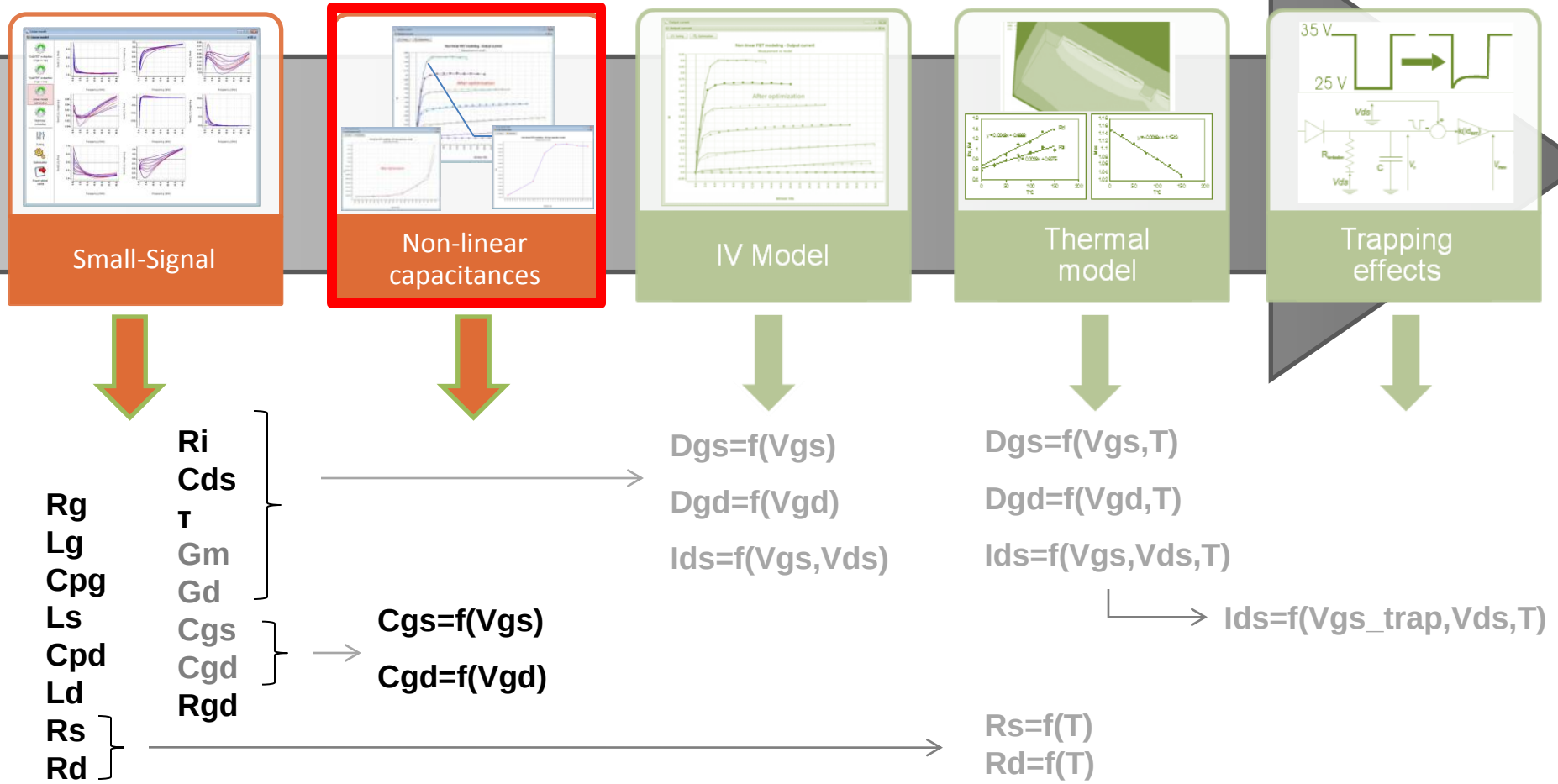
Step 3 – Linear model optimization

- The selection of several plots enable to get rid off unrealistic solutions (Resistance ≥ 0 only)
- Optimization can be launched simultaneously for all the points selected
- Each linear model has the same extrinsic values





Compact FET model extraction flow



Various effects are successively added



Nonlinear capacitances

- 1 dimension capacitances extracted along optimal load-line are preferred due to simplicity. 1D capacitance models with equations based on hyperbolic tangents are naturally charge conservatives
- Output Capacitance C_{ds} is linear – no voltage dependence (weak anyway)

$$\rightarrow C_{gd} = f(V_{gd})$$

+ Modeling simplicity. Very good convergence

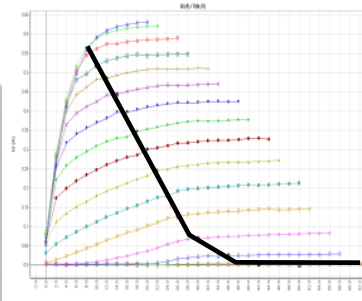
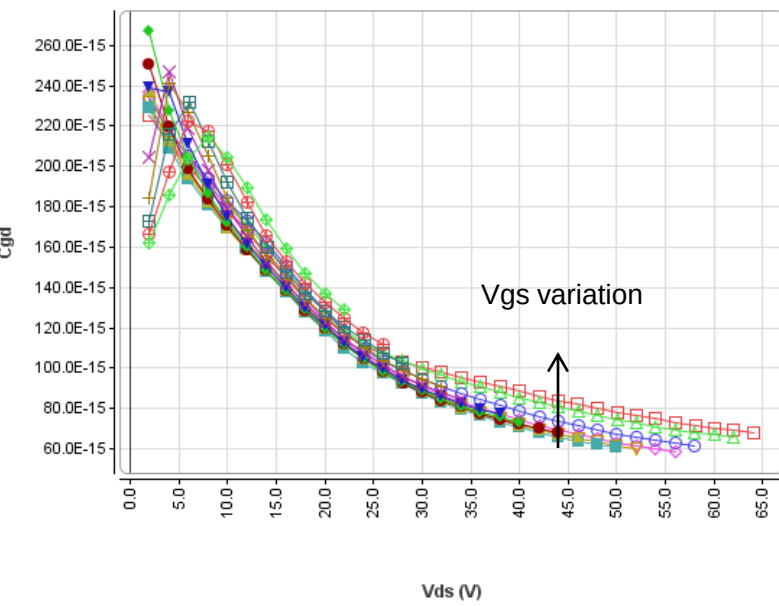
$$\rightarrow C_{gs} = f(V_{gs})$$

$$\frac{\partial C_{gs}}{\partial V_{gd}} = \frac{\partial C_{gd}}{\partial V_{gs}}$$

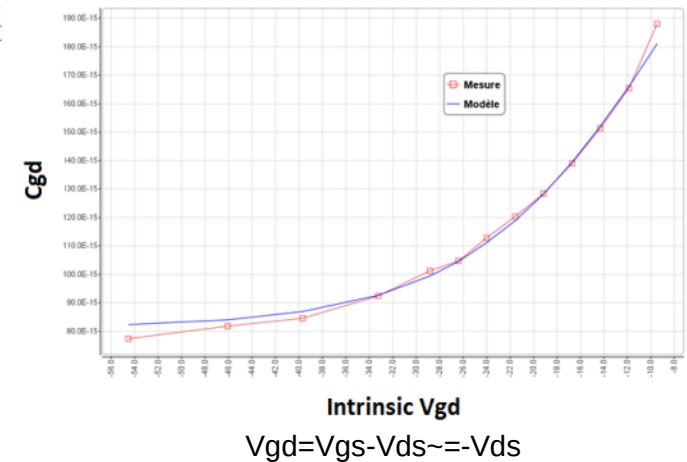
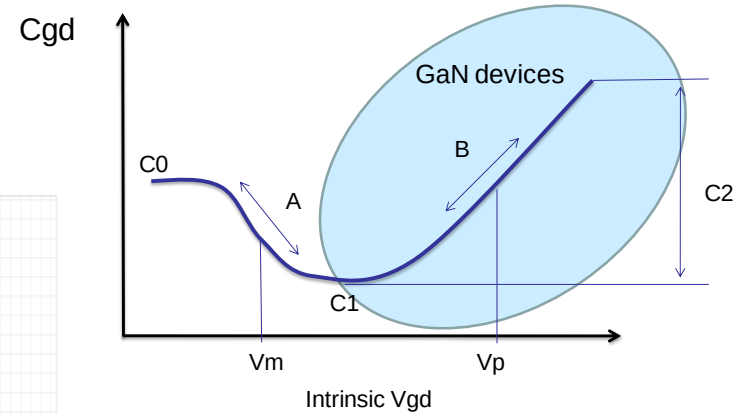
The charges in the transistor are conservatives.

Nonlinear capacitances

Cgd • Feedback capacitance Cgd is a strong function of drain voltage.



Non linear capacitances



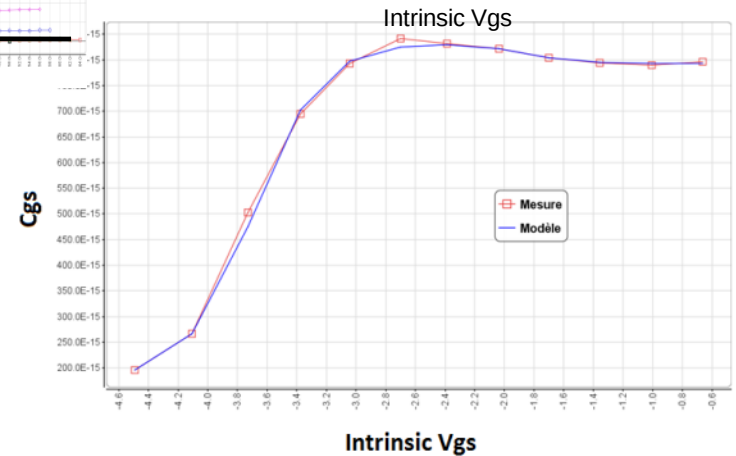
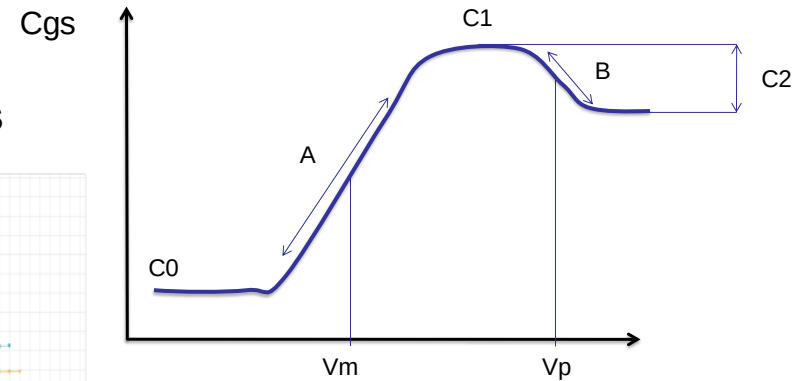
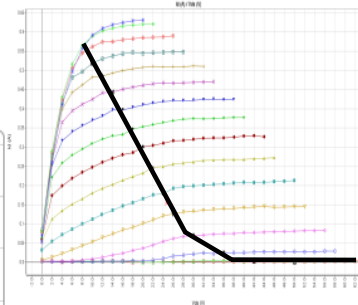
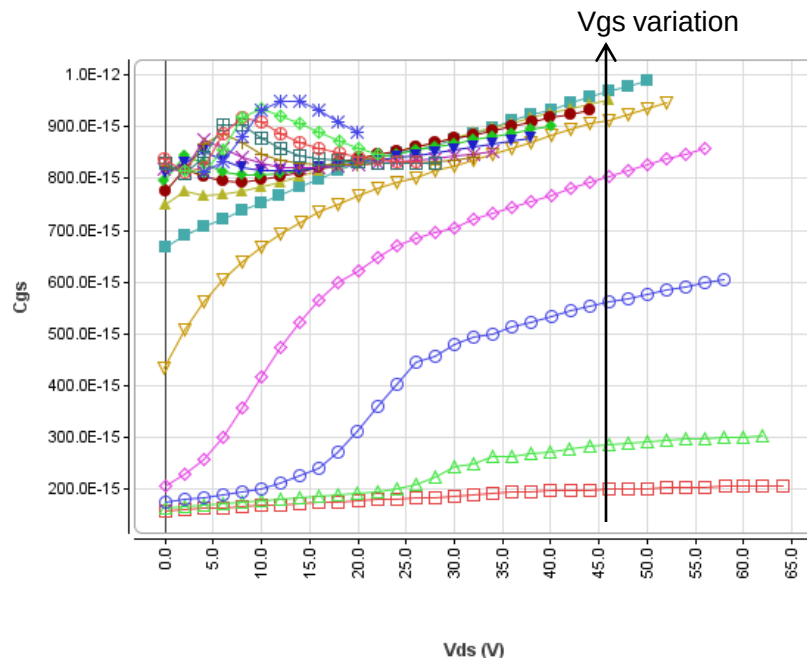
Cgd capacitance extracted along optimal load-line for power amplification



Nonlinear capacitances

Cgs Input capacitance Cgs is a strong function of gate voltage.

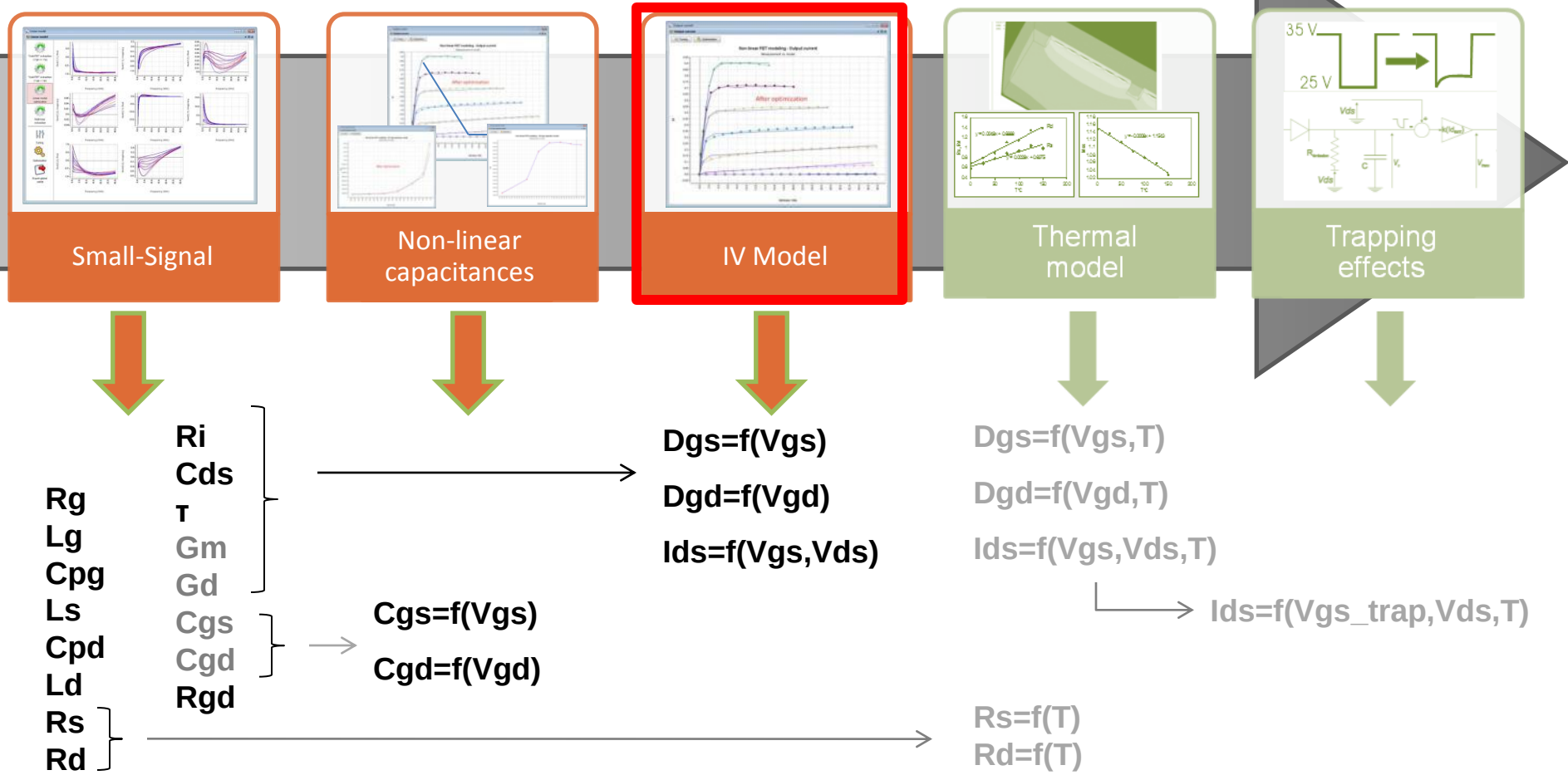
The gate-voltage non-linearity also effects model's harmonic generation



Cgs capacitance extracted along optimal load-line for power amplification



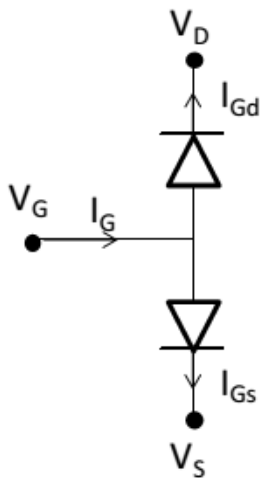
Compact FET model extraction flow



Various effects are successively added

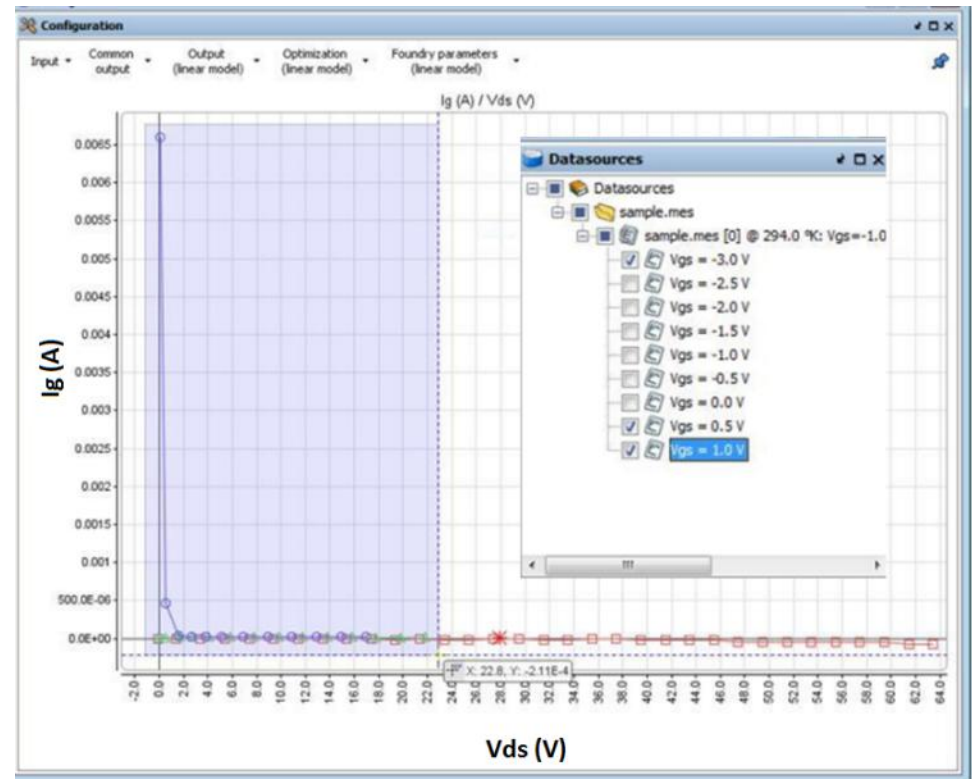
Diodes

Selection of curve without Gate Current + 1 or 2 curves With Gate current



$$I_{Gd} = I_{SGd} \cdot \left(e^{\left(\frac{G}{N_{gd} K T} V_d \right)} - 1 \right)$$

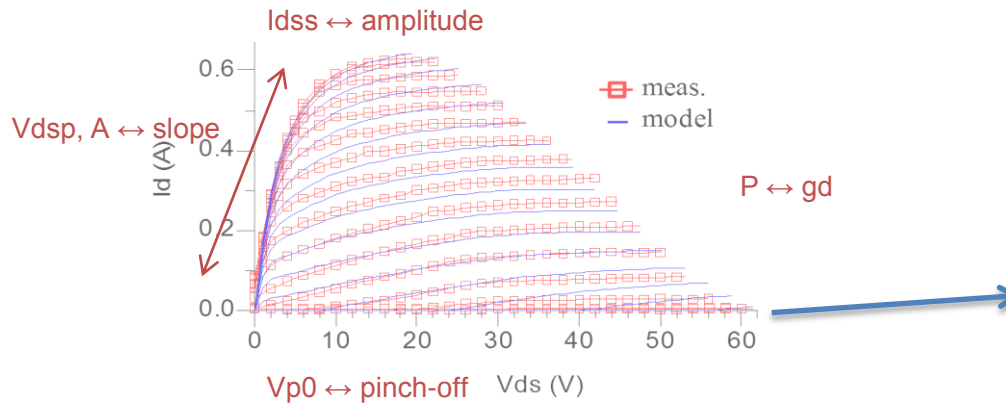
$$I_{Gs} = I_{SGs} \cdot \left(e^{\left(\frac{G}{N_{gs} K T} V_d \right)} - 1 \right)$$



- Gate-drain and gate-source diode equations include forward conduction of gate current

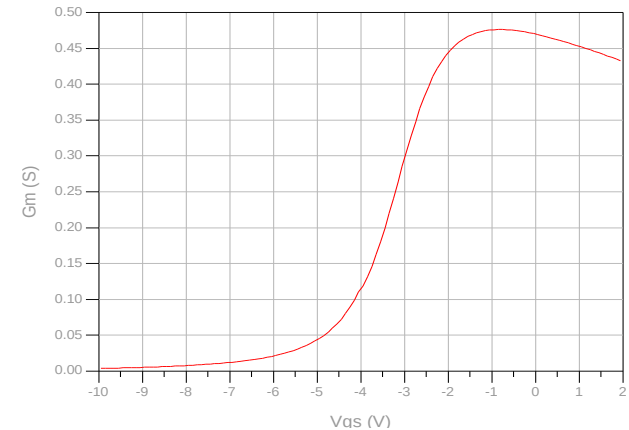
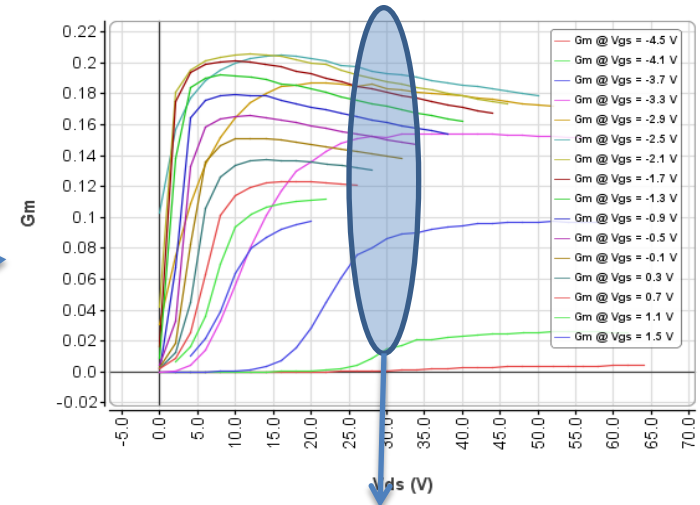


Output current source



M, P ↔ fitting parameters

AlphaGm, Vgm, BetaGm, Vdm ↔ gm (derivative)



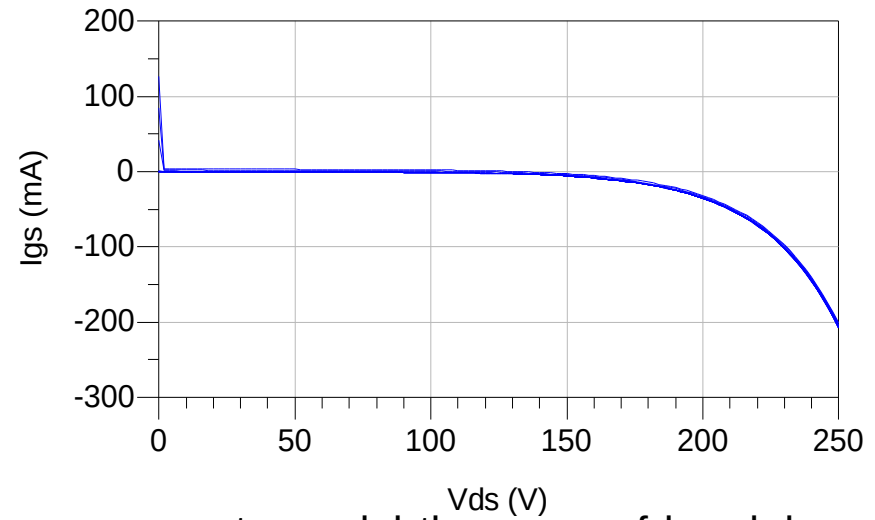
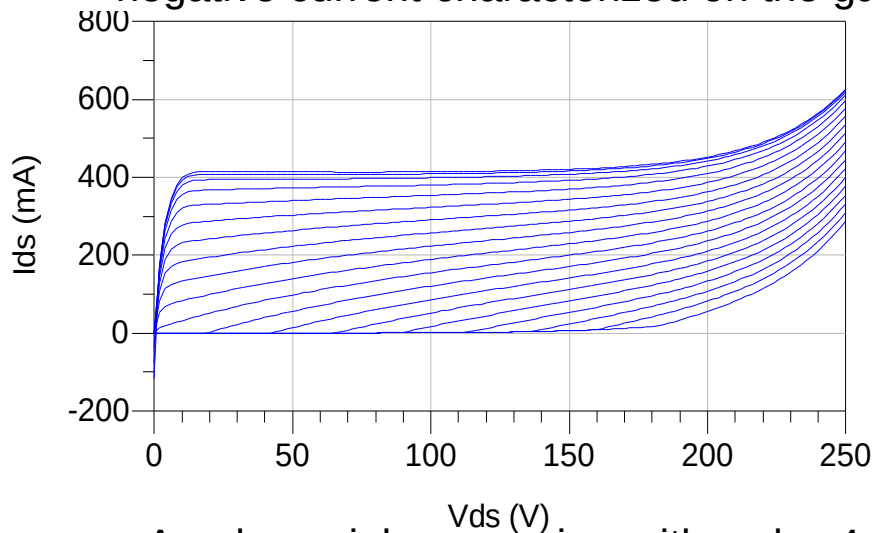
- AMCAD drain current model formulation allows to predict very accurately the I-V curves, the partial derivatives gm and gd, the knee voltage and the transconductance decrease at high current.



Breakdown generator

- Gate-drain Breakdown generator

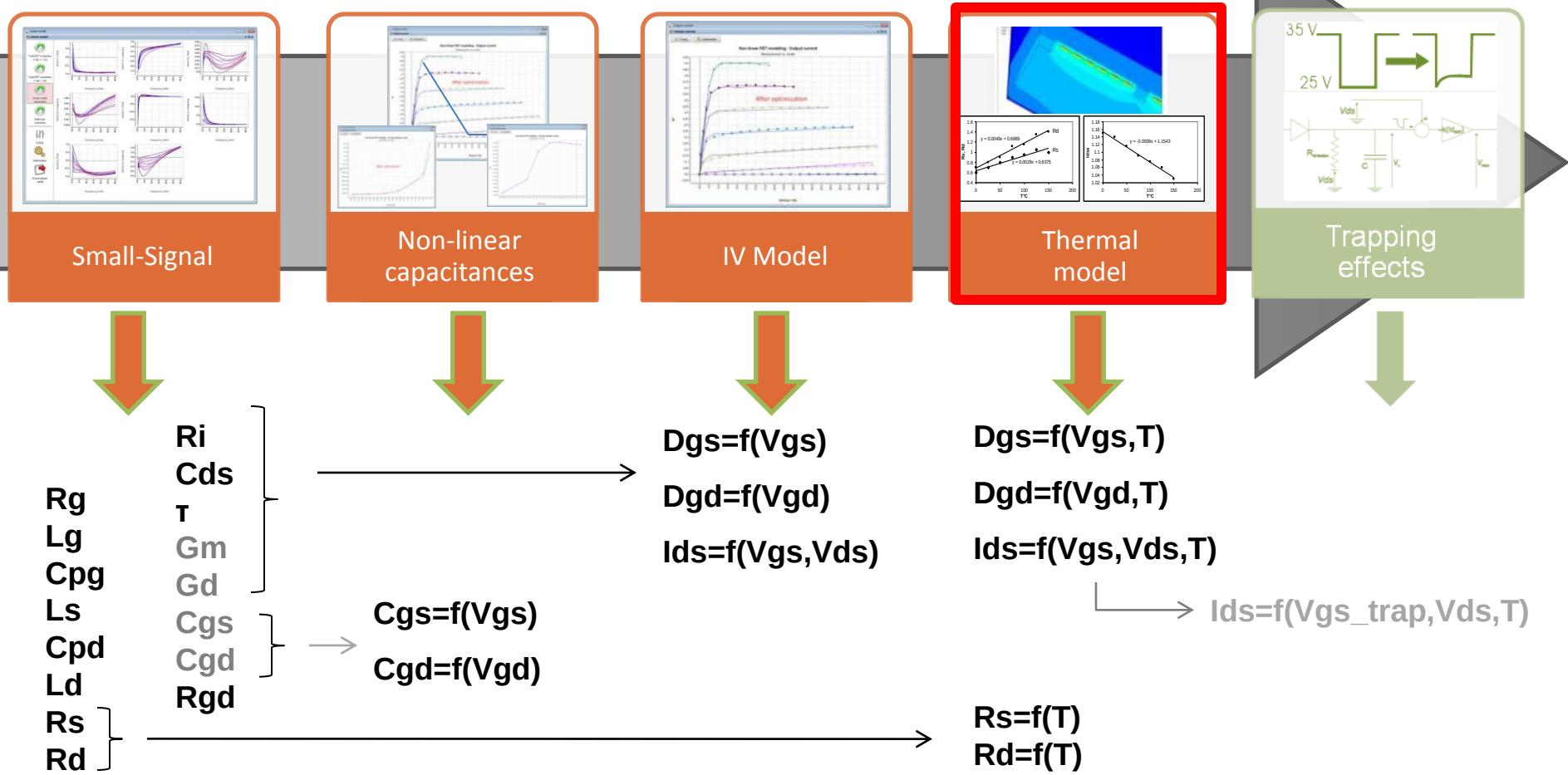
The breakdown phenomena leads to a current from the drain to the gate when the device is pinched-off and for high values of V_{ds} voltage. In this case, the whole negative current characterized on the gate is seen in positive on the drain



A polynomial expression with order 4 is necessary to model the cross of breakdown curves, with varies depending on the process



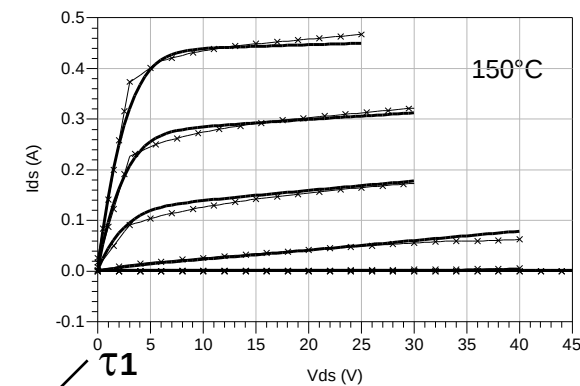
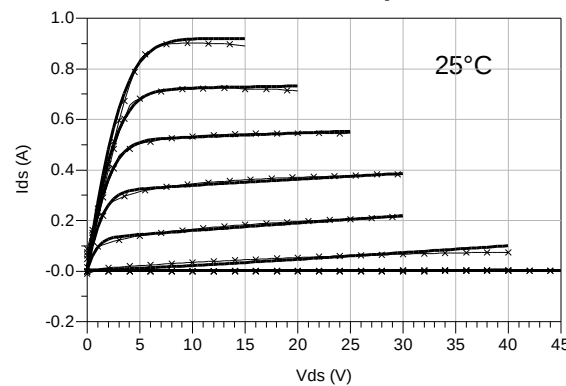
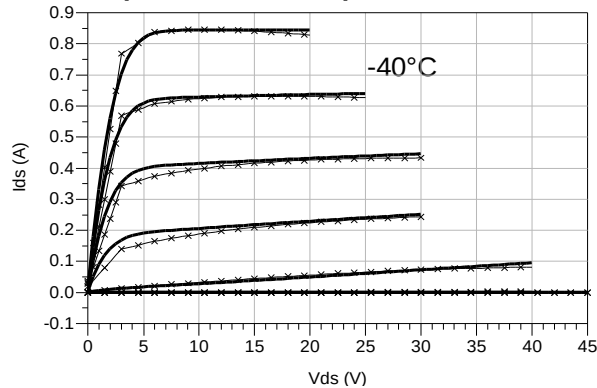
Compact FET model extraction flow



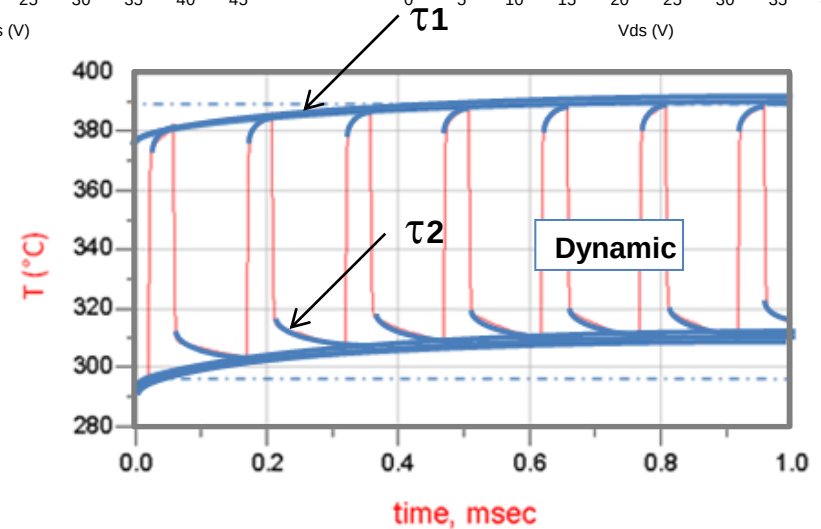
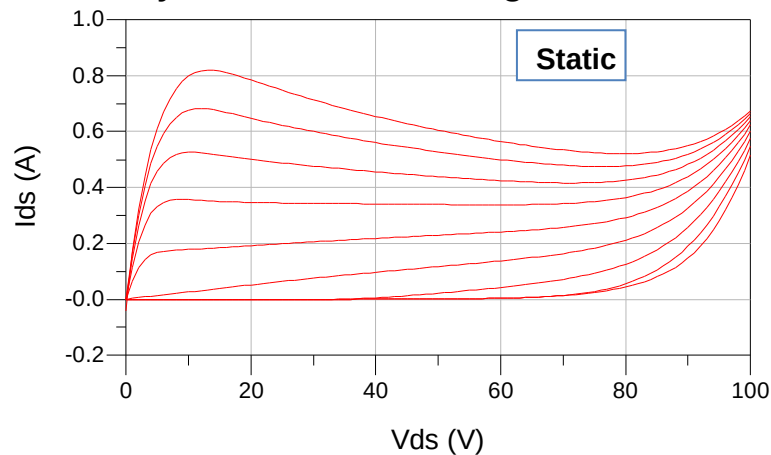
Various effects are successively added

Thermal effects

- Temperature dependence with ambient or chuck temperature



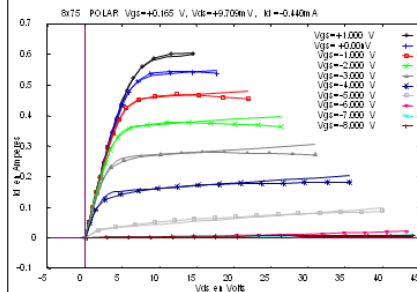
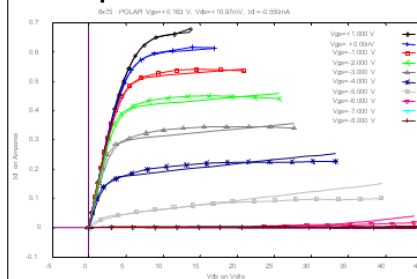
- Static and Dynamic self-heating effects



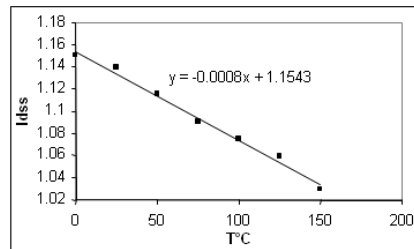
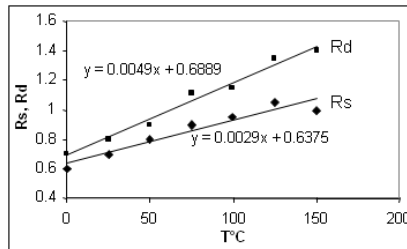


Thermal effects

IV Modeling @ several temperatures



IV parameters versus temperature



Temperature dependence

Equations

Thermal parameters

- Access Resistances
- Current Source
- Diodes

$$R_s = R_{s0} + \alpha_{R_s} \cdot T$$

$$R_d = R_{d0} + \alpha_{R_d} \cdot T$$

$$I_{dss} = I_{dss0} + I_{dss_t} \cdot T$$

$$P = P_0 + P_t \cdot T$$

$$N_{gs} = N_{gs0} + N_{gs_t} \cdot T$$

$$N_{gd} = N_{gd0} + N_{gd_t} \cdot T$$

$$I_{sgs} = I_{sgs0} + I_{sgs_t} \cdot e^{(T/T_{sgs})}$$

$$I_{sgd} = I_{sgd0} + I_{sgd_t} \cdot e^{(T/T_{sgd})}$$



Thermal effects

Thermal resistance extraction → coincidence method



— DC, $T_{chuck1} = 25^\circ\text{C}$

— Pulsed from (0,0), $T_{chuck2} = 100^\circ\text{C}$

DC curve

$$T_{j1} = T_{chuck1} + R_{th} \cdot P_{diss1}$$

Pulsed curve

$$T_{j2} = T_{chuck2} + R_{th} \cdot P_{diss2}$$

$\rightarrow = 0$

At intersection point

$$T_{j1} = T_{j2}$$

$\Rightarrow$

$$T_{chuck1} + R_{th} \cdot P_{diss1} = T_{chuck2}$$

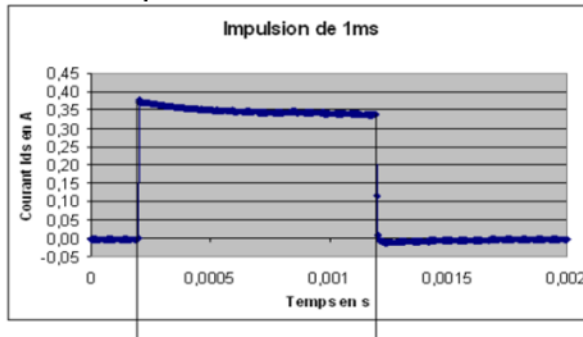
$$R_{th} = (T_{chuck2} - T_{chuck1}) / P_{diss1}$$



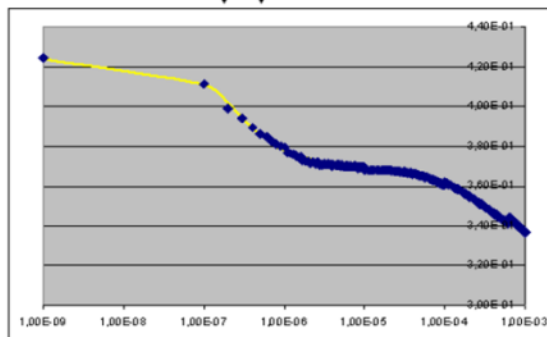
Thermal effects

Thermal impedance extraction – *by measurements*

Wide current pulse characterization



in log scale to determine time constants

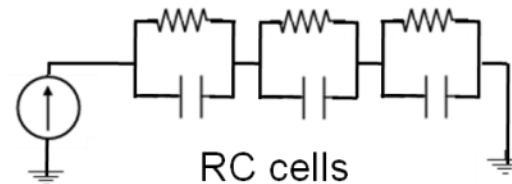


Self Heating Extraction = $f(t)$

$$i(t) = I_0 - \sum_{i=1}^n I_i \cdot \left(1 - \exp\left(\frac{-t}{\tau_i}\right) \right)$$



$$Z_{th}(t) = \sum_{K=1}^n R_{kth} \left(1 - e^{-\frac{t}{R_{kth} C_{kth}}} \right)$$



Device dynamical self-heating

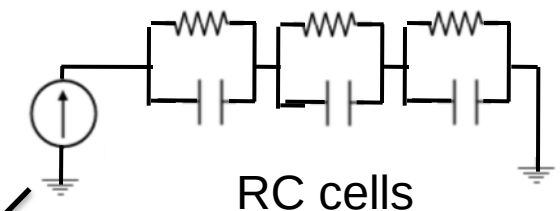
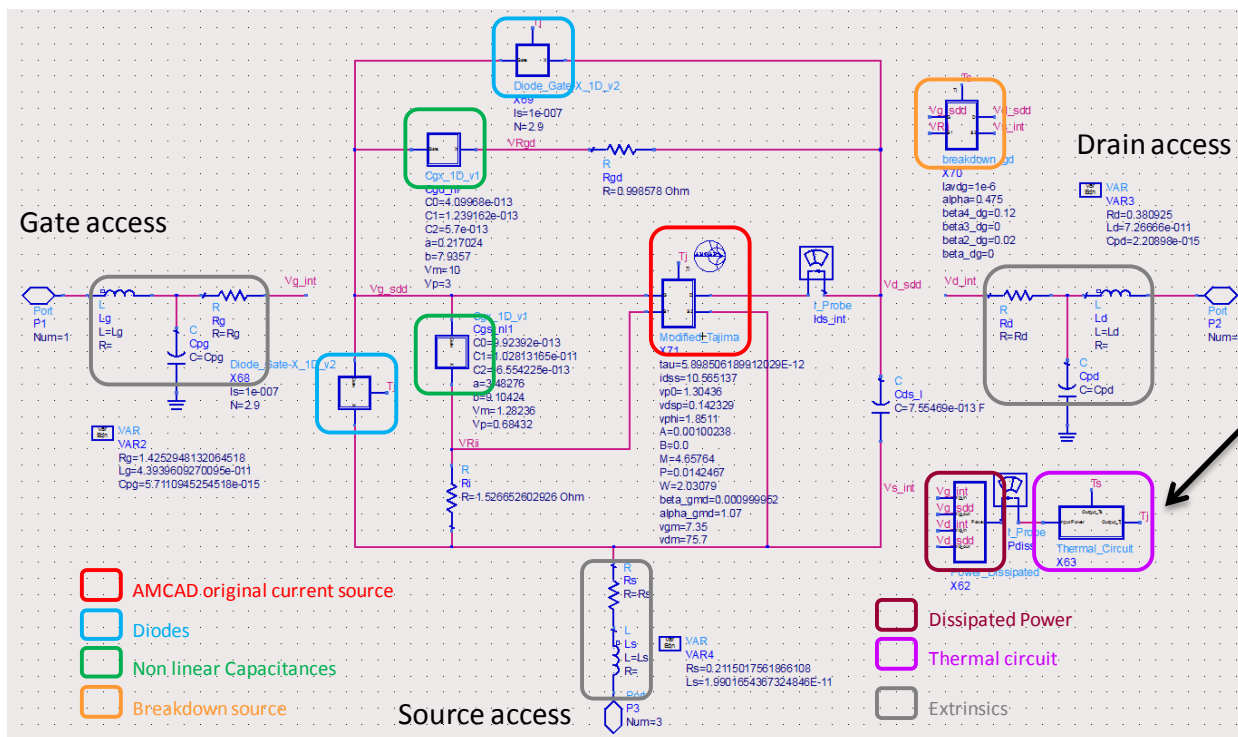


Thermal effects

- Drain current is only temperature dependent model element

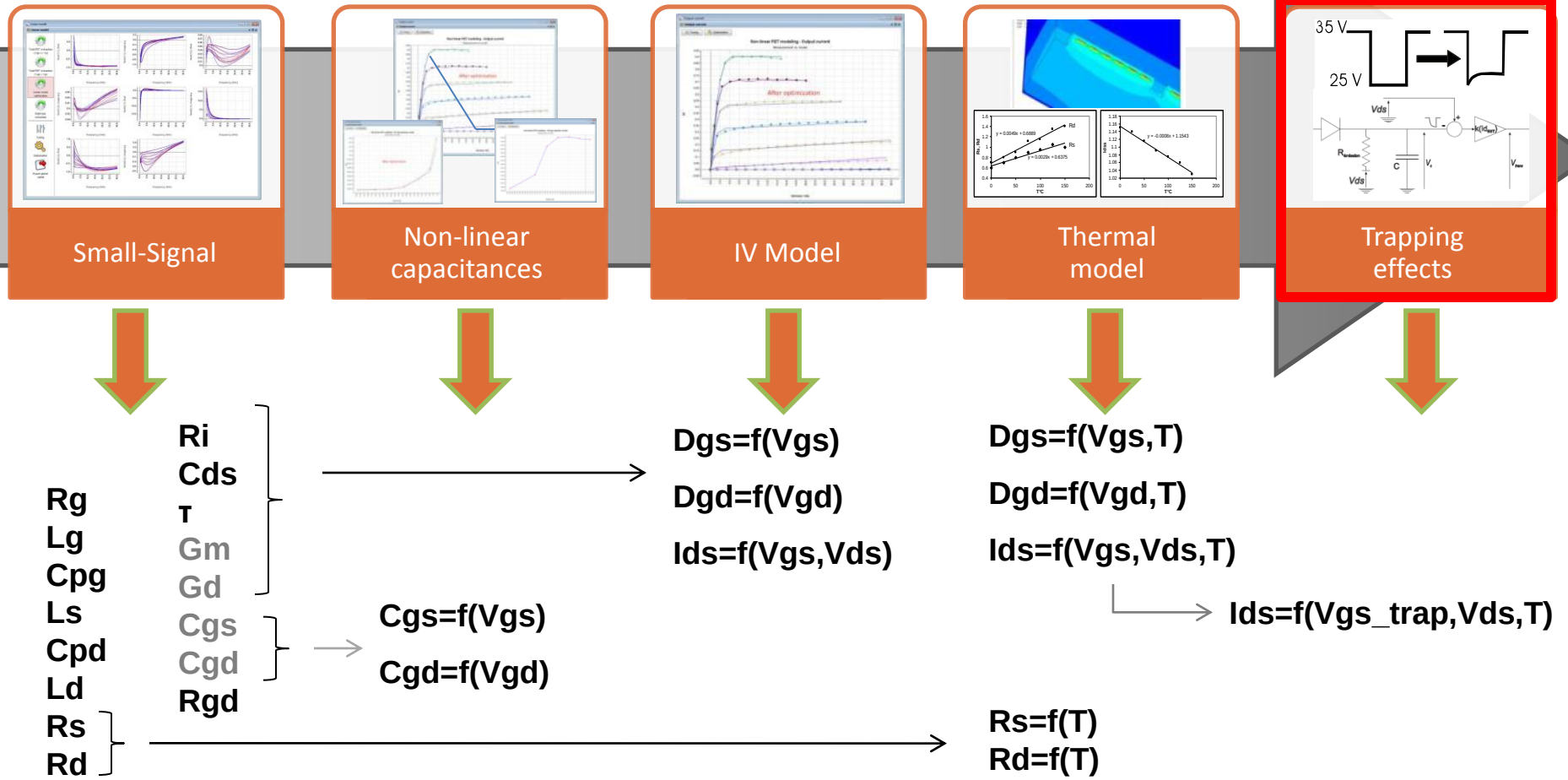
Takes into account ambient temperature and self-heating effects

Thermal analog circuit to model self-heating and elevated heat sink temperatures





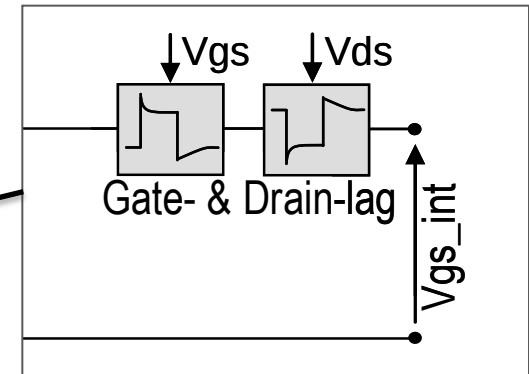
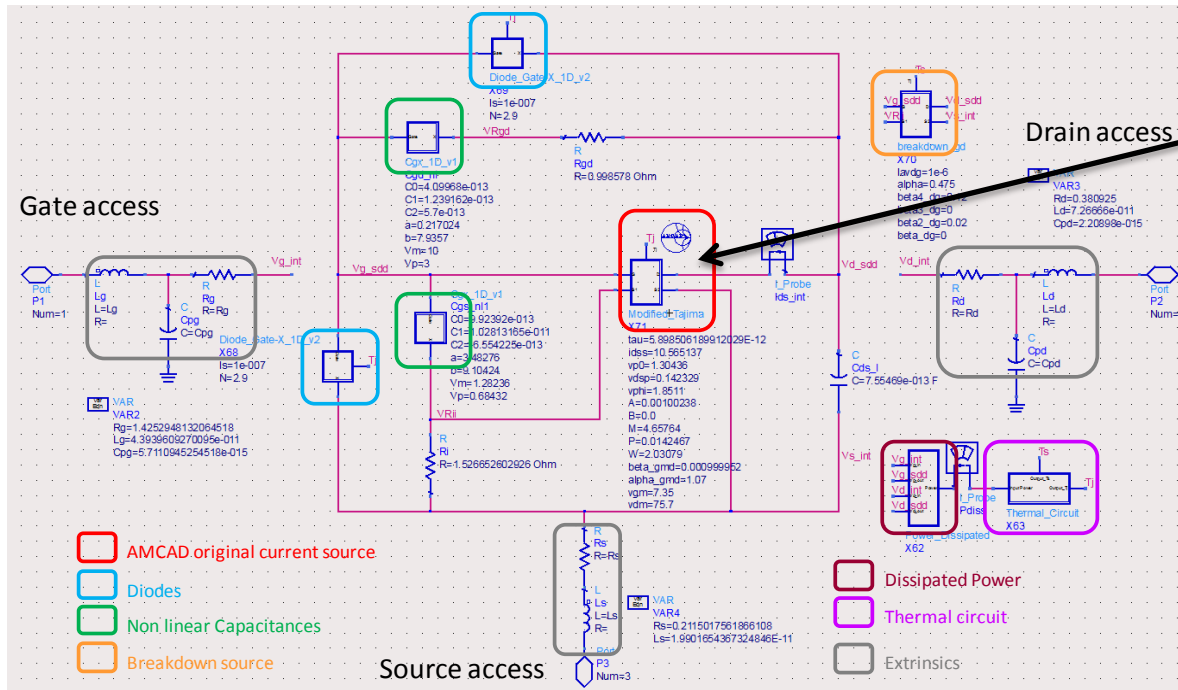
Compact FET model extraction flow





Trapping effects

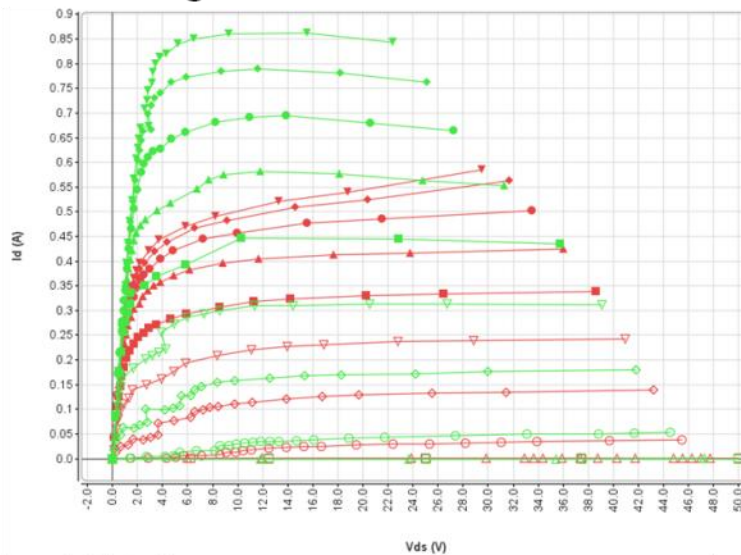
- Charging and discharging of traps has influence on I_{ds} and leads to current collapse. This is described in the model by trapping effects modifying the gate command and separated into gate and drain lag sub-circuits





Trapping effects

Gate-lag: decrease of drain current

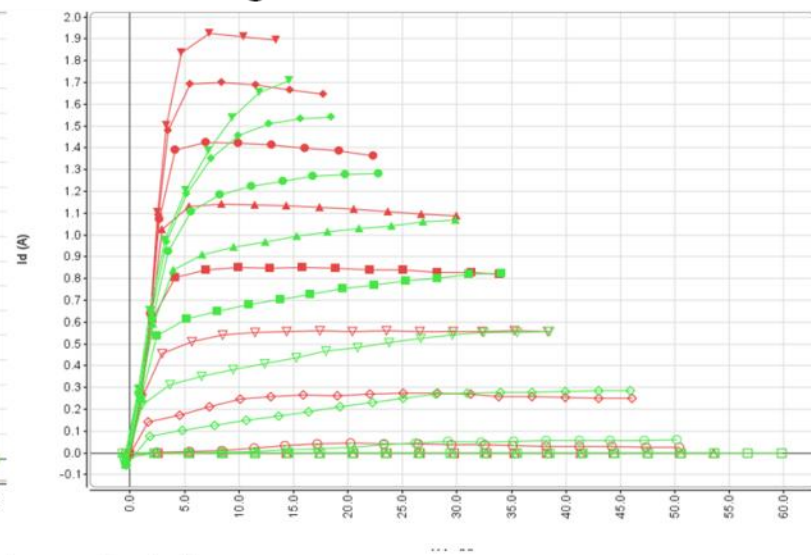


Gate-lag

green ($V_{gs0}=0V, V_{ds0}=0V$)

red ($V_{gs0}=-4V, V_{ds0}=0V$)

Drain-lag: increase of V_{knee}



Drain-lag

red ($V_{gs0}=-3V, V_{ds0}=0V$)

green ($V_{gs0}=-3V, V_{ds0}=30V$)

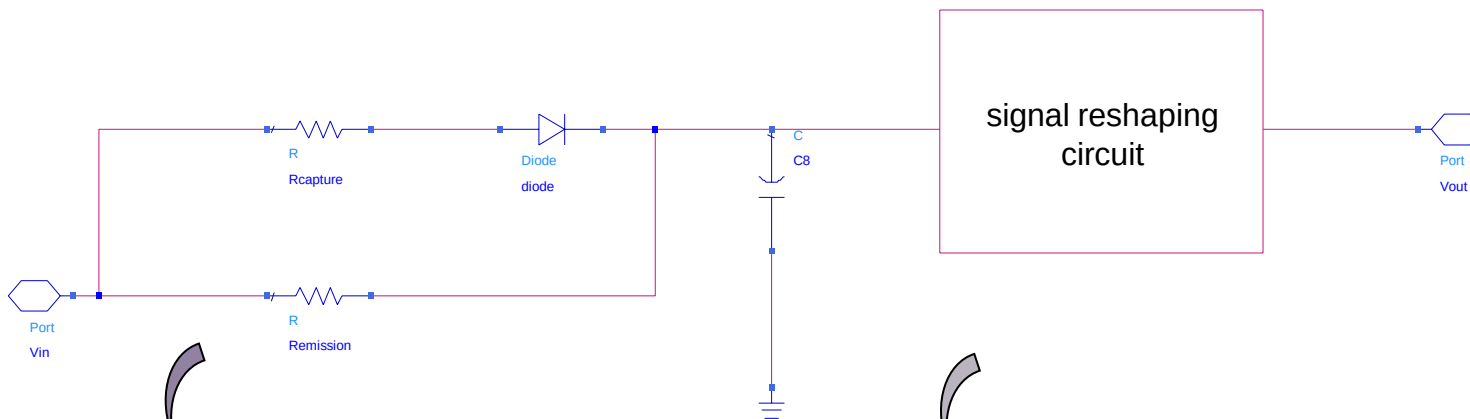
$$T_{\text{capture}} \ll t_{\text{IMPULSION}} \ll T_{\text{émission}}$$

During the pulses capture takes place, emission freed

Trapping effects

- Charge of the capacitance = Ionized traps

Charge through R_{capture} , Emission through R_{emission}



Diode = dissymmetry of the capture and emission process

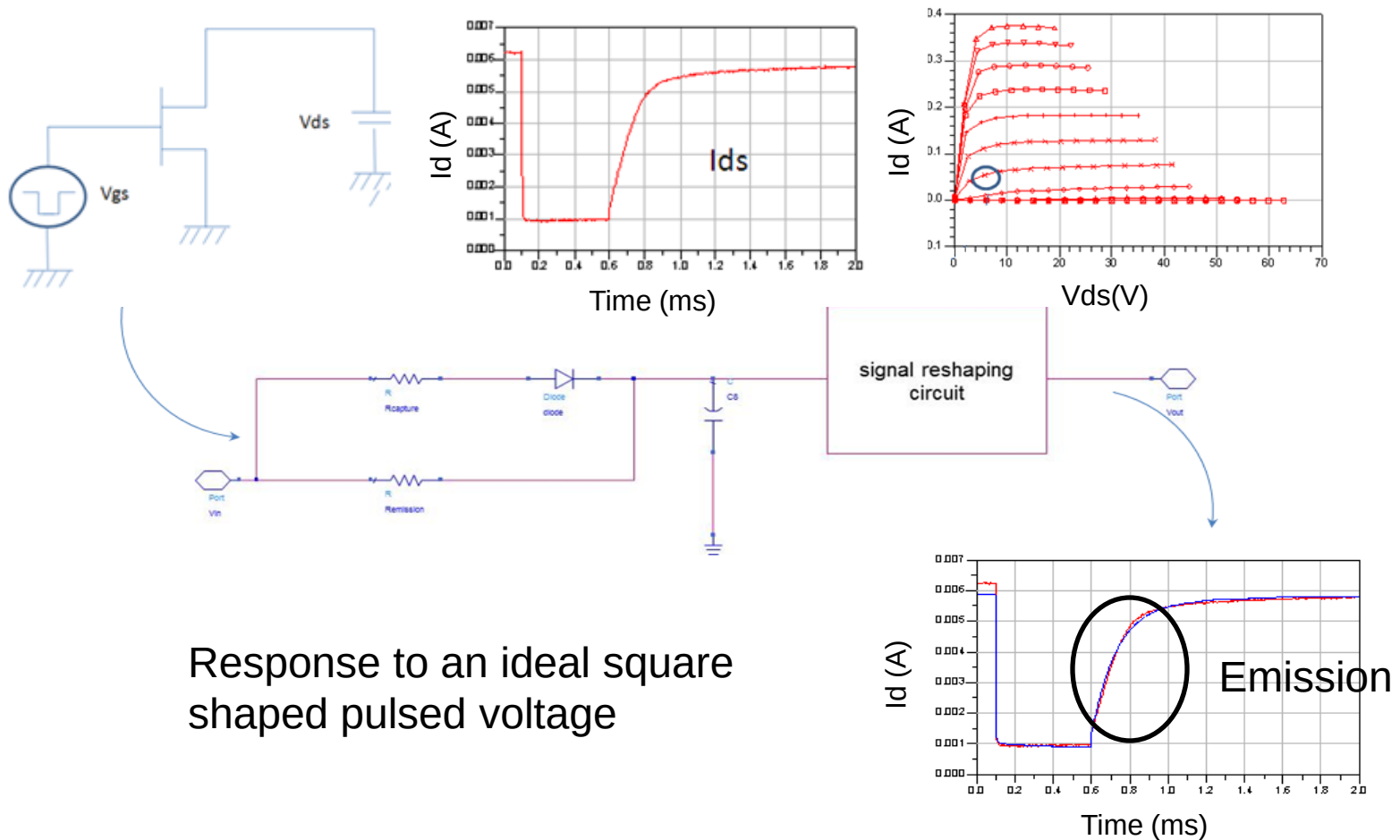
Tuning of the magnitude of the trapping effects

Fundamental assumption : dissymmetry of the capture and emission process



Trapping effects

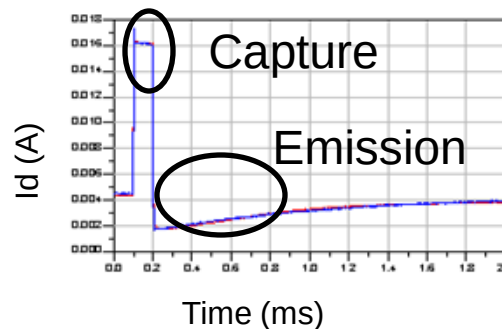
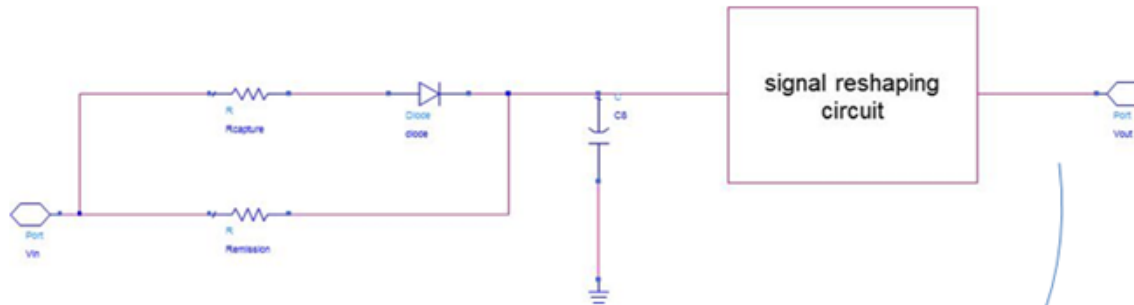
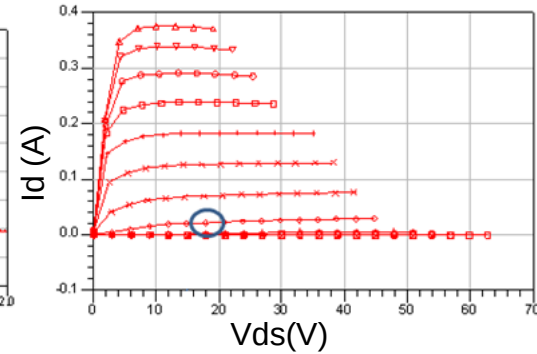
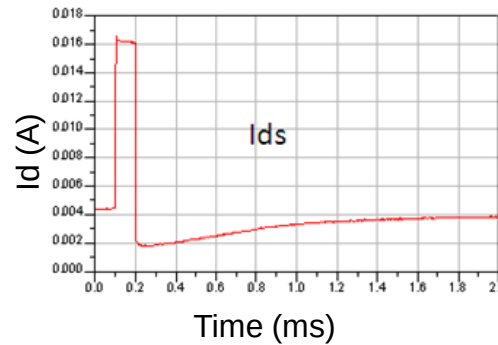
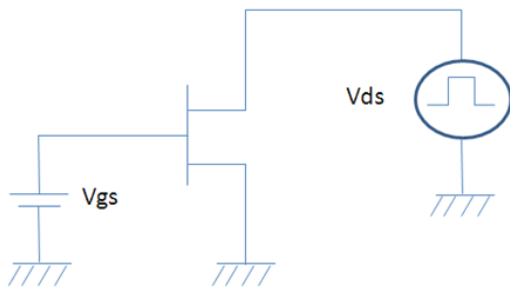
Measurements to extract Gate-lag at very low dissipated power





Trapping effects

Measurements to extract Drain-lag at very low dissipated power



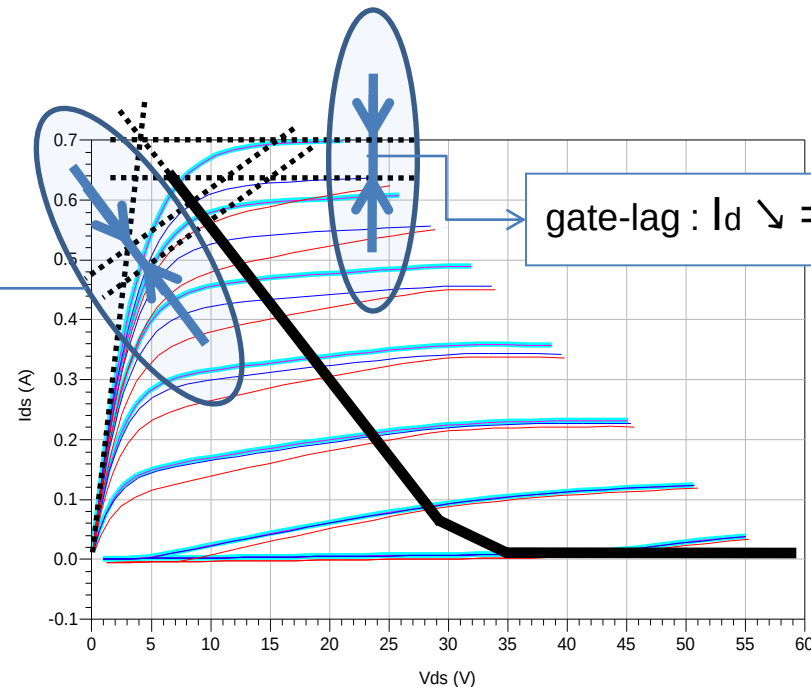


Trapping effects

- Bias dependant gate lag -> current reduction over the entire characteristic

Bias dependent drain lag -> current reduction and shifts the knee-voltage to a higher V_{ds}

Model covers knee walkout to avoid errors in calculation of output power.



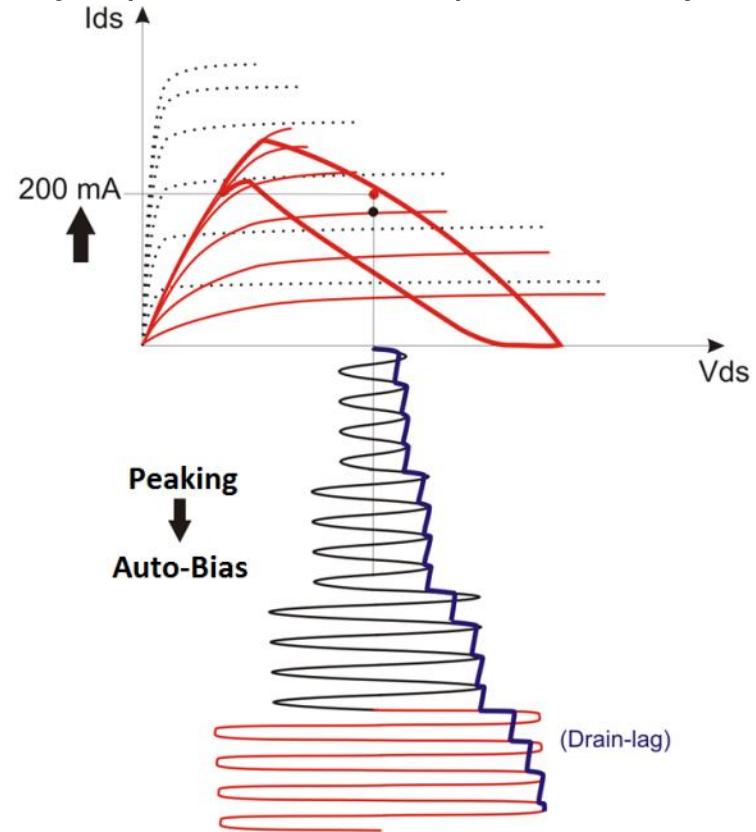
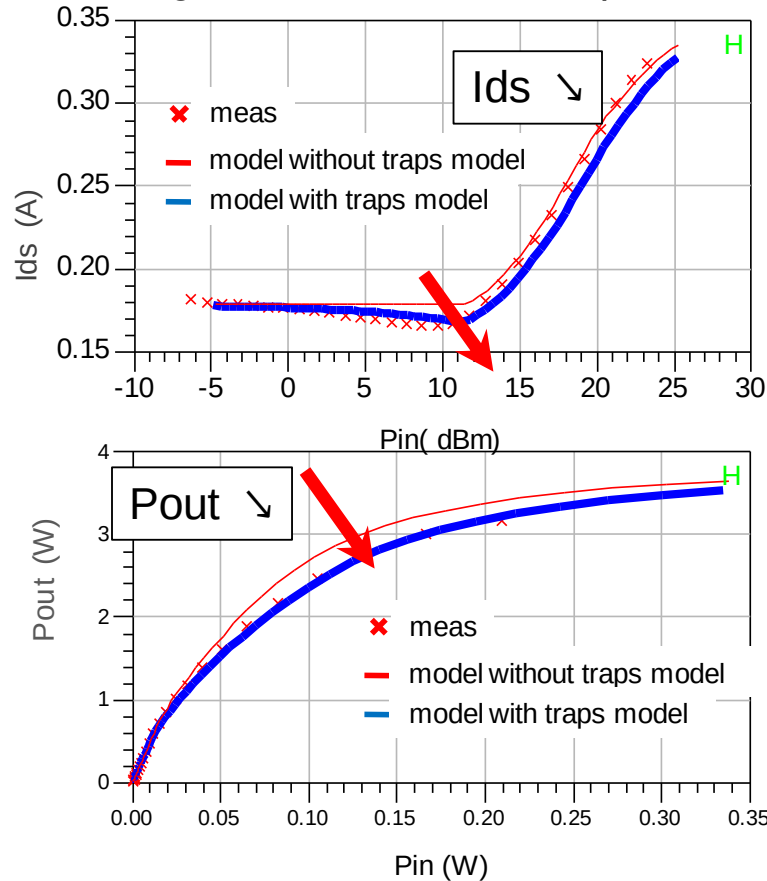
gate-lag : $I_d \searrow \Rightarrow P_{out} \searrow$

drain-lag : $I_d \searrow$
 $V_{knee} \nearrow$ } $\Rightarrow P_{out} \searrow$



Trapping effects

- Decreasing form of the mean output current only reproduced with traps accurately modeled



A Drain-Lag Model for AlGaIn/GaN Power HEMTs, Jardel, O. ; Charbonniaud, Microwave Symposium, 2007. IEEE/MTT-S International 2007



Trapping effects

- **A Non Linear Electrothermal Model of AlGaIn/GaN HEMT for Switch Applications**

Charbonniaud, C. and al. (AMCAD) ; Compound Semiconductor Integrated Circuit Symposium (CSICS), 2012 IEEE , Page(s): 1 - 4

- **A distributed electro-thermal model of AlGaIn/GaN HEMT power-bar derived from the elementary cell model**

Xiong, A. and al. (AMCAD) ; Microwave Integrated Circuits Conference (EuMIC), 2012 7th European, Page(s): 64 - 67

- **A non linear power HEMT model operating in multi-bias conditions**

Charbonniaud, C and al. (AMCAD) ; Microwave Integrated Circuits Conference (EuMIC), 2010 European, Page(s): 134 - 137

- **A Scalable and Distributed Electro-Thermal Model of AlGaIn/GaN HEMT Dedicated to Multi-Fingers Transistors**

• Xiong, A. and al. (AMCAD) ; Compound Semiconductor Integrated Circuit Symposium (CSICS), 2010, Page(s): 1 - 4



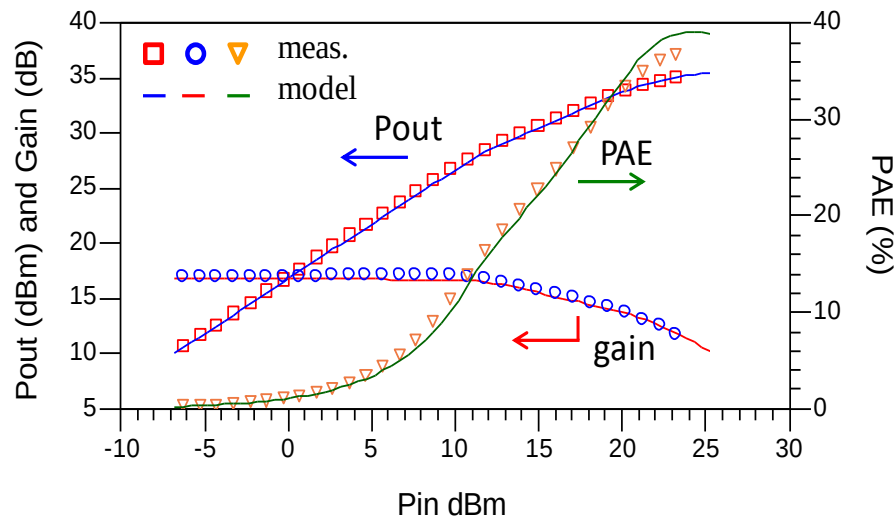
- Instructor Introduction
- Pulsed IV/RF and Compact Modeling
- Load Pull
- Amplifier Design in ADS
- Stability Analysis
- X-Parameters
- System-level Simulations



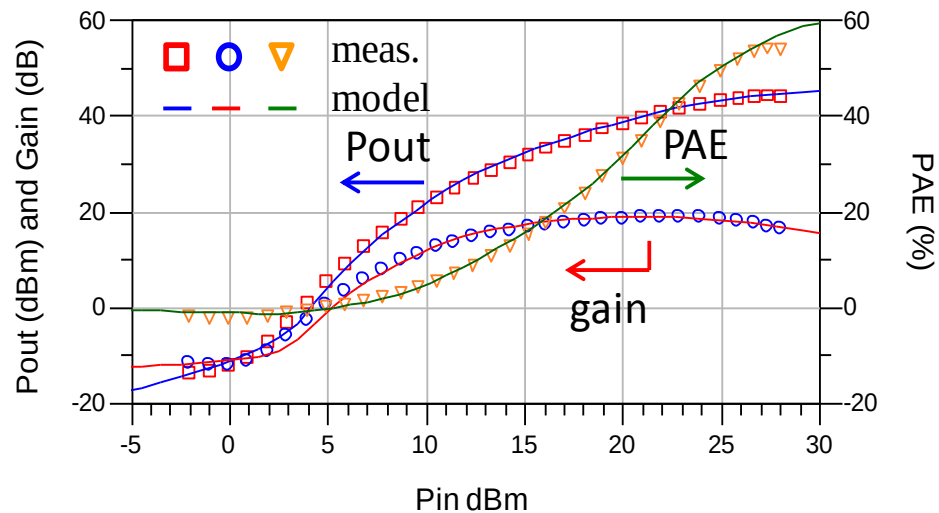
Model Validation

- Large-signal

Model validation of a 8x75 μm GaN HEMT with load-pull measurements performed at 6 GHz for optimum PAE load impedance in class-AB

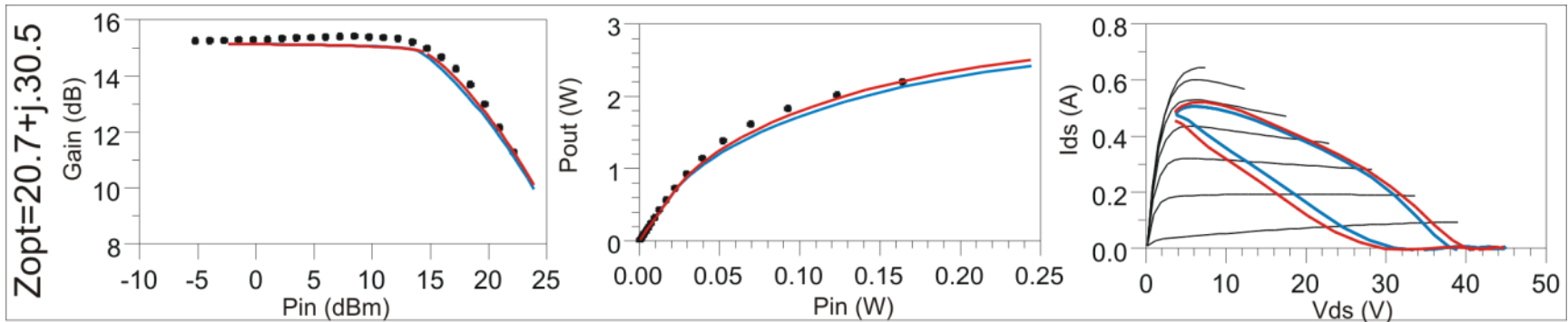


Model validation of a 8x400 μm GaN HEMT with load-pull measurements performed at 3 GHz for the optimum Pout load impedance in class-B

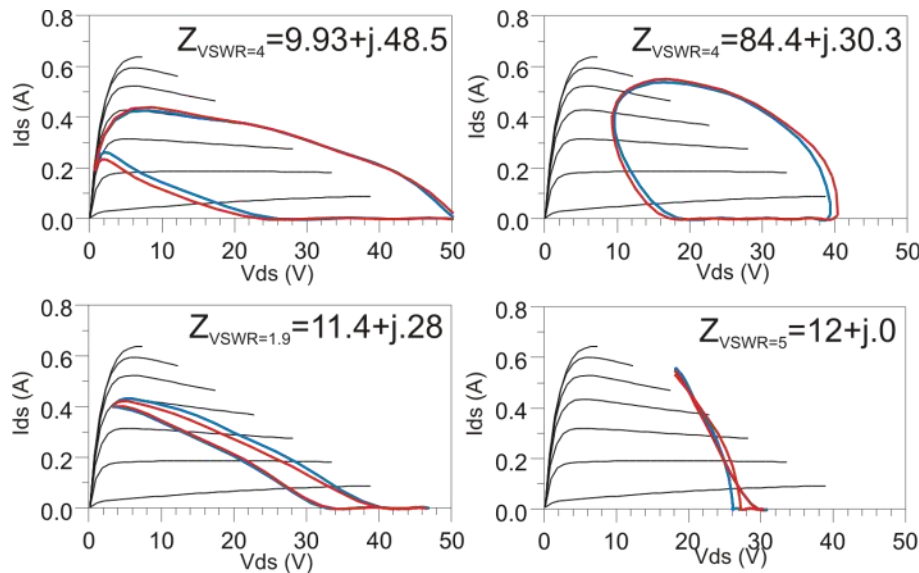




Model Validation



With non optimal loads :



→ Time domain load pull measurements

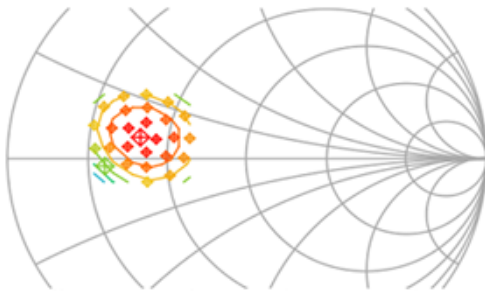
→ Deembedding in the intrinsic reference plane

→ Parasitic extrinsic elements must be accurately extracted by previous S parameter measurements



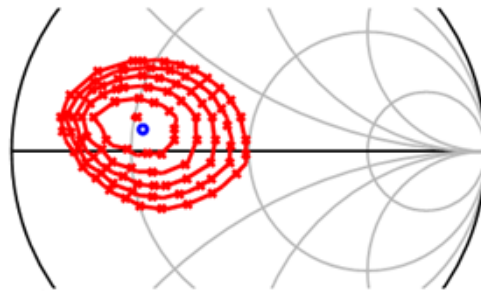
Model Validation

Iso Pout contours
measured @ 1.85GHz

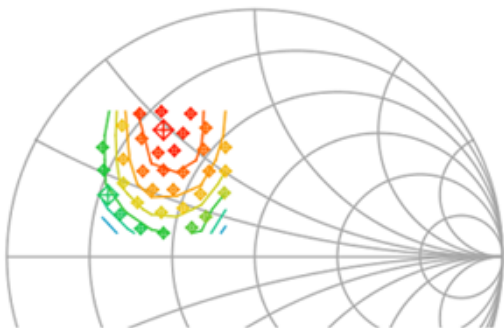


Iso Output Power contour measured for a gain compression of 3dB

Iso Pout contours
simulated @ 1.85GHz

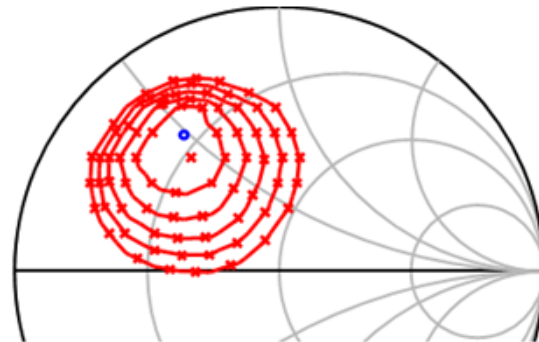


Iso PAE contours
measured @ 1.85GHz



Iso PAE contour measured for a gain compression of 3dB

Iso PAE contours
simulated @ 1.85GHz



What are the unique or specific requirements for load pull with regards to model validation?

-Independent powers at each frequency (f_0 , $2f_0$...)

-Independence of source impedance match



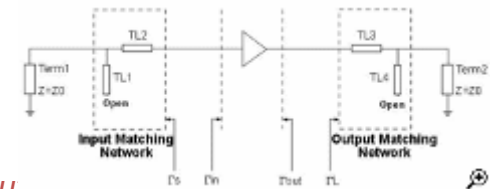
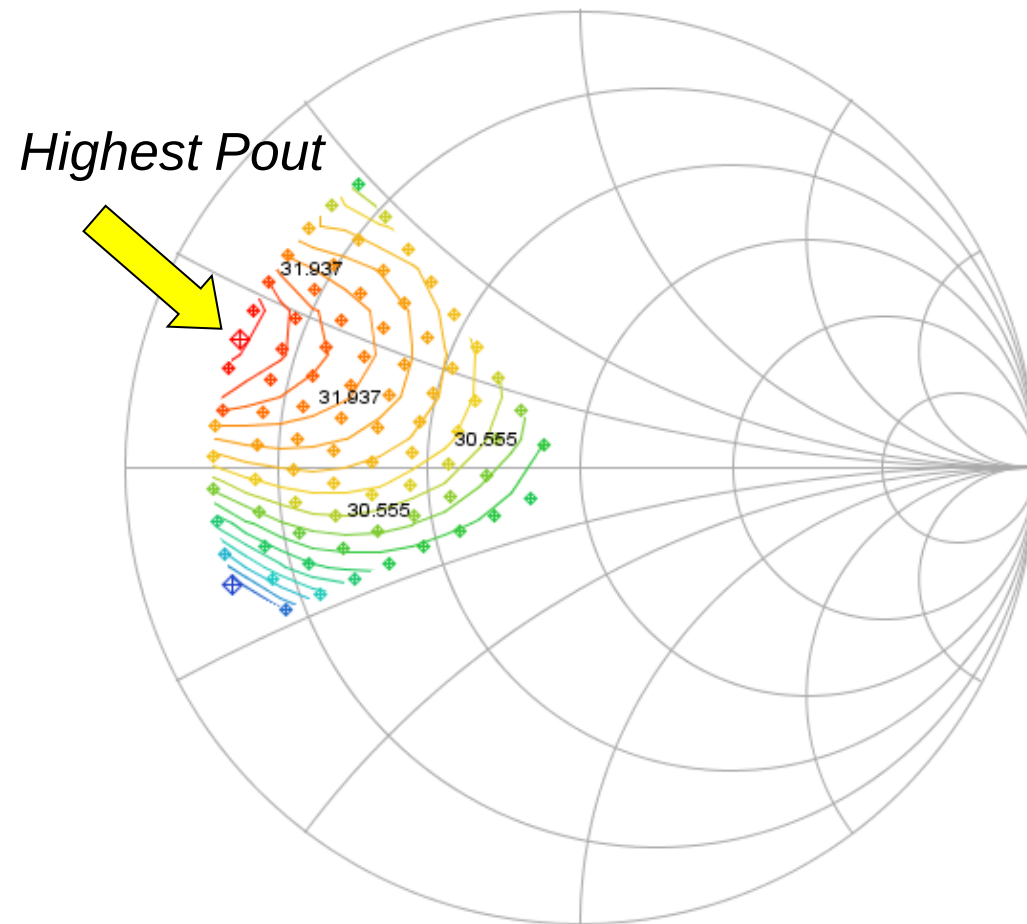
Introduction to load pull

1) Vary impedance presented to DUT (active device, transistor)

2) Measure Pout, Gain, Efficiency...

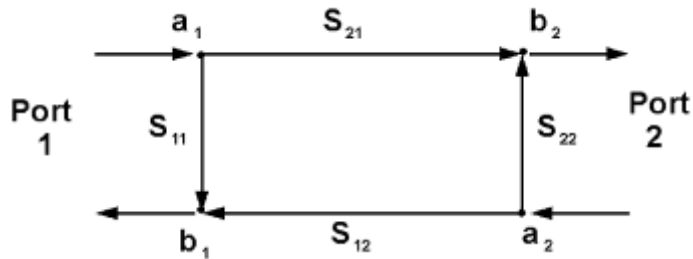
3) Determine best matching impedance

4) Design matching network (EEsof ADS)





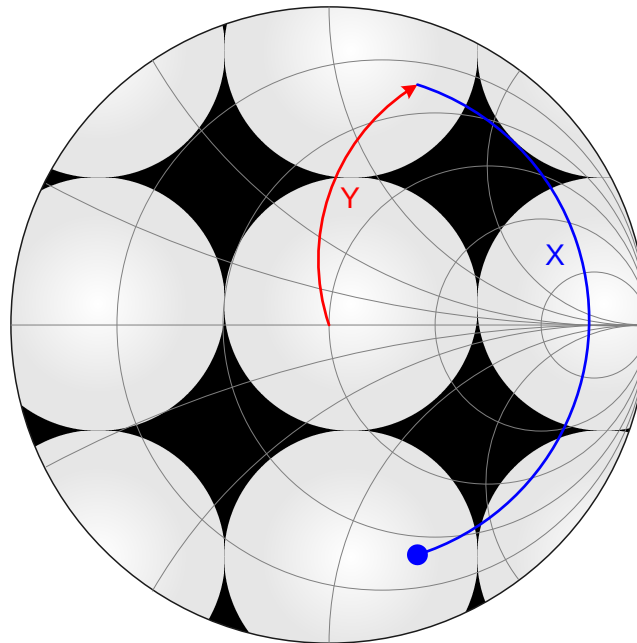
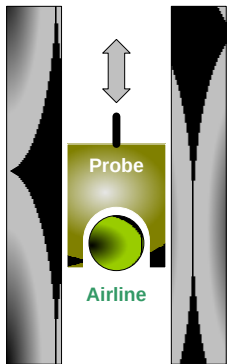
Impedances and impedance tuners



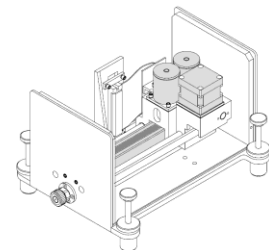
$$\Gamma_{x,n}(f_n) = \frac{a_{x,n}(f_n)}{b_{x,n}(f_n)}$$

$VSWR \propto \Gamma \propto 1/\Omega$
 10:1 VSWR = $\Gamma=0.82 = 5\Omega$
 20:1 VSWR = $\Gamma=0.9 = 2.5\Omega$

$$\Gamma = a/b$$

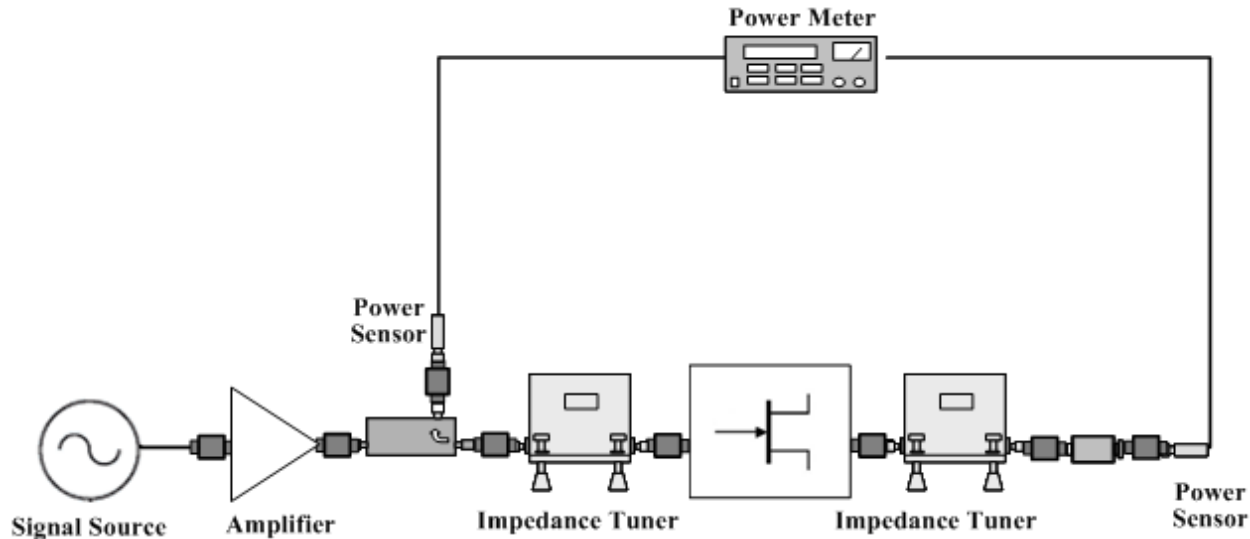


Mechanical Tuner
 Gamma comes from probe
 (slug) inserted into airline





Traditional load pull measurements



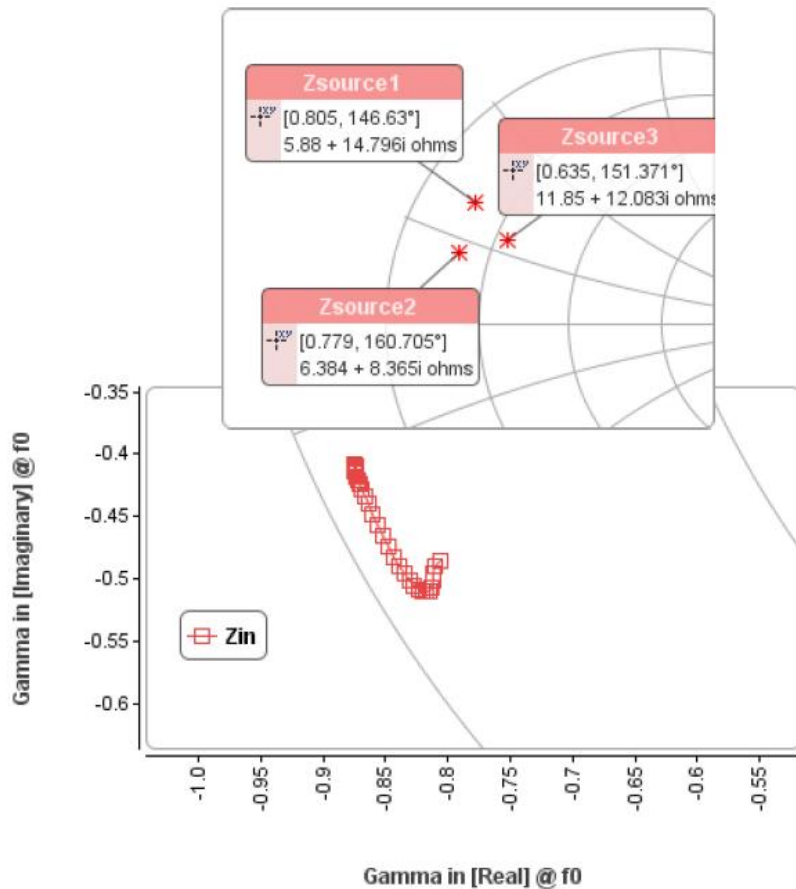
In **Traditional Load Pull**, delivered output power is calculated from Power Meter de-embedded through S-Parameter block and Impedance Tuner

Available input power is calculated from gain lookup table created during power calibration or from input Power Meter and then de-embedded through S-Parameter block and Impedance Tuner

A reflect power meter can be used to calculate delivered input power by measuring the reflected power through a reverse coupler, however the accuracy decreases as the mismatch between source impedance and device input impedance increases.



What about large-signal Z_{in} ?



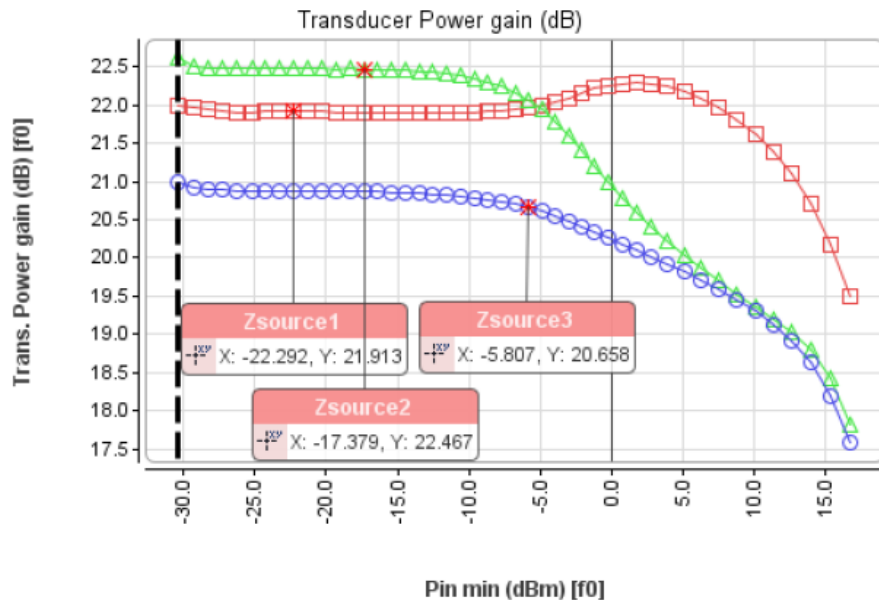
Large signal input impedance, Z_{in} , changes as function of:

- Drive power
- Zload

Traditional load pull matches source impedance at single power, not taking into account varying Z_{in} during power sweep



What about large-signal Z_{in} ?

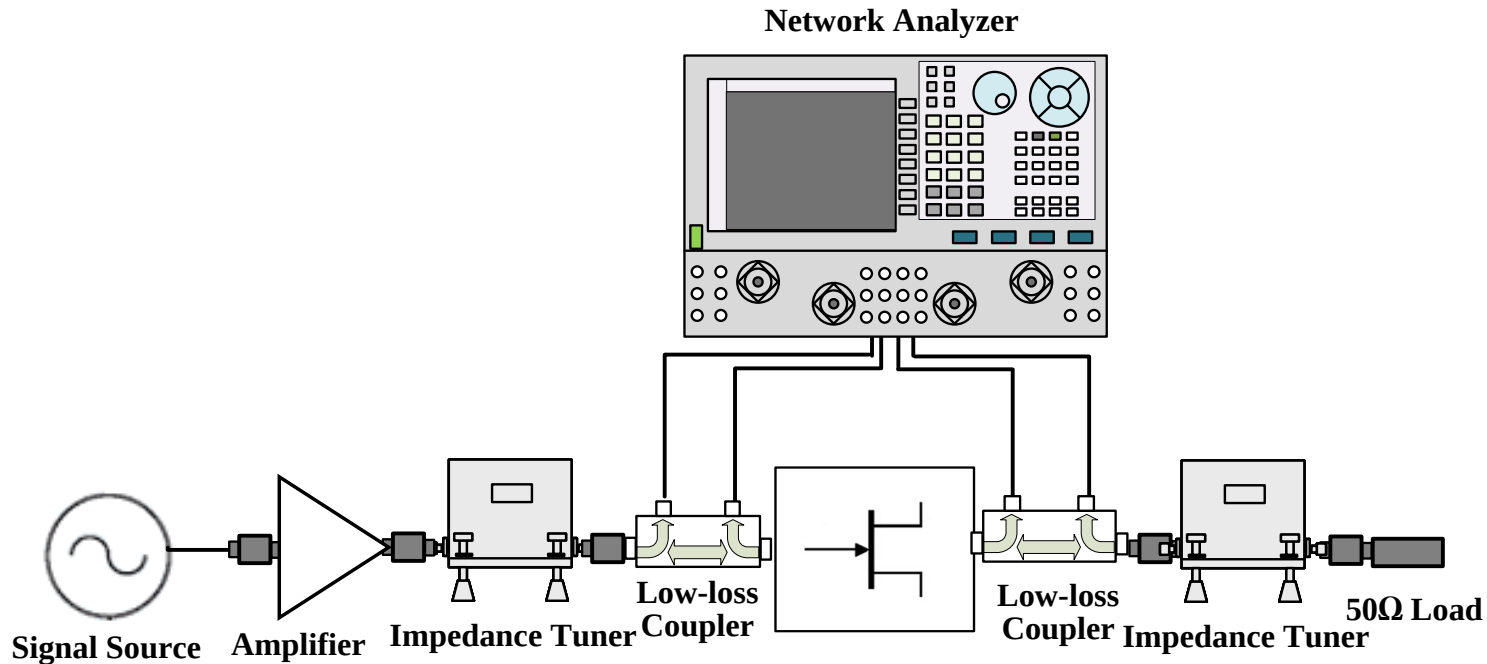


Gain values look low because only P_{in} , available is used... reflected power due to mismatch is not taken into account

Traditional load pull only reports Transducer Gain



Vector-receiver (real-time) load pull



$$P_{out} = \frac{1}{2} (|b_2|^2 - |a_2|^2) = \frac{1}{2} |b_2|^2 * (1 - |\Gamma_{load}|^2)$$

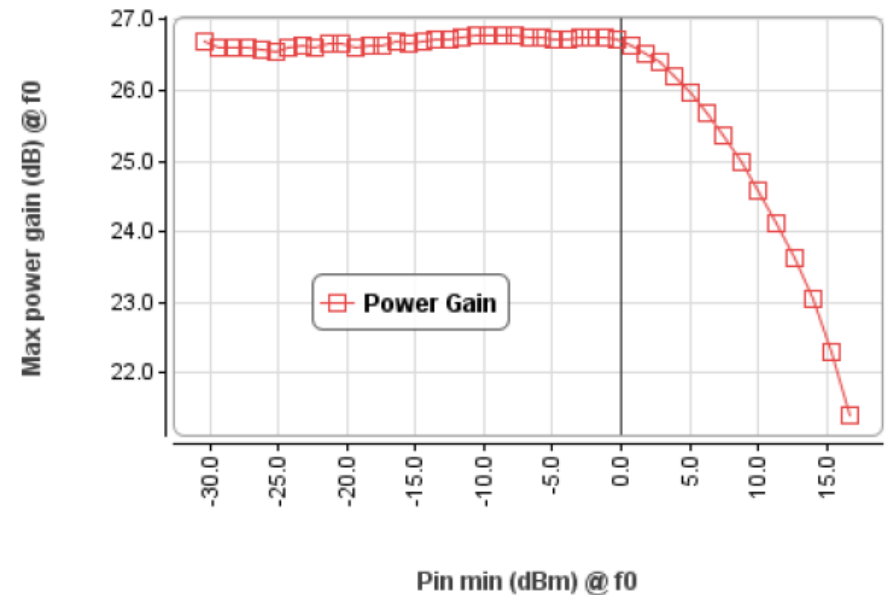
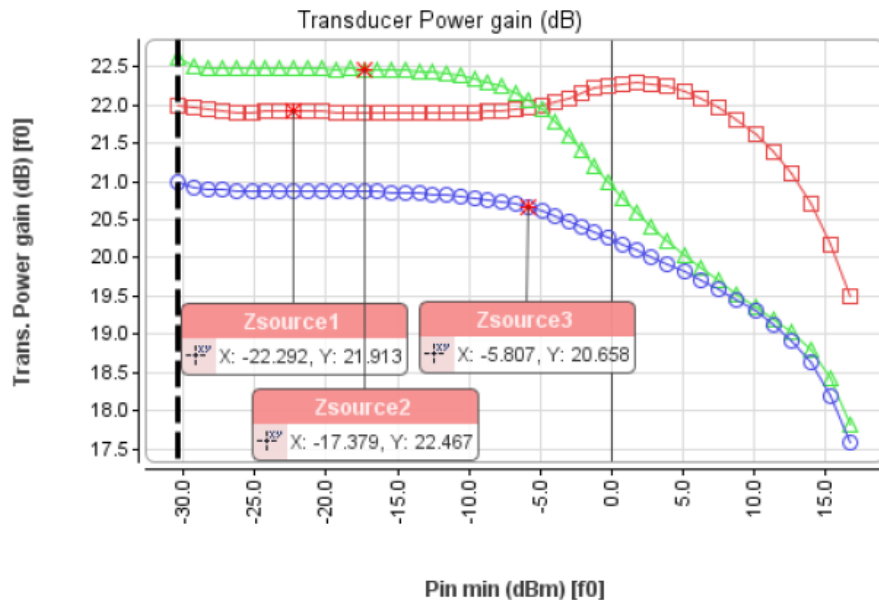
$$G_p = \frac{P_{out}}{P_{in,del}} = \frac{|b_2|^2 * (1 - |\Gamma_{load}|^2)}{|a_1|^2 * (1 - |\Gamma_{in}|^2)}$$

$$P_{in,del} = \frac{1}{2} (|a_1|^2 - |b_1|^2) = \frac{1}{2} |a_1|^2 * (1 - |\Gamma_{in}|^2)$$

$$PAE = \frac{P_{out} - P_{in,del}}{P_{DC}}$$



Vector-receiver (real-time) load pull

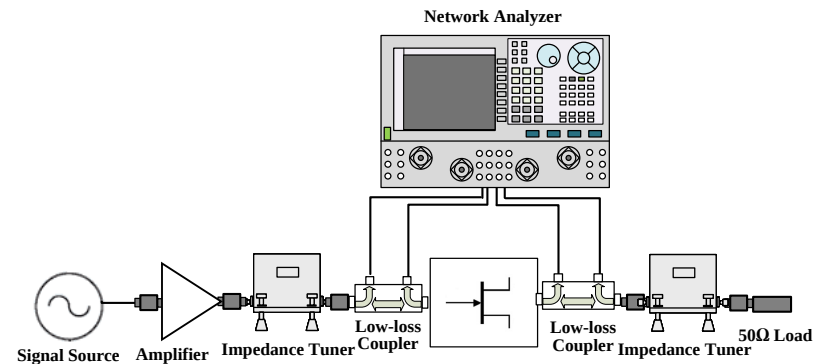
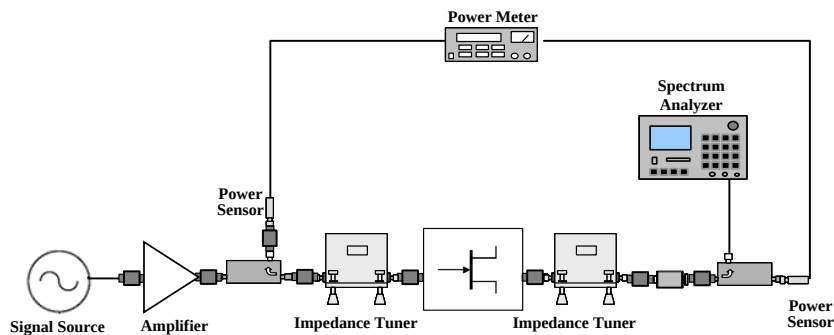


Gain values look low because only $P_{in,available}$ is used... reflected power due to mismatch is not taken into account

Knowing Z_{in} allows us to calculate Power Gain, taking into account mismatch thereby showing true gain potential of device

Vector-receiver (real-time) load pull

	Traditional LP	Vector Receiver LP
Pre-Characterization	Required	Recommended (not required)
Number of Points	More points = greater accuracy (even with interpolation)	Minimum points required (no impact on accuracy)
Tuner De-embedding	Critical! (Accuracy relies on de-embedding)	No tuner de-embedding





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Vector-receiver (real-time) load pull

	Traditional LP	Vector Receiver LP
Verification Procedure	ΔG_t complex conjugate matched verification	Zin vs. Zload comparison ΔG_t complex conjugate matched verification



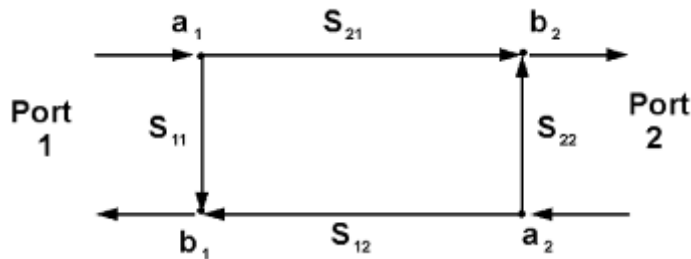
Vector-receiver (real-time) load pull

Measurement Parameter	Traditional Load Pull	Vector-Receiver Load Pull
Input Reflection Coefficient (Z_{in})	✗	✓
Available Input Power ($P_{in,avail}$)	✓	✓
Delivered Input Power ($P_{in,del}$)	✗ *	✓
Output Power (P_{out})	✓	✓
Power Gain (G_p)	✗	✓
Transducer Gain (G_t)	✓	✓
Power Added Efficiency (PAE)	✗	✓
Efficiency (Eff)	✓	✓
AM/PM	✗	✓
Calibrated Harmonic Power	Spectrum analyzer required	✓
Multi-tone Measurements	Spectrum analyzer required	✓
Modulated Measurements	Spectrum analyzer required	✗
Power Sweep Speed (for 25 power levels)	~20 seconds	~1 second

*A reflect power meter can be used to calculate delivered input power by measuring the reflected power through a reverse coupler, however the accuracy decreases as the mismatch between source impedance and device input impedance increases.



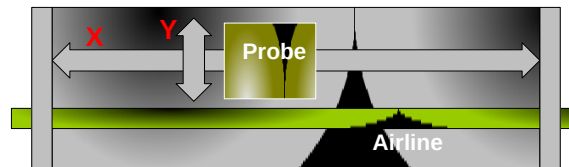
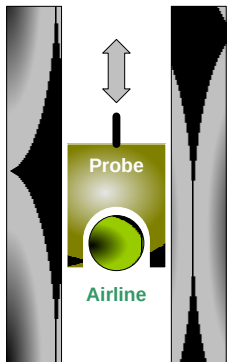
Active and hybrid-active load pull



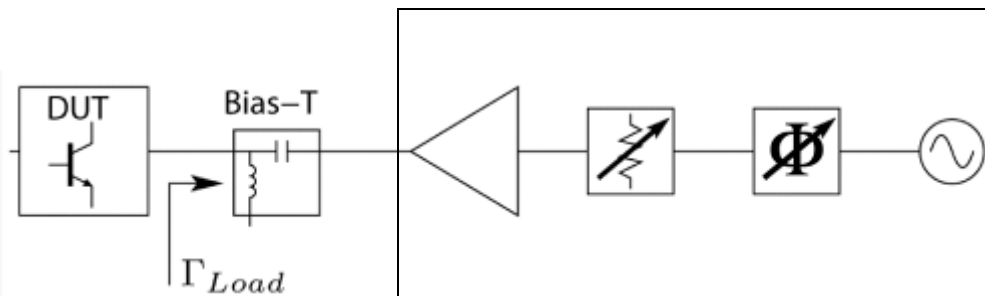
$$\Gamma_{x,n}(f_n) = \frac{a_{x,n}(f_n)}{b_{x,n}(f_n)}$$

$VSWR \propto \Gamma \propto 1/\Omega$
 10:1 VSWR = $\Gamma=0.82 = 5\Omega$
 20:1 VSWR = $\Gamma=0.9 = 2.5\Omega$

$$\Gamma = a/b$$



Mechanical Tuner
 Gamma comes from probe
 (slug) inserted into airline
 $\Gamma < 1$

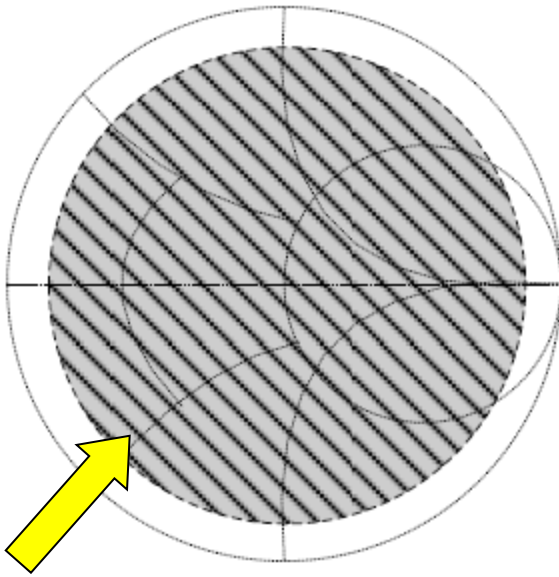


Active Tuner
 Gamma comes from signal
 generator and amplifier
 $\Gamma = 1$ or $\Gamma > 1$

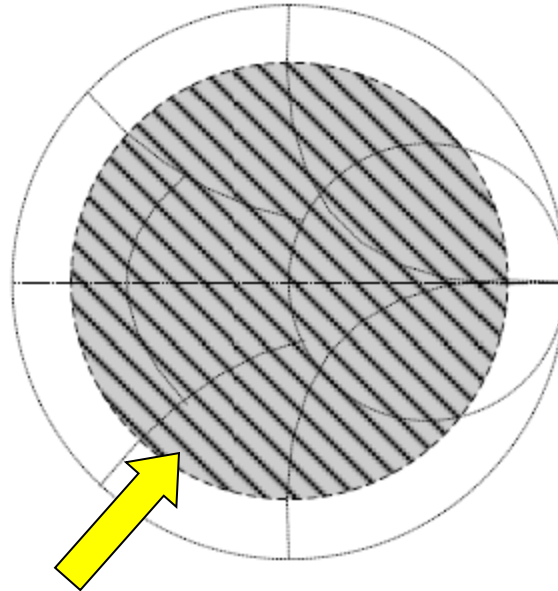


Active and hybrid-active load pull

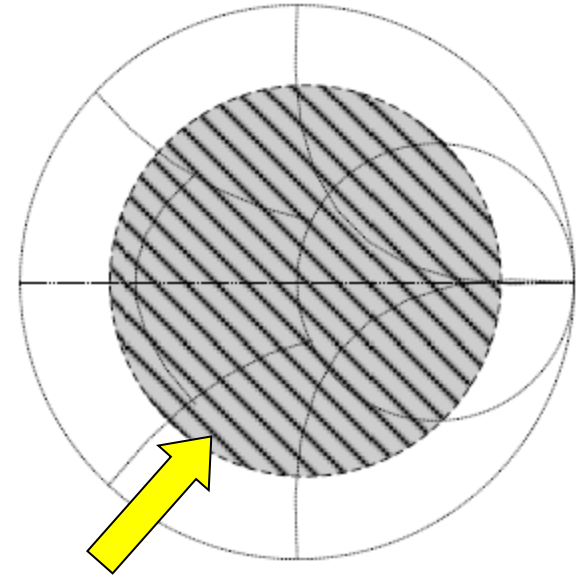
➔ Maximum Tuning Range (*exaggerated for effect*)



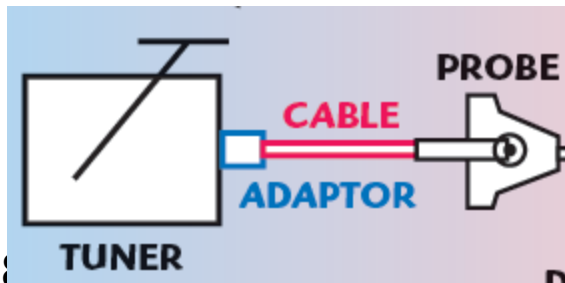
Tuner



Tuner +
Cable



Tuner + Cable + Probe



Losses of cables, probes, test fixtures reduces tuning range and cannot be overcome using traditional load pull methods



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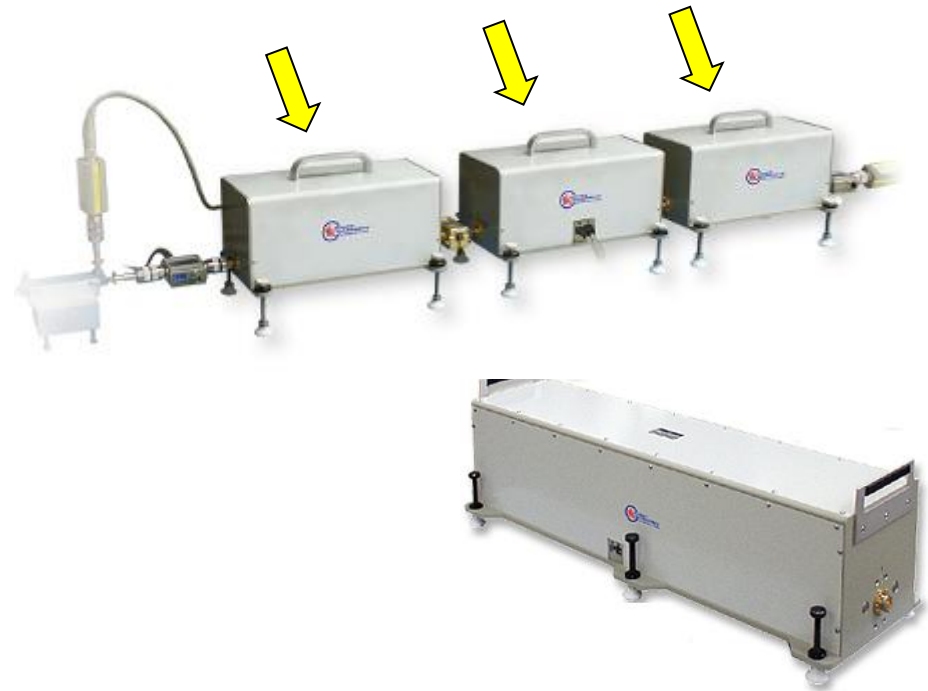
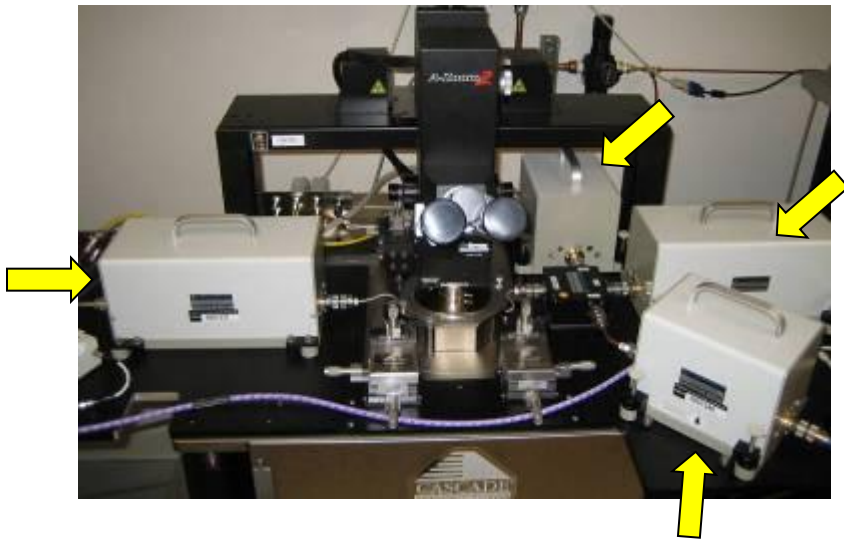
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Active and hybrid-active load pull

➔ External Tuners



For Harmonic Load Pull, Traditional Load Pull systems require one mechanical tuner per frequency per DUT side

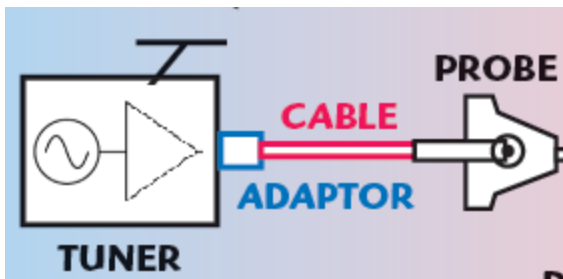
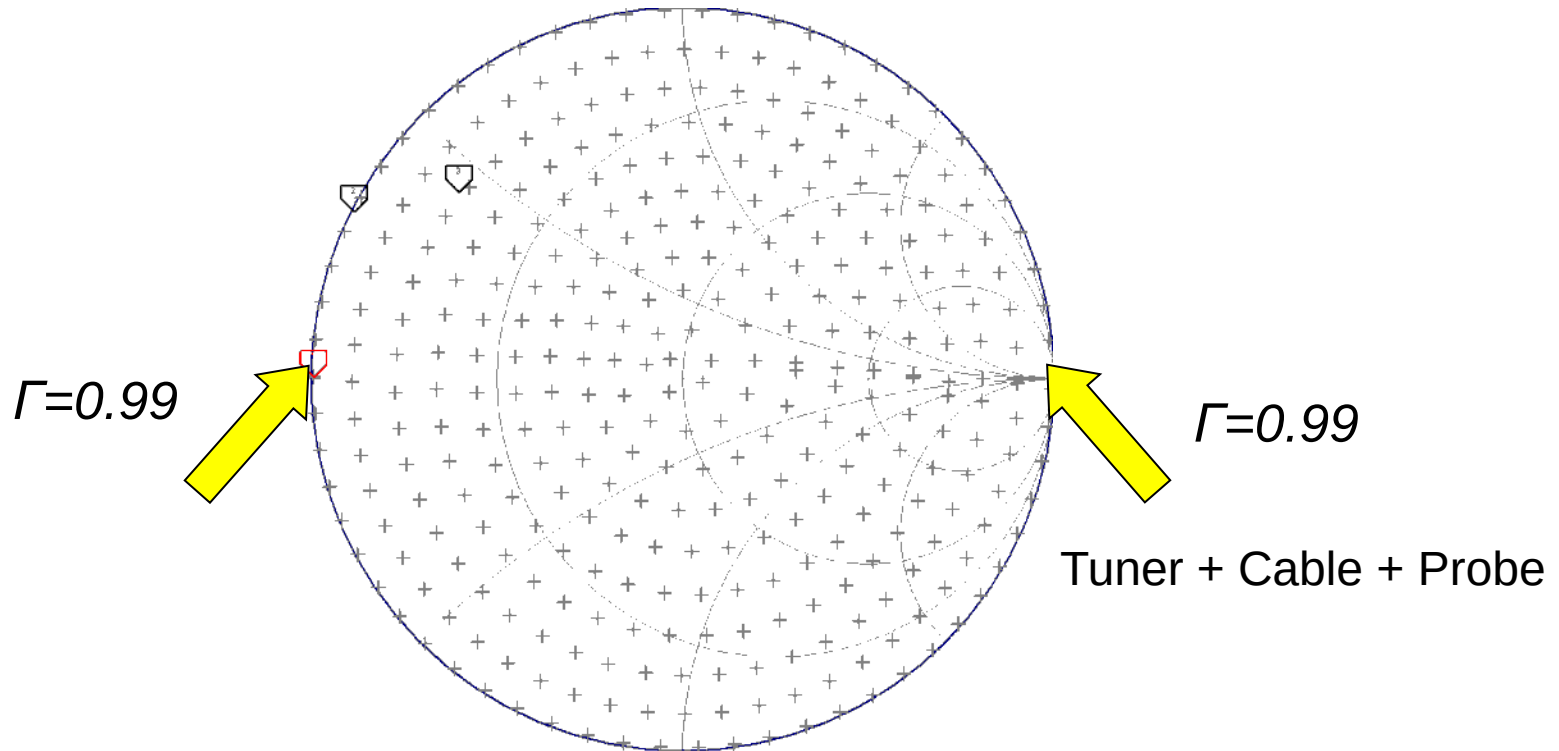
To tune F_0 , $2F_0$ and $3F_0$ at the same time requires 3 tuners (using multiplexer or cascaded methods)

It is possible to build 3 tuners in 1 box, but it becomes 2-3x longer and 2-3x more expensive

Your Calibration, Measurement & Modeling Solutions Partner!



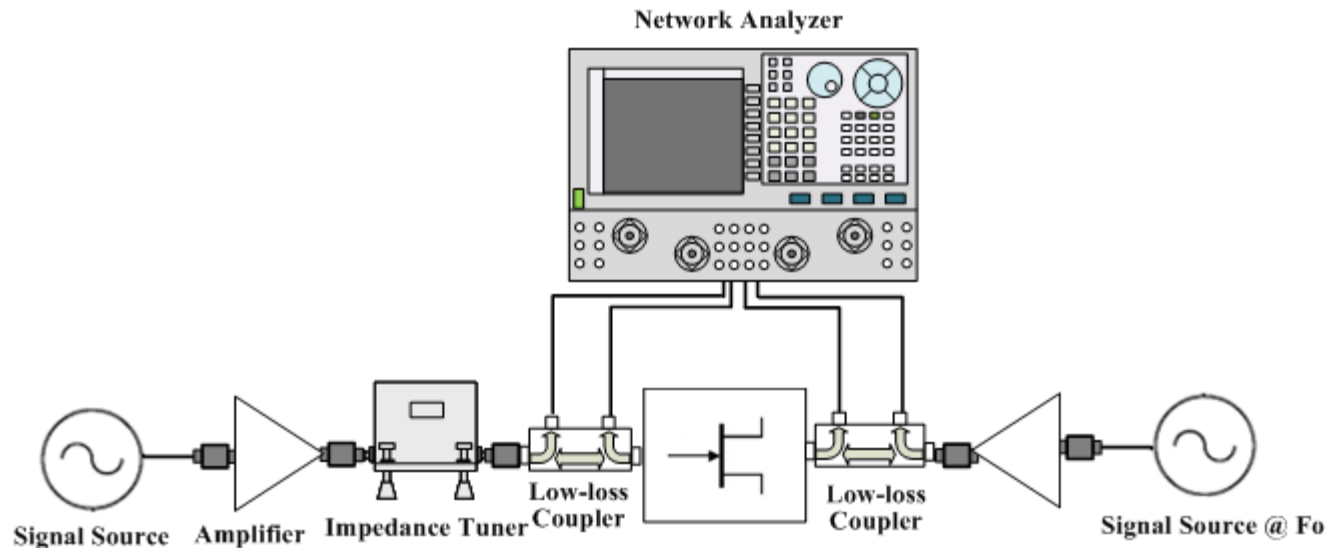
Active and hybrid-active load pull



Gamma advantage of Active Load Pull $\rightarrow$ Losses of cables, probes, test fixtures reduces tuning range, and can be overcome using larger amplifiers

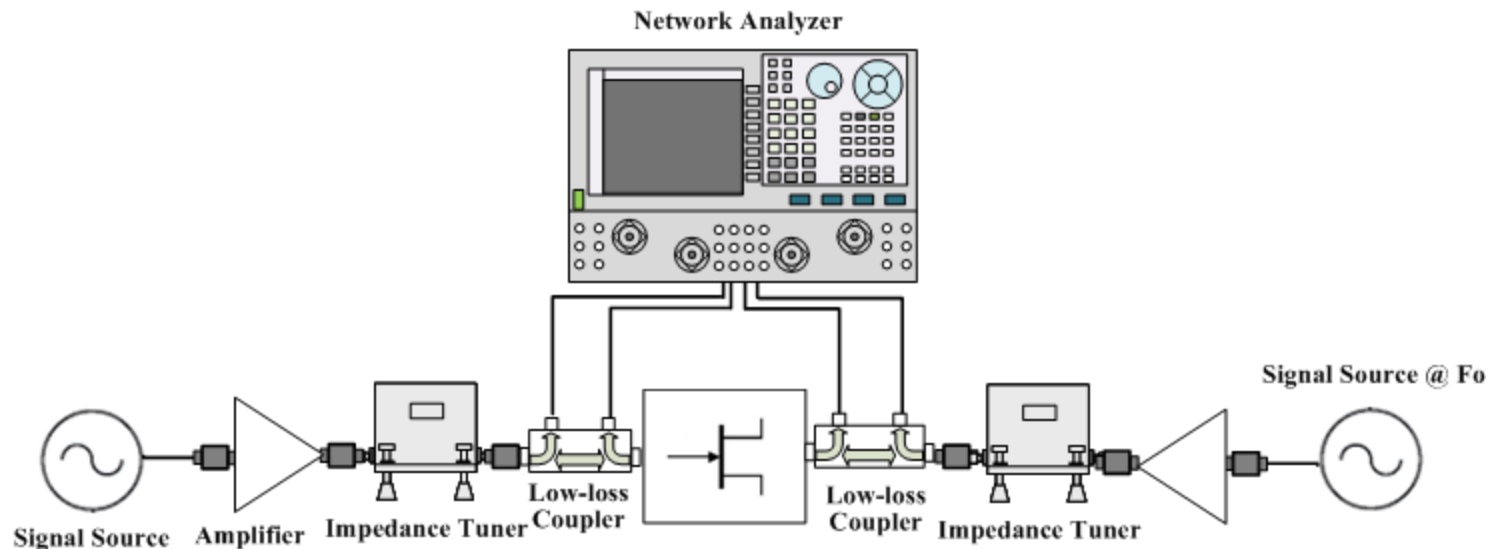
Active and hybrid-active load pull

Active Fo Load Pull



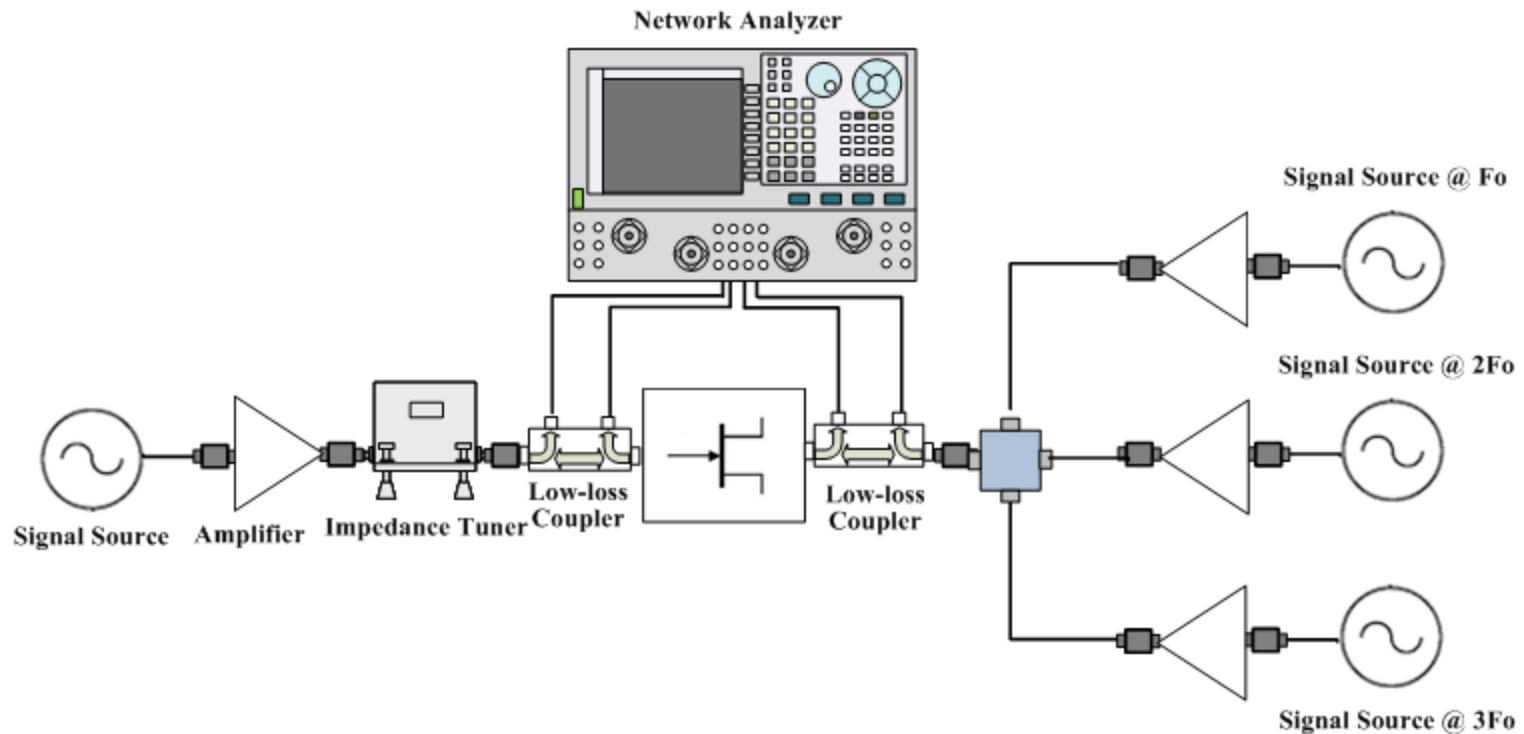
Active and hybrid-active load pull

Hybrid-Active Fo Load Pull



Active and hybrid-active load pull

Active F_o , $2F_o$, $3F_o$ Load Pull

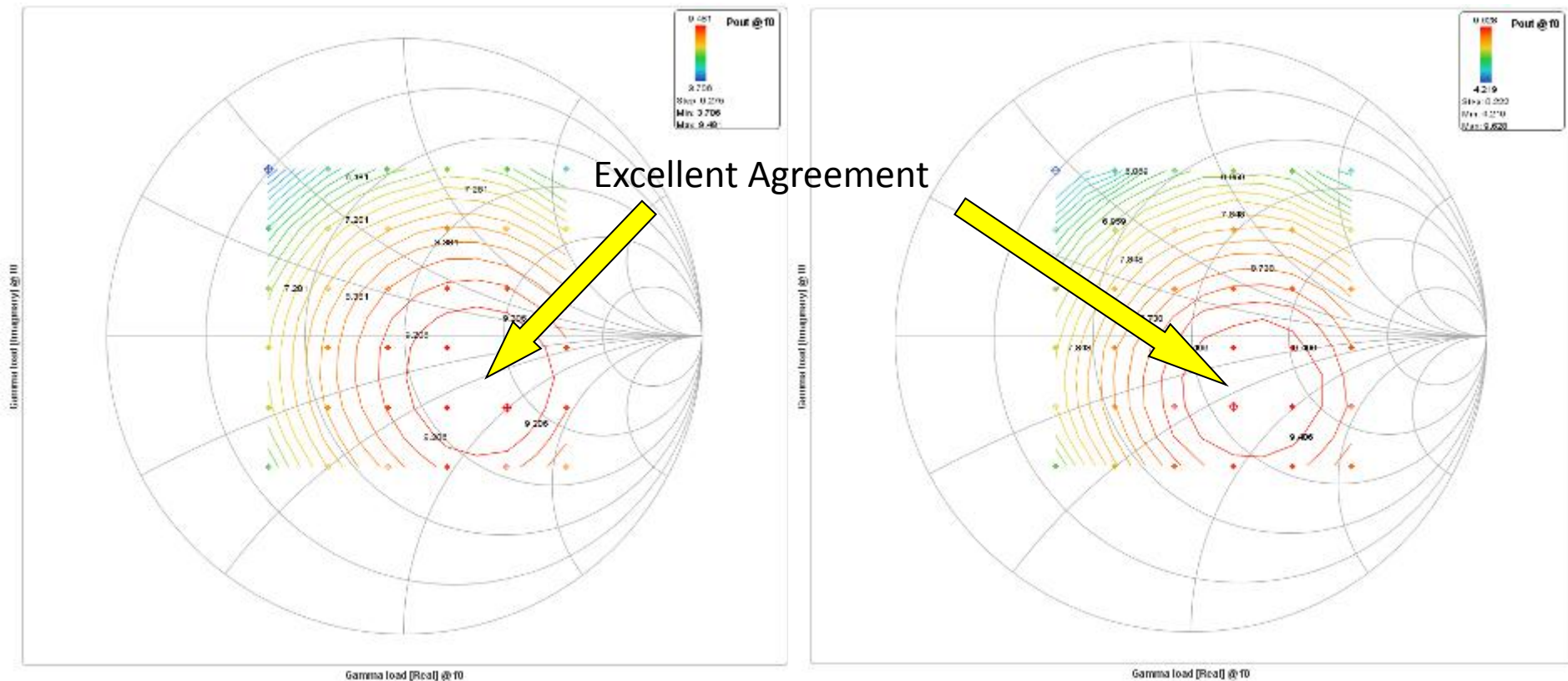


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Active and hybrid-active load pull

Measured Data – Passive VS Active

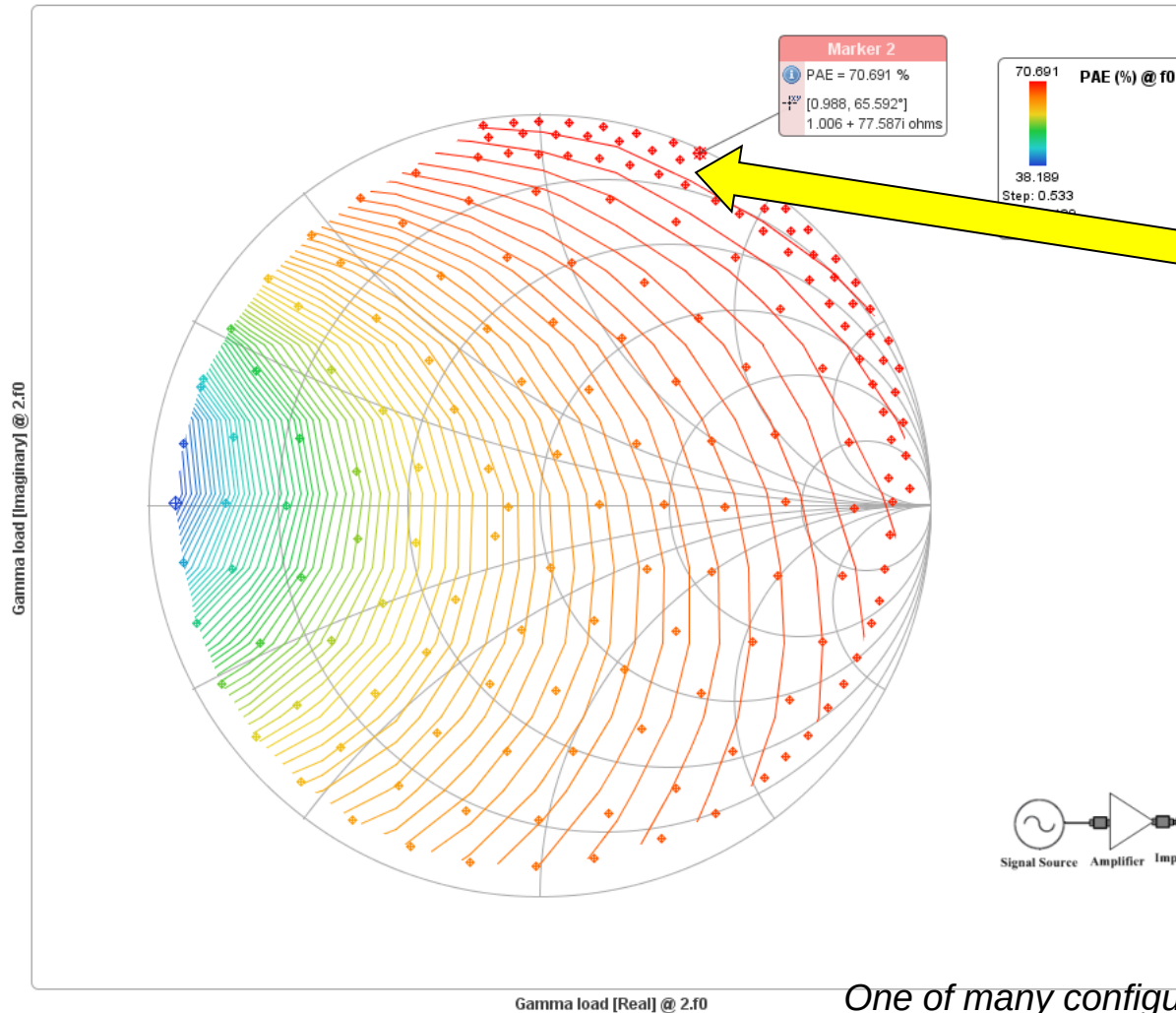


Traditional Load Pull

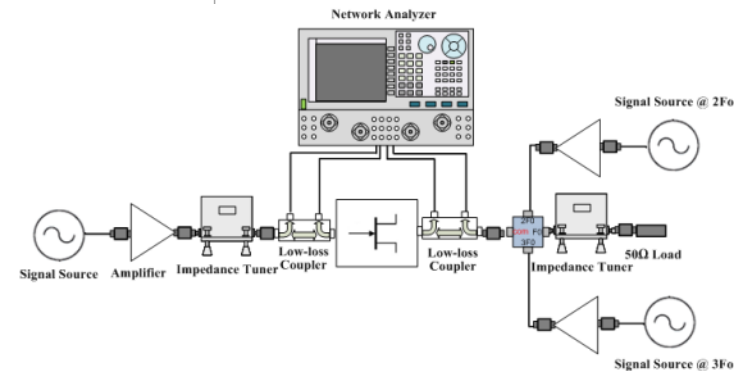
Active Load Pull



Active and hybrid-active load pull



Passive Fo
Active 2Fo, 3Fo
 $\Gamma_{2Fo} = 0.988$ @ DUT
on-wafer!



One of many configurations of hybrid/active load pull



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Vector-receiver (real-time) load pull

Load pull for model validation

Compact model → ADS

Load pull for measurements

Measurements → ADS



- Instructor Introduction
- Pulsed IV/RF and Compact Modeling
- Load Pull
- Amplifier Design in ADS
- Stability Analysis
- X-Parameters
- System-level Simulations



Which Type Are You?

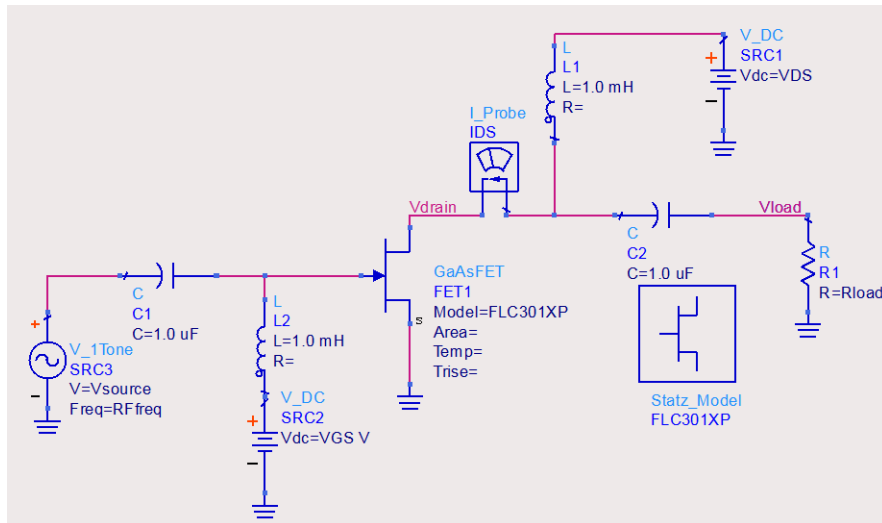
Designers usually fall into one of two camps:

Compact or X-parameter models	Measured LP data
Use any of the setups in the <i>Load Pull Design Guide</i>	Must use a “Data-based LP” component
HB	S-parameter analysis
• Can sweep • Can optimize	• Can sweep • Can optimize
A wide variety of simulations possible; <i>great</i> data displays	Good for designing matching networks

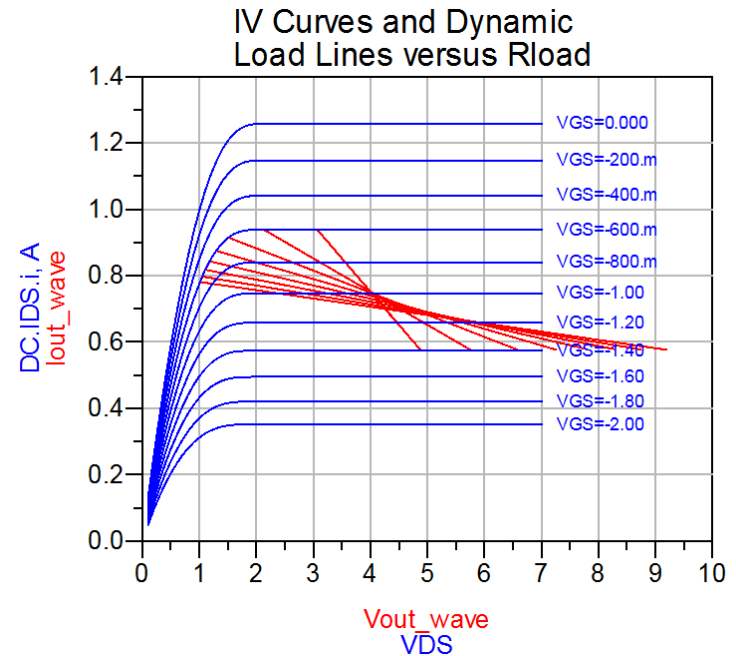
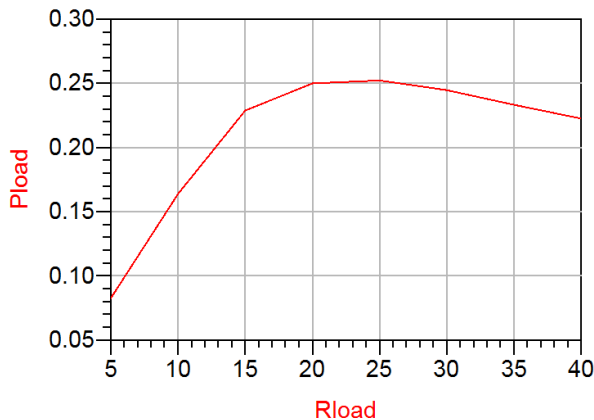
ADS is set up to handle any case.



Simple load pull – introduction to concepts



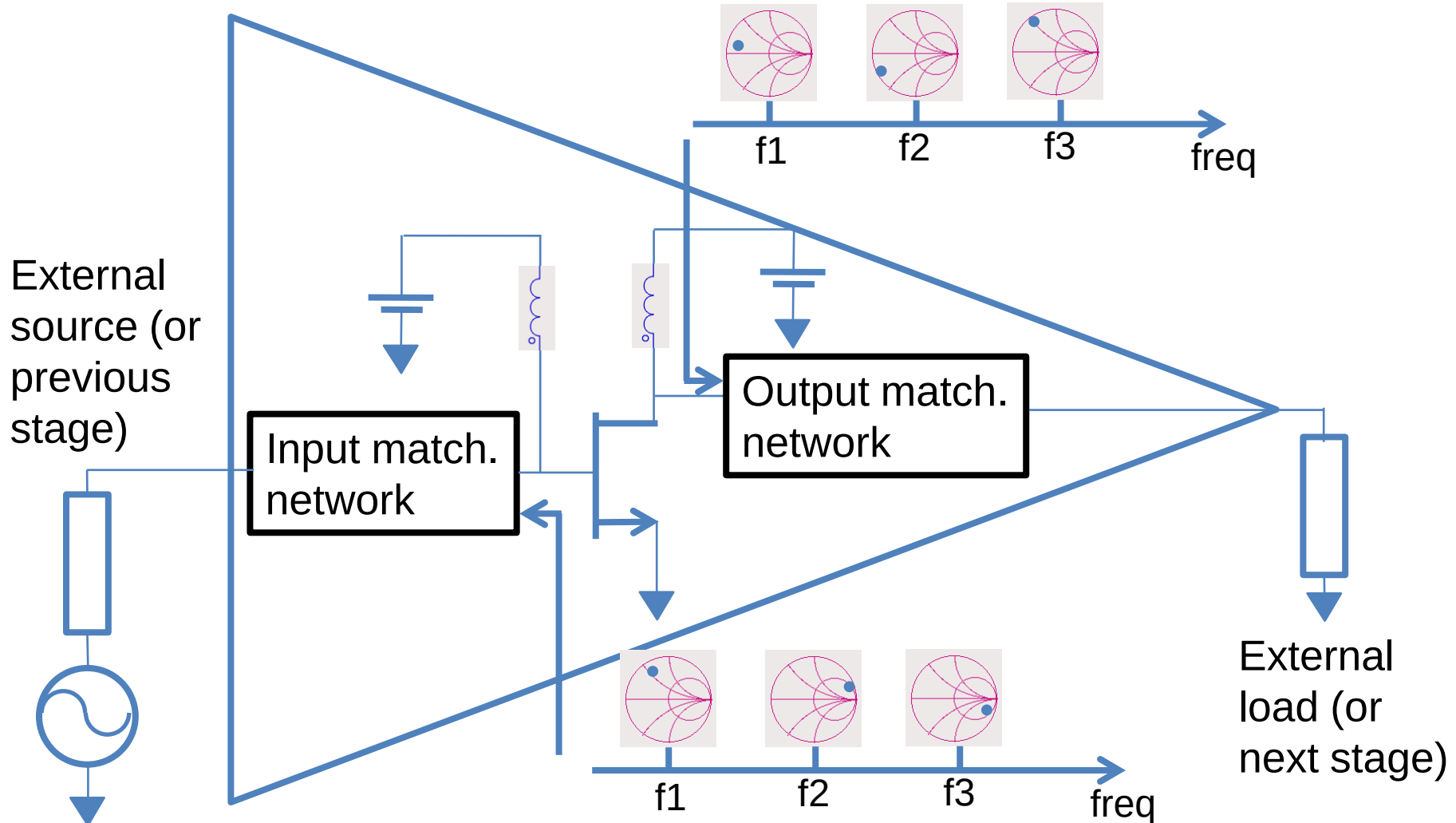
Fundamental Load Power (Watts)
versus Load Resistance



Which Impedance should I present the Device at the in- and output (over a broad frequency range to over the higher harmonics) to have a maximal Pdel, PAE and Gain with minimal distortion (XdB-compression, EVM, ACLR, etc.)?



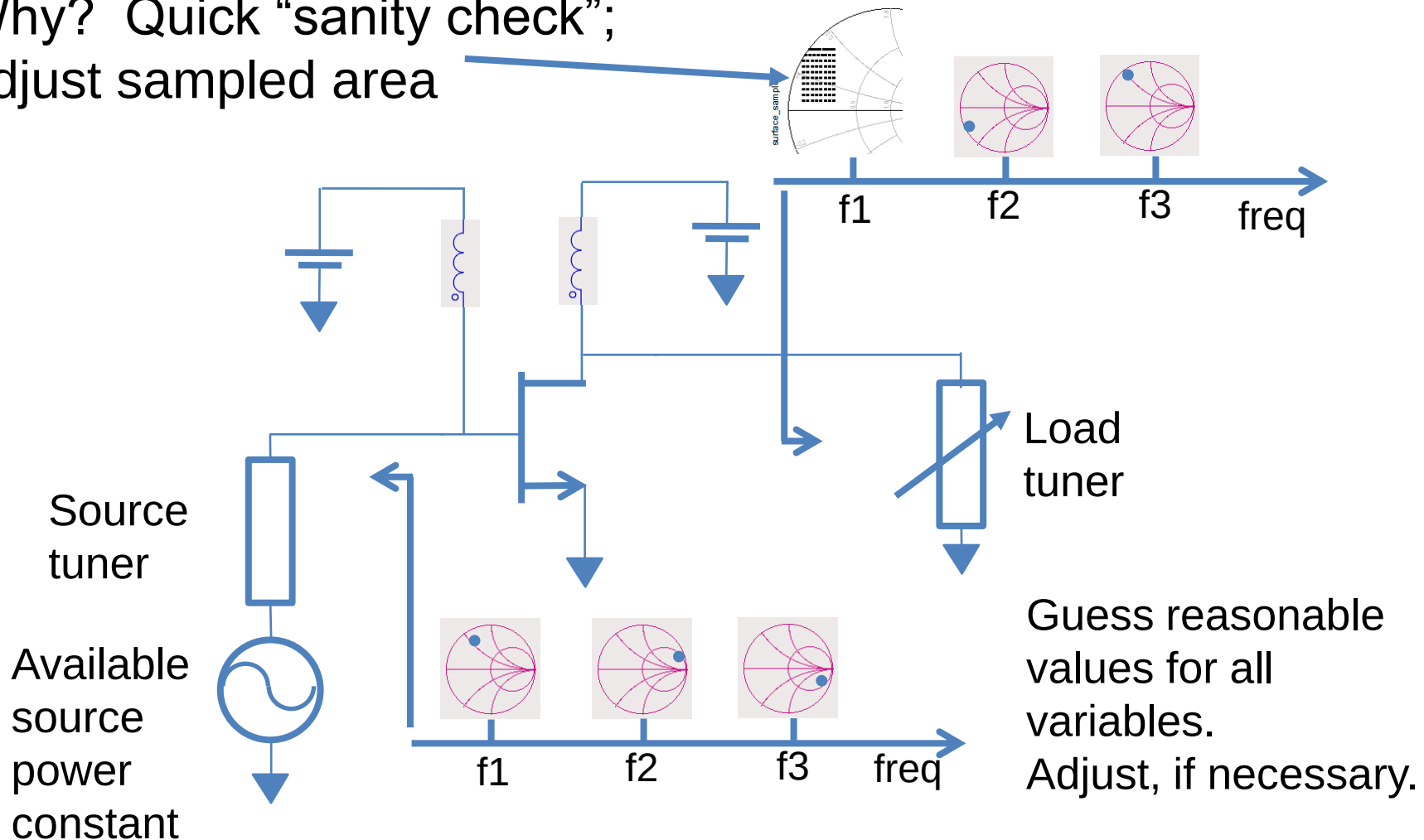
Device performance due to Z_i and Z_s





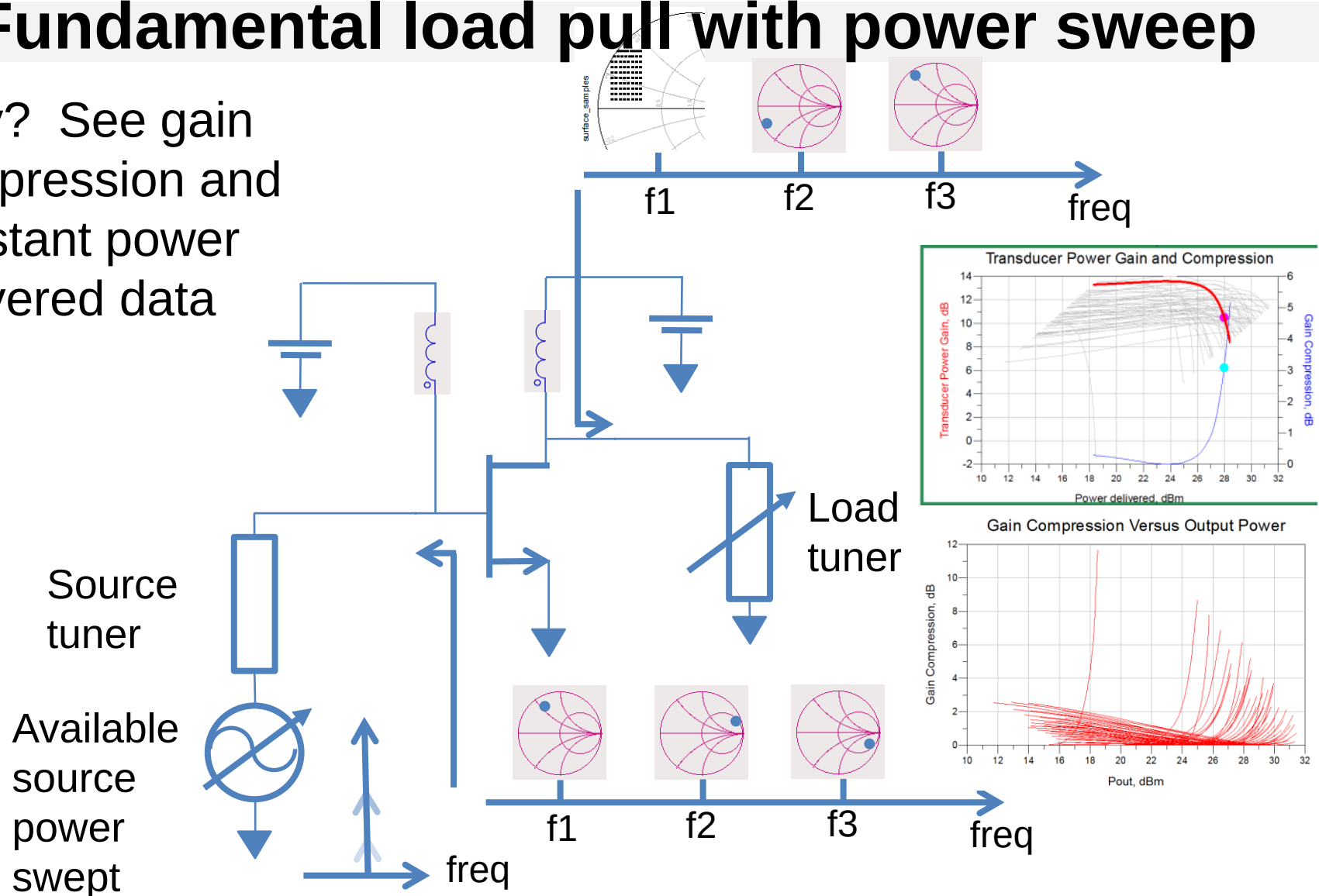
Fundamental load pull

Why? Quick “sanity check”;
adjust sampled area



Fundamental load pull with power sweep

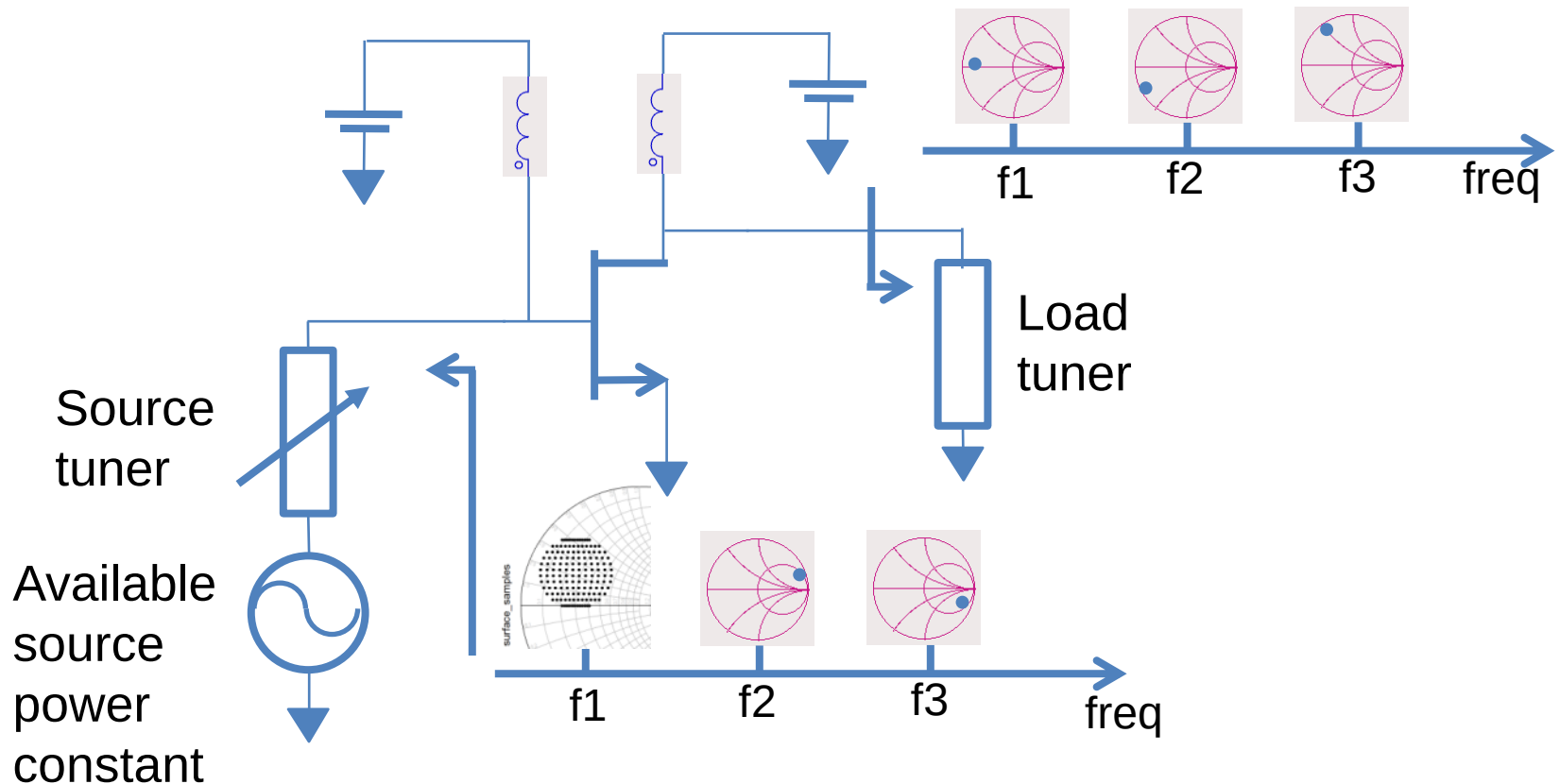
Why? See gain compression and constant power delivered data





Fundamental source pull

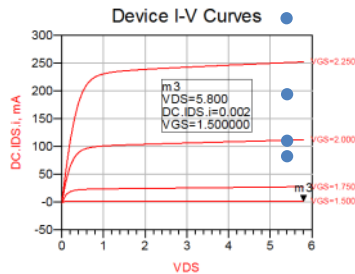
Why? Source impedances affect gain primarily, but also PAE



Fundamental load pull with parameter sweep

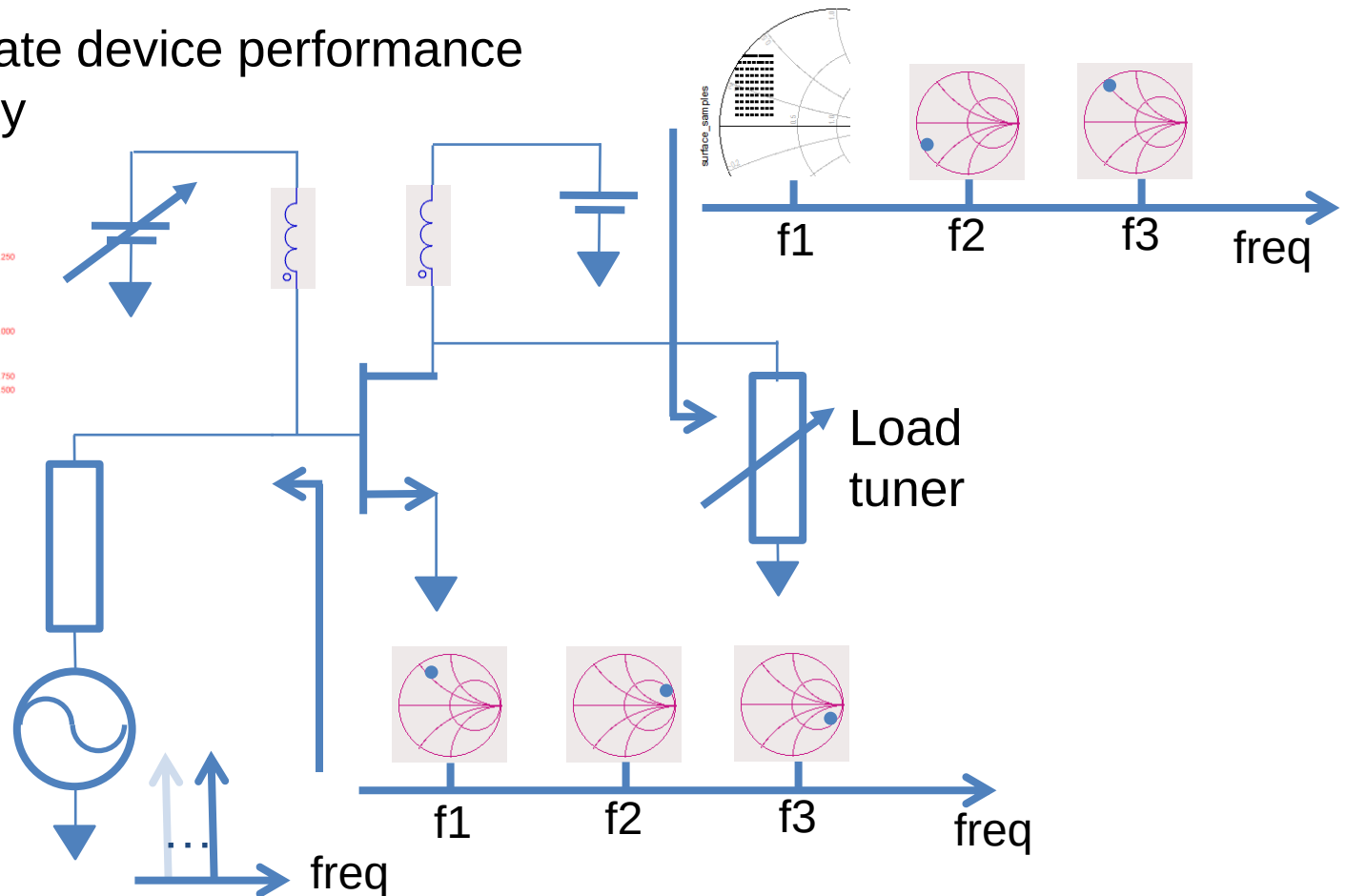
Sweep any parameter - source frequency, bias, stability network parameter values, etc.

Why? Investigate device performance more thoroughly



Source tuner

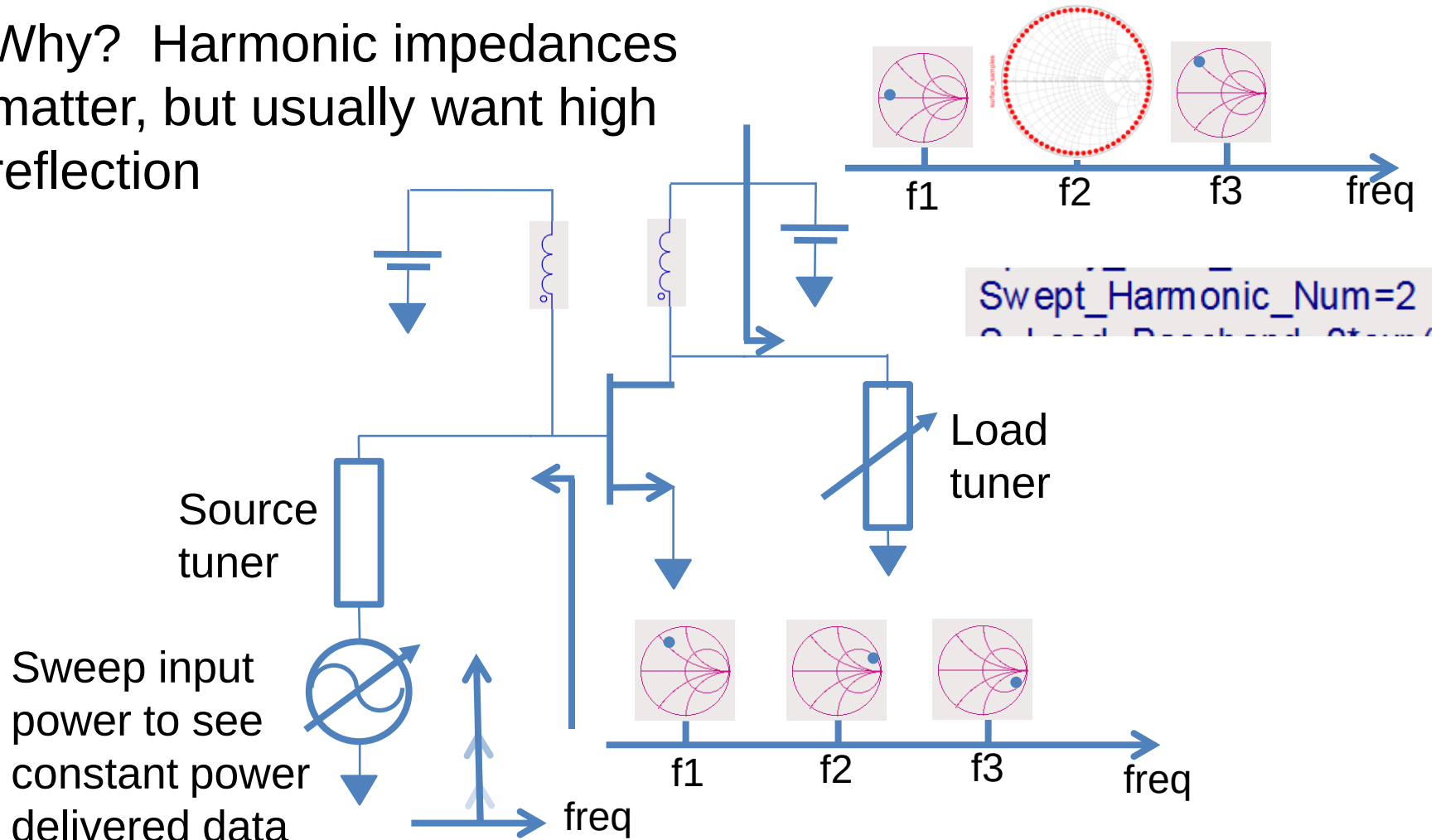
Available source power constant





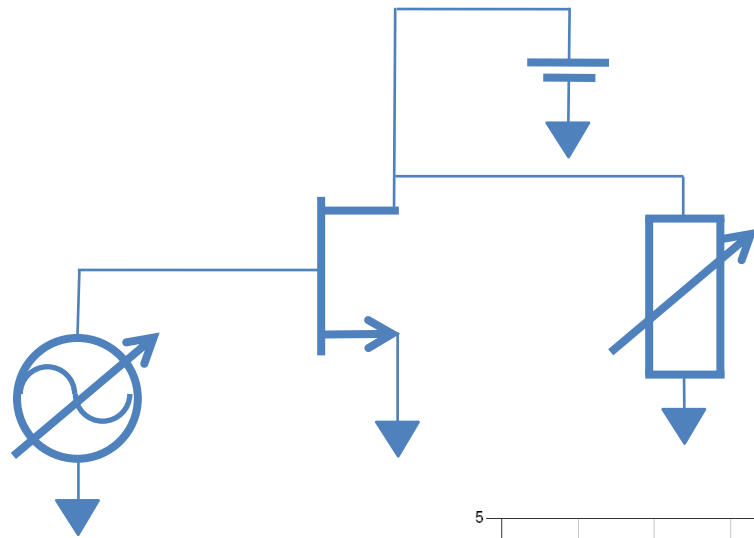
Harmonic load phase sweep

Why? Harmonic impedances matter, but usually want high reflection

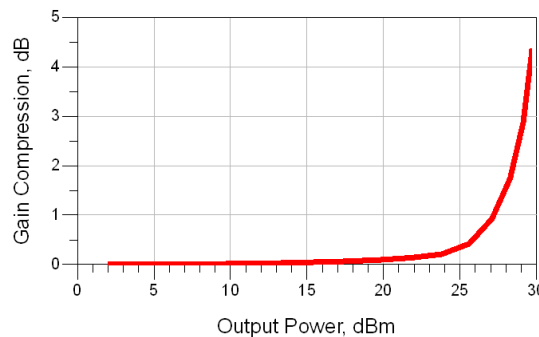




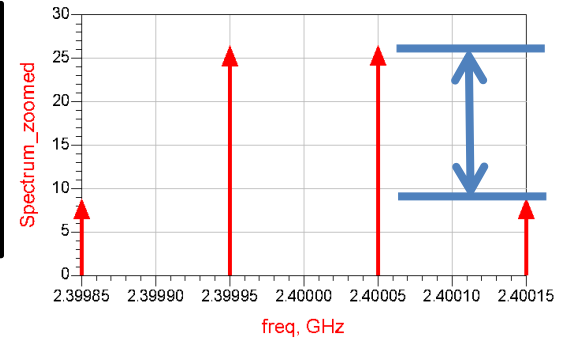
Source stimulus responses



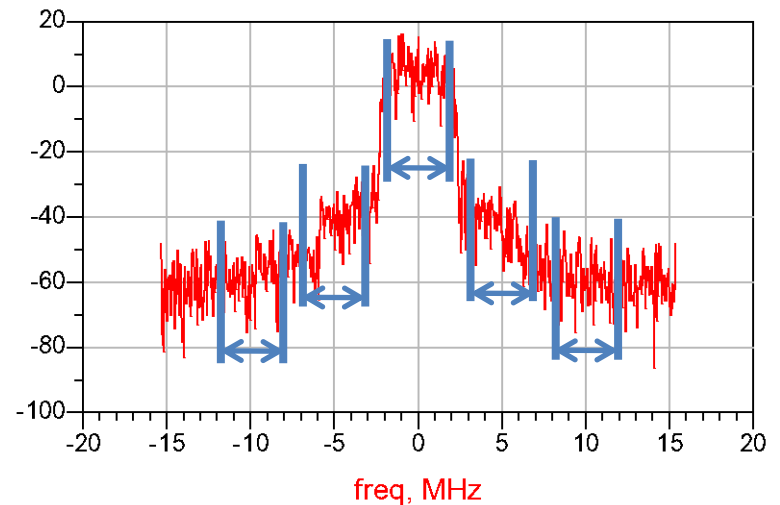
Gain comp.
curves from
source power
sweep



IMD from
2-tone
source



ACLR from
modulated source





Amplifier design in ADS

What is available for the non-linear device?

Model → run load pull simulations to determine optimal matching and biasing conditions for amplifier design

Measured Load Pull Data → analyze measured data and determine optimal matching and biasing conditions for amplifier design



Start with fast, simple load pull

Most parameters are passed to tuner inside “instrument” subcircuit

One Tone Load Pull Simulation; output power and PAE found at each fundamental or harmonic load

Load_Pull_Instrument1_r1

X1

V_Bias1=2 V

V_Bias2=5.8 V

RF_Freq=850 MHz

Pavs_dBm=15

Z0=50+j*0

Specify_Load_Center_S=1

Sweep_Rectangular_Region=1

Swept_Harmonic_Num=1

S_Load_Baseband=0*exp(j*0*pi)

S_Load_Center_Fund=0.6*exp(j*0.85*pi)

S_Load_Center_2nd=1*exp(j*0*pi)

S_Load_Center_3rd=1*exp(j*0*pi)

S_Load_Radius=0.3

S_imag_min=0.1

S_imag_max=0.6

S_imag_num_pts=10

S_real_min=-0.85

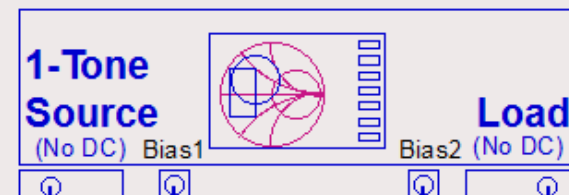
S_real_max=-0.55

S_real_num_pts=8

Z_Source_Fund=5+j*0

Z_Source_2nd=1000

Load Pull Instrument 1



Device Model
from Design Kit

HP_MOS

HPMOS1

Model=hpm os

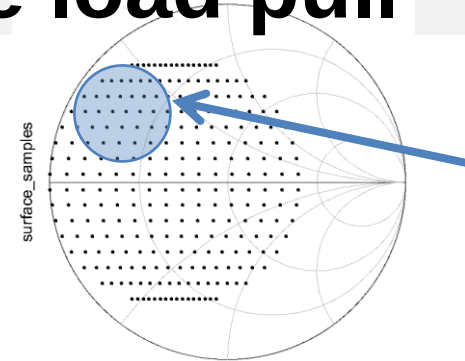
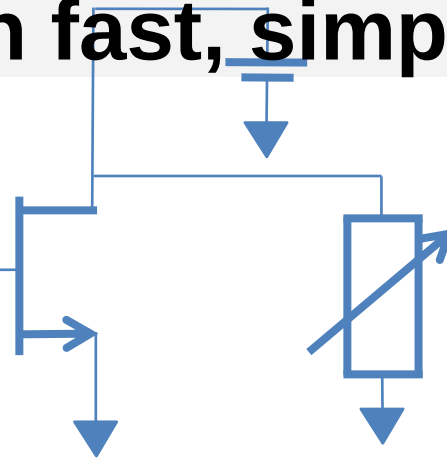
Wtot=(704e-6)*cells

N=8*cells

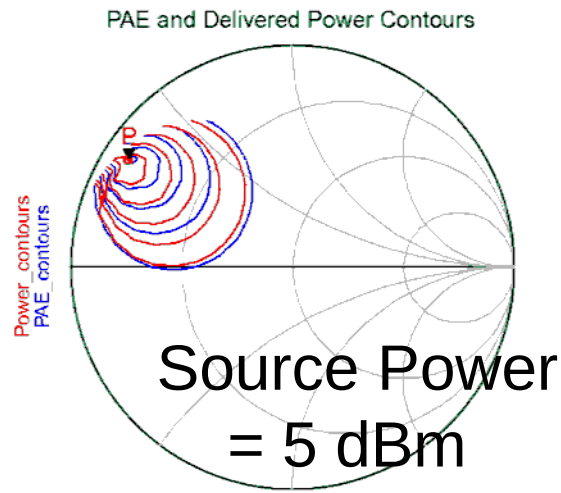


Start with fast, simple load pull

- Available source power held constant
- Guess optimal Z_{source} and harmonic Z_s



Refine sample space



(PAE, %) level=25.417926, number=34
(Power Delivered, dBm) level=22.510307, number=31

Maximum Power and contour levels, dBm:

22.54

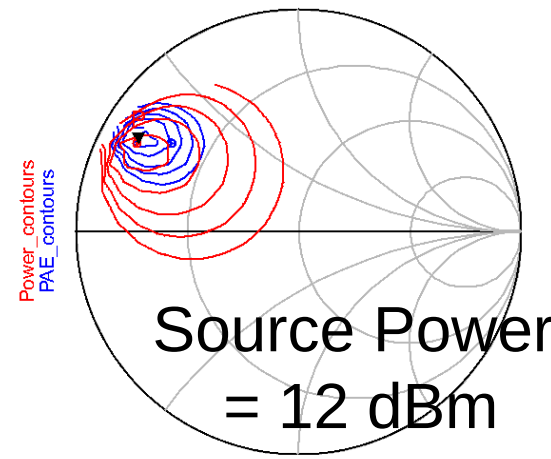
22.00
21.00
20.00
19.00
18.00

Maximum PAE and contour levels, %:

25.518

24.000
20.000
16.000
12.000
8.000

PAE and Delivered Power Contours



(PAE, %) level=58.308996, number=34
(Power Delivered, dBm) level=27.504476, number=3

Maximum Power and contour levels, dBm:

27.53

27.00
26.00
25.00
24.00
23.00

Maximum PAE and contour levels, %:

58.409

56.000
52.000
48.000
44.000
40.000



Load pull with power sweep

Load_Pull_Instrument_PSweep1_r1

X1

V_Bias1=1.75 V

V_Bias2=5.8 V

RF_Freq=850 MHz

Pavs_dBm_Start1=5

Pavs_dBm_Stop1=10

Pavs_dBm_Step1=1

Pavs_dBm_Start2=11

Pavs_dBm_Stop2=20

Pavs_dBm_Step2=0.5

Z0=50+j*0

Specify_Load_Center_S=yes

Sweep_Rectangular_Region=no

Swept_Harmonic_Num=1

S_Load_Baseband=0*exp(j*0*pi)

S_Load_Center_Fund=0.75*exp(j*0.85*pi)

S_Load_Center_2nd=1*exp(j*0*pi)

S_Load_Center_3rd=1*exp(j*0*pi)

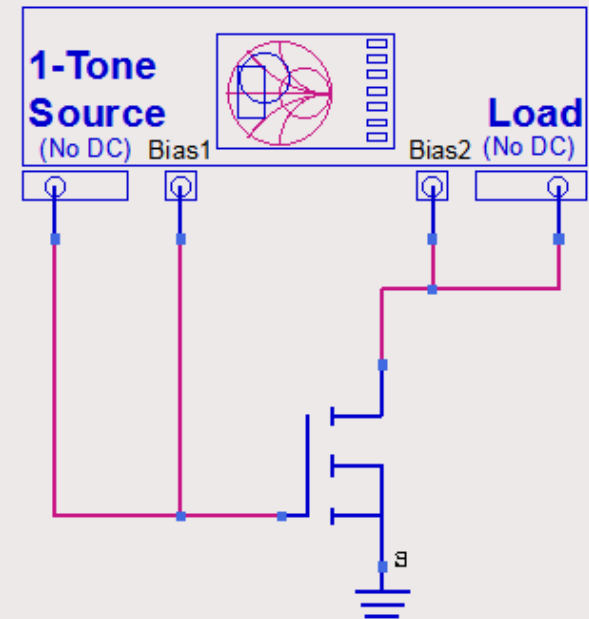
S_Load_Radius=0.2

Num_Points=100

Z_Source_Fund=10+j*0

Z_Source_2nd=1000

Load Pull Instrument
w/ Source Power Sweep





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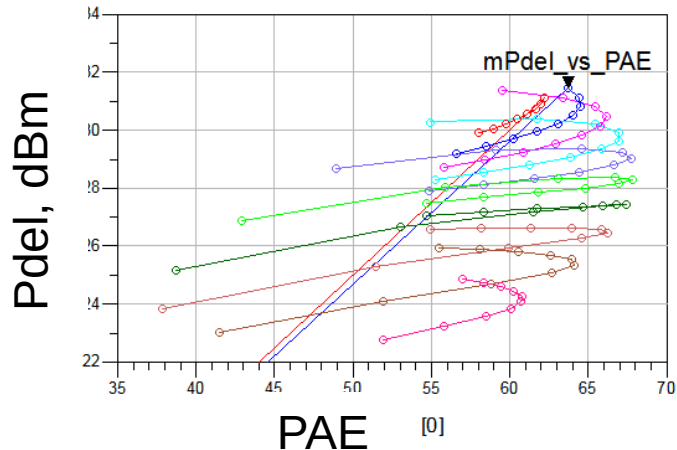


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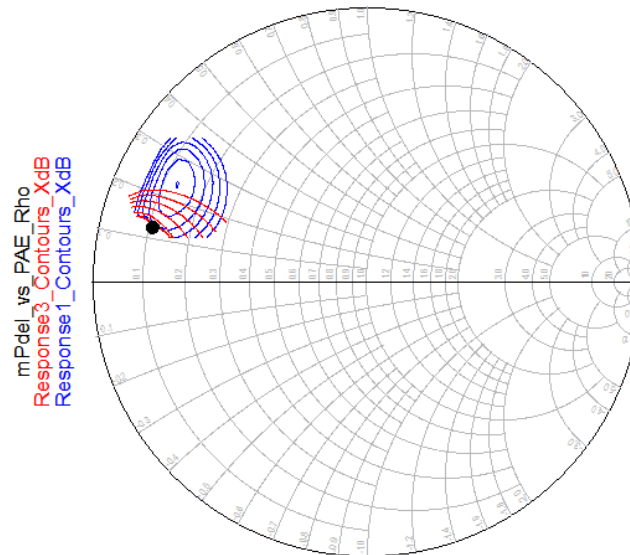
Select load for highest Pdel or highest PAE

Data and Contours at 1.000 dB Gain Compression

Choose desired point on power-delivered-versus-power-added-efficiency curves. Tabular data, black dot on Smith Chart and traces below will be updated.



Reference Z0 = 50.000 / 0.000



Maximum PAE
and contour
levels, %:

67.742
66.000
64.000
62.000
60.000
58.000

Maximum Power
Delivered and
contour levels, dBm:

31.415
31.000
30.500
30.000
29.500
29.000

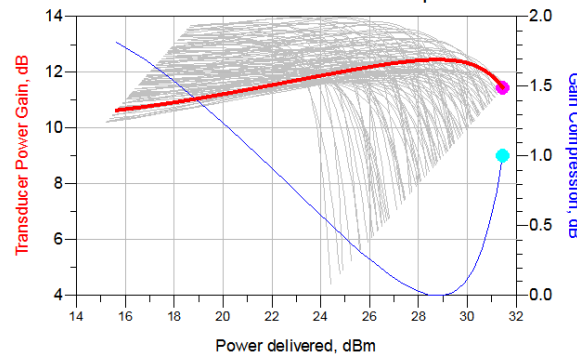
Data corresponding to marker mPdel_vs_PAE:

Zload_at_mPdel_vs_PAE	mPdel_vs_PAE_Rho
5.539 + j6.218	0.803 / 165.651

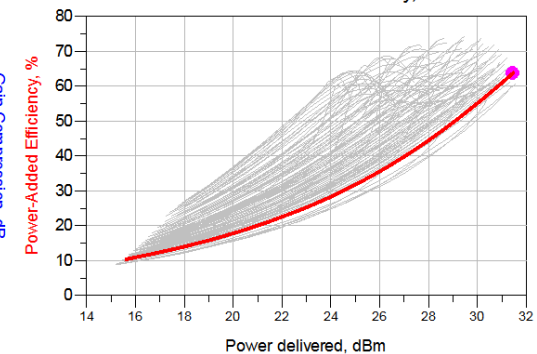
PAE_at_mPdel_vs_PAE	GainComp_at_mPdel_vs_PAE
63.731	1.001

Pdel_dBm_at_mPdel_vs_PAE	Gain_at_mPdel_vs_PAE
31.446	11.456

Transducer Power Gain and Compression

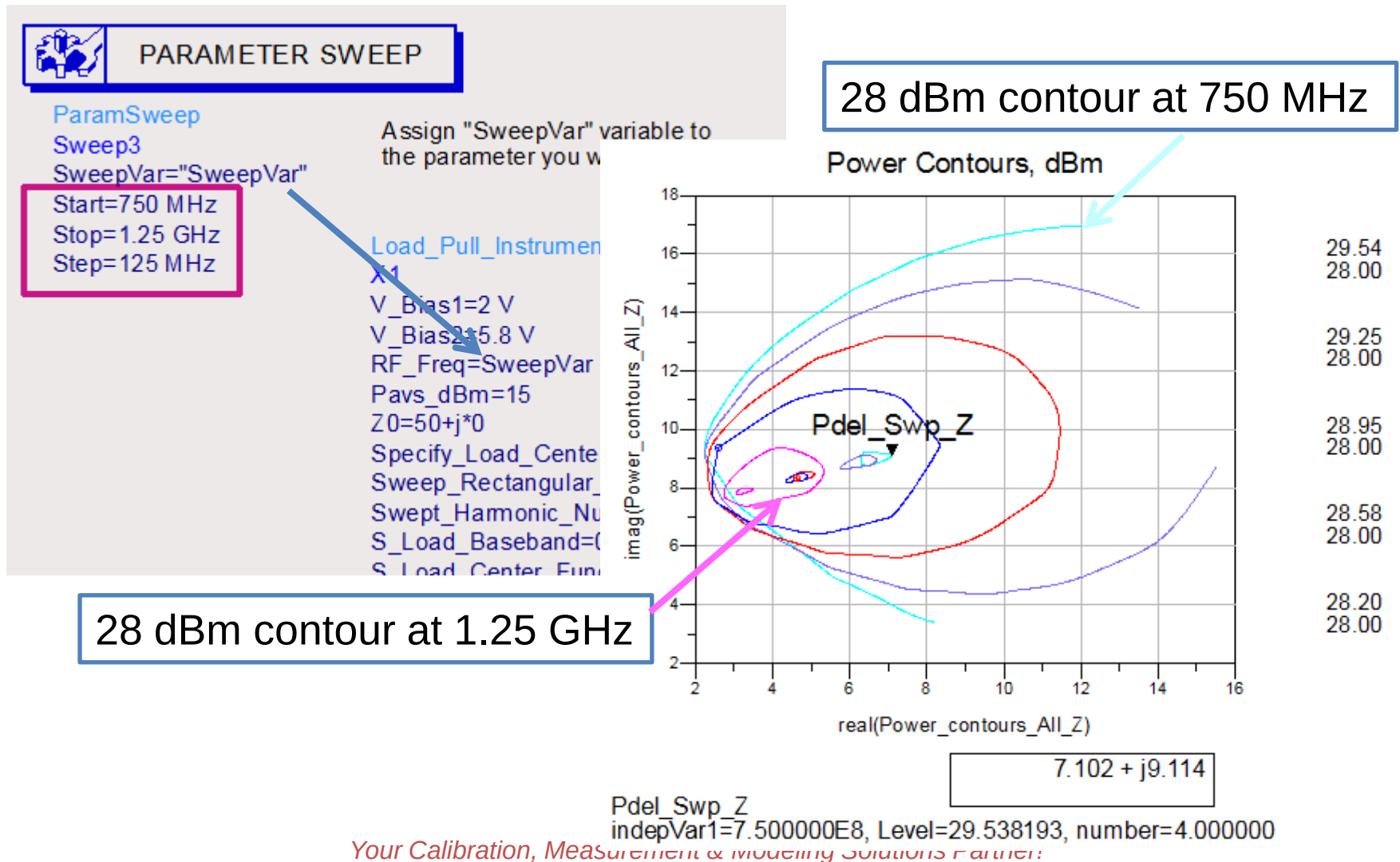


Power-Added Efficiency, %



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Contours versus swept parameter (frequency)



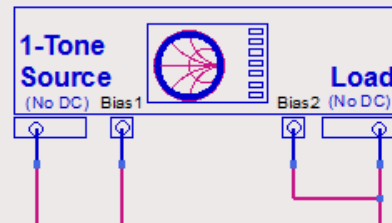
Dependency on phase of gamma at harmonic

Harmonic_Load_Phase_Sweep_Instrument

X6
 V_Bias1=2 V
 V_Bias2=5.8 V
 RF_Freq=850 MHz
 Pavs_dBm=17
 Rho_Phase_Degrees_Start=0
 Rho_Phase_Degrees_Stop=355
 Rho_Phase_Degrees_Step=5
 Z0=50+j*0
 Specify_Load_S=1
 Swept_Harmonic_Num=2
 S_Load_Baseband=0*exp(j*0*pi)
 S_Load_Fund=0.6*exp(j*0.85*pi)
 S_Load_2nd=0.95*exp(j*135*pi/180)
 S_Load_3rd=0.95*exp(j*0*pi/180)
 Rho_Mag=0.95

Z_Source_Fund=5+j*0
 Z_Source_2nd=1000

Load Phase Sweep Instrument



Bias 1 Voltage = 2.000
 Bias 2 Voltage = 5.800
 Fundamental RF Frequency = 850000000.000
 Swept Load Harmonic Number = 2.000
 Available Source Power = 17.000 dBm

Data Display: HB1Tone_Load_Harmonic_Phase_Sweep
 Dataset Date/Time: Aug 09, 2011 09:37:15 AM
 Dataset: HB1Tone_Load_Harmonic_Phase_Sweep

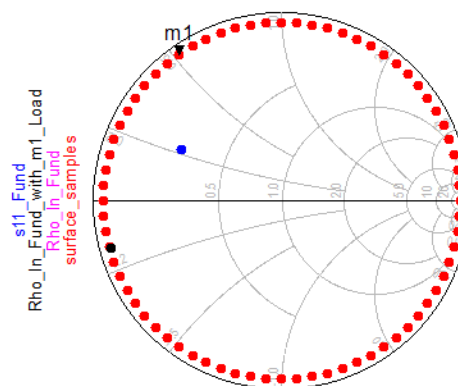
Fundamental Source Z

5.000 + j0.000

Fundamental Load Z

13.173 + j11.213

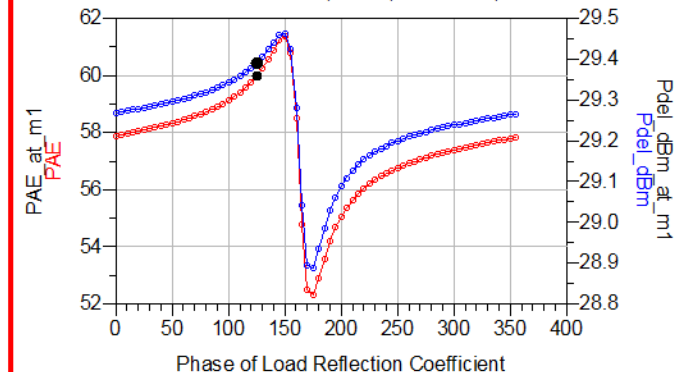
Simulated load reflection coefficients



- Blue dot is load reflection coefficient at fundamental.
- Black dot is input reflection coefficient with harmonic load selected by marker m1.

The data inside the red polygon corresponds to the load selected by the m1 marker and will only change if you move the marker to a different load.

Power Delivered, dBm, and PAE, %



Sweep Gate Bias

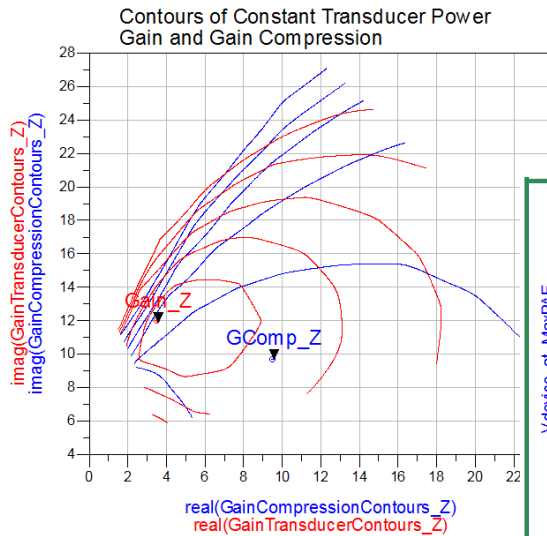
GComp_Z
level=0.319038, number=33

9.578 + j9.697

Gain_Z
level=18.880092, number=33

3.587 + j11.926

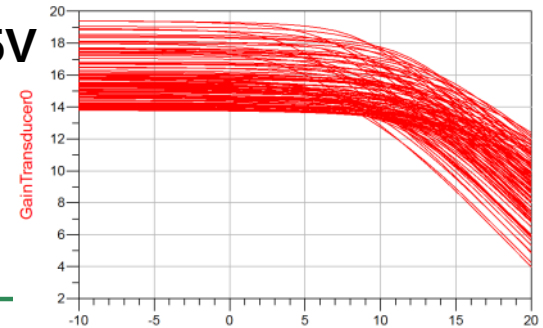
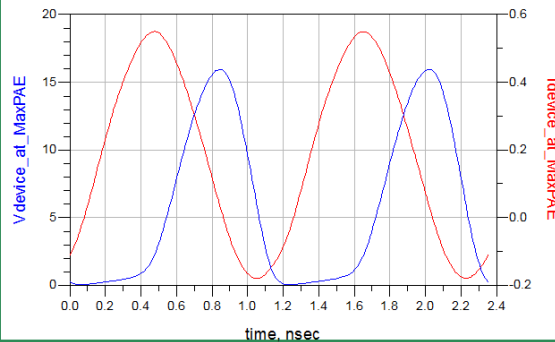
Results with gate bias = 2.25V



Maximum Transducer Power Gain and contour levels, dB: 18.880, 18.000

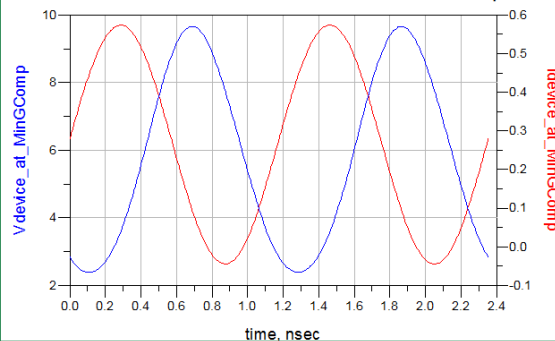
Minimum Gain Compression and contour levels, dB: 0.319, 0.500

Device I and V Waveforms at Max PAE



BiasCurrent_at_MaxPAE	Zload_at_MaxPAE	MaxPAE_Rho
0.182	4.935 + j21.290	0.846 / 133.5...
PAEmax	GainComp_at_MaxPAE	
28.398	7.116	
Z_In_at_MaxPAE	Gain_at_MaxPAE	
1.791 - j4.679	8.678	
Pdel_dBm_at_MaxPAE		
25.089		

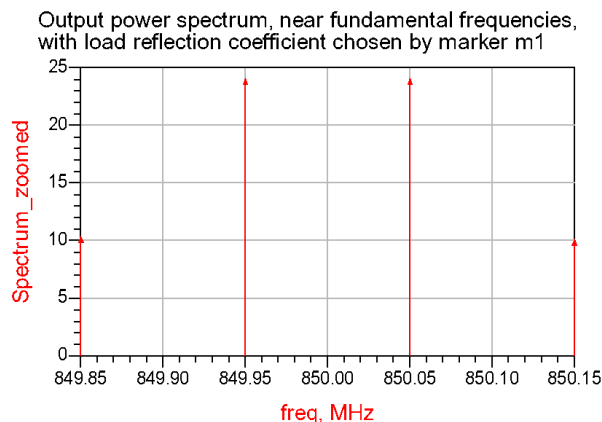
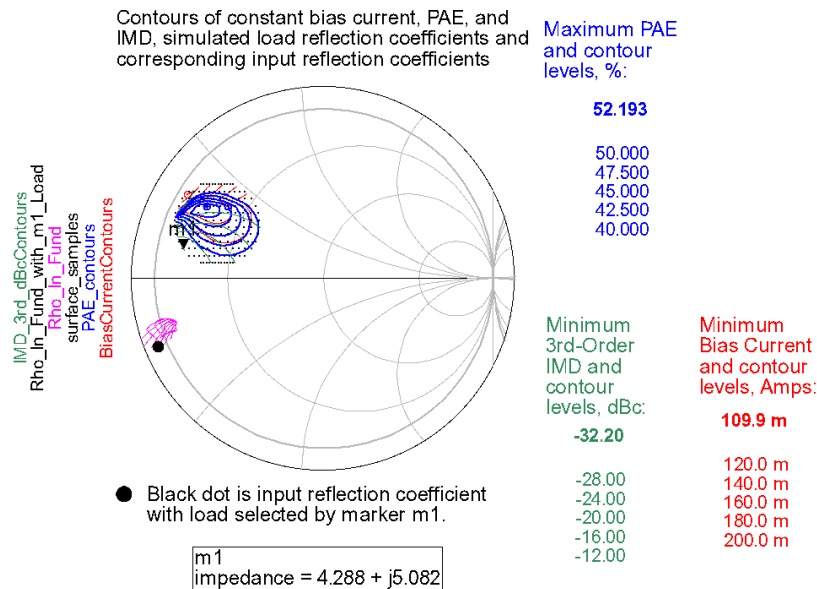
Device I and V Waveforms at Min Gain Comp.



At load that gives minimum gain compression:

BiasCurrent_at_MinGainComp	Zload_at_MinGainComp	MinGainCompRho
0.265	6.808 + j9.631	0.768 / 157.808
PAE_at_MinGainComp	GainCompMin	
21.165	0.326	
Z_In_at_MinGainComp	Gain_at_MinGainComp	
-0.598 - j5.302	18.222	
Pdel_dBm_at_MinGainComp		
25.100		

Constant power del. load pull with two tones



At load that gives minimum 3rd-order IMD:

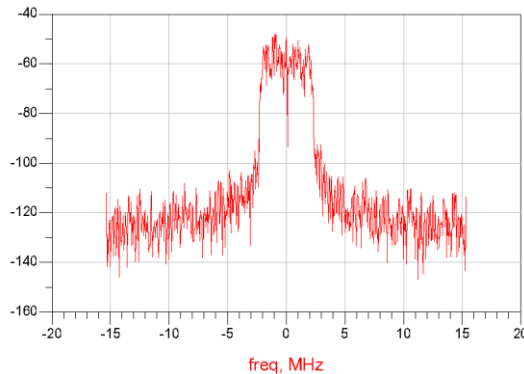
BiasCurrent_at_MinIMD3	Zload_at_MinIMD3	MinIMD3_Rho
0.270	$4.288 + j5.082$	0.844 / 168.308
PAE_at_MinIMD3		GainComp_at_MinIMD3
33.205		1.714
Z_In_at_MinIMD3		Gain_at_MinIMD3
$-1.370 - j10.004$		12.232
Pdel_dBm_at_MinIMD3		IMD_3rd_dBcMin
27.096		-32.198

At load that gives maximum PAE:

BiasCurrent_at_MaxPAE	Zload_at_MaxPAE	MaxPAE_Rho
0.166	$5.816 + j13.997$	0.805 / 148.345
PAEmax		GainComp_at_MaxPAE
52.193		8.283
Z_In_at_MaxPAE		Gain_at_MaxPAE
$0.606 - j7.123$		9.428
Pdel_dBm_at_MaxPAE		IMD_3rd_dBc_at_MaxPAE
27.076		-13.733



Load pull with WCDMA signal



ViDataset

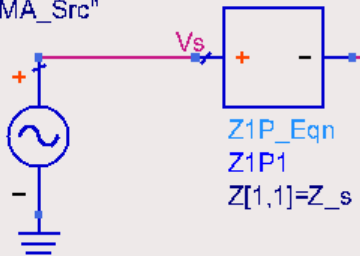
SRC9

Dataset="WCDMA3G_PA_UE_ACLR.ds"

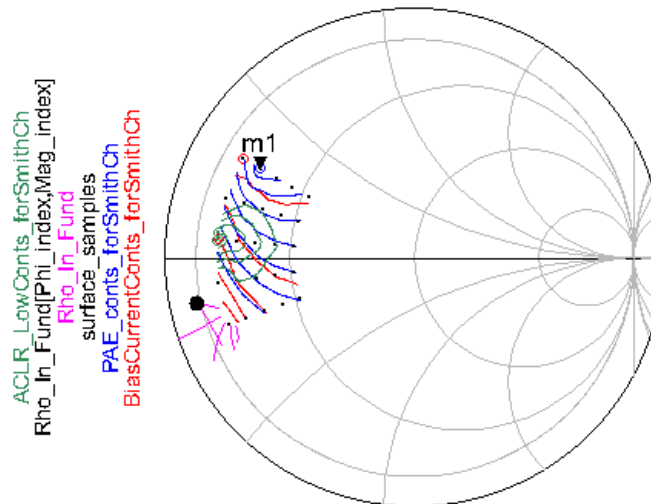
Expression="WCDMA_Src"

Freq=RFfreq

Gain=10**(SFexp)



Contours of constant bias current, PAE, and ACLR, simulated load reflection coefficients and corresponding input reflection coefficients



● Black dot is input reflection coefficient with load selected by marker m1.

m1
impedance = 5.540 + j13.244

Maximum PAE and contour levels, %:

54.181

50.000
45.000
40.000
35.000
30.000
25.000
20.000

Minimum Bias Current and contour levels, Amps:

140.1 m

200.0 m
300.0 m
400.0 m
500.0 m
600.0 m

Minimum Lower ACLR and contour levels, dBc:

-33.411

-33.000
-32.000
-31.000
-30.000
-29.000

Read modulated data from file. Scale signal amplitude by optimizing "SFexp" variable.



Maury measured data

- Examine contours and make trade-offs for optimal load condition
- Use measured data files directly in impedance matching network design and optimization

```
! Power sweep plan data file
!
!
Number of Frequencies = 1
Number of Variables = 2
VAR=<F1 Load Gamma>, Units=<>
VAR=<Pin_avail>, Units=<dBm>
! Freq Points per VAR
4.7500 61 21
User params= 3
User names: I_out_raw I_in_raw Drain_eff
User units: mA mA %

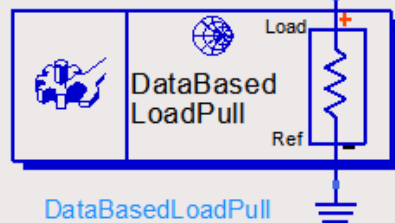
Freq = 4.7500 GHz
num_src_harmonics = 1, num_ld_harmonics = 1
valid gamma_src1 gamma_ld1 Freq Pin_avail_dBm Pout_dBm Gt_dB Vq_out_v Iq
1 -0.90605 -0.19391 -0.93757 -0.08513 4.7500 19.99412 39.06655 19.07
1 -0.90605 -0.19391 -0.93757 -0.08513 4.7500 22.00718 41.27562 19.26
1 -0.90605 -0.19391 -0.93757 -0.08513 4.7500 23.99421 43.44201 19.44
1 -0.90605 -0.19391 -0.93757 -0.08513 4.7500 26.00875 45.60356 19.59
1 -0.90605 -0.19391 -0.93757 -0.08513 4.7500 26.98651 46.63387 19.64
1 -0.90605 -0.19391 -0.93757 -0.08513 4.7500 27.99266 47.66695 19.67
1 -0.90605 -0.19391 -0.93757 -0.08513 4.7500 29.01515 48.68948 19.67
1 -0.90605 -0.19391 -0.93757 -0.08513 4.7500 29.99830 49.61656 19.61
1 -0.90605 -0.19391 -0.93757 -0.08513 4.7500 30.98119 50.46132 19.48
```



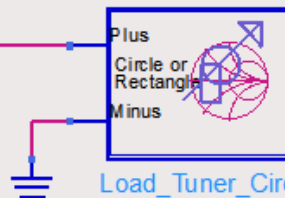
Performance contours from Load Pull Data

- 1) Reads LP data file
- 2) Simulates S-parameters of network
- 3) Gets corresponding performance data

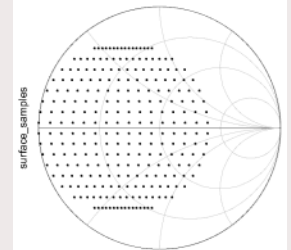
Tuner generates loads in region you specify



```
DataBasedLoadPull
DBLP
FileType=Maury
FileName[1]="Maury_Loadpull_data.lp"
Freq=2 GHz
FreqVarName=Freq
IndepVarName[1]=Pin_avail
IndepVarValue[1]=dbmtow( 1 )
```



```
Load_Tuner_Circular_or_Rect_wHarm_Zs_1
X3
Z0=50+j*0
Fund_Freq=2 GHz
Sweep_Rectangular_Region=1
Specify_Load_Center_S=1
Swept_Harmonic_Num=1
S_imag_min=-0
S_imag_max=0.7
S_imag_num_pts=10
S_real_min=0
S_real_max=0.7
S_real_num_pts=10
Simulation_1="DBLP"
```



Set to 2 if load pull data file contains a 2nd harmonic load pull, or to 3 if a 3rd harmonic load pull.

Indep. variables and performance parameters

Data Based Load Pull Simulation:4

ads_simulation:DataBasedLoadPull Instance Name

DBLP

Load pull file type: Maury

File Variables Interpolation Analysis Term Output Display

Independent Variables Performance Parameters

Reflection coefficients

	Name	Type	Harm	Domain	Min	Max	Points
1	gamma_id_1_x	Load	1	Rect-Real	-0.8944	0.87418	279
2	gamma_id_1_y	Load	1	Rect-Imag	-0.88305	0.88461	279

Frequency and input power constant

Other independent variables

	Name	Value	Unit	Min	Max	Points
1	Freq	freq	GHz	2	2	1
2	Pin_avail	1	dBm	1	1	1

Data Based Load Pull Simulation:4

ads_simulation:DataBasedLoadPull Instance Name

DBLP

Load pull file type: Maury

File Variables Interpolation Analysis Term Output Display

Independent Variables Performance Parameters

	Name	Min	Max	Output	Auto Display in DDS
1	C_2nd	-44.5432	-19.0437	☒	☐
2	C_3rd	-65.1157	-20.572	☒	☐
3	Carrier	-6.3792	19.0161	☒	☐
4	Eff	3.24607	59.1155	☒	☐
5	Gp	16.0834	41.9565	☒	☐
6	Gt	-1.07703	11.5361	☒	☐
7	I3	-107.94	-30.5327	☒	☐
8	I5	-209.088	-80.1838	☒	☐
9	I7	-310.236	-129.835	☒	☐
10	I_in	0	0	☒	☐
11	I_out	10	10	☒	☐
12	Ip3	43.7696	45.2542	☒	☐
13	Ip5	43.7696	45.0203	☒	☐
14	Ip7	43.7696	44.9423	☒	☐
15	Pin deliv	-29.4803	-4.53746	☒	☐
16	Pout	-0.07703	12.5361	☒	☐



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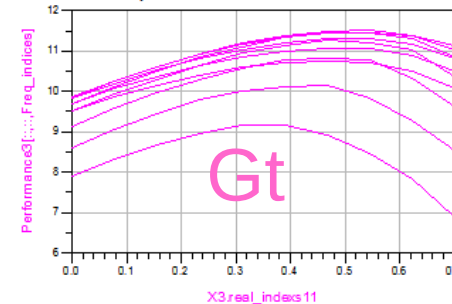
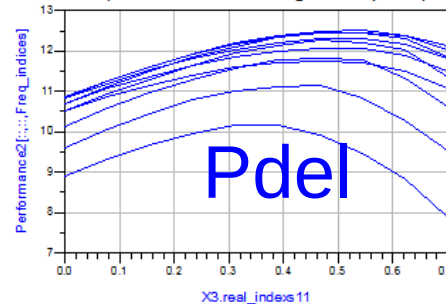
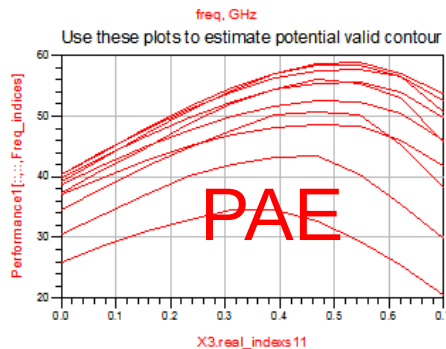
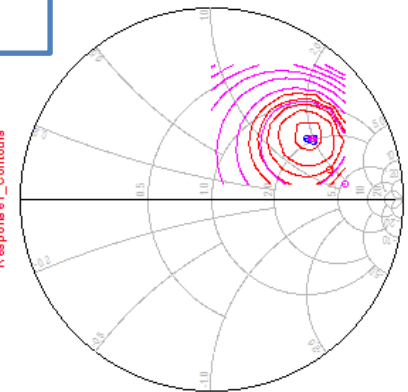
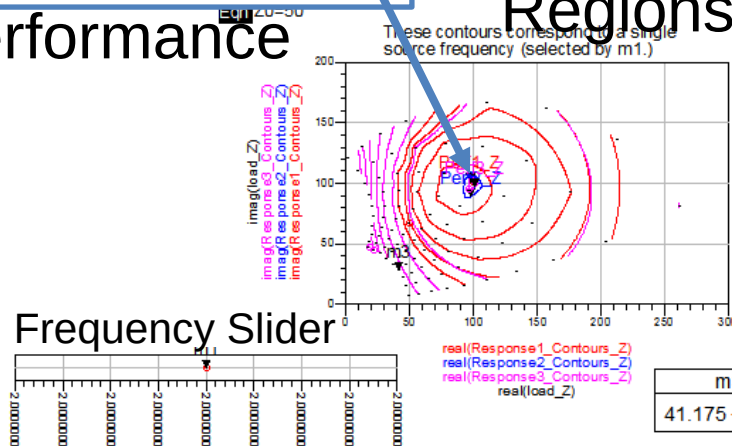


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Plot performance contours from LP Data

Load giving
best
performance

Check the Contours,
Rectangular or Circular
Regions



Use these sliders to limit the range of gamma_x gamma_y values used when computing c

Eqn Gamma_sliders_enable=1

gamma_x_limit1 indep(gamma_x_limit1)=0.000 gamma_y_limit1 indep(gamma_y_limit1)=1.000

gamma_x_limit1

gamma_y_limit1

gamma_x_limit1

gamma_y_limit1

gamma_x_limit1

gamma_y_limit1

gamma_x_limit1

gamma_y_limit1

gamma_x_limit1

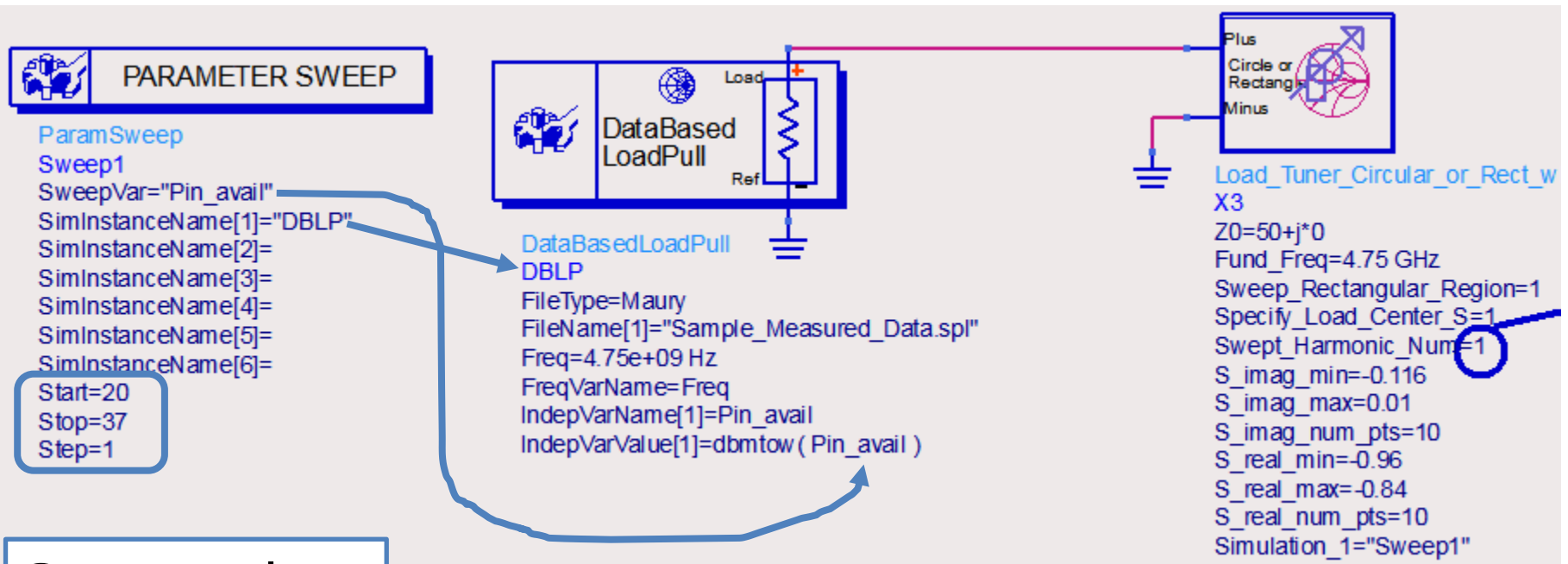
gamma_y_limit1

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Using power sweep of Load Pull data

Why sweep power? See gain compression data.



Sweep values
within range
of those in file

Sweep based on
 γ_x , γ_y
values in file



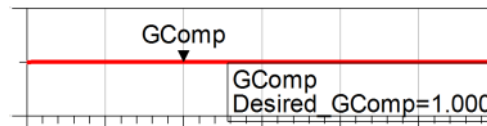
Contours at specified gain compression

EqnPerformance1=Eff

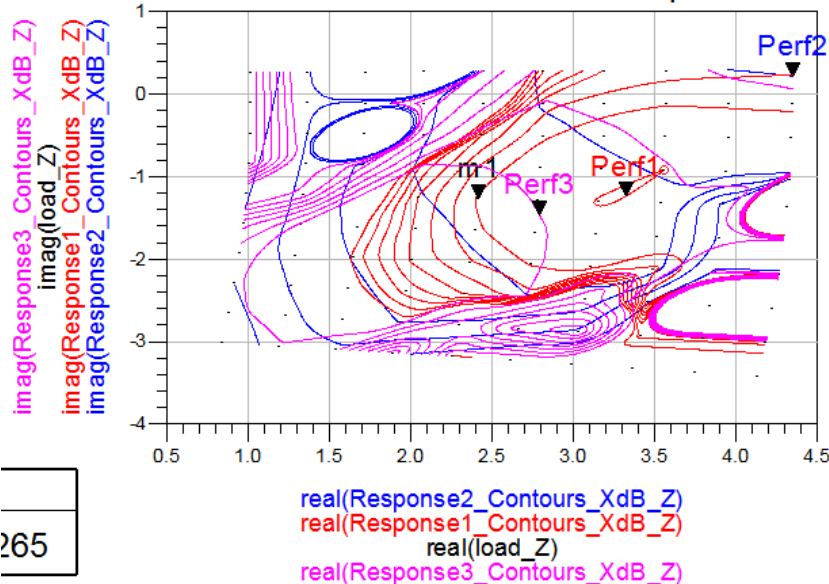
EqnPerformance2=Gain

EqnPerformance3=Pout

Specify Amount of Gain Compression



Contours at 1.000 dB Gain Compression



63.770
62.000
61.000
60.000
59.000
58.000
57.000
56.000
55.000

21.496
21.000
20.000
19.000
18.000
17.000

52.786
51.000
50.000
49.000
48.000
47.000
46.000
45.000

Perf1
level=63.769686, number=83

3.321 - j1.239

Perf2
level=21.496489, number=9

4.347 + j0.219

Perf3
level=51.000000, number=10

2.790 - j1.466

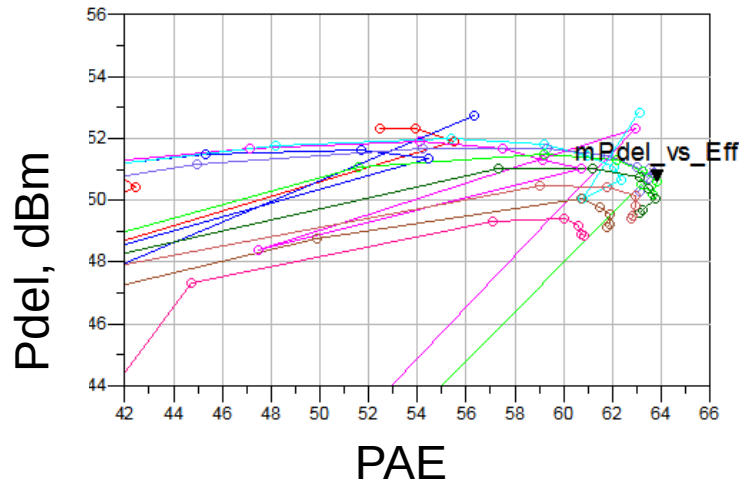
Why do contours look strange?
Measurements at some loads were not valid.

Choosing load: high efficiency or high power

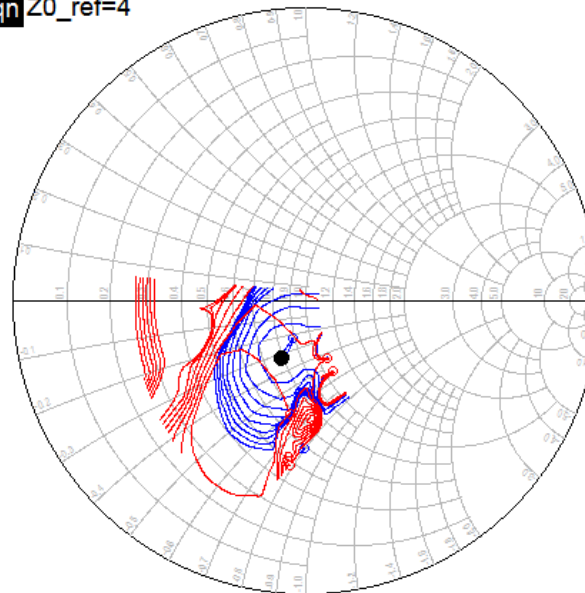
Data and Contours at 1.000 dB Gain Compression

Eqn Z0_ref=4

Choose desired point on power-delivered-versus-efficiency curves. Tabular data, black dot on Smith Chart and traces below will be updated.



mPdel_vs_Eff_Rho_wZ0_ref
 Response3_Contours_XdB_wZ0_ref
 Response1_Contours_XdB_wZ0_ref



Response1
 contour
 levels:

63.770
 63.000
 62.000
 61.000
 60.000
 59.000
 58.000
 57.000

Response3
 contour
 levels:

52.786
 51.000
 50.000
 49.000
 48.000
 47.000
 46.000
 45.000

Data corresponding to marker mPdel_vs_Eff:

Zload_at_mPdel_vs_Eff

3.160 - j1.301

mPdel_vs_Eff_Rho

0.881 / -177.008

Eff_at_mPdel_vs_Eff

63.834

GainComp_at_mPdel_vs_Eff

1.000

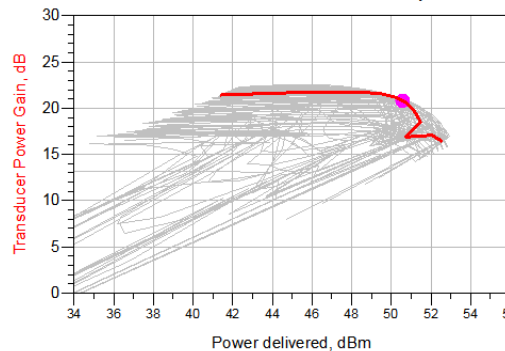
Pdel_dBm_at_mPdel_vs_Eff

50.593

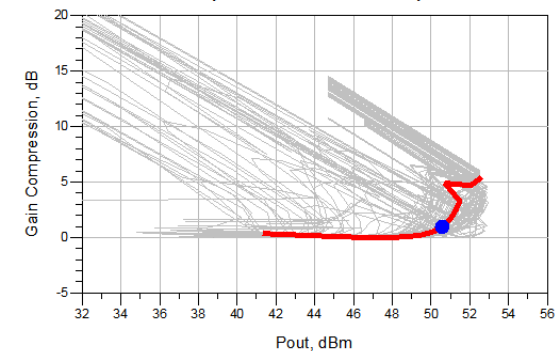
Gain_at_mPdel_vs_Eff

20.765

Transducer Power Gain Versus Output Power

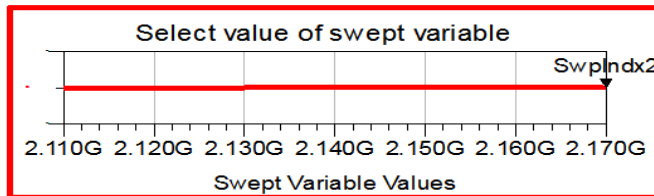


Gain Compression Versus Output Power

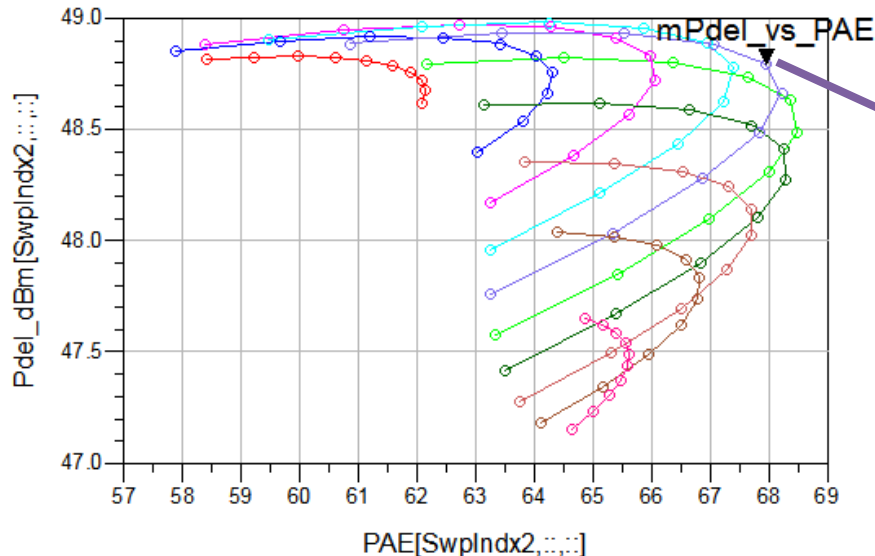




Choosing optimal load at 2.17 GHz



Choose desired point on power-delivered-versus-power-added-efficiency curves. Tabular data, black dot on Smith Chart and traces below will be updated.



Reference Z0

3.000 + j2.000

PAE and Delivered Power Contours

Maximum
Power
and contour
levels, dBm:

48.93
48.50
48.00
47.50

Maximum PAE
and contour
levels, %:

68.421
68.000
66.000
64.000
62.000
60.000

m6

level=68.421001, number=34

3.108 - j1.884

m7

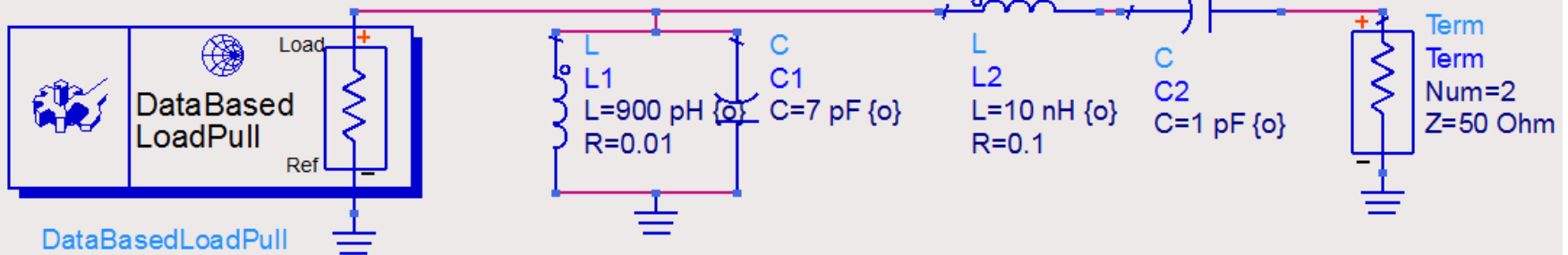
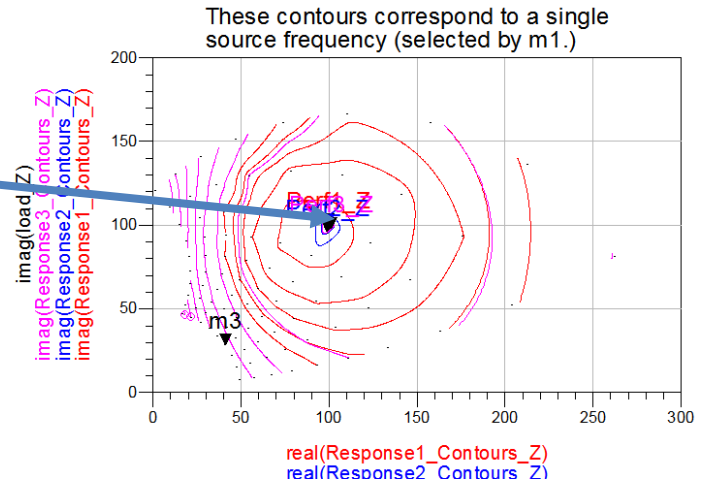
level=48.930662, number=100

2.525 - j2.304



Use measured data directly in optimization

This impedance should be the same as this.



DataBasedLoadPull

DBLP

FileType=Maury

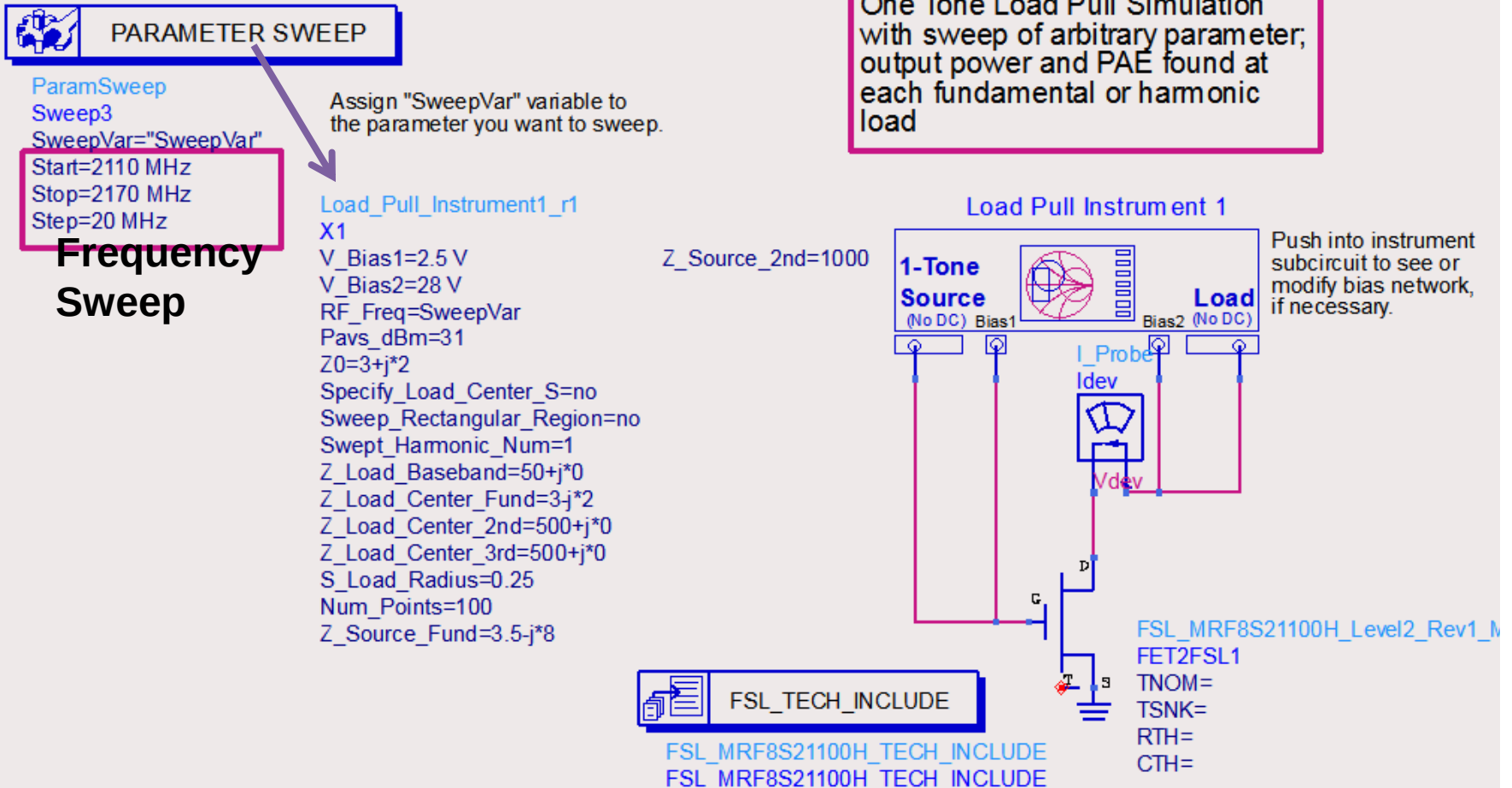
FileName[1]="Maury_Loadpull_data.lp"

Freq=2 GHz

FreqVarName=Freq



Load Pull delivers the Impedance for the Matching Network Design





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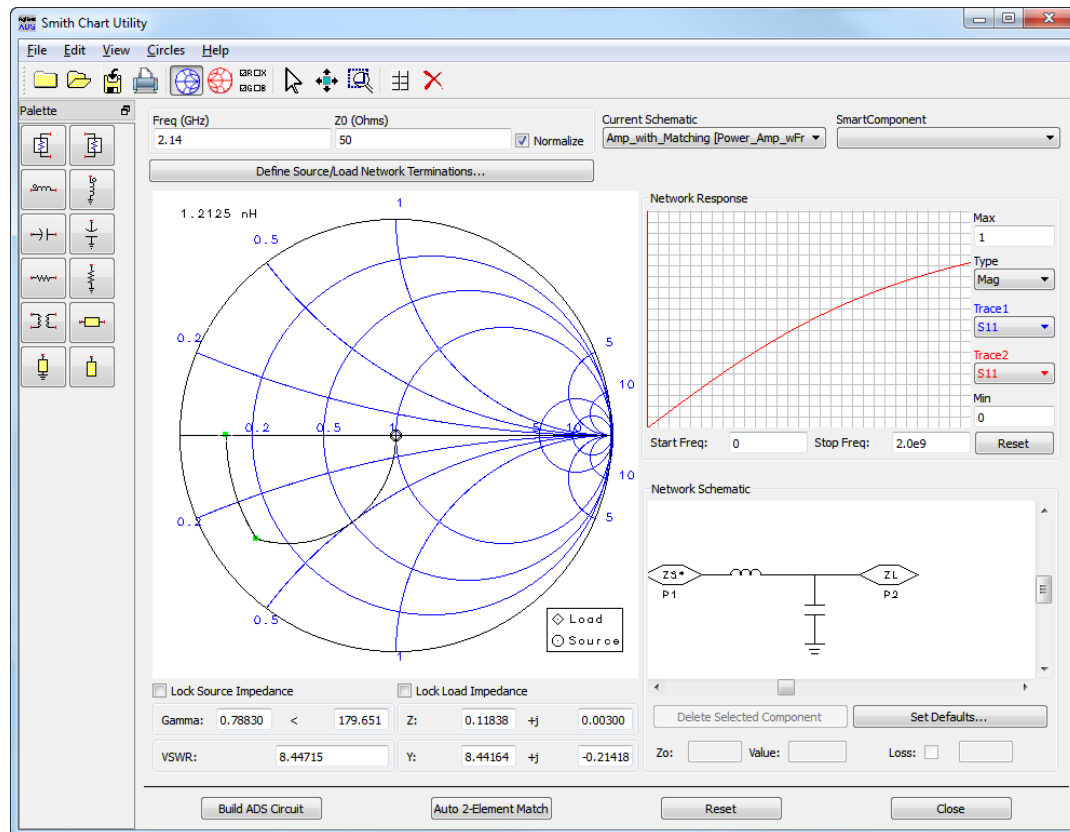


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Matching Network Design

Smith Chart Utility

Design impedance matching network(s) using existing techniques, or optimization



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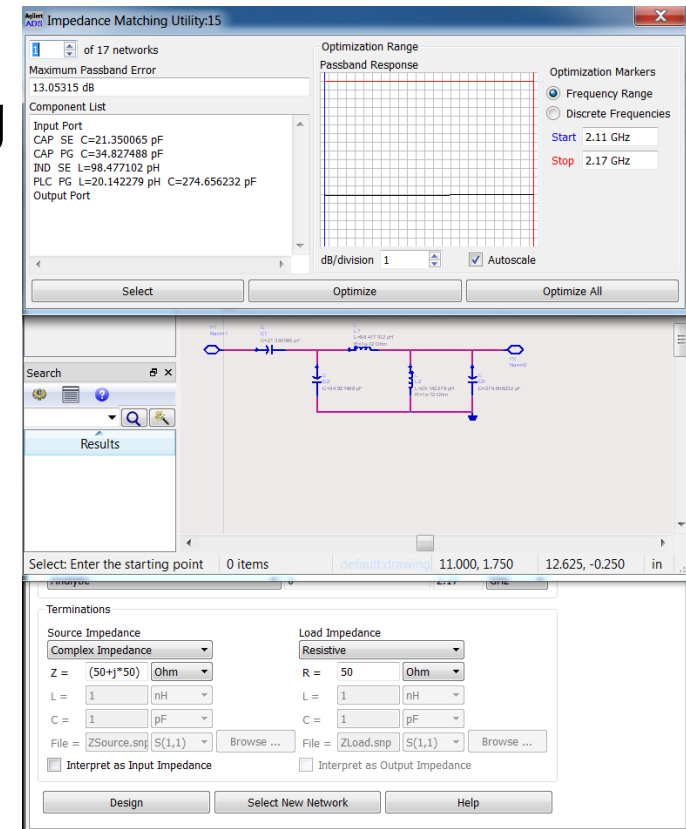
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Matching Network Design

Matching Utility (Broad Band)

ADS Impedance Matching Utility –

- Low-pass, high-pass, and band-pass, lumped element matching
- Multi-section quarter-wave matching
- Tapered-line impedance matching
- Single-stub impedance matching
- Several others

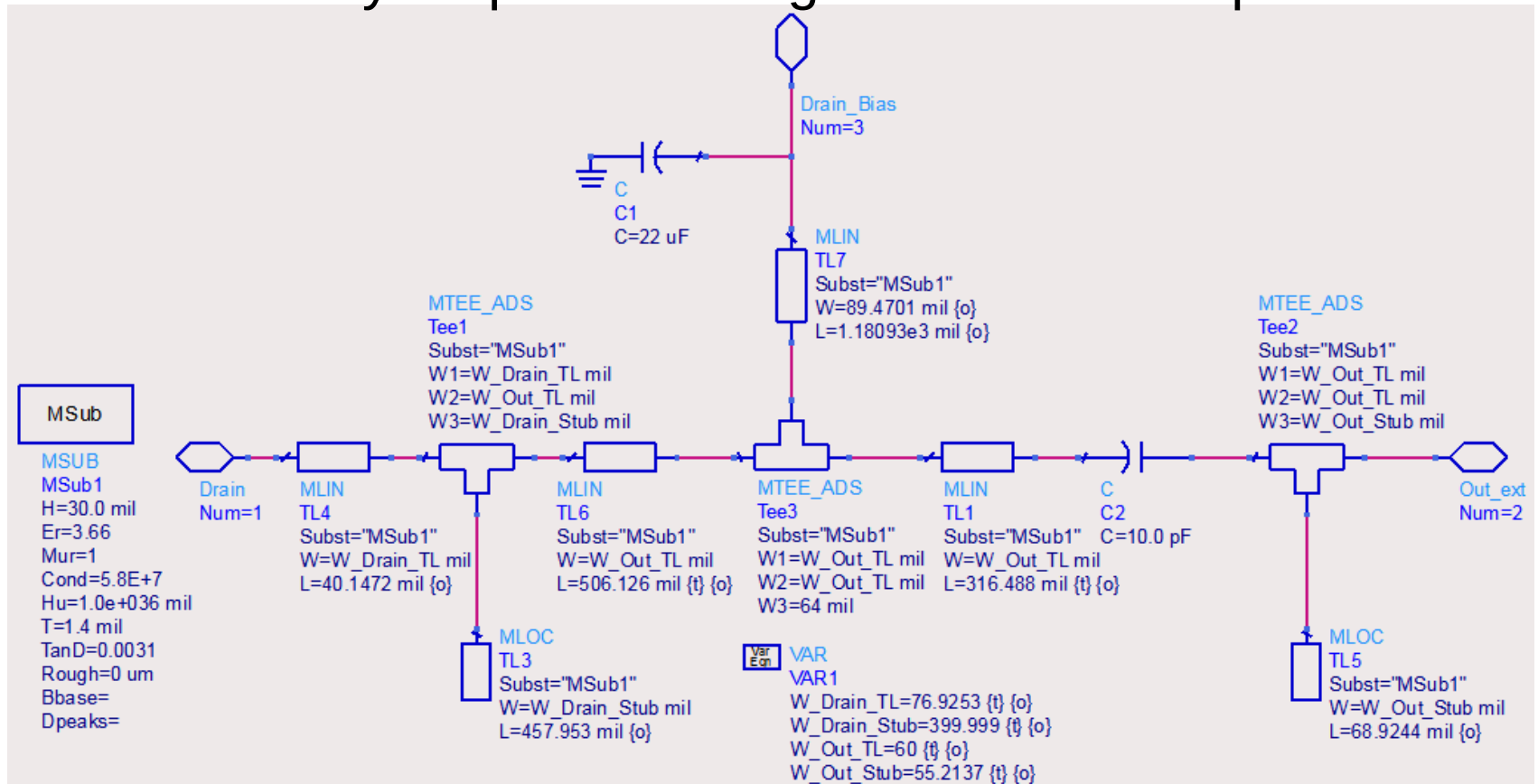


Your Calibration, Measurement & Modeling Solutions Partner!



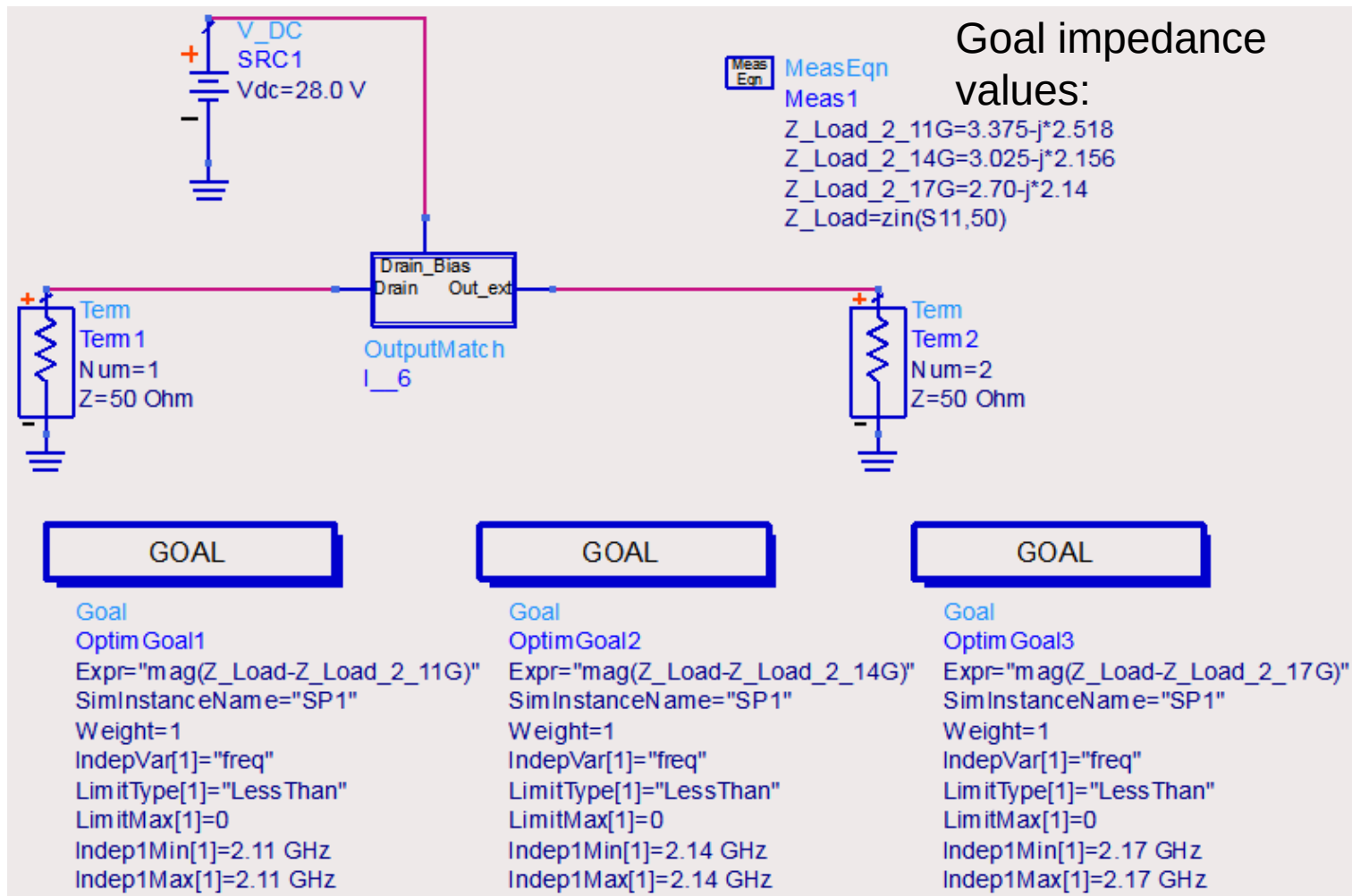
Using optimization to adjust parameter values

Preliminary output matching network to be optimized





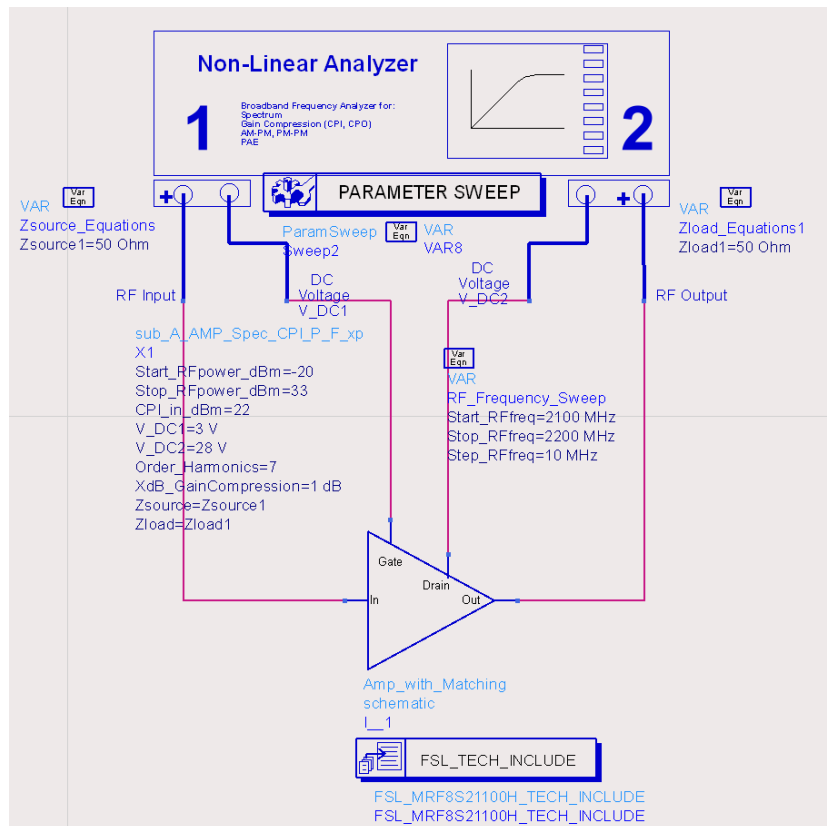
Impedance optimization at 3 frequencies



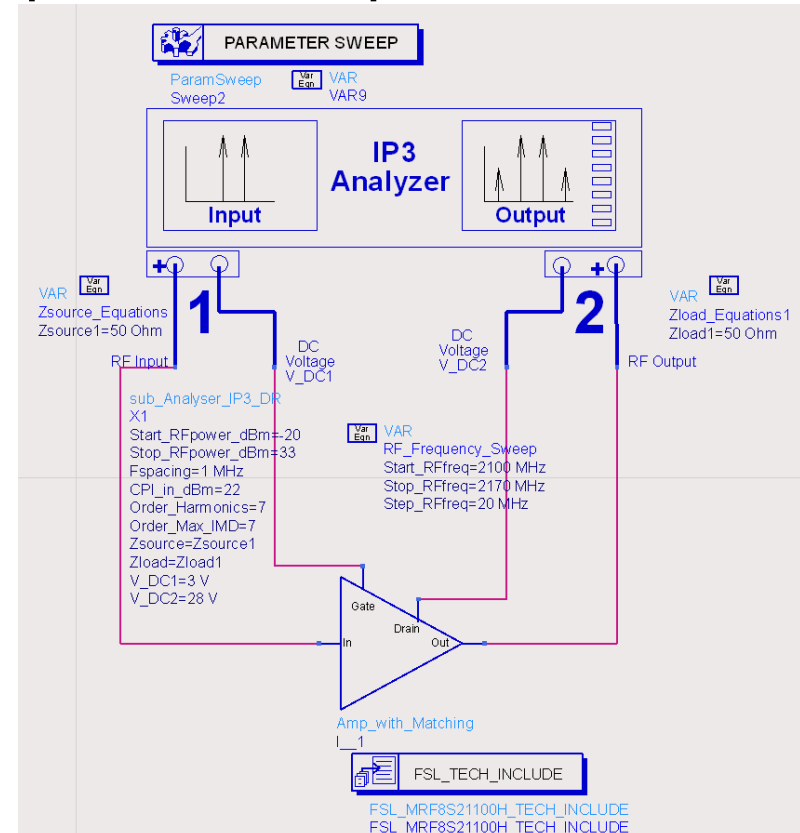


Testing performance of completed amplifier

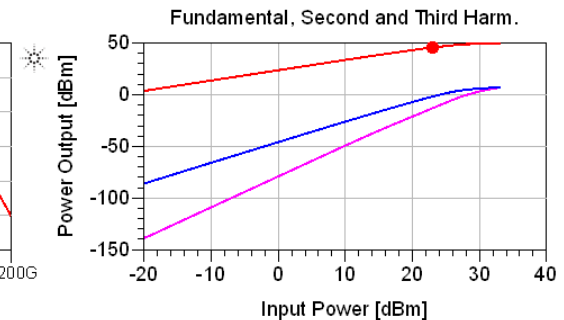
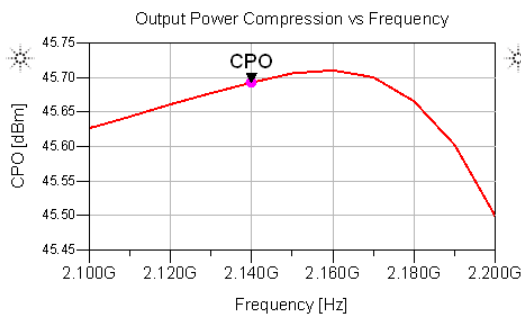
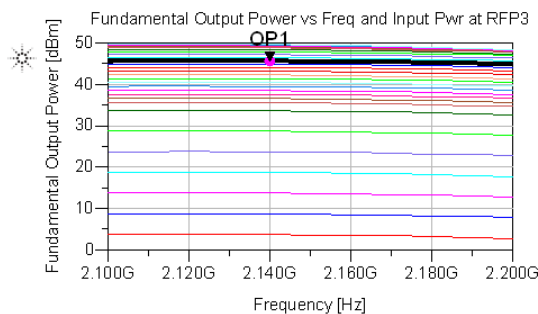
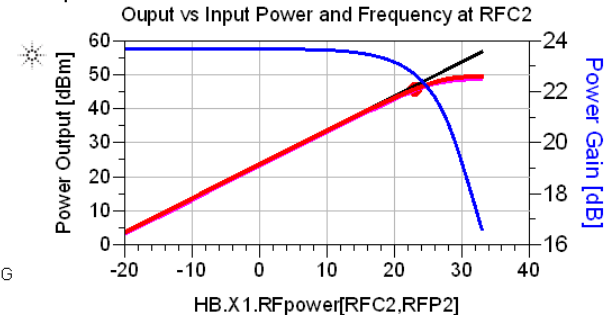
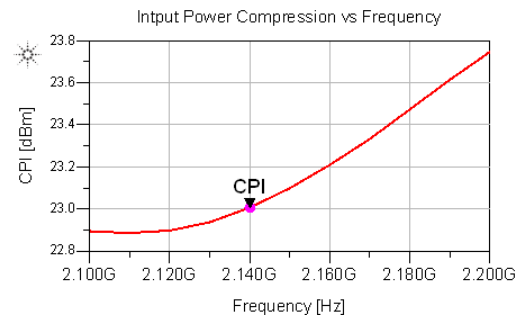
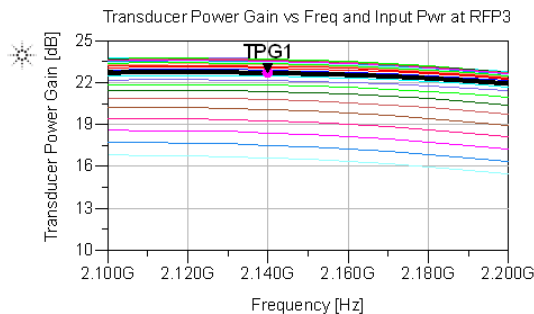
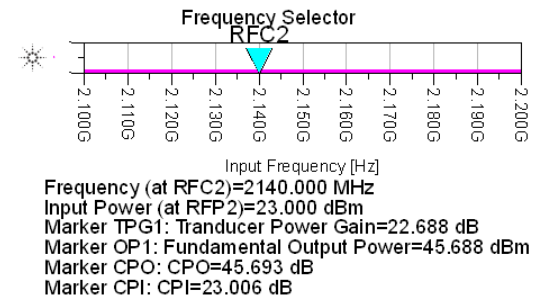
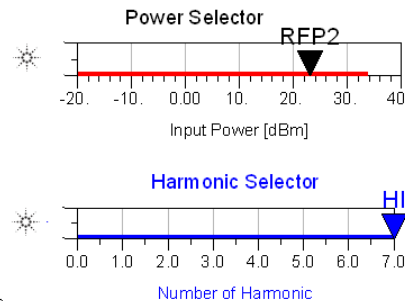
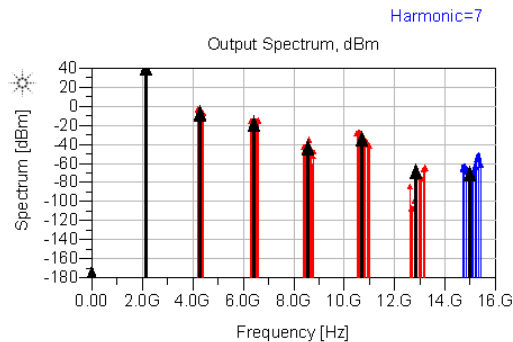
One-tone harmonic balance frequency and power sweep



Two-tone harmonic balance frequency and power sweep



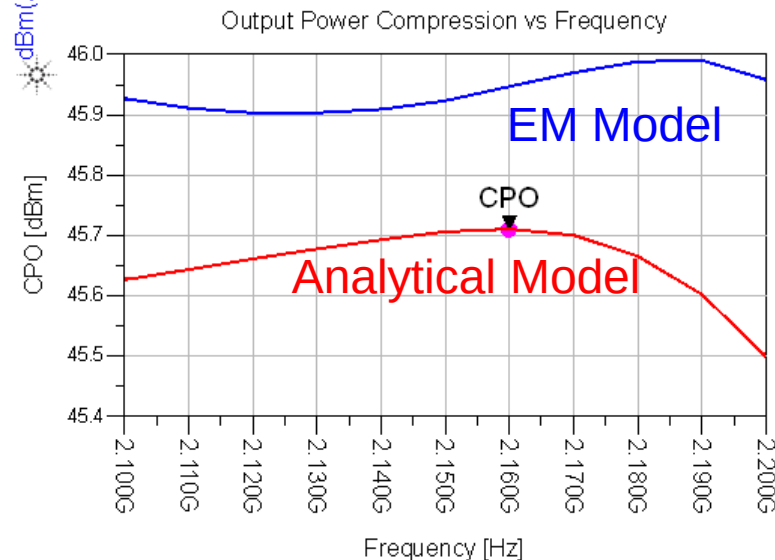
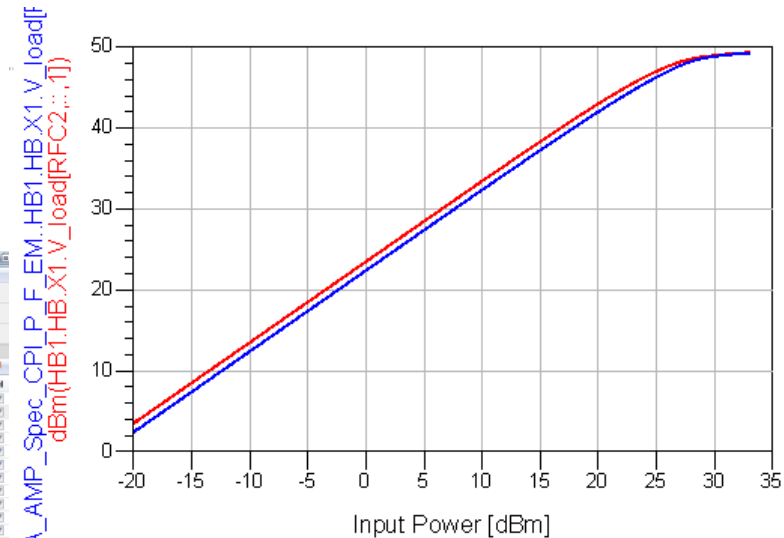
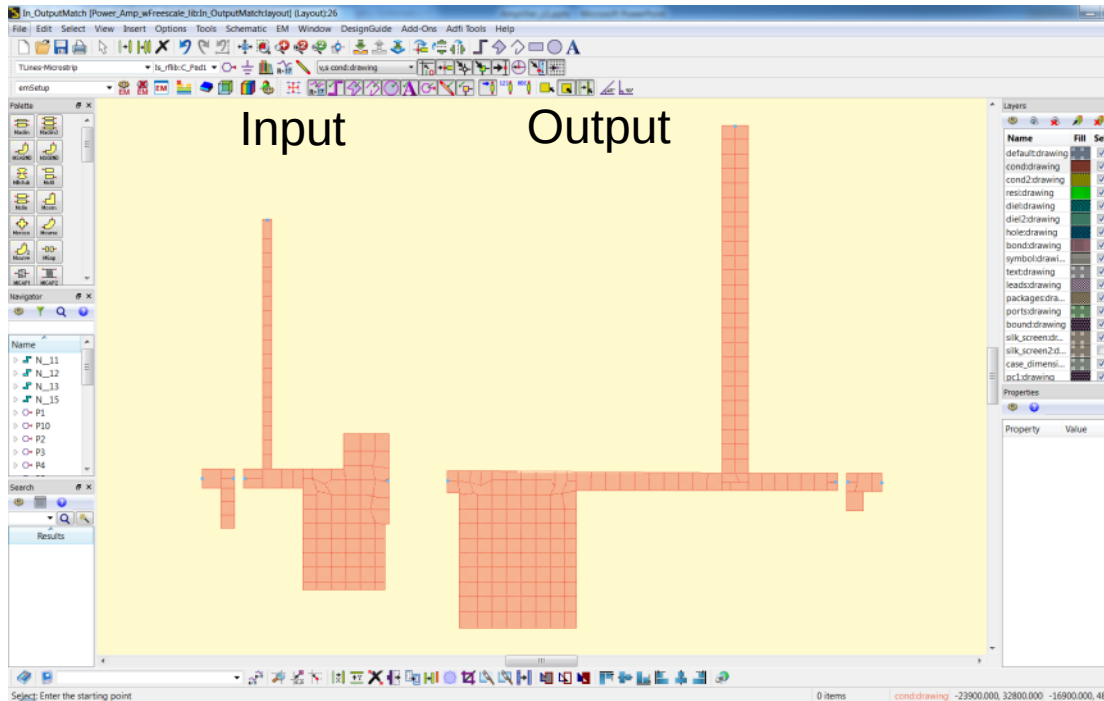
Testing performance of completed amplifier





Verification of the of the Layout – EM Cosim

Run EM to obtain more accurate results





PA Design Workflow

- 1) Run load pull simulation on the active device model or load pull measured data
 - a. 1-tone, 1 input power load pull
 - b. Power sweep to see gain compression
 - c. Frequency or bias sweep
 - d. Harmonic load phase sweep
 - e. Constant output power with swept var
 - f. Source pull
 - g. 2-tones to see IMD
 - h. Modulated signal to see ACLR
- 1) Choose optimal load impedances across frequency band
- 2) Use Smith Chart Utility or favorite matching tool to design preliminary matching network
- 3) Use optimization to adjust values
- 4) Use EM simulation and/or optimization to obtain more accurate results
- 5) Repeat steps 1-5 for to design source matching network
- 6) Test final design, including matching networks



- Instructor Introduction
- Pulsed IV/RF and Compact Modeling
- Load Pull
- Amplifier Design in ADS
- **Stability Analysis**
- X-Parameters
- System-level Simulations



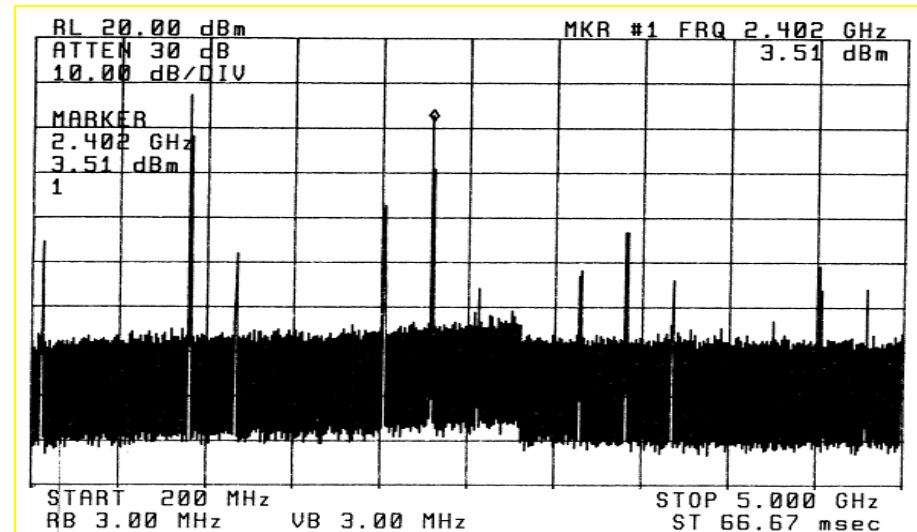
Introduction to stability analysis

Stability analysis is a critical step of RF design flow

Classical methods are either not complete or too complex...

Stability analysis need to be efficient (especially in large signal)

- Rigorous
- Fast
- User-friendly
- Compatible with commercial CAD softwares





Existing methods

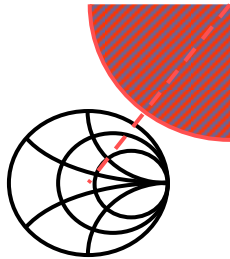
- Linear analysis “small signal”
 - K factor
 - Normalized Determinant Function (NDF)
 - Stability envelope
- Non-linear analysis “large signal”
 - Nyquist criterion
 - NDF
 - Bolcato, Di Paolo & Leuzzi, Mochizuki, ...



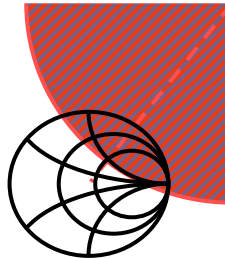
Existing methods – linear analysis

Widely used: K factor (also μ and μ' now)

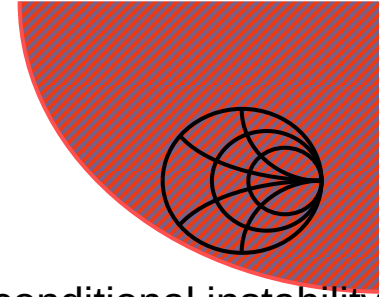
- $K > 1$ & $|\Delta| < 1$: unconditional stability of two port network
- $K < 1$: conditional stability $\rightarrow$ stability circles



Unconditional stability



Conditional stability



Unconditional instability

Limitations:

Only indicates that a stable circuit will continue to be stable when loading it with passive external loads at the input or output

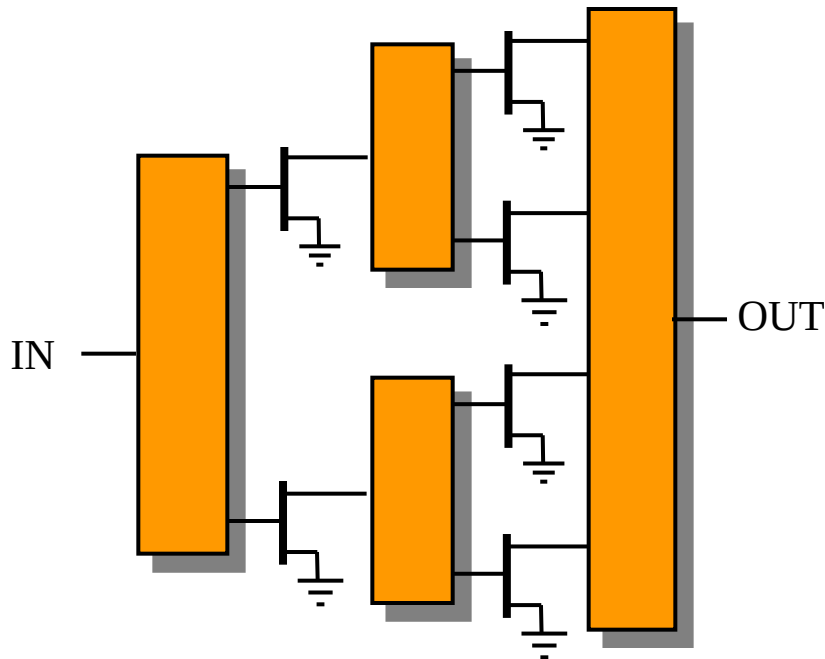
Do not guarantee the internal stability of the circuit !



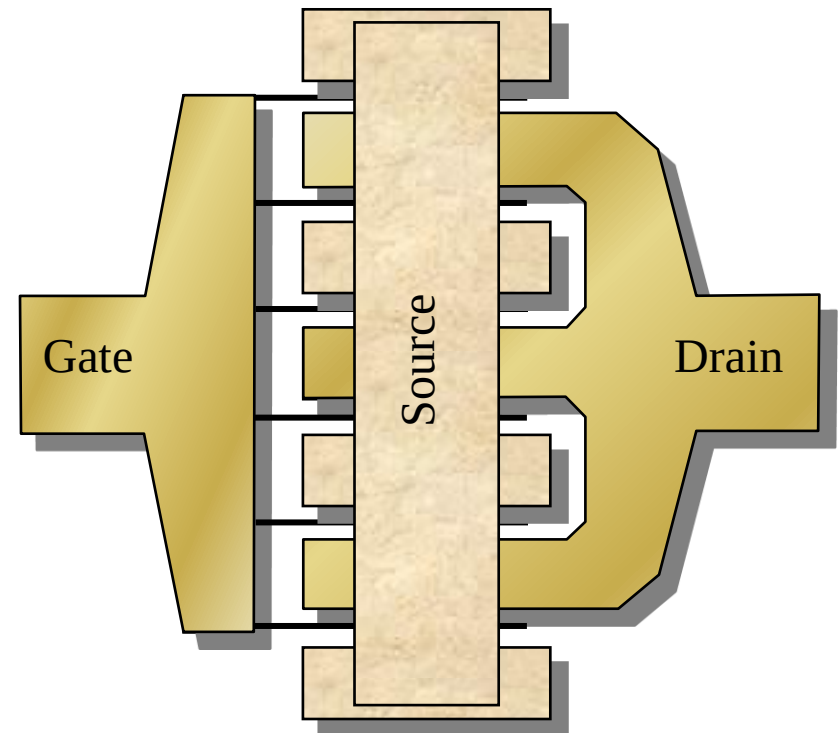
Existing methods – linear analysis

Potentially instable architectures for which K factor is not enough

Multi-stage power amplifier



Multi-fingers transistor



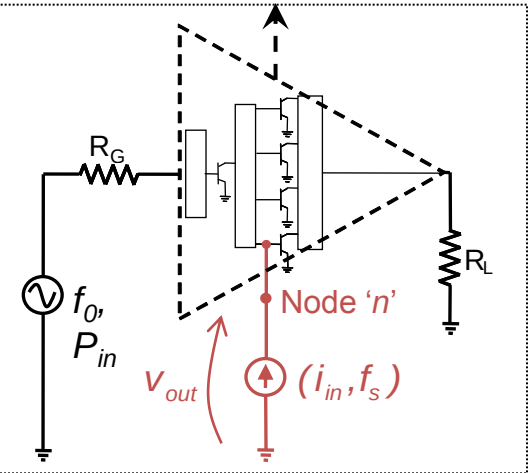
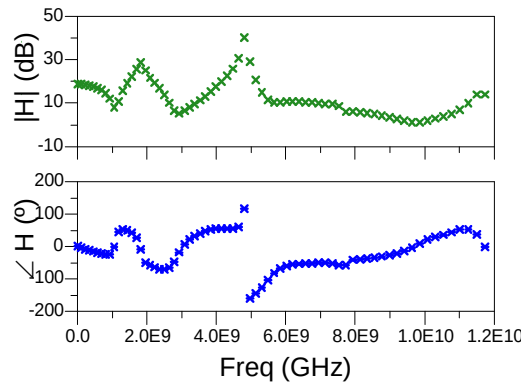


Pole-zero identification principle

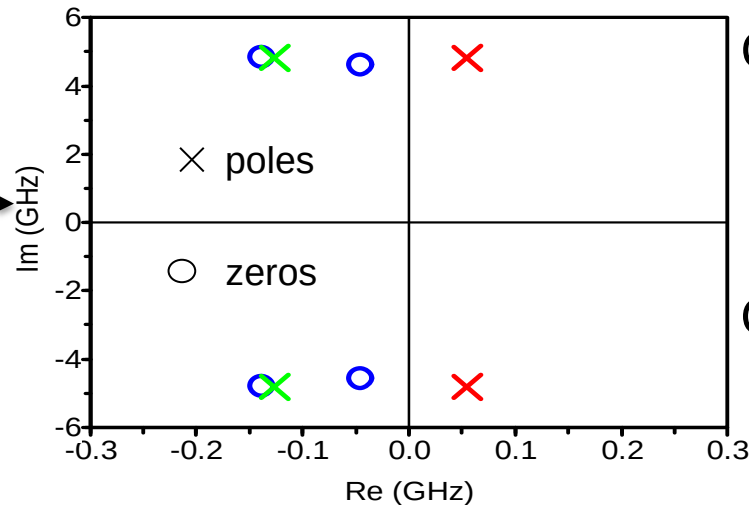
$H(j\omega)$

Frequency
domain
Identification
techniques

$$\hat{H}(s) = \frac{\prod_{i=1}^n (s - z_i)}{\prod_{j=1}^p (s - \lambda_j)}$$



Pole-zero plot



Complex conjugate
poles with positive
real part -> start-up
of an oscillation

Oscillation frequency
= Module of the
imaginary part



Agilent Technologies



Maury Microwave



AMCAD Engineering
Advanced Modeling for Computer-Aided Design

New stability analysis method - STAN

Suitable for both linear and non-linear stability analysis

Very easy to use with any CAD tool

Very easy to analyze results

Relative stability information delivered

Oscillation mode knowledge -> Help to find the suitable stabilization strategy

Parametric Analysis implemented

Monte-Carlo Analysis



Agilent Technologies

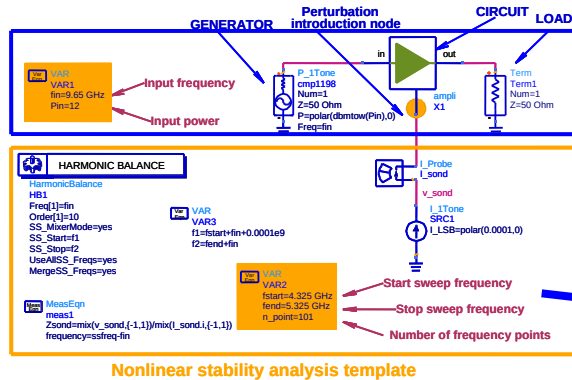


Maury Microwave



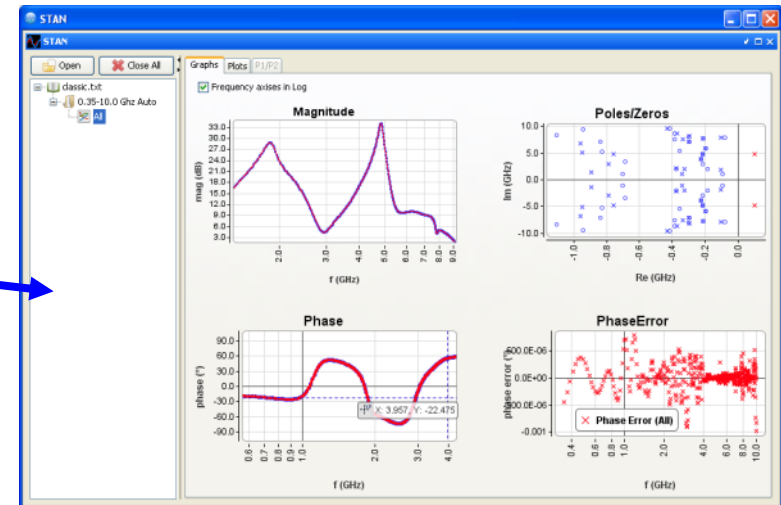
AMCAD Engineering
Advanced Modeling for Computer-Aided Design

STAN – Integration with CAD



EDA Tool

Templates for Agilent ADS
AC simulation for linear
HB simulation for non-linear



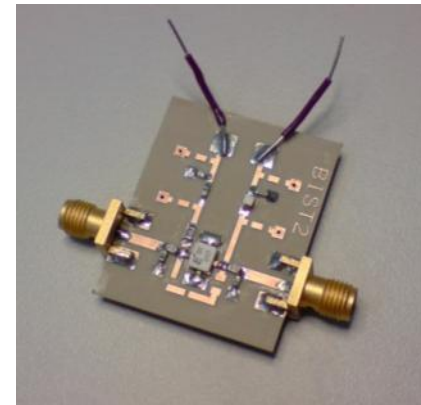
STAN tool

integrated in IVCAD software
User-friendly GUIs

STAN – Integration with CAD

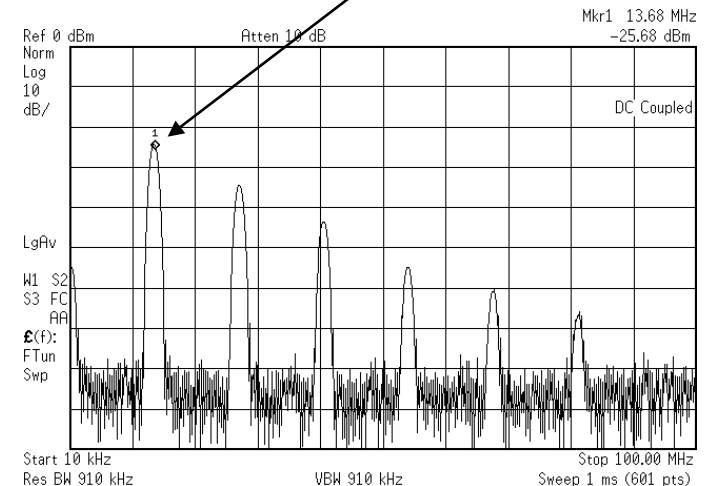
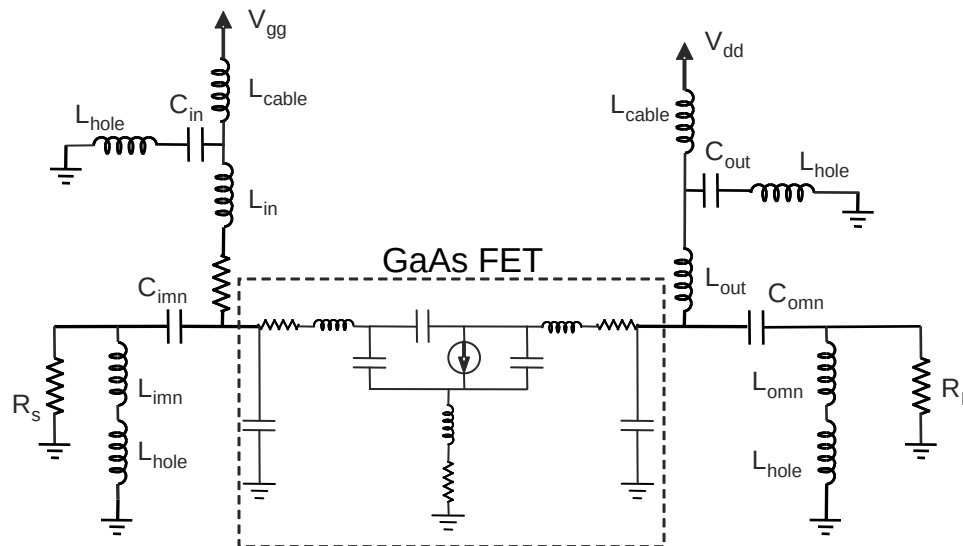
Example 1 (Linear analysis)

Low frequency instability of a medium power 1.2 GHz FET amplifier built in hybrid microstrip technology



Low frequency oscillation

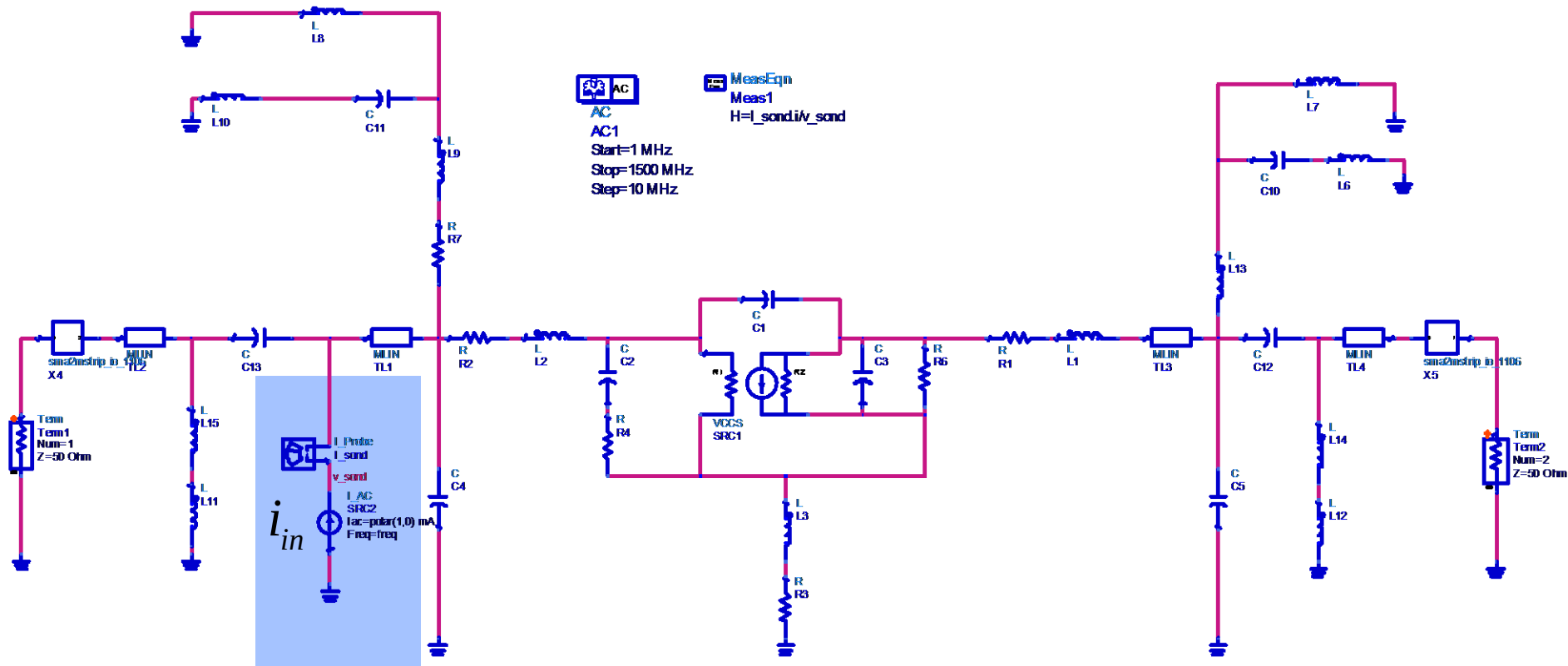
~ 14MHz





STAN – Integration with CAD

ADS schematic



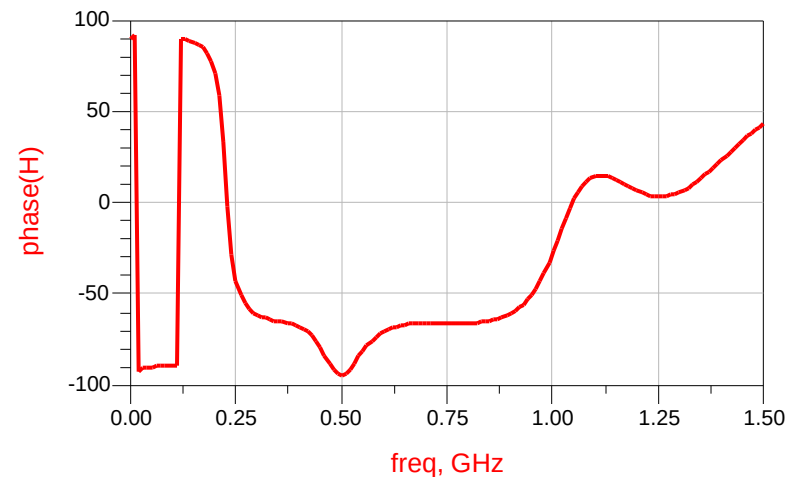
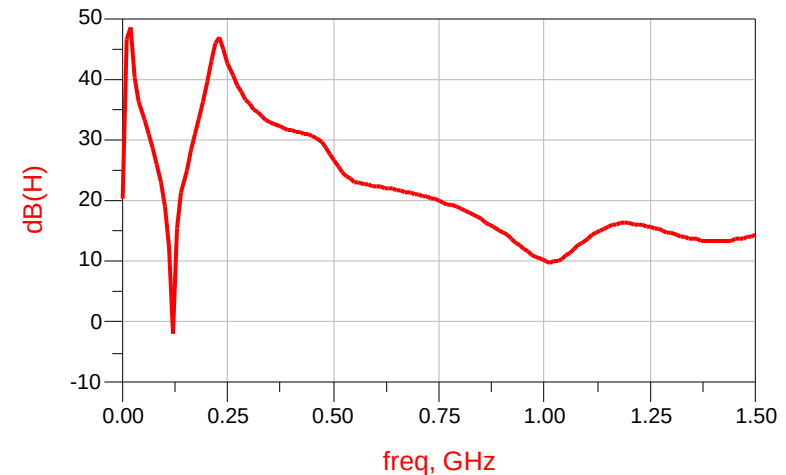
Note: It can also be done with a **voltage probe** connected in series at a circuit branch and observing the total admittance



STAN – Integration with CAD

Simulated frequency response $Z(j\omega)$

freq	mag(Z)	phase(Z)
1.000 MHz	10.332	90.003
11.00 MHz	216.618	91.941
21.00 MHz	270.189	-92.672
31.00 MHz	104.612	-90.829
41.00 MHz	65.997	-90.299
51.00 MHz	47.193	-89.998
61.00 MHz	35.371	-89.798
71.00 MHz	26.831	-89.667
81.00 MHz	20.060	-89.595
91.00 MHz	14.297	-89.585
101.0 MHz	9.095	-89.647
111.0 MHz	4.145	-89.794
121.0 MHz	0.795	89.951
131.0 MHz	5.958	89.554
141.0 MHz	11.599	88.969
151.0 MHz	18.047	88.123
161.0 MHz	25.766	86.902
171.0 MHz	35.480	85.120
181.0 MHz	48.404	82.446
191.0 MHz	66.727	78.252
201.0 MHz	94.580	71.219
211.0 MHz	138.977	58.377
221.0 MHz	199.477	33.980
231.0 MHz	219.444	-1.348
241.0 MHz	176.951	-28.478
251.0 MHz	135.681	-42.995
261.0 MHz	108.466	-50.829
271.0 MHz	90.570	-55.440
281.0 MHz	78.205	-58.363
291.0 MHz	69.241	-60.317





Agilent Technologies



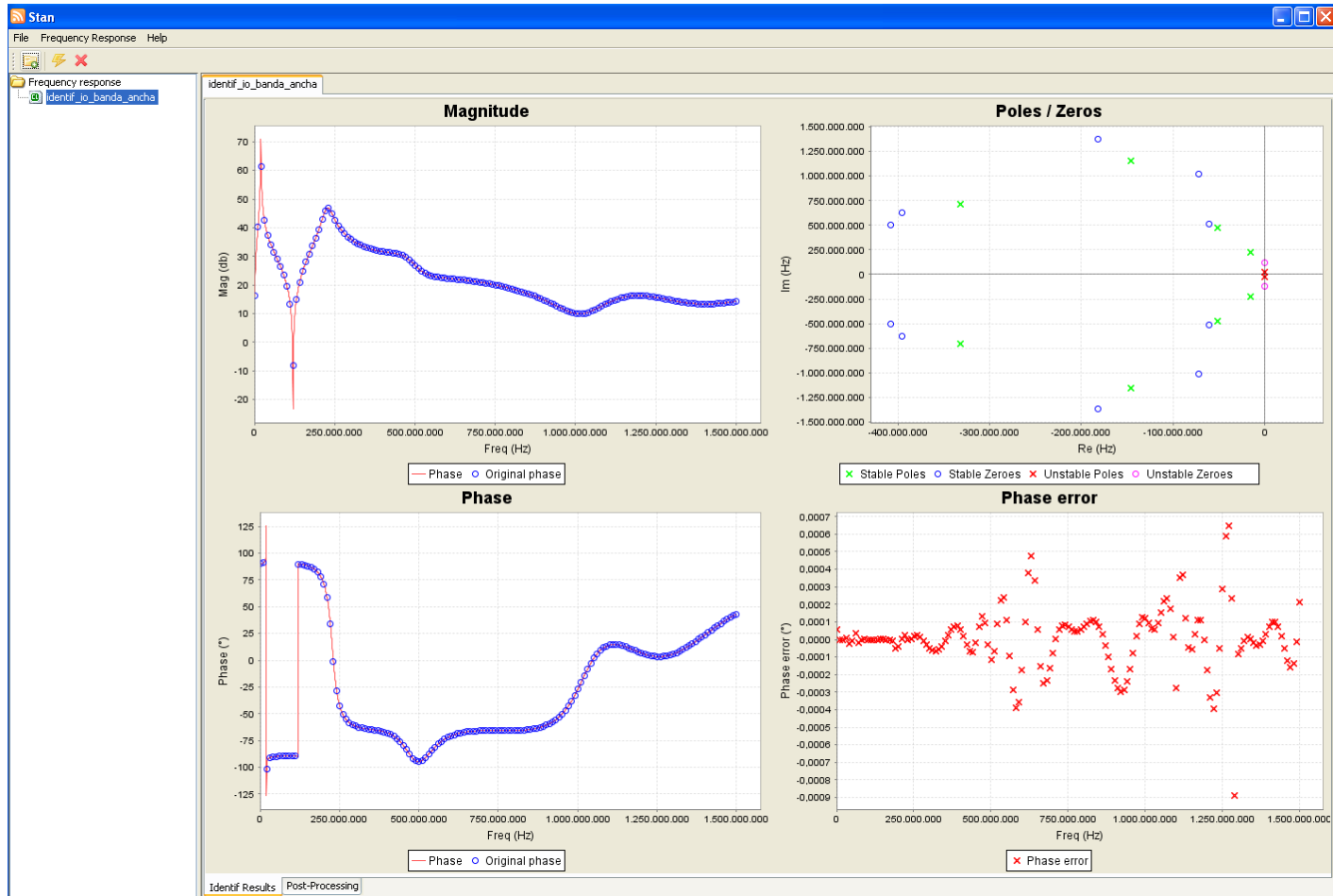
Maury Microwave



AMCAD Engineering
Advanced Modeling for Computer-Aided Design

STAN – Integration with CAD

Identification results



Your Calibration, Measurement & Modeling Solutions Partner!



Agilent Technologies



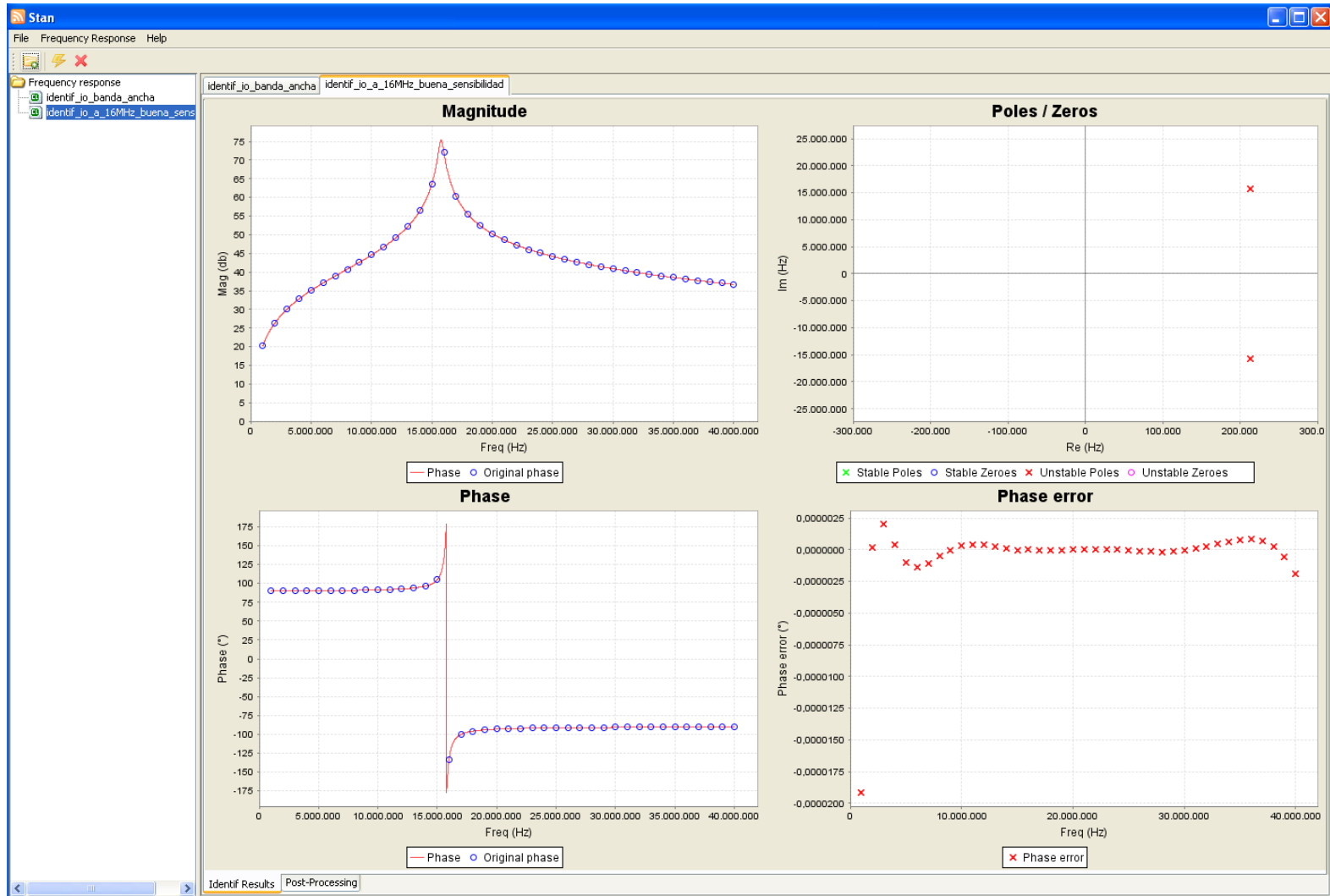
Maury Microwave



AMCAD Engineering
Advanced Modeling for Computer-Aided Design

STAN – Integration with CAD

Centered
on 20 MHz

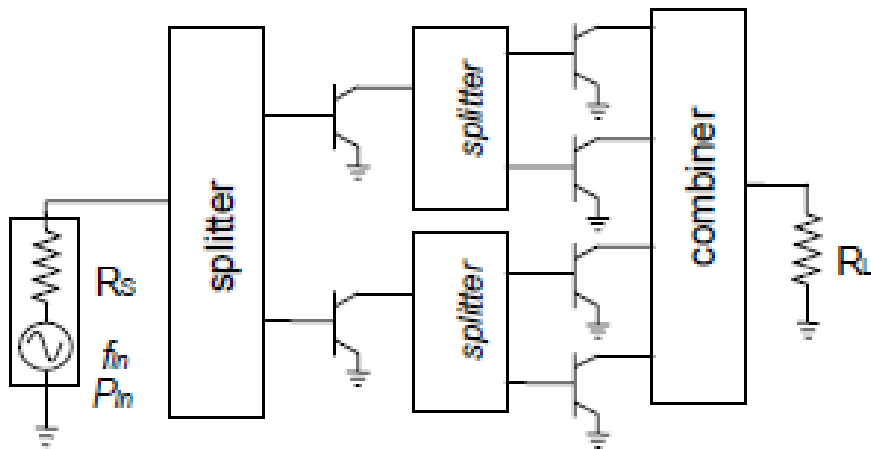


Your Calibration, Measurement & Modeling Solutions Partner!



STAN - Selecting the node

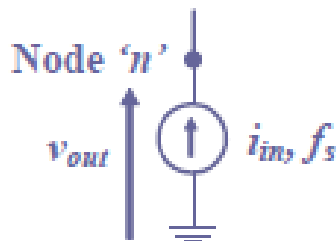
“All nodes are equal, but some nodes are more equal than others”



SISO transfer function → exact pole/zero cancellations are possible

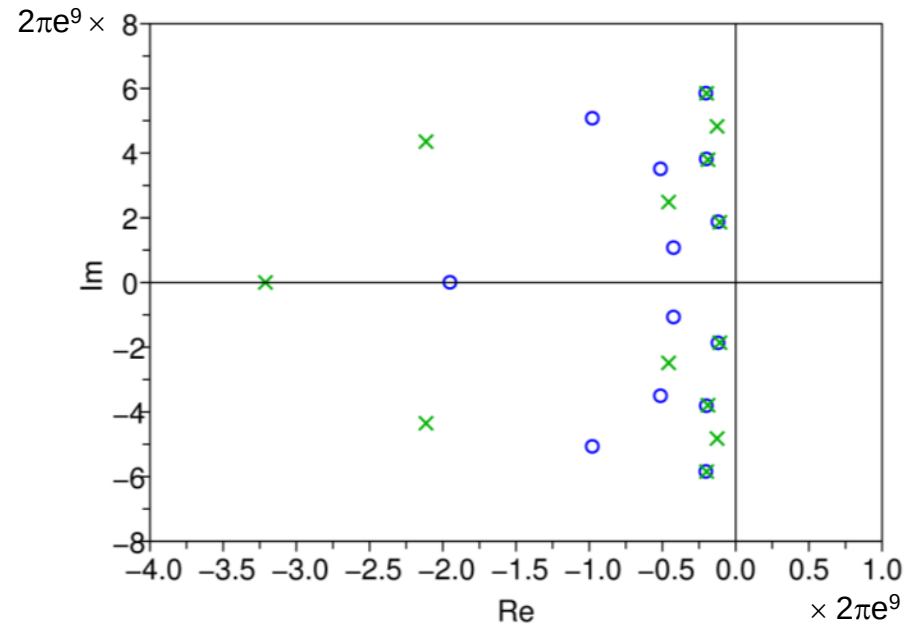
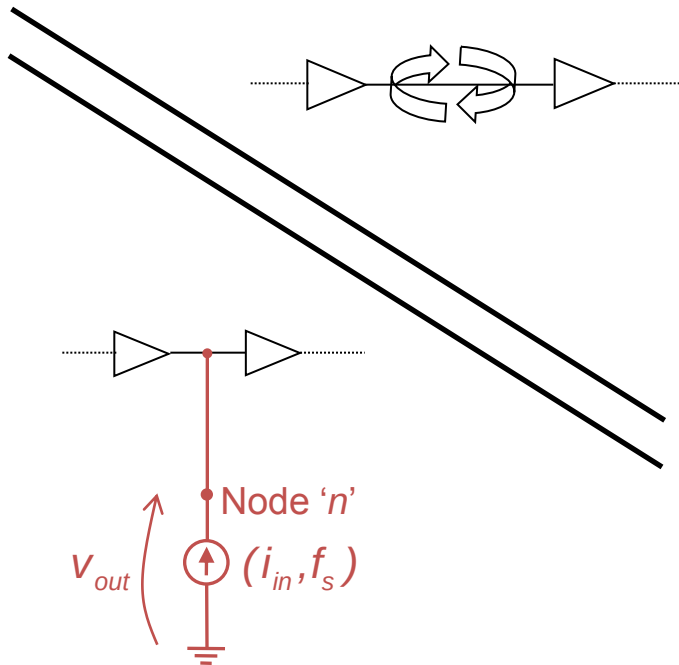
Pole/zero cancellations are associated with the lack of *controllability* and/ or *observability* in the system

???



STAN - Selecting the node

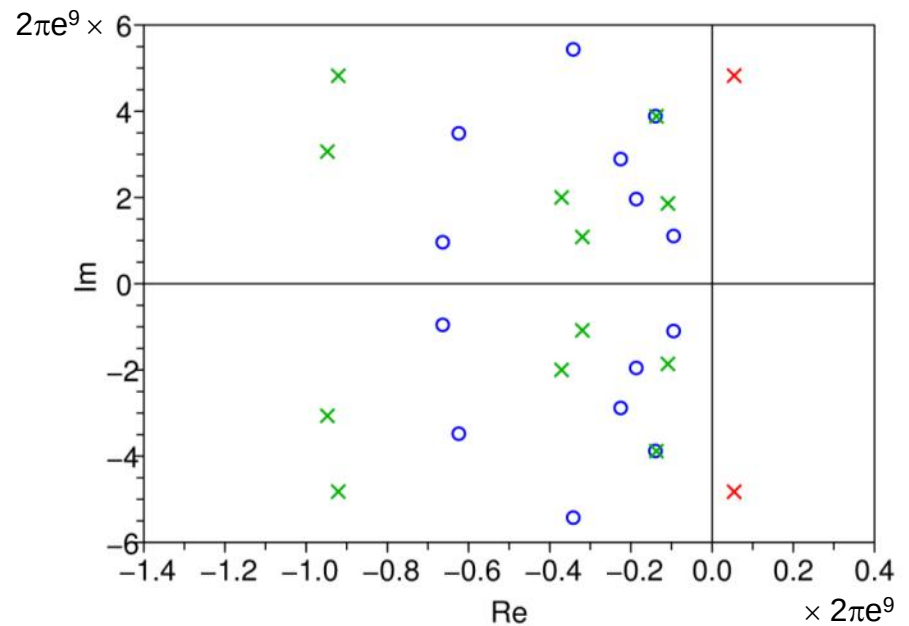
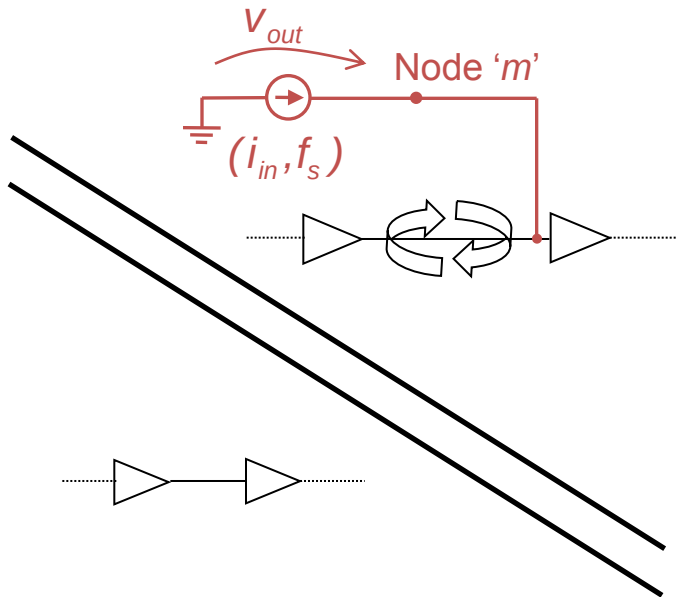
Example: let us consider a complex circuit in which the oscillation is taking place in a part that is totally isolated from the node selected to perform the analysis





STAN - Selecting the node

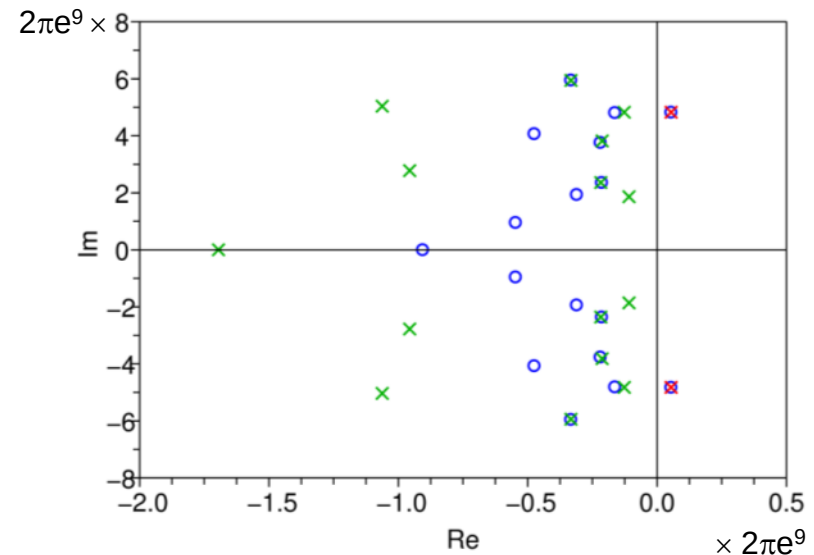
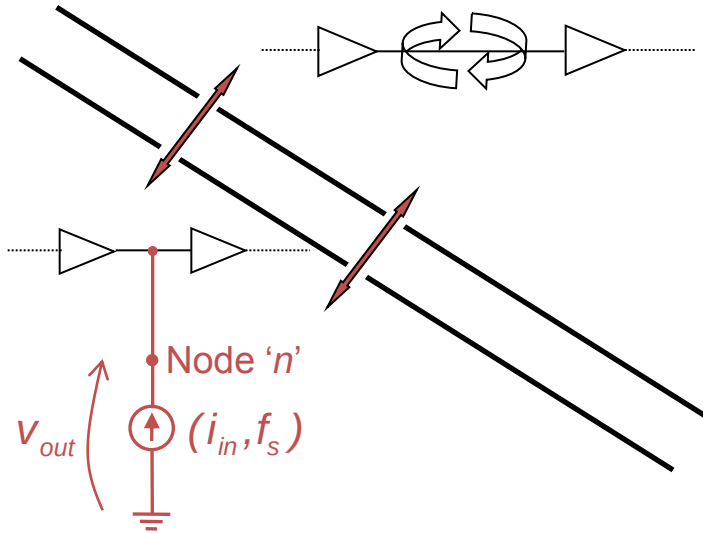
However, the oscillation can be predicted if the node is connected in the part of the circuit that is not isolated





STAN - Selecting the node

- The analysis is performed at a node that is not “completely” isolated from the part where the oscillation is taking place, i.e. there is a weak electrical link between parts



Quasi-cancellation $\rightarrow$ poles and zeroes almost cancelled $\rightarrow$ low degree of controllability/observability in the selected node $\rightarrow$ we are (electrically) far from the place where the oscillation is being generated $\rightarrow$ useful information for circuit stabilization

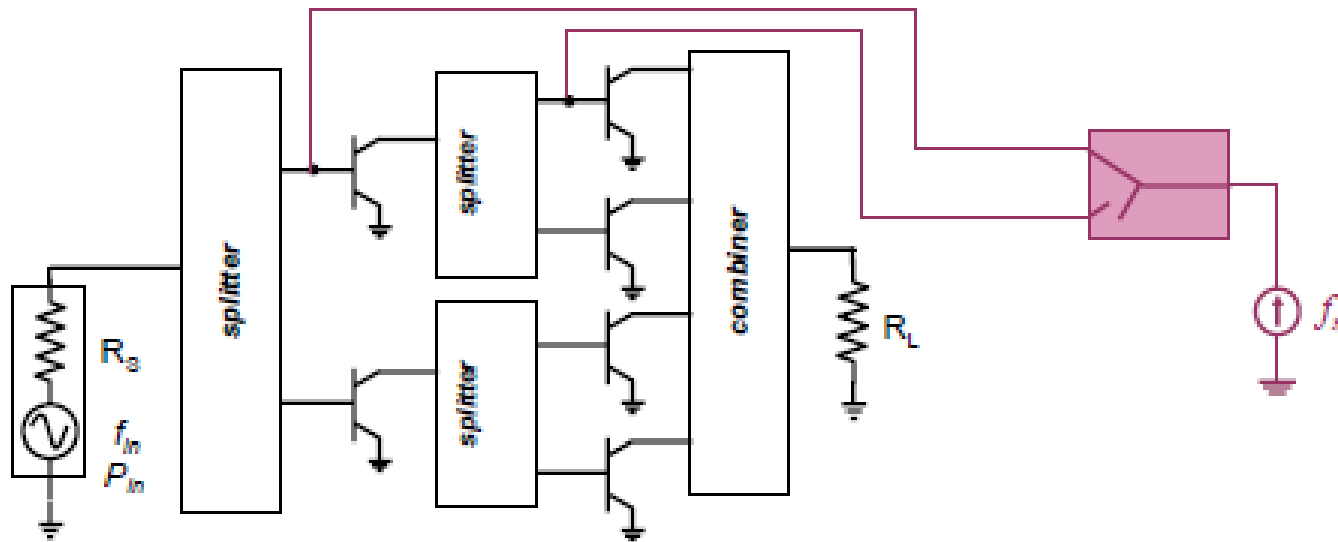


STAN - Selecting the node

Recommendations

In simple circuits with a clear feedback structure any node should serve for the analysis

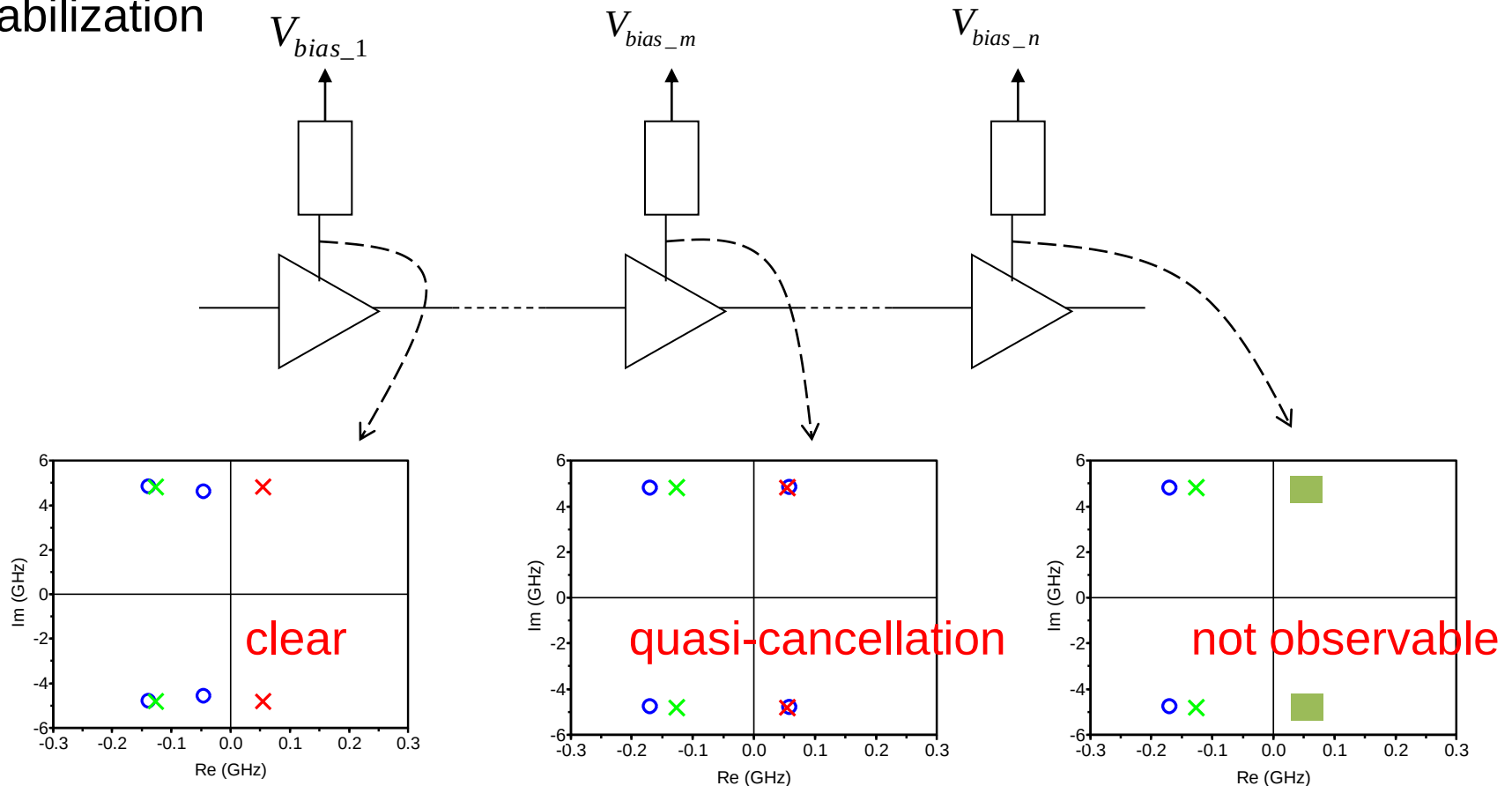
Multistage power amplifiers → At least one analysis per stage





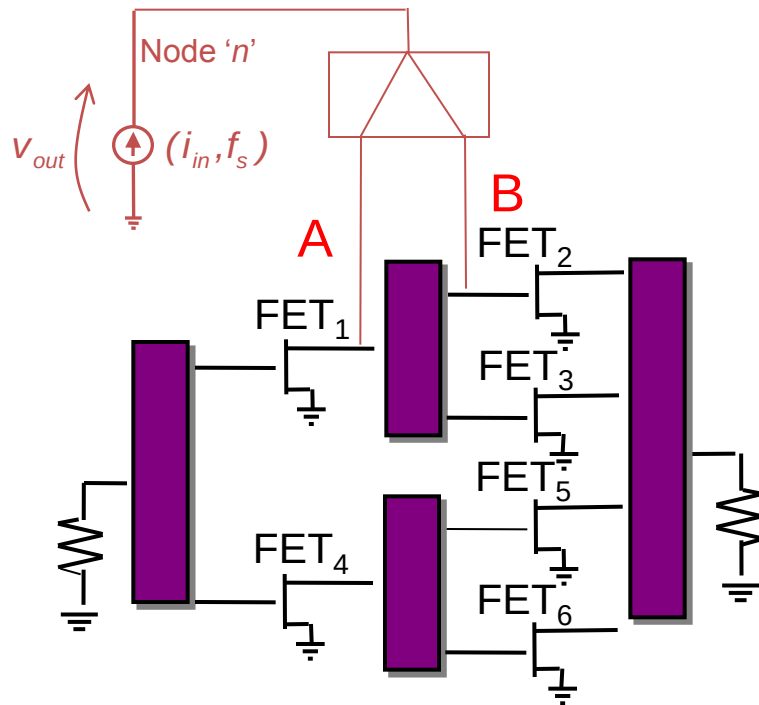
STAN - Selecting the node

→ Relevant information about the nature of the oscillation and the place in which it is being generated can be extracted → extremely useful for circuit stabilization





STAN - Selecting the node

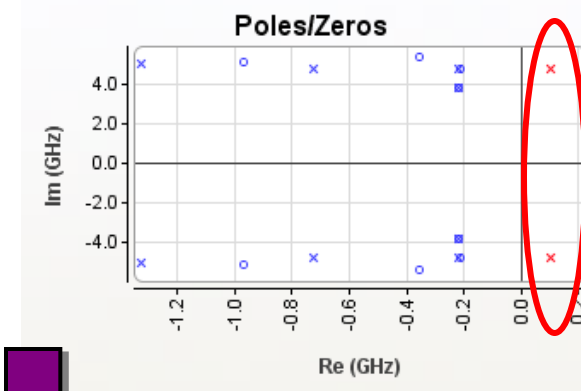
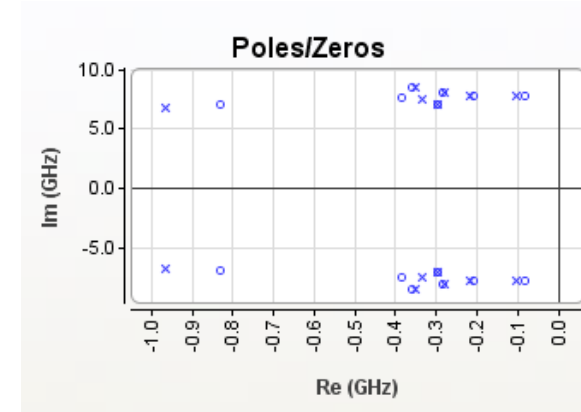
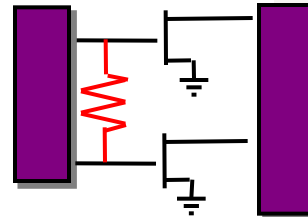


A- No oscillation detected in the common node

B- Oscillation detected in the transistor node

⇒ Odd mode (parametric frequency division)

⇒ will determine the stabilization strategy





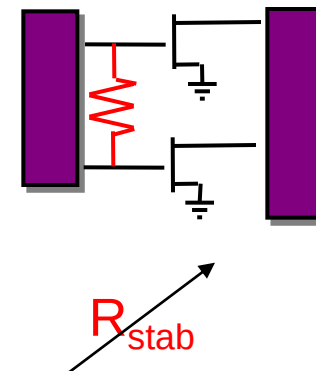
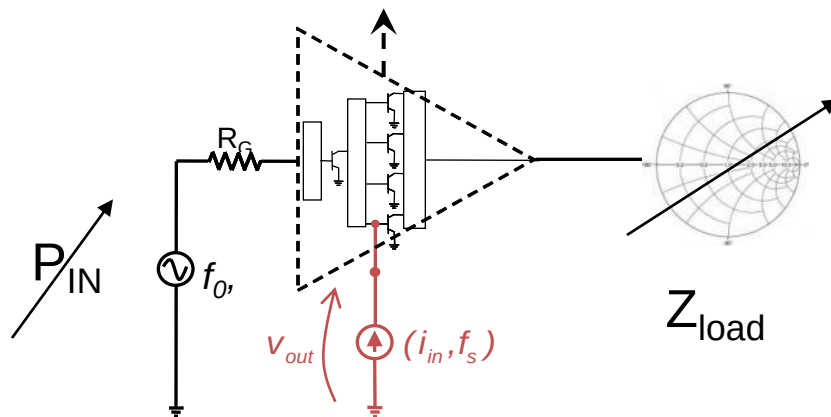
STAN – Multi parameters

Analysis with swept parameter(s)

Verification for various conditions (P_{in} , Z_{load} , ...)

Checking of critical resonances

Optimization of stabilization networks



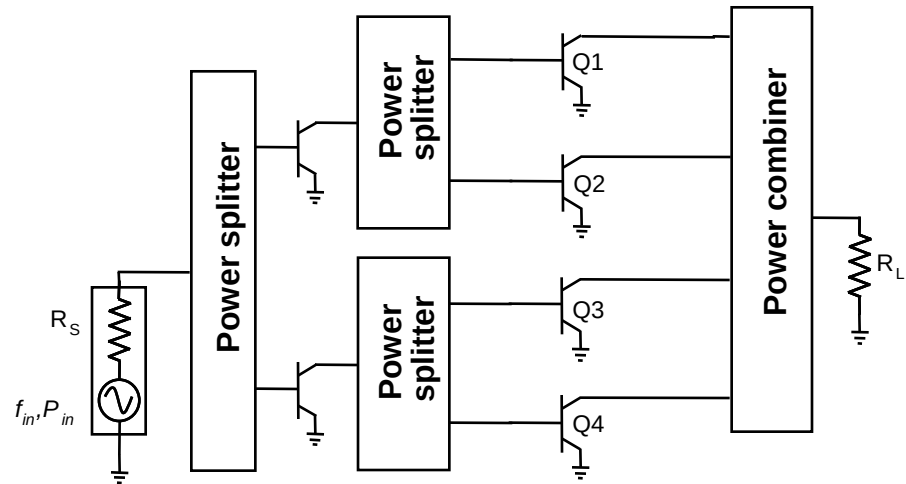
We might increase the stability margin of critical resonances when part of the system dynamics is not correctly modeled (or is likely to change).



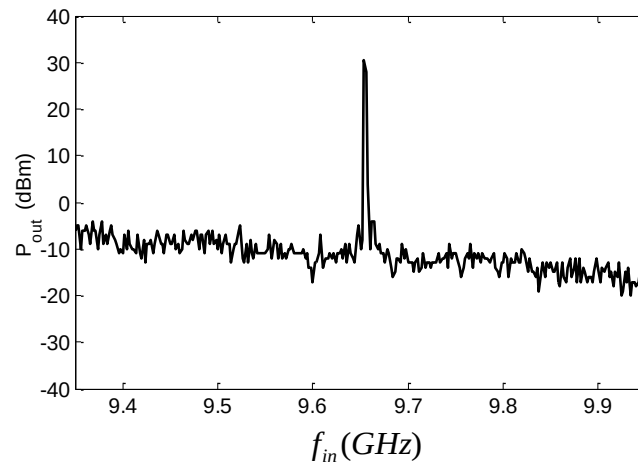
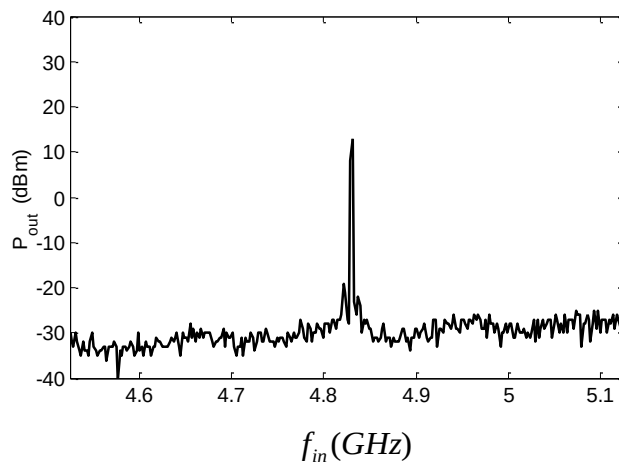
STAN – Multi parameters

Example 2 (HB analysis)

Two-stage X-Band MMIC power amplifier built in HBT technology based on AsGa/GaInP process

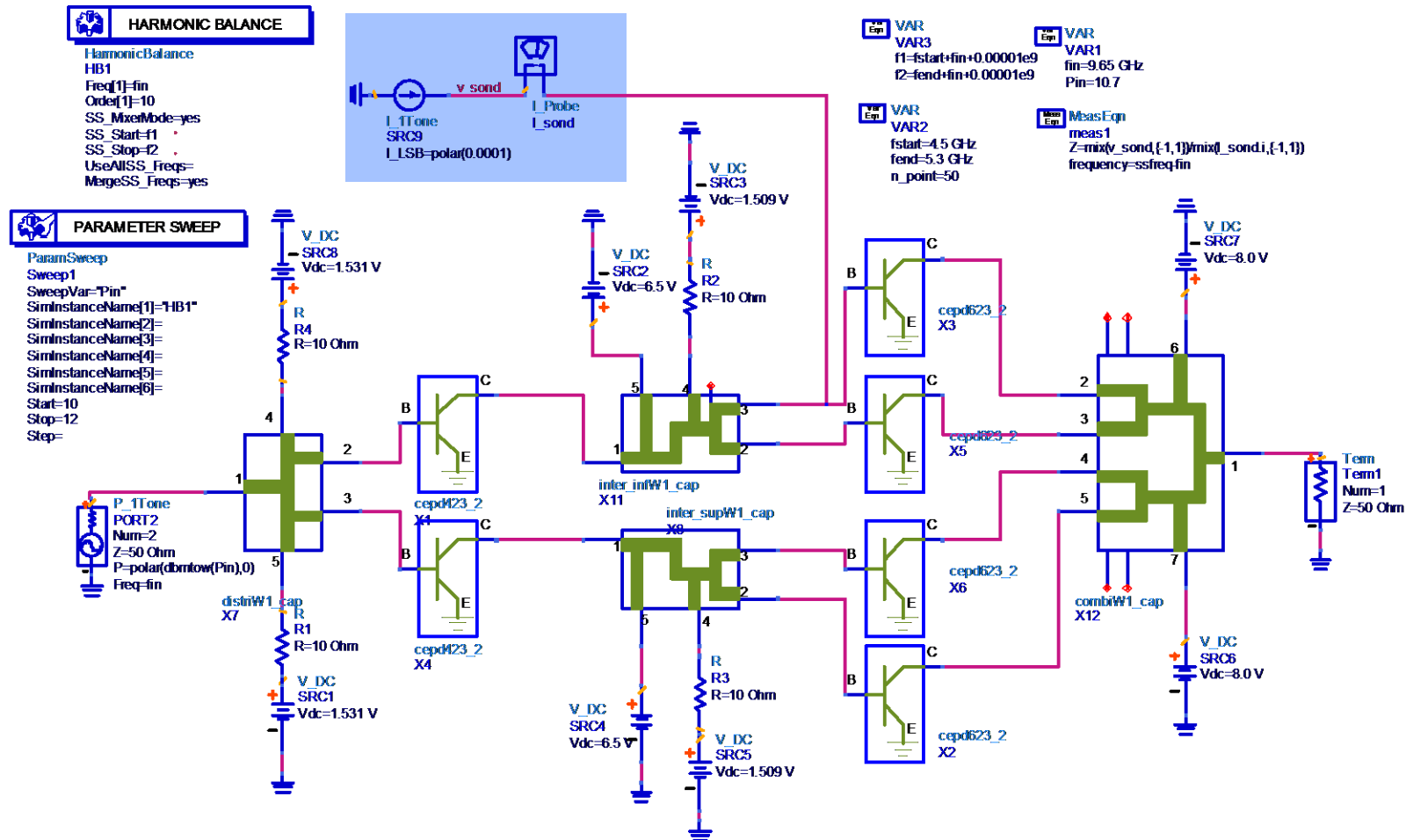


Parametric frequency division measured for $P_{in}=13.8$ dBm and $f_{in}=9.65$ GHz



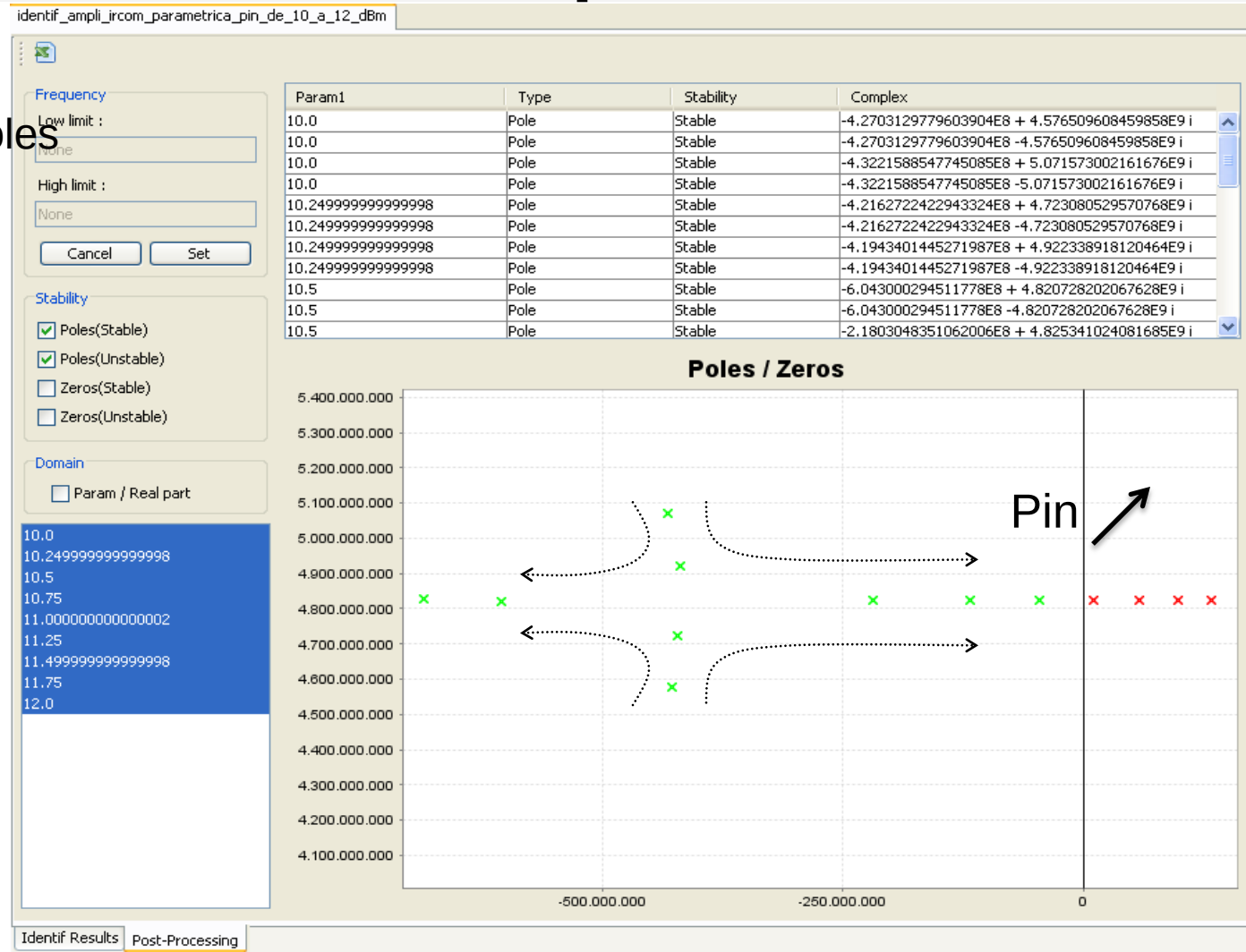
STAN – Multi parameters

ADS schematic (parametric analysis on P_{in})



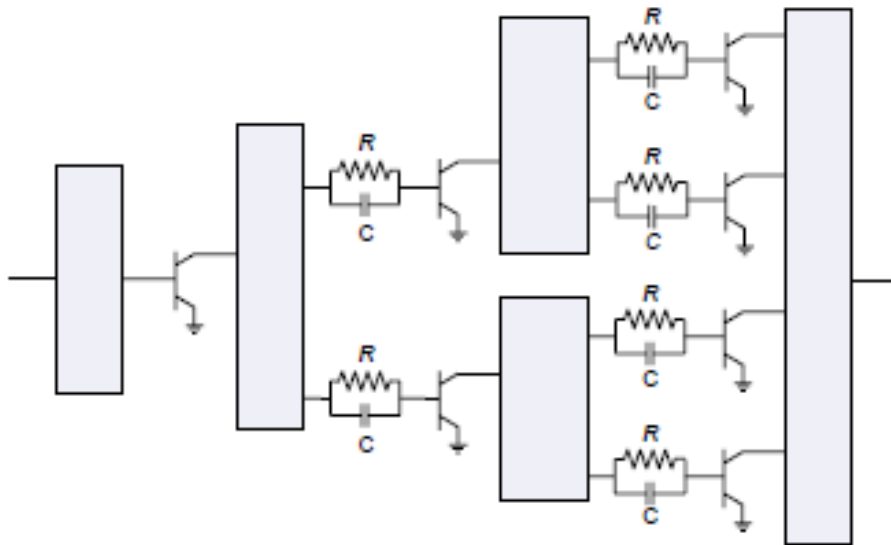
STAN – Multi parameters

Identification results:
evolution of critical poles



STAN – Multi parameters

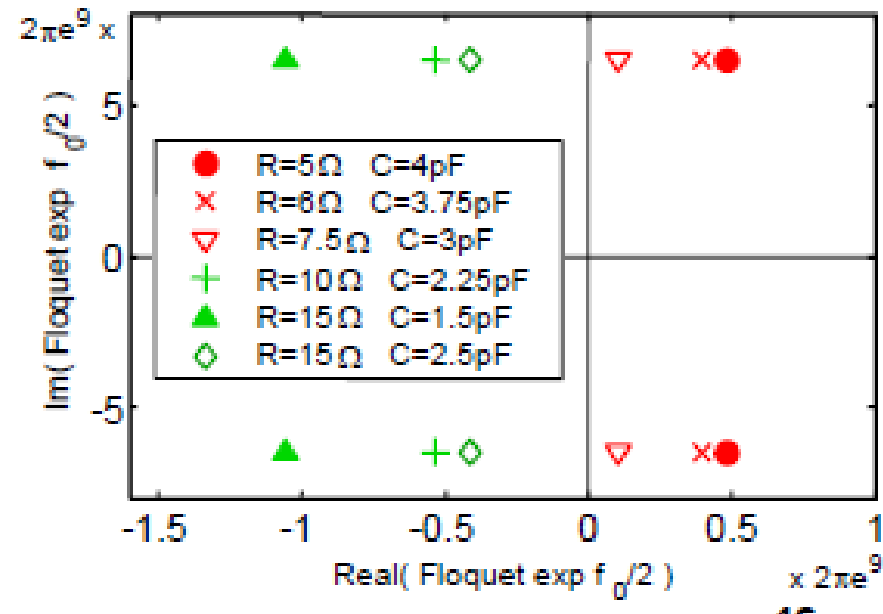
Example 3: RC network in series at the base of the transistors trying to increase resistance at $f_0/2$ without significantly degrading performances at f_0



Guaranteeing sufficient stability margin considering technological dispersion

Selected circuit: $R=15\Omega$ $C=2.5\text{pF}$

Large signal stability analysis to find suitable values for RC



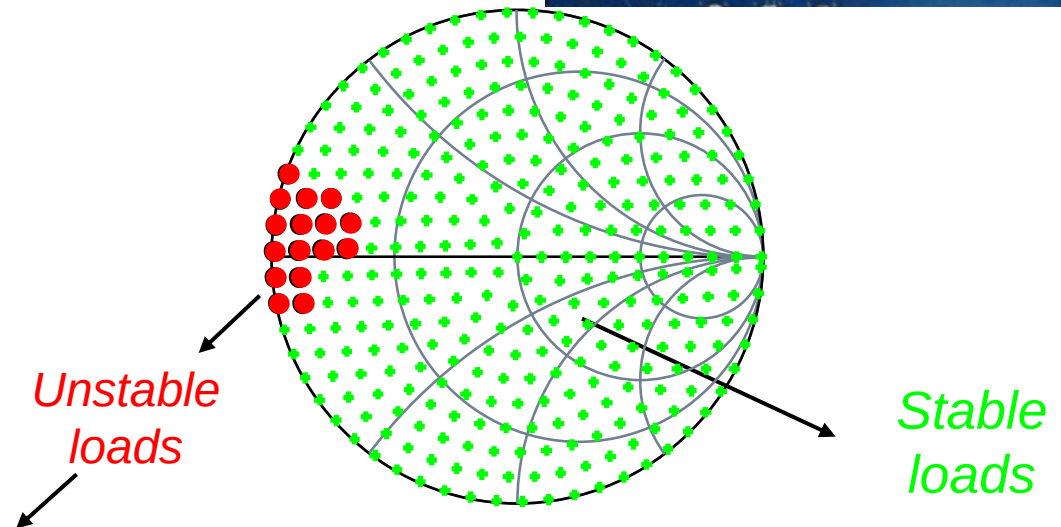
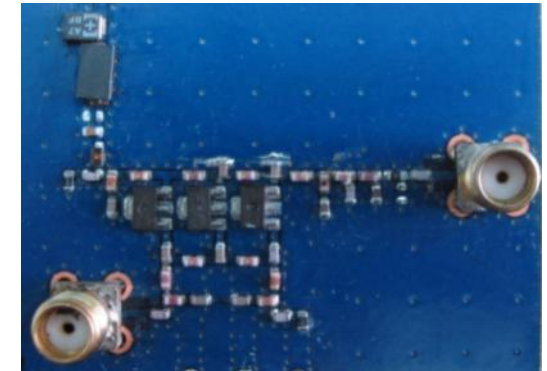


STAN – Multi parameters

Example: 3-stage LDMOS DPA for SDR applications

Application requires absence of spurious for a wide range of operating conditions

Multivariable large-signal stability analysis versus input frequency, input power and real and imaginary parts of load termination Z_L .



Frequency division ($f_{in}/2$) detected

Stable and unstable regions in the Γ_L plane for $f_{in}=500$ MHz and $P_{in}=17.1$ dBm

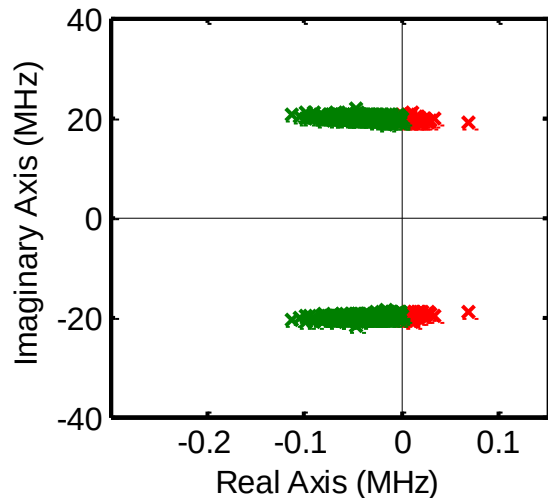
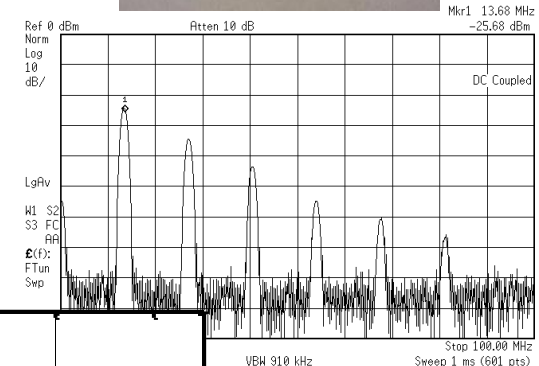
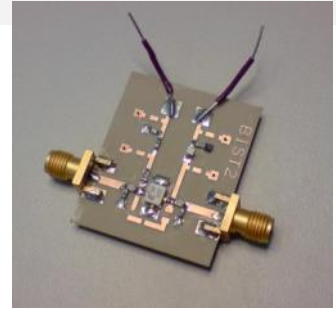
STAN – Monte Carlo

Example: L-Band medium power FET amplifier

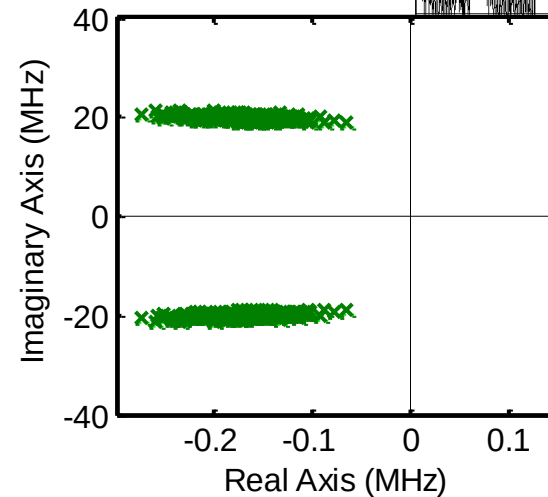
Low frequency instability related to the input bias network

Stabilization by the inclusion of a gate-bias resistor R_{STAB}

Monte Carlo sensitivity analysis for different R_{STAB} (5 % dispersion in all circuit parameters)



✗ $R_{STAB} = 44 \Omega$



✓ $R_{STAB} = 70 \Omega$

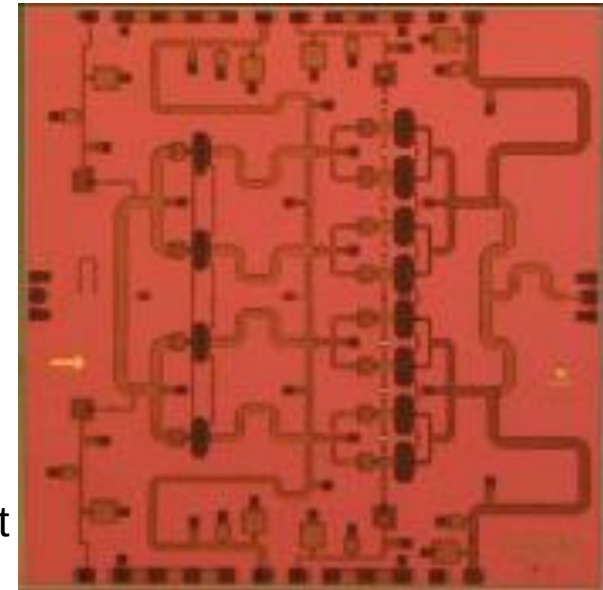
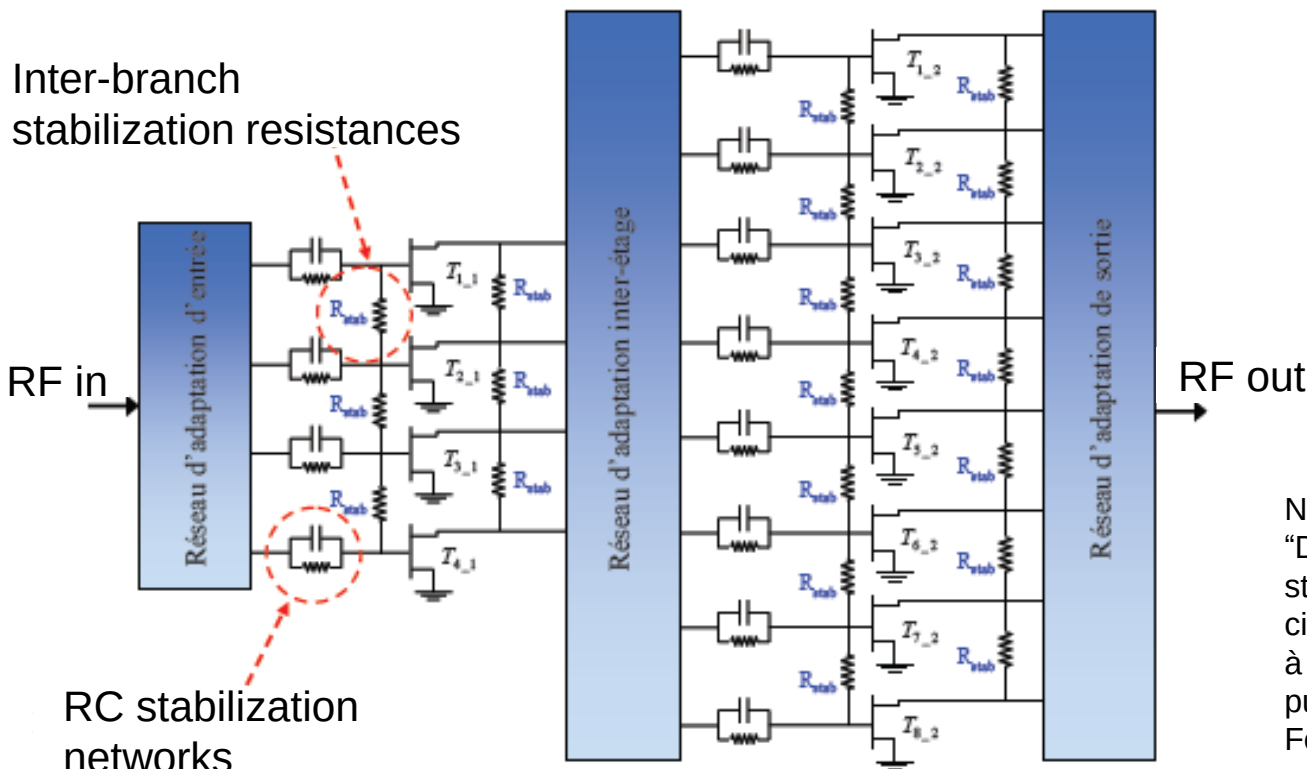


STAN – Performance optimization

Example: Ku-Band MMIC PA for active space antenna

Stable original circuit

Inter-branch
stabilization resistances



Natanael Ayllón Rozas
"Développement des méthodes de
stabilisation pour la conception des
circuits hyperfréquences : Application
à l'optimisation d'un amplificateur de
puissance spatiale.", *PhD Thesis*,
February 2011.

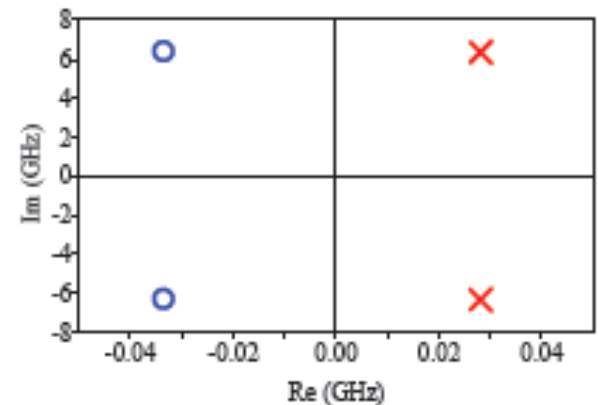
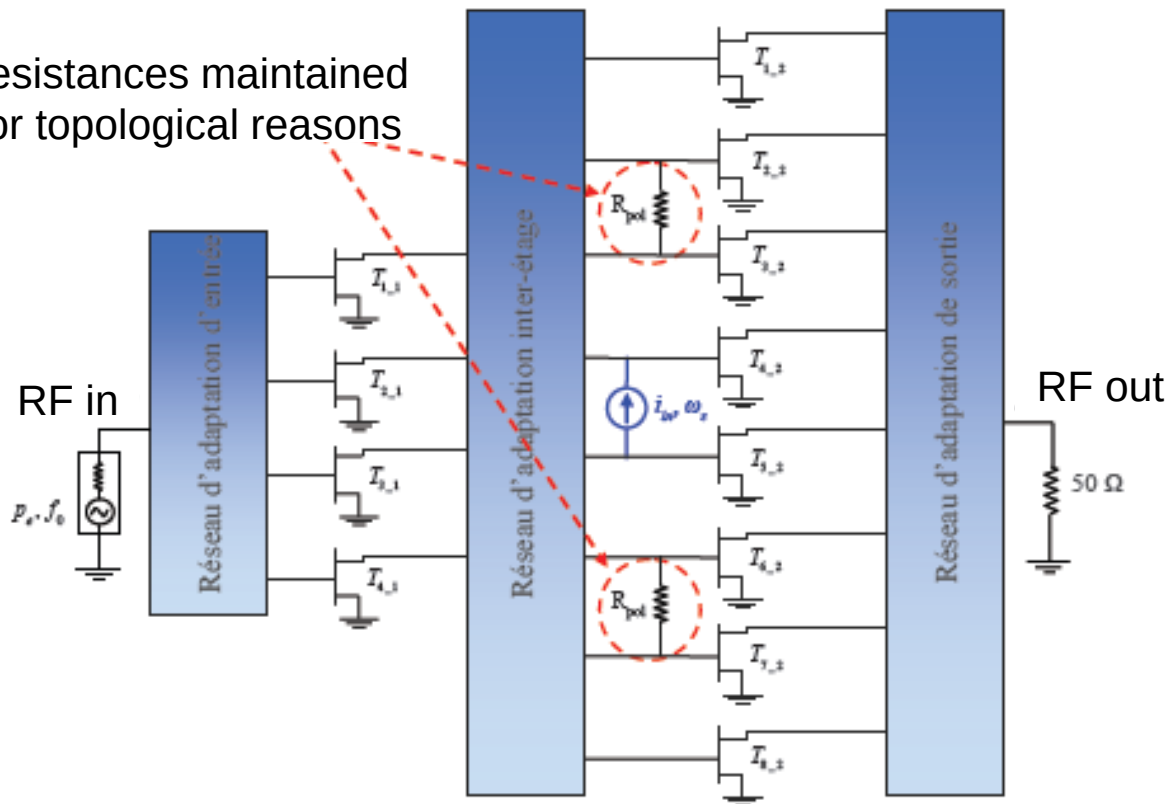


STAN – Performance optimization

Example: Ku-Band MMIC PA for active space antenna

All stabilization networks removed

resistances maintained
for topological reasons



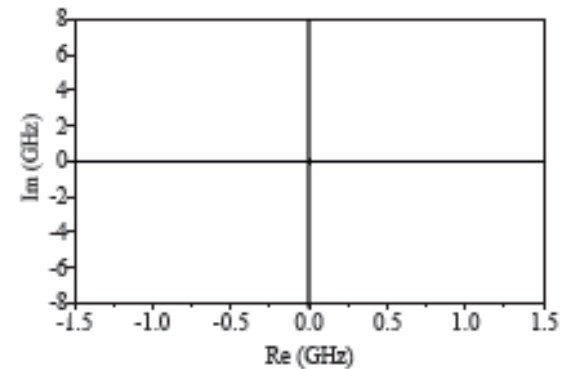
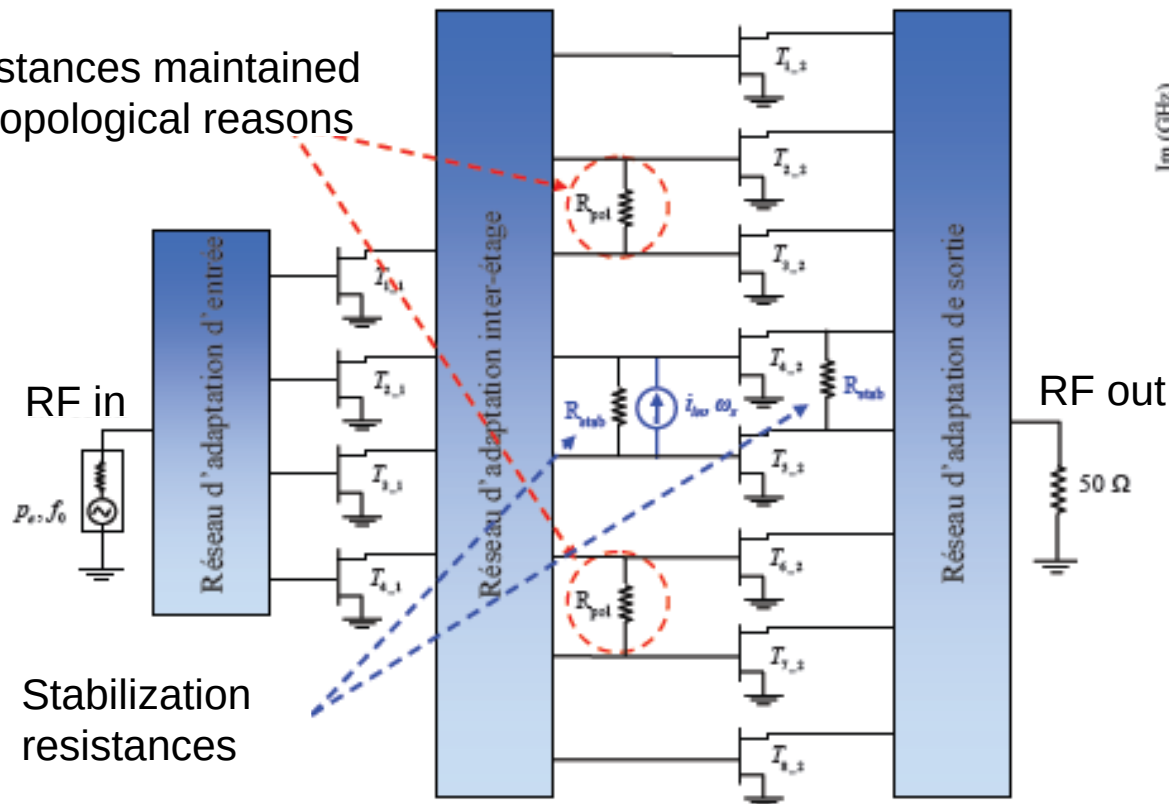
*Parametric frequency
division / 2 instability*

STAN – Performance optimization

Example: Ku-Band MMIC PA for active space antenna

Optimized version

resistances maintained
for topological reasons



No oscillation detected,
especially around $F_0/2$

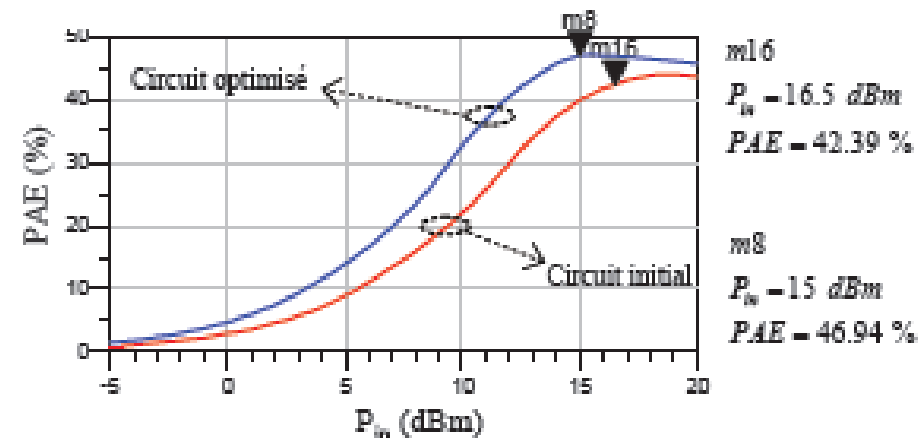
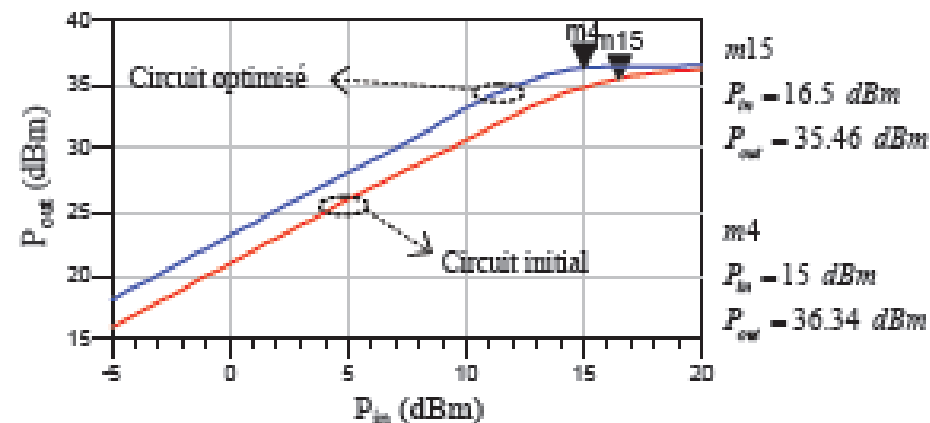
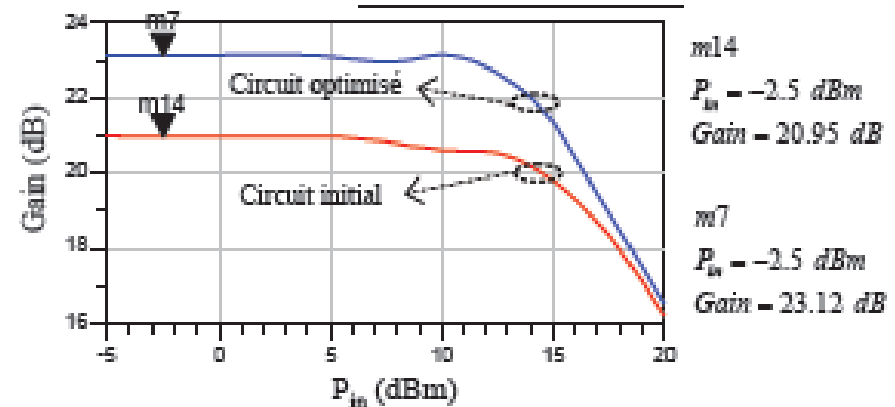
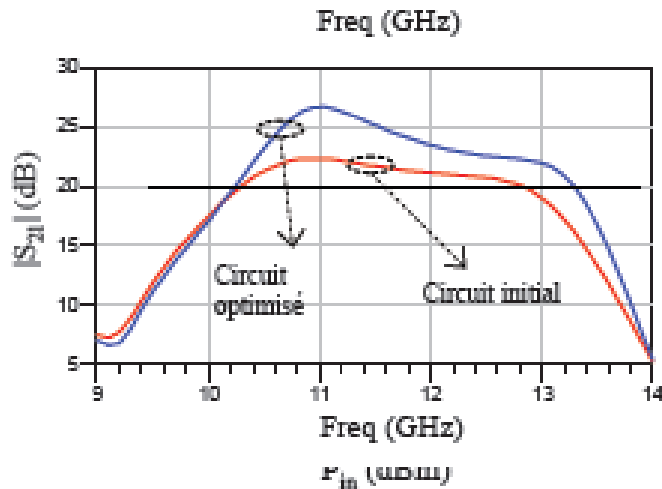
STAN – Performance optimization

Example: Ku-Band MMIC PA for active space antenna

Results comparison

— Original

— Optimized







Agilent Technologies



Maury Microwave



AMCAD Engineering
Advanced Modeling for Computer-Aided Design

- Instructor Introduction
- Pulsed IV/RF and Compact Modeling
- Load Pull
- Amplifier Design in ADS
- Stability Analysis
- X-Parameters
- System-level Simulations

Your Calibration, Measurement & Modeling Solutions Partner!

What Exactly Are X-Parameters???

- Two words: behavioral models!
- Provide a comprehensive description of a device's nonlinear performance under varying conditions
- Include the **magnitude and phase** of the fundamental signal, all of its **harmonics** and **intermodulation** products, and all of their dependence on source and load **impedance**, **bias**, etc.
- Are **cascadable** like S-parameters



Why Are X-Parameters Revolutionary?

- Provide predictable measurement-based nonlinear design
- Generate nonlinear models much faster than traditional methods
- X-parameters, ADS, and NVNA are used to:
 - Reconstruct time-domain waveforms
 - Calculate performance parameters such as ACPR, EVM, and PAE
 - Design multi-stage amplifiers and sub-systems
 - Optimize nonlinear system performance
- Less design iterations required, resulting in shorter design cycle

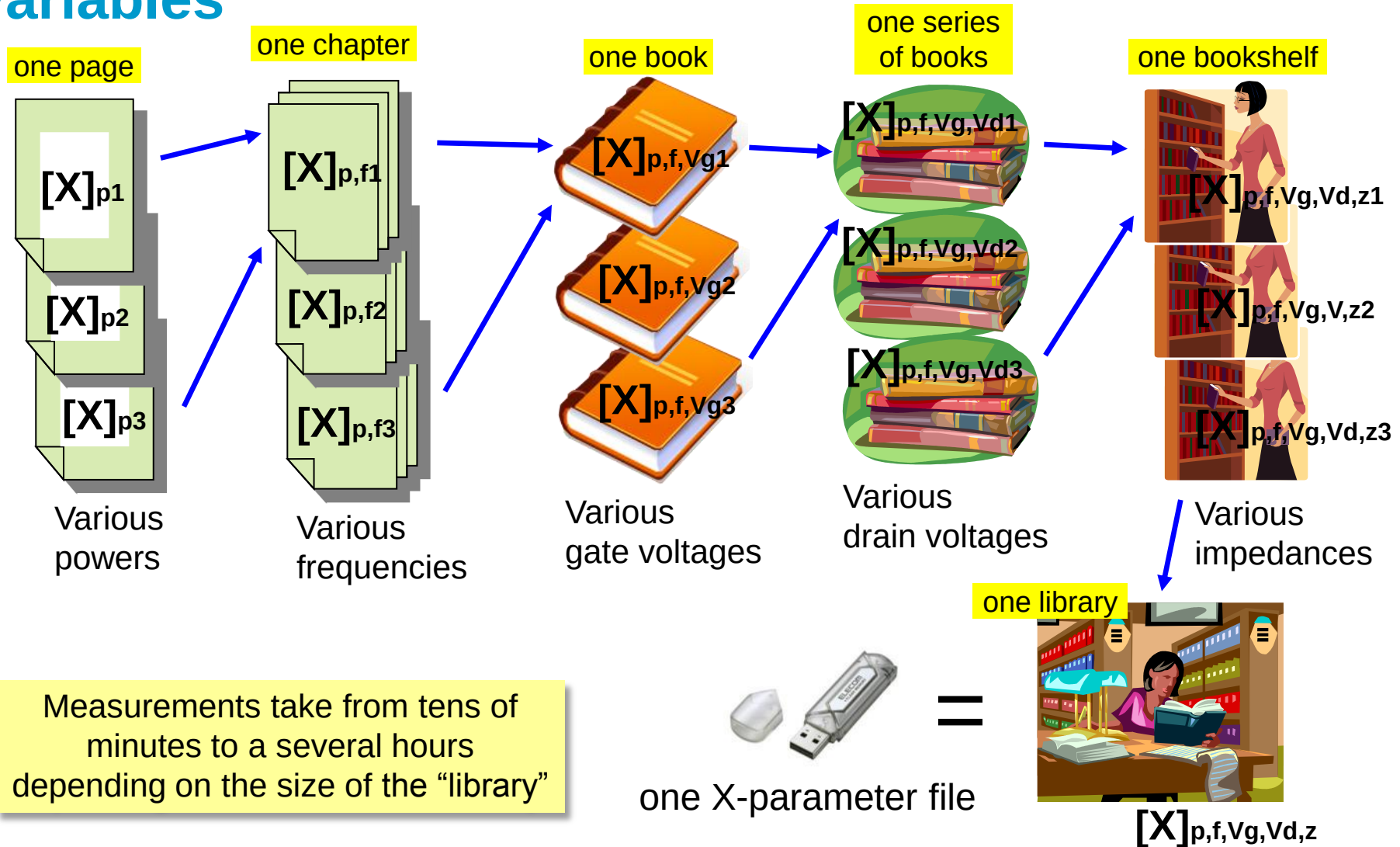


Business value: faster time-to-market!

- Protect intellectual property

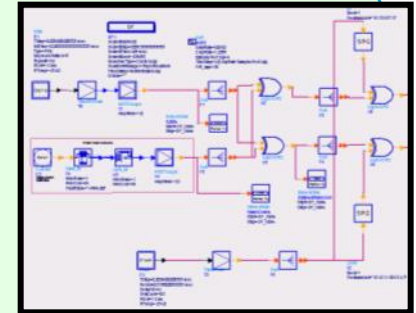


X-Parameters: Large Data Library with Many Variables

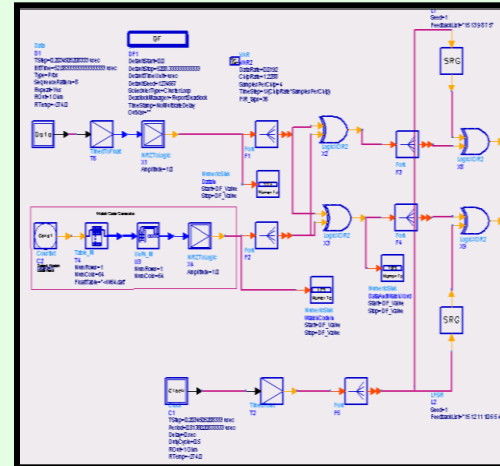


X-Parameters Span Range of Component Complexity

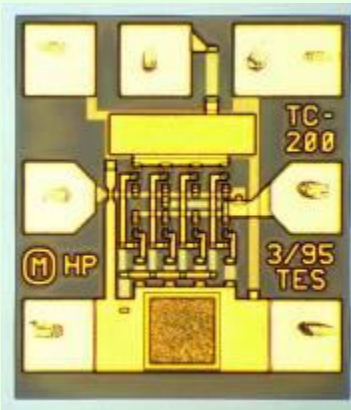
Systems



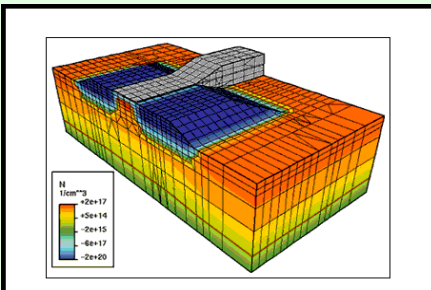
Modules



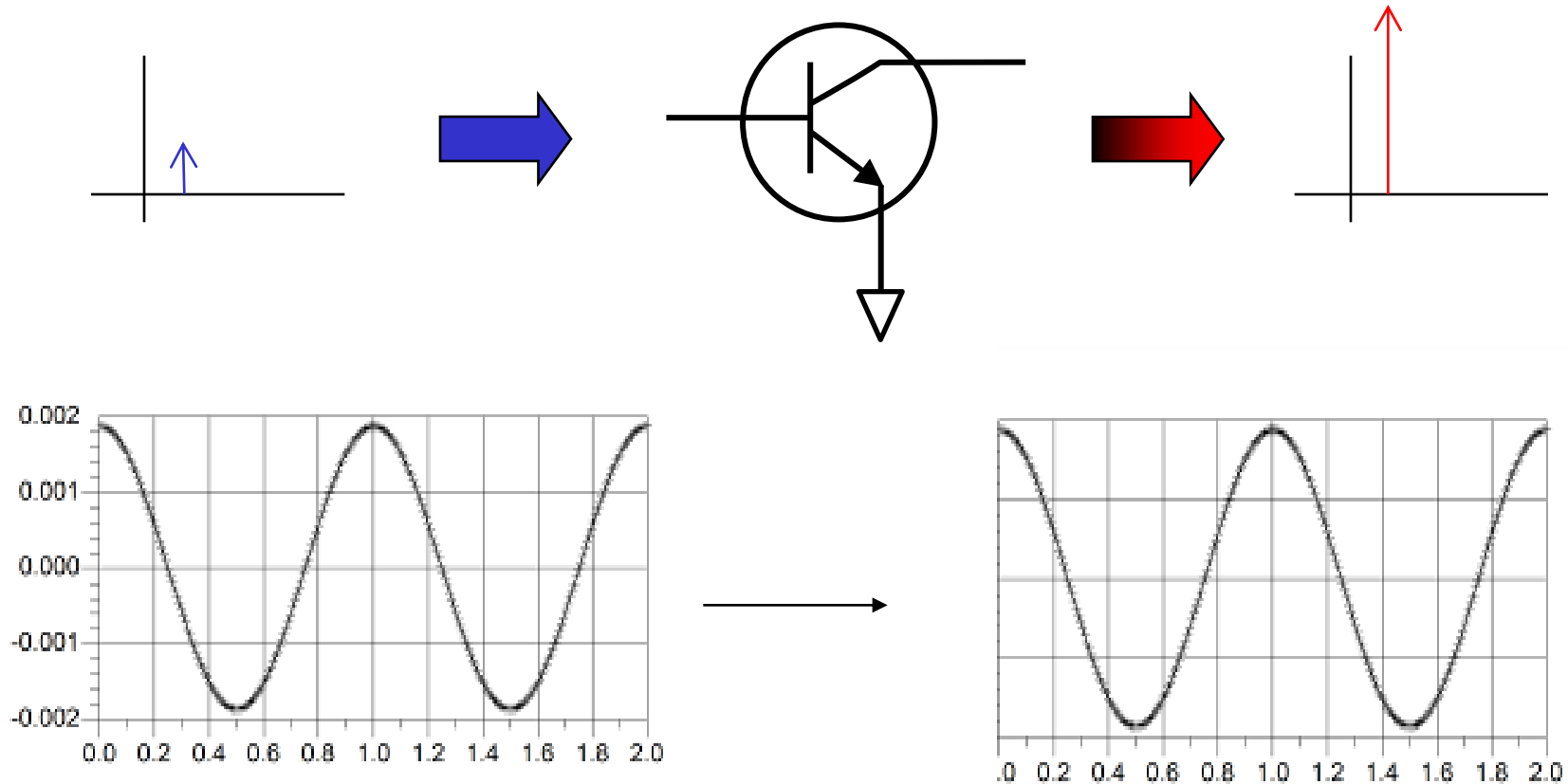
Integrated Circuits



Devices

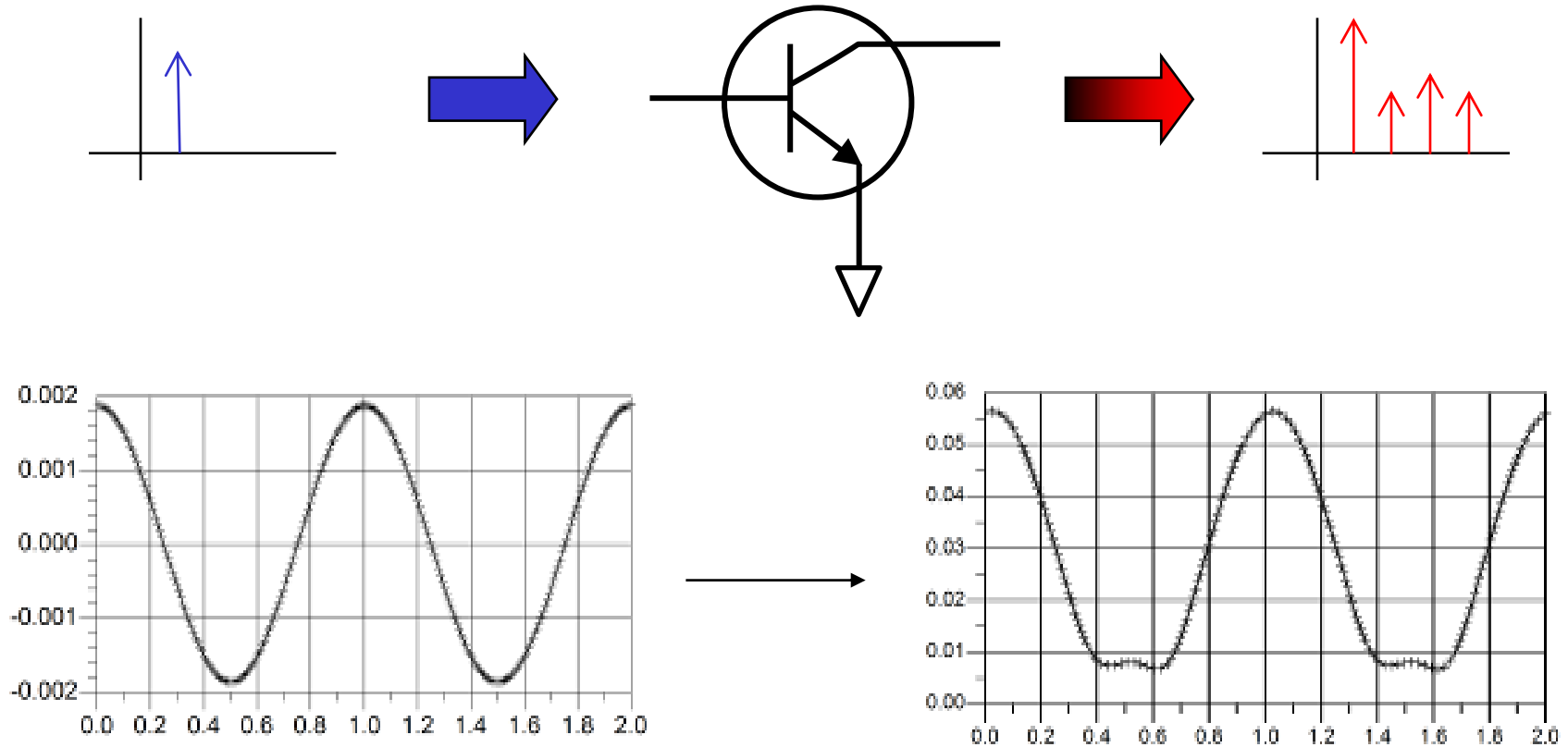


Measurements on Linear Components



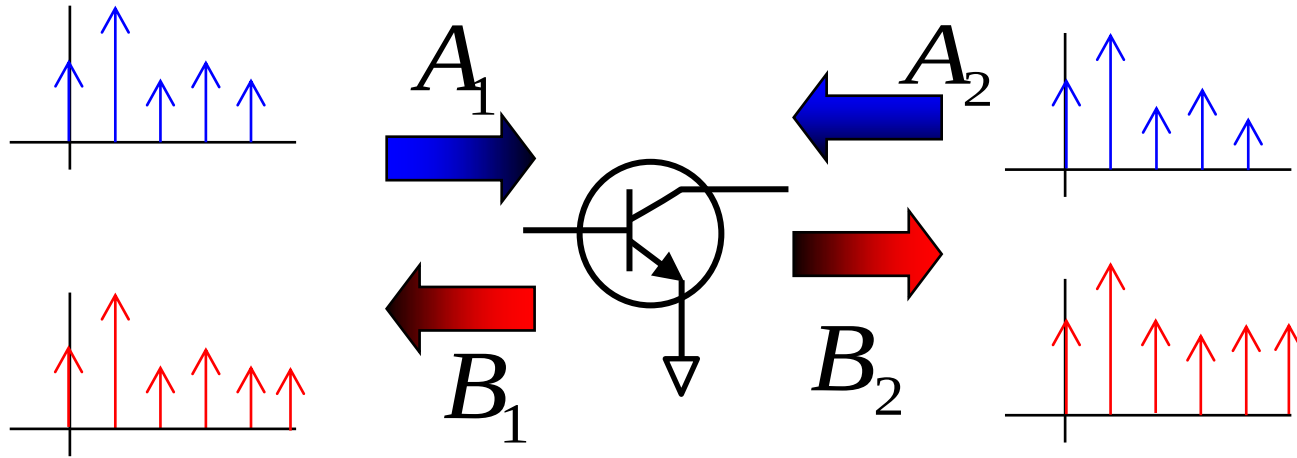
All measurements at same frequency as stimulus

Measurements on Nonlinear Components

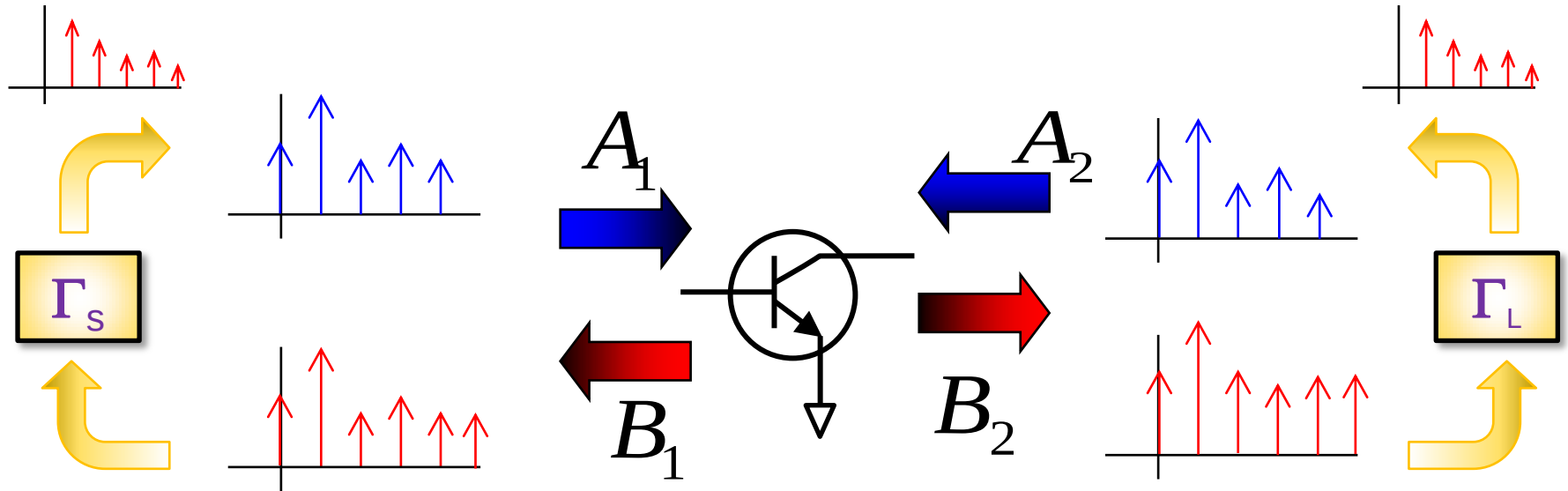


Measurements at many frequencies

It's Really Even More Complex...

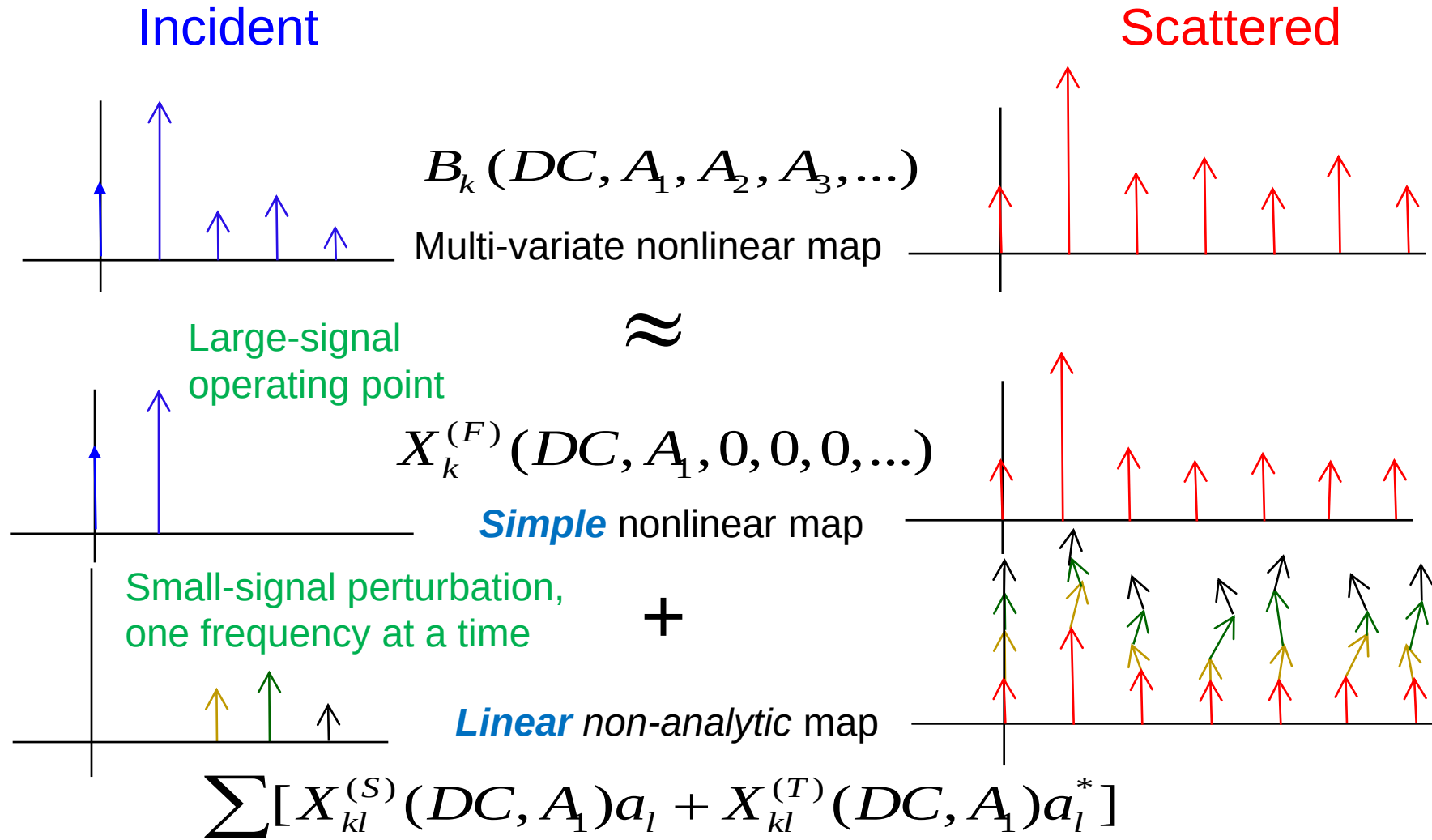


Now Add Mismatch at Input and Output

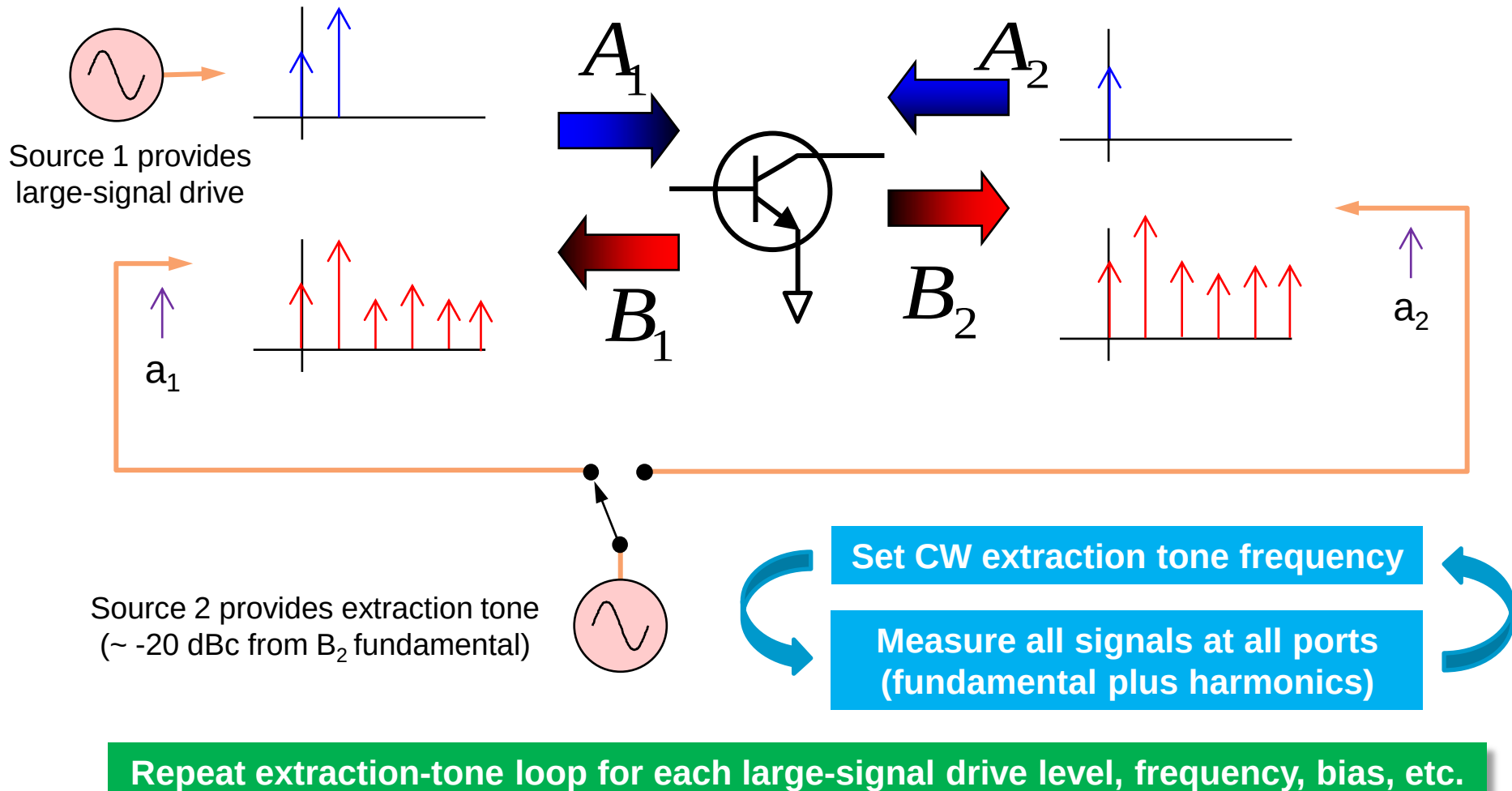


How do we practically measure the mismatch interaction between all of the signals?

X-parameter Concept:



Extraction Tone Provides Small-Signal Perturbation For Each Harmonic



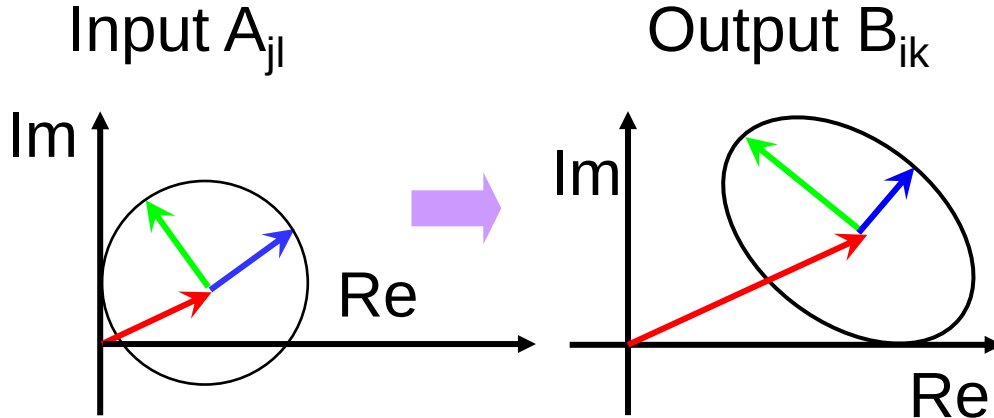
X-Parameter Extraction: Form of Active Load Pull

Large-signal 50 Ω
operating point

Extraction tone measures match
dependency using two phase conditions

$$B_{ik} = \underbrace{X_{ik}^{(F)}(|A_{11}|)P^k}_{\text{red line}} + \underbrace{X_{ik,jl}^{(S)}(|A_{11}|)P^{k-l}a_{jl} + X_{ik,jl}^{(T)}(|A_{11}|)P^{k+l}a_{jl}^*}_{\text{green and blue lines}}$$

Perform 3 independent experiments with fixed A_1 using orthogonal phases of a_1, a_2



$$B_{ik}^{(0)} = X_{ik}^{(F)}(|A_{11}|)P^k$$

$$B_{ik}^{(1)} = X_{ik}^{(F)}(|A_{11}|)P^k + X_{ik,jl}^{(S)}(|A_{11}|)P^{k-l}A_{jl}^{(1)} + X_{ik,jl}^{(T)}(|A_{11}|)P^{k+l}A_{jl}^{(1)*}$$

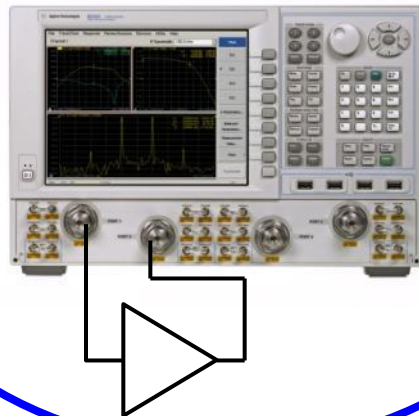
$$B_{ik}^{(2)} = X_{ik}^{(F)}(|A_{11}|)P^k + X_{ik,jl}^{(S)}(|A_{11}|)P^{k-l}A_{jl}^{(2)} + X_{ik,jl}^{(T)}(|A_{11}|)P^{k+l}A_{jl}^{(2)*}$$

For output port i , output harmonic k , input port j , input harmonic l

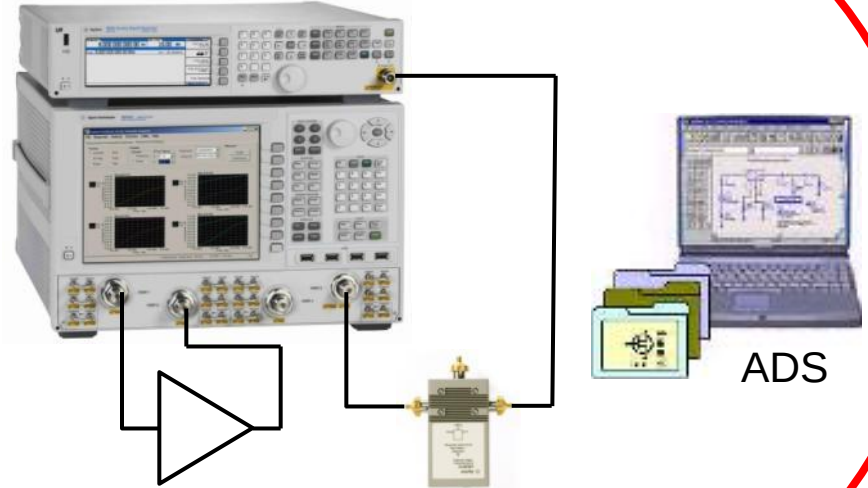


Comparing PNA-X and NVNA X-Parameter System

PNA-X



X-parameters



(Plus software options)

It's easy and relatively cheap to
turn a 4-port PNA-X into an NVNA

Configuring a PNA-X for X-Parameters

MXG (to drive comb generators)



PNA-X with NVNA options

**Power sensor
(for amplitude calibration)**



**ECal module
(for vector calibration)**



**Comb generator 1
(phase reference)**

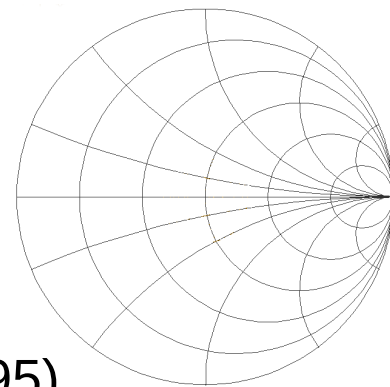


**Comb generator 2
(for phase calibration)**



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Why Add Tuners to an NVNA?

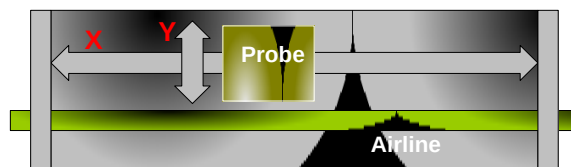
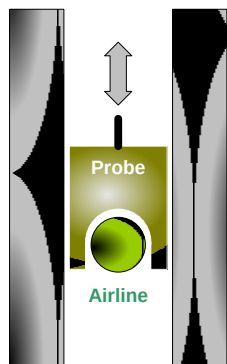


- With NVNA only, X-parameters are valid for mismatched conditions near central portion of Smith chart ($\Gamma \sim < 0.5$)
- With NVNA and tuners, X-parameters can also be measured in highly mismatched conditions (e.g. $\Gamma = 0.95$)
- Useful for high-power, multi-stage amplifiers, and power transistors that are designed to work far from a 50-ohm environment
- Load-dependent X-parameters includes magnitude and phase of all harmonics as functions of power, device bias, and load impedances
- Data can be immediately used in a nonlinear simulator as a large signal model for complex microwave circuit analysis and design

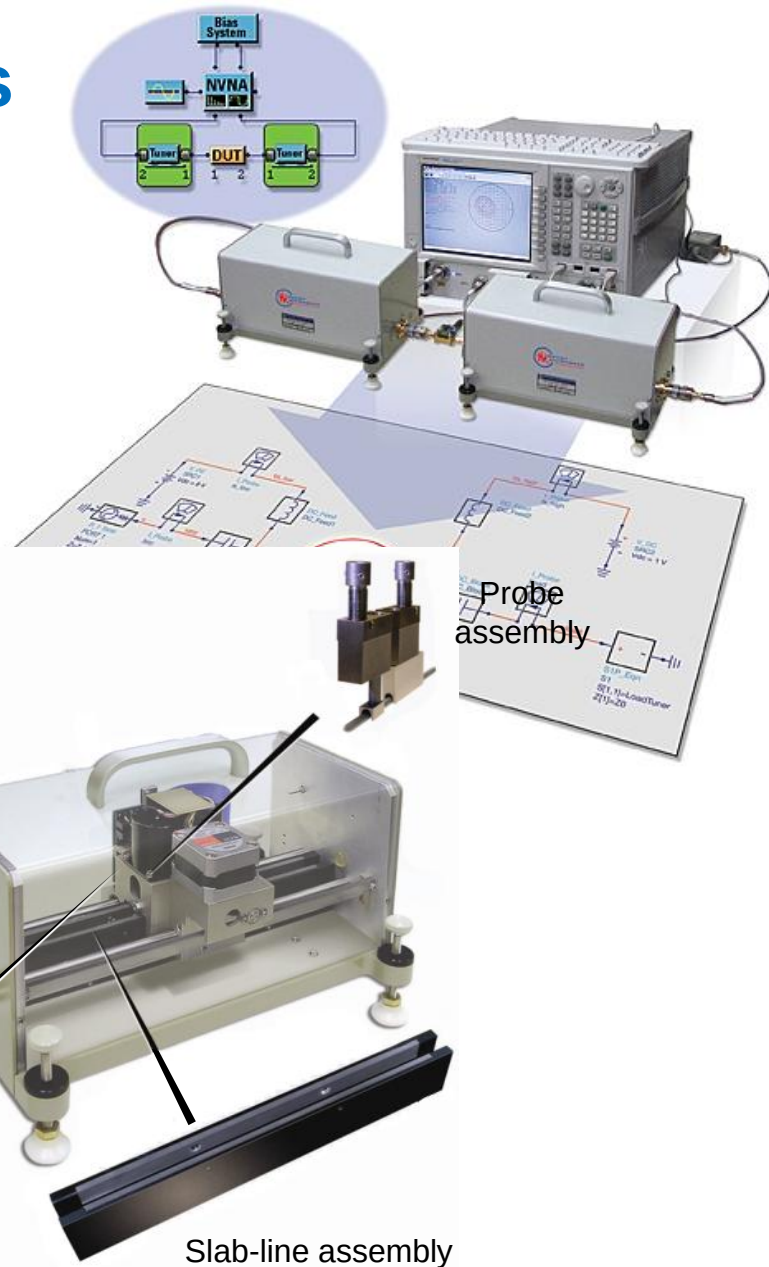


Load-Dependent X-Parameters

- Extend X-parameters over entire Smith chart for high-gamma devices
- Requires impedance tuners, which can be electromechanical, active, or a combination



- Magnitude - move probe up / down
- Phase - move probe horizontally

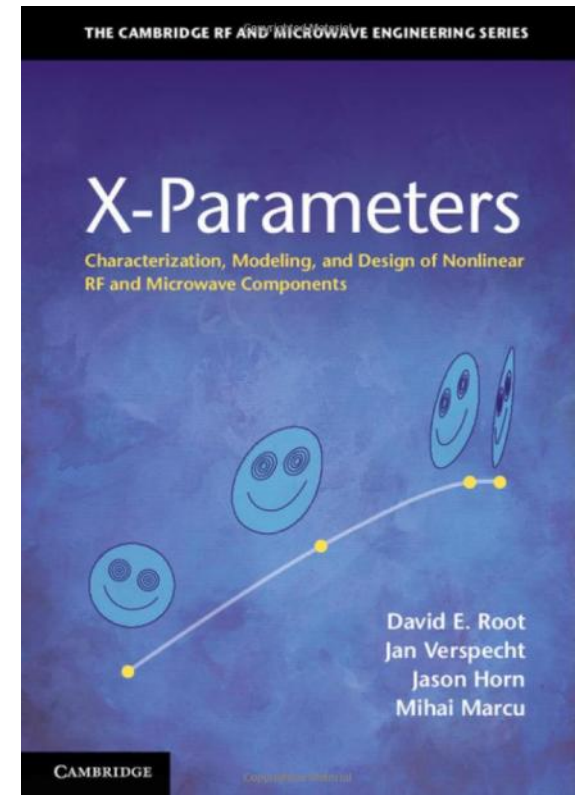
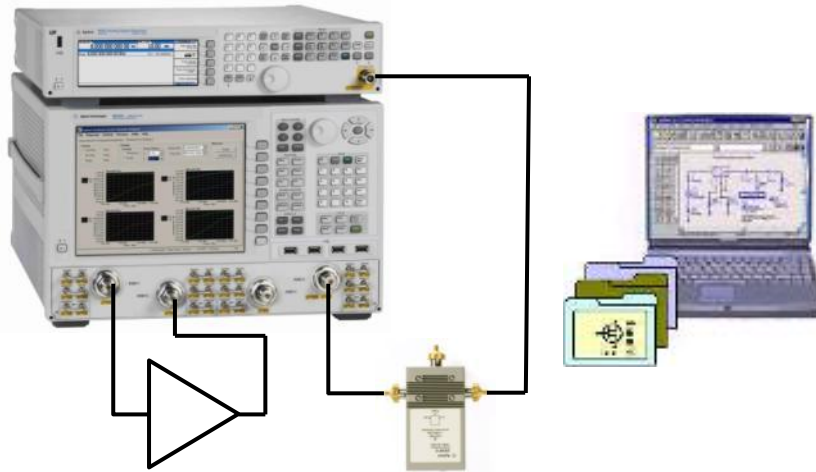


Summary

X-parameters go beyond S-parameters

- Comprehensive non-linear behavioral model
- Easily measured with VNA hardware and accessories
- Less design iterations provides faster time-to-market
- For more theory, read new book “X-Parameters”

$$B_{ik} = X_{ik}^{(F)}(|A_{11}|)P^k + X_{ik,jl}^{(S)}(|A_{11}|)P^{k-l}a_{jl} + X_{ik,jl}^{(T)}(|A_{11}|)P^{k+l}a_{jl}^*$$



Introduction to X-Parameters

X-Parameters can be generated from circuits inside of ADS

X-Parameters can be measured using PNA-X with NVNA

X-Parameters from circuits

YouTube videos on X-parameters:

Part 1 of 4, Generating X-Parameter Models from Circuits:

<http://www.youtube.com/watch?v=FP54LL8C2rQ>

Part 2 of 4, Generating Load-Dependent X-Parameter Models:

http://www.youtube.com/watch?annotation_id=annotation_760993&feature=iv&src_vid=FP54LL8C2rQ&v=xSpKIBNEedY

Part 3 of 4, Generating Two-Tone X-Parameter Models:

http://www.youtube.com/watch?annotation_id=annotation_379020&feature=iv&src_vid=xSpKIBNEedY&v=X0fJMQzzEjk

Part 4 of 4, Using X-Parameter Models in ADS for Wireless Verification:

http://www.youtube.com/watch?annotation_id=annotation_942060&feature=iv&src_vid=X0fJMQzzEjk&v=jiti6Plullo

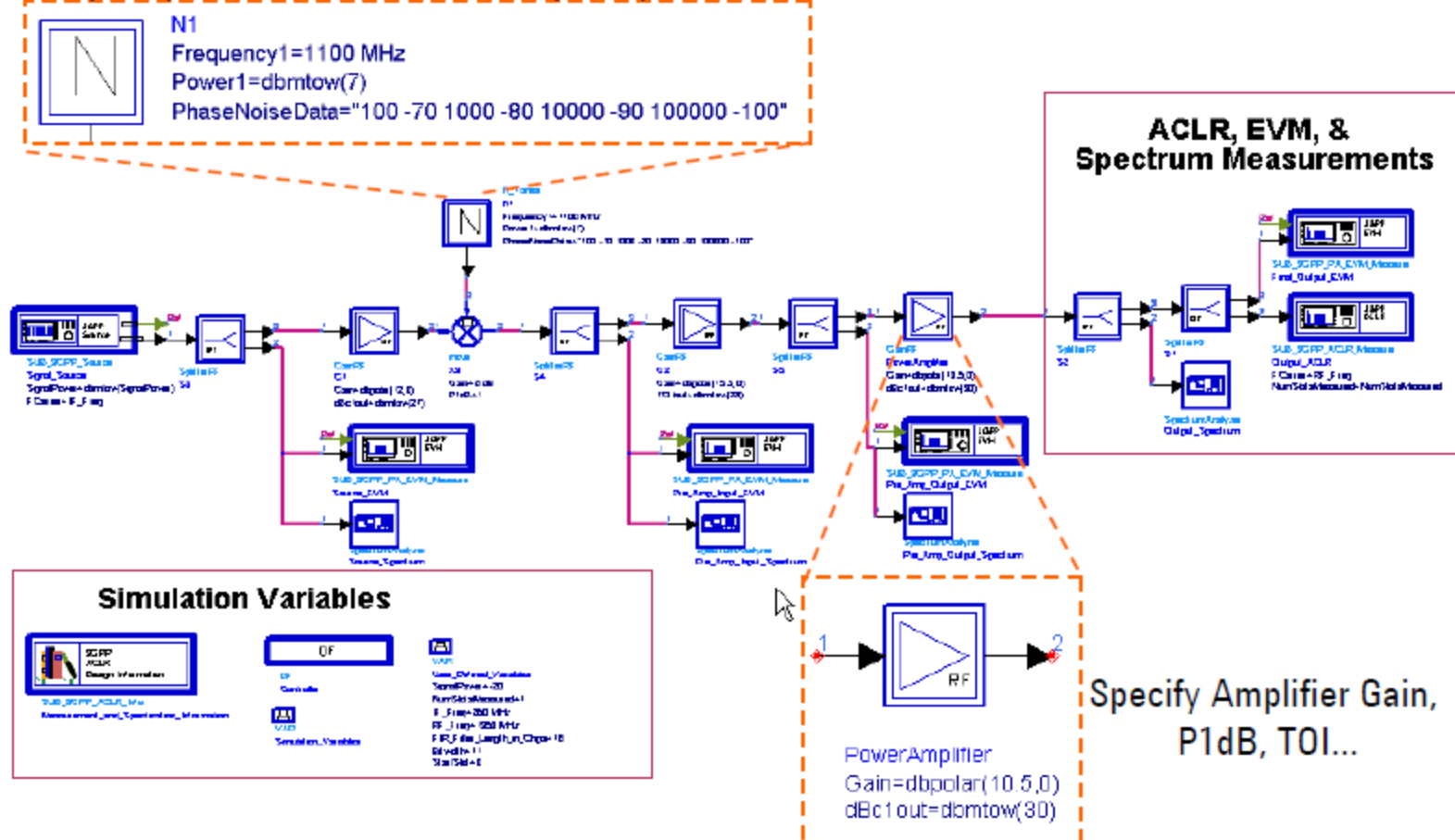




- Instructor Introduction
- Pulsed IV/RF and Compact Modeling
- Load Pull
- Amplifier Design in ADS
- Stability Analysis
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System-level schematic

Specify LO Phase Noise vs. Frequency Offset





System-level simulations

Meets +24 dBm Output Power Specification

3GPP Specification TS 25.101 section 6.6.2.2

Main, Upper, and Lower Channel Powers (dBm)

Main_Ch_Pwr	U10_Ch_Pwr	U5_Ch_Pwr	L5_Ch_Pwr	L10_Ch_Pwr
24.390232	-40.057538	-9.482697	-9.126897	-40.012311

ACLR (dB)

ACLR_U10	ACLR_U5	ACLR_L5	ACLR_L10
64.387770	33.812929	33.457129	64.342543

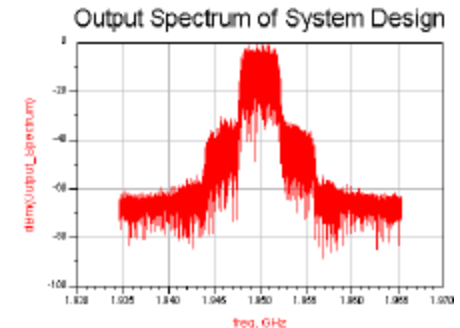
Specification requirements

+5 MHz or -5 MHz 33 dB
+10 MHz or -10 MHz 43 dB

Meets ACLR Specifications
at 5 & 10 MHz Offsets

Test Results

Passed





System-level simulations

3GPP Error Vector Magnitude (EVM)

3GPP Specification TS 25.101 section 6.8.2

Source_EVM.W6.EVM	Pre_Amp_Input_EVM.W6.EVM	Pre_Amp_Output_EVM.W6.EVM	Final_Output_EVM.W6.EVM
1.92E-9	1.70	2.44	2.81

Specification requirements

The modulation accuracy EVM shall not exceed 17.5% at the maximum output power

Test results

EVM at Output of 1st Behavioral Amplifier, Behavioral Pre-Amplifier, and Final Output- meets 17.5% Specification



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AMCAD Engineering
Advanced Modeling for Computer-Aided Design

System level simulations

YouTube videos on system-level design using X-parameters:

Using Analog/RF X-Parameter Models in System-Level Design:

<http://www.youtube.com/watch?v=CV5s1ZhPPx4>